Gmail

Hindvee Vispute <hindveevispute@gmail.com>

AI assisted edge detection

74 messages

Kishor Durve <kishor.durve@outlook.com> Wed, May 7, 2025 at 9:31 AM
To: "hindveevispute@gmail.com" <hindveevispute@gmail.com>

Hello Hindvee,

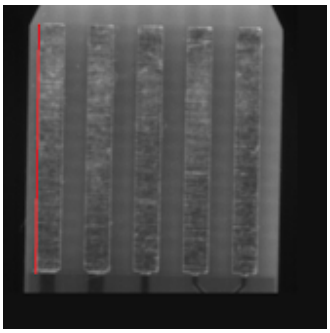
I have gone thru the document uploaded by you.

It looks ok to me.

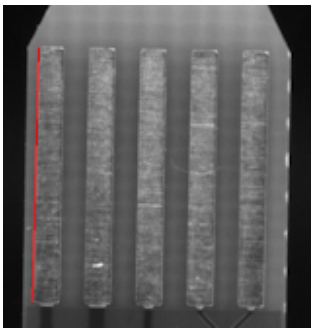
I can have attached a few images so you will get a better idea of the requirements.

Kishor Durve

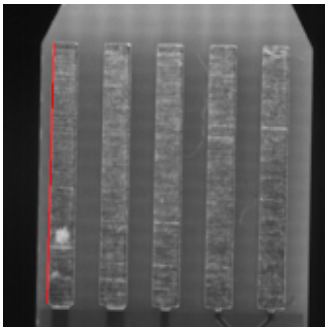
6 attachments



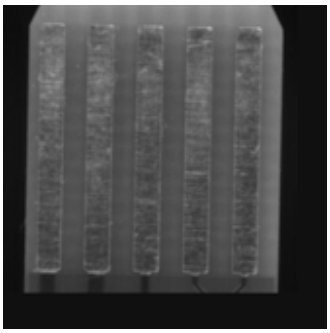
1 Annot.png
148K



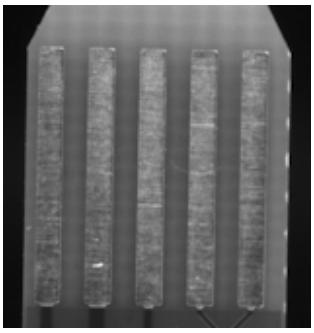
18 Annot.png
164K



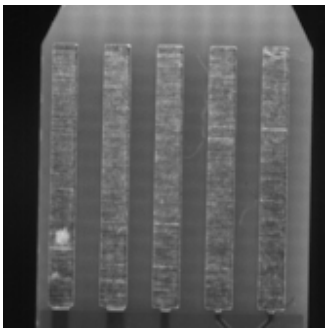
20 Annot.png
164K



1.png
148K



18.png
217K



20.png
210K

Hindvee Vispute <hindveevispute@gmail.com> Wed, May 7, 2025 at 12:45 PM
To: Kishor Durve <kishor.durve@outlook.com>

Greetings, Mr. Durve,
Thank you for reviewing my proposal and for sharing the sample images. By looking at the images I came across the specific challenges faced by the inspection system, like detecting faint lines, subtle surface variations, and fine defects that can be missed by traditional image processing. Based on these observations I suggest that we can do the following:-

1. **Hybrid approach**: Objective - Combine the speed of OpenCV line detection (example Hough Transform) with deep learning's pattern recognition capability.
Approach
- Use OpenCV to pre-locate candidate regions and feed them into a deep model for verification.
- This can speed up inference and increase resistance to false positives.

2. **Deep learning-based fine defect detection**: Objective: Improve the detection of faint, low contrast line defects and localized surface anomalies.
Approach
- Train a model, such as YOLOv8 or U-Net, tailored to your defect patterns using the annotated images.
- To improve feature visibility, add contrast enhancement.
- Assess using detection accuracy and intersection over union (IoU) to compare to baselines based on OpenCV.

I can begin with a proof-of-concept model using the images shared by you. Thank you for sharing the opportunity. I am excited to get a chance to contribute to the advancement in inspection technology.

Warm regards
Hindvee Vispute
[Quoted text hidden]

Kishor Durve <kishor.durve@outlook.com> Thu, May 8, 2025 at 7:18 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

Your analysis seems good.

At this point we are not interested in detecting defects (though at a later stage we may be), and want to detect the faint lines only.

Note that though I have annotated only a single line in the image I sent to you yesterday, we need to detect edge lines of all vertical stripes.

I have attached 2 sets of 20 images each named 'Good Images' and 'Bad Images' - images in which it is mostly easy to detect lines by the conventional image processing method and those in which lines are faint and hard to detect. These images may help you get a better understanding of the problem.

Please also check the following issues:
1. How much time will you be able to devote to this task?
2. Will you be able to complete the task within the available time?

There are other issues like time taken by the AI route, integration of AI line detect code in VC++ project, further training / fine tuning of model, applicability of the method to detect faint circles etc. But we will get into them later once the basic method is proven.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, May 7, 2025 3:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

[Quoted text hidden]

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2 attachments

Bad Images.zip
3924K

Good Images.zip
3583K

Hindvee Vispute <hindveevispute@gmail.com> Thu, May 8, 2025 at 2:45 PM
To: Kishor Durve <kishor.durve@outlook.com>

Greetings Mr Durve,
I have the problem now so I will focus on faint line detection for the first step. I can proceed using Sobel, Scharr, and Canny edge detectors provided by OpenCV. First I will preprocess the images and then move to edge detection using the above 3 detectors. I will carry out line detection using Hough Transform, and further we will filter out short or irrelevant lines based on length and their angle of orientation. Finally, we will draw the detected lines on the images using cv.line()

I can devote 14 hrs per week to this task for 2 months, we can extend the time if required. Also, I wanted to ask what stipend you can provide me. I believe I can complete the task in the above time frame. Thank you for your time.

Warm regards
Hindvee

[Quoted text hidden]

Kishor Durve <kishor.durve@outlook.com> Fri, May 9, 2025 at 6:03 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

I think there is a mismatch in understanding. We are already using conventional line detection techniques (various edge detection operators and Hough Transform and other methods for line fitment) .

What we are looking to get from you is AI based / AI assisted line detection. Since we know the probable position of the line beforehand we can use this information to find a line. For example,

say we know that the line end points are expected to be (x1, y1) and (x2, y2), we can construct a small rectangle (ROI) using these points and search for a line within this ROI.

As for the stipend it depends both on the time you spend on the project and the results you produce. It will comprise of an hourly rate and a result linked amount. I suggest we have an hourly rate of Rs. 200/hour and a matching amount linked to the results at the end of the project. So, if you are able to produce satisfactory results it will come to Rs. 400/hour. Also, if the results are satisfactory, we can take this forward next year and work on other issues like surface defect detection, detection of shape anomalies etc.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 8, 2025 5:15 PM
[Quoted text hidden]

[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Sat, May 10, 2025 at 1:16 AM
To: Kishor Durve <kishor.durve@outlook.com>

Greetings Mr. Durve,

Thank you for your detailed response and clarification. I now clearly understand that the goal is to enhance line detection using AI within known ROIs, leveraging the expected position of endpoints.

To support this objective, I've explored and implemented several **AI-based approaches**, any of which can be used to complement or improve upon conventional methods:

1. **CNN-Based Line Presence Classifier (NB-1):** Identifies whether a line exists within a given patch/ROI—useful for filtering candidate regions before applying traditional detection.
2. **CNN Regression Model (NB-2):** Predicts line endpoints directly from the image, offering robust localization even in low-contrast or noisy scenarios.
3. **Denoising Autoencoder (NB-5):** Enhances faint or broken lines by cleaning up noise, thereby boosting the effectiveness of existing edge or Hough-based methods.
4. **Vision Transformer Model (NB-4):** Designed for fine-grained classification tasks and can be adapted for future objectives like surface defect or anomaly detection.

These methods are independently deployable and can be integrated selectively depending on the specific inspection challenge or ROI structure. I'd be happy to align with your ROI-driven strategy and customize accordingly.

Regarding the stipend, I agree with the proposed Rs. 200/hour base rate and performance-based bonus, and I'm enthusiastic about the opportunity to collaborate further on defect detection and shape anomaly tasks in the future.

I have attached screenshots of the code files I executed today. I have the names of the notebooks (NB) with NB-1, NB-2,NB-4 and NB-5 respectively.

NB-1
NB-2
NB-4
NB-5

Please let me know if you'd like me to share these outputs or run a quick live demonstration.

Warm regards,
Hindvee

[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Mon, May 12, 2025 at 12:29 AM
To: Kishor Durve <kishor.durve@outlook.com>

Greetings Mr. Durve,
Please go through the above mail once and let me know your views on the same.

Warm regards
Hindvee
[Quoted text hidden]

Kishor Durve <kishor.durve@outlook.com> Tue, May 13, 2025 at 5:38 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

I haven't received your mail. Can you please resend it?

Get Outlook for Android

From: Kishor Durve <kishor.durve@outlook.com>
Sent: Friday, May 9, 2025 8:33:05 AM
To: Hindvee Vispute <hindveevispute@gmail.com>
[Quoted text hidden]

[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Tue, May 13, 2025 at 10:21 AM
To: Kishor Durve <kishor.durve@outlook.com>

[Quoted text hidden]

Kishor Durve <kishor.durve@outlook.com> Thu, May 29, 2025 at 8:21 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

Hello Hindvee,

I have attached the on-boarding letter.

Please let me know if you have any queries.

Regards,

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Saturday, May 10, 2025 3:46 AM
[Quoted text hidden]

[Quoted text hidden]

Hindvee Vispute Internship.pdf
229K

Hindvee Vispute <hindveevispute@gmail.com> Thu, May 29, 2025 at 2:42 PM
To: Kishor Durve <kishor.durve@outlook.com>

Greetings Mr. Durve,

Thank you very much for giving me the opportunity to contribute to Guangdong Longtian Smart Instruments Co. Ltd. through this exciting project. I am pleased to accept the offer and look forward to working on the development of the AI-assisted line detection tool.

With your permission, I would like to begin the internship from the coming Monday, i.e., 2nd June 2025.

Additionally, I wanted to ask about the preferred method for tracking and sharing the time I spend on the project. Would you like me to maintain a shared timesheet (e.g., a Google Sheet or Excel file), or would you prefer weekly time summaries via email or any other format?

Looking forward to your guidance on this.

Thank you once again for this opportunity.

Warm regards,
Hindvee Rajesh Vispute

[Quoted text hidden]

Kishor Durve <kishor.durve@outlook.com> Thu, May 29, 2025 at 2:53 PM
To: Hindvee Vispute <hindveevispute@gmail.com>

Hello Hindvee,

Welcome aboard!

You can keep the time data on Google Sheet.

Regards,

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 29, 2025 5:12 PM
[Quoted text hidden]

[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Thu, May 29, 2025 at 2:58 PM
To: Kishor Durve <kishor.durve@outlook.com>

Dear Sir,

Thank you for the confirmation. I will maintain the time data on a Google Sheet and ensure it is updated regularly.

Looking forward to starting from Monday, 2nd June.

Best regards,
Hindvee

[Quoted text hidden]

Kishor Durve <kishor.durve@outlook.com> Fri, May 30, 2025 at 6:11 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

We have found that using a super resolution Pytorch model to get a larger image results in significant noise removal. Perhaps it is a by-product.

You can try it out. I have the model and can share it on Google Drive if you want to try it out. We have tried the models available in OpenCV DNN but they do not seem to give the same results.

Regards,

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 29, 2025 5:28 PM
[Quoted text hidden]

[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Fri, May 30, 2025 at 10:20 AM
To: Kishor Durve <kishor.durve@outlook.com>

Greetings sir!
Pls do share the model over drive I will try it out. Also I request you to share more image database so that I can start working on it.
I will start work from today itself then.

Regards
Hindvee
[Quoted text hidden]

Kishor Durve <kishor.durve@outlook.com> Fri, May 30, 2025 at 11:41 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

You can download the model, a working VC++ (VS 2022) project and image here.

https://drive.google.com/drive/folders/1mLBDjJR3fFKs8Brr_nNj2z9YyI89nruE?usp=sharing

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, May 30, 2025 12:50 PM
[Quoted text hidden]

[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Fri, May 30, 2025 at 4:04 PM
To: Kishor Durve <kishor.durve@outlook.com>

Greetings sir!
I went through the google drive folder shared by you. It would be really helpful if you could walk me through it.
Best regards
Hindvee
[Quoted text hidden]

Kishor Durve <kishor.durve@outlook.com> Sat, May 31, 2025 at 5:43 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

Sure. You need to do the following to make the project work.

1. Download and install VS 2022 Community edition
2. Download and install Anaconda. Install an additional environment Python 3.8.16. Use this version of Python for the project.

The project should run at this point.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, May 30, 2025 6:34 PM
[Quoted text hidden]

[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Sat, May 31, 2025 at 12:46 PM
To: Kishor Durve <kishor.durve@outlook.com>

Thankyou sir! I will try this right away.
Also I hope you received this google sheet, I have shared edit access to you on your gmail account

Hindvee Vispute - Time data - Intern

Regards
Hindvee
[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Mon, Jun 2, 2025 at 4:45 PM
To: Kishor Durve <kishor.durve@outlook.com>

Greetings Mr. Durve,
I am facing an issue again an again while running the VS 2022 code, pls help me with this, we have the module named as model_zoo not models. Pls tell me how to proceed with this.



Warm Regards
Hindvee
[Quoted text hidden]

Kishor Durve <kishor.durve@outlook.com> Tue, Jun 3, 2025 at 9:23 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

I have file network.py in folder 'models'. This file has the RRDBNet implementaion.

Kishor Durve
Duravision Systems
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Monday, June 2, 2025 7:15 PM
[Quoted text hidden]

[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Tue, Jun 3, 2025 at 4:41 AM
To: Kishor Durve <kishor.durve@outlook.com>

I've reviewed the project directory and noticed that there isn't a folder named `models` — only `model_zoo` and `utils` are present. As a result, the import statement `from models.network import RRDBNet as net` is causing a `ModuleNotFoundError`.

Would you prefer that I rename the `model_zoo` folder to `models` and move `network.py` there, or should I update the import statement to reference `model_zoo` instead? Please let me know how you'd like to proceed, and I'll make the changes accordingly.

Warm regards
Hindvee
[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Wed, Jun 4, 2025 at 1:34 AM
To: Kishor Durve <kishor.durve@outlook.com>

Greetings Mr. Durve,

Thanks for sharing the super-resolution model. I've set it up and tested it, the results inspiring. It does a good job cleaning up noise and improving contrast, especially in the faint line areas. As we discussed, I'll now add this to the bigger pipeline along with the AI modules (NB-1 to NB-5). The super-resolution output can help improve both the CNN classifier and regression models. I'll also check how it works together with NB-5, the denoising autoencoder, to see if we get even better results.

Please let me know if you want me to run it on any specific ROI samples or start putting together comparison results.

Warm regards
Hindvee

[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Wed, Jun 4, 2025 at 1:56 AM
To: Kishor Durve <kishor.durve@outlook.com>

Additionally, Sir, I would like to confirm if the current format of the Google Sheet meets your expectations. Please let me know if you would prefer any modifications, such as a more detailed breakdown of the work completed.

Warm Regards
Hindvee
[Quoted text hidden]

Kishor Durve <kishor.durve@outlook.com> Wed, Jun 4, 2025 at 6:42 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

Yes, it looks ok to me. We will take a relook once you make more progress.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, June 4, 2025 4:26 AM
[Quoted text hidden]

[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Fri, Jun 6, 2025 at 2:54 AM
To: Kishor Durve <kishor.durve@outlook.com>

Greetings sir,
I hope you're doing well. I've just finished training the CNN-based line presence classifier, and I'm attaching the loss graph for your review. The model seems to have learned quite quickly, with both training and validation losses staying very close to zero throughout. There was a small spike in training loss around epoch 80, but validation stayed stable, so it might just be a minor fluctuation.

Since the dataset is quite small and the model converged very early, I'm unsure how best to move forward.According to you should I try to get more data to avoid overfitting. Please guide me towards a direction in which I can make this model more effective.

Warm regards

Hindvee

[Quoted text hidden]

NB_1_CNN_Based_line_presence_classifier (1).ipynb
335K

Kishor Durve <kishor.durve@outlook.com> Fri, Jun 6, 2025 at 11:31 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

I am not going to be of much help to you as I have limited experience in this field. Also, what little I know was gained by applying available models in OpenCV's DNN module.

There are two images available in the images folder. You may crop out patches from them for the Training / test sets. We use ImageJ and IrfanView for such work as MS Paint cannot handle such big images. If you need more images please let me know.

Kishor Durve
Duravision Systems
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, June 6, 2025 5:24 AM
[Quoted text hidden]

[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Sat, Jun 7, 2025 at 2:06 AM
To: Kishor Durve <kishor.durve@outlook.com>

Greetings sir,
It would be great if I get more images, that would expand my dataset. Thank you for your previous email.
I also wanted tk inform you that I am travelling and will be able to resume work by Tuesday. I request you to kindly allow my leave.

Warm Regards
Hindvee
[Quoted text hidden]

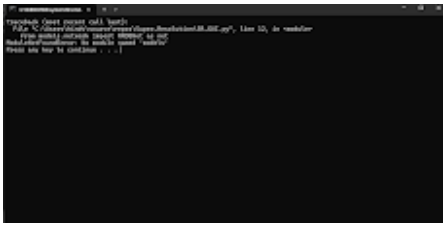


image.png
30K

Kishor Durve <kishor.durve@outlook.com> Sat, Jun 7, 2025 at 3:22 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

Sure, no problem.

Get Outlook for Android

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Saturday, June 7, 2025 4:36:27 AM
[Quoted text hidden]

[Quoted text hidden]

Kishor Durve <kishor.durve@outlook.com> Sat, Jun 7, 2025 at 2:51 PM
To: Hindvee Vispute <hindveevispute@gmail.com>

I have uploaded more images. You can crop out patches from them. One of the images has many boards which will give many useful patches.

Kishor Durve
Duravision Systems
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Saturday, June 7, 2025 4:36 AM
[Quoted text hidden]

[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Tue, Jun 10, 2025 at 11:07 PM
To: Kishor Durve <kishor.durve@outlook.com>

Thank you for sharing more images. I have updated the work on the sheets.

Warm regards
Hindvee
[Quoted text hidden]

Kishor Durve <kishor.durve@outlook.com> Fri, Jun 13, 2025 at 5:33 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

The images I sent are big. OpenCV will not be able to handle them as the max size allowed is 32K x 32K. The crash you are facing could be due to this reason.

You can crop out small patches from the big images and use them for trials.

Kishor Durve
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, June 11, 2025 1:37 AM
[Quoted text hidden]

[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Tue, Jun 17, 2025 at 12:36 AM
To: Kishor Durve <kishor.durve@outlook.com>

Hello sir,
I have completed the training and evaluation of the denoising autoencoder using grayscale image patches taken from the `IMG` folder. The model was trained for 20 epochs on 64×64 patches with binary cross entropy loss and a 10% validation split. During training, the loss reduced steadily and stabilized, with the final reconstruction loss reaching 0.3688. The results show that the model can effectively denoise images and recover clean patterns like faint lines. In the next step, I plan to use the output of the Super-Resolution model as a base to evaluate the accuracy and quality of the denoising process, to better understand how both methods contribute to image enhancement. I am attaching the .ipynb file for your reference and also the IMG Folder which I just mentioned.

Warm regards
Hindvee

IMG FOLDER

Accuracy results:-




```
[13] loss = autoencoder.evaluate(x_train_noisy, x_train_clean)
     print(f"Reconstruction Loss (Binary Crossentropy): {loss:.4f}")

16/16 ━━━━━━━━━━━━━━━━━━━━ 2s 154ms/step - loss: 0.3720
Reconstruction Loss (Binary Crossentropy): 0.3688
```

[Quoted text hidden]

Copy_of_NB_5_Deionising_autoencoder_code.ipynb
452K

Kishor Durve <kishor.durve@outlook.com> Tue, Jun 17, 2025 at 7:16 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

Good to know that you are making progress at a nice clip.

I saw the images and the difference between before / after can be seen. Can you please process the images attached to this mail and send me the results? The images are similar to those I had sent earlier, the exception being 5.png and 6.png. In these last two images the challenge is to detect circles whose edges are not well defined and the edge quality may vary from image to image. This is to test the effectiveness of noise removal / sharpening.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, June 17, 2025 3:06 AM
[Quoted text hidden]

[Quoted text hidden]

Kishor Durve <kishor.durve@outlook.com> Tue, Jun 17, 2025 at 7:21 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

Sorry! Forgot the attachment.

Kishor Durve

From: Kishor Durve <kishor.durve@outlook.com>
Sent: Tuesday, June 17, 2025 9:46 AM
To: Hindvee Vispute <hindveevispute@gmail.com>
[Quoted text hidden]

[Quoted text hidden]

Set2.zip
1454K

Hindvee Vispute <hindveevispute@gmail.com> Tue, Jun 17, 2025 at 4:30 PM
To: Kishor Durve <kishor.durve@outlook.com>

Thank you for sharing your views on my work sir. I will share the results for the 2 images which u have shared with me by today eod.

Warm Regards
Hindvee
[Quoted text hidden]

2 attachments

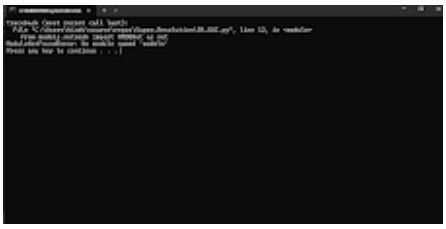


image.png
30K

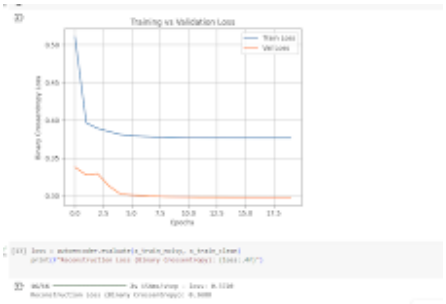


image.png
50K

Hindvee Vispute <hindveevispute@gmail.com> Wed, Jun 18, 2025 at 1:56 AM
To: Kishor Durve <kishor.durve@outlook.com>

Hello sir,
I have processed the 5.png and 6.png, I faced lower accuracy rate earlier. After fixing the earlier issues with mismatched data sizes and high Binary Crossentropy loss (~0.6098), the model is now working smoothly. It reached a much lower loss of just 0.0528 in 5 epochs, and both training and validation losses settled nicely by the 2nd or 3rd epoch without signs of overfitting. The denoised images look very clean and closely match the original ones, even when the input images had different levels of noise. I am attaching the results in the attached file.
Warm regards
Hindvee
[Quoted text hidden]

For_5_png_&_6_png_Deionising_autoencoder_code (1).ipynb
391K

Hindvee Vispute <hindveevispute@gmail.com> Wed, Jun 18, 2025 at 11:14 AM
To: Kishor Durve <kishor.durve@outlook.com>

Sir, please let me know your views on the above .ipynb file.
Warm Regards
Hindvee
[Quoted text hidden]

Kishor Durve <kishor.durve@outlook.com> Wed, Jun 18, 2025 at 12:32 PM
To: Hindvee Vispute <hindveevispute@gmail.com>

I have checked the file. These two images contain circles as primary features. Also, the patches in the file seem to be for a small part of the original image. If you have processed the original images (5.png and 6.png) please send me the processed images on mail. If you have processed only parts of these images, I suggest you process the full images and send me the results.

Similarly, in the previous exercise please process complete images. Most of the set are of size 500x500 pixel, so it should not consume too much time.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, June 18, 2025 1:44 PM
[Quoted text hidden]

[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Thu, Jun 19, 2025 at 12:22 PM
To: Kishor Durve <kishor.durve@outlook.com>

Hello sir,

I've completed the full-image denoising as discussed for the images `5.png` and `6.png`. This time, the model was trained and applied directly on the complete images (cropped to a common size of **570×588**) instead of small patches, as per your suggestion. The denoising autoencoder was able to reduce background noise and enhance structural features, particularly the edges of the circles, which were initially faint or inconsistent. The model output shows a noticeable improvement in clarity while preserving the original patterns.

I've attached the processed result image along with this mail. Please let me know your thoughts on this file.

Warm regards,
Hindvee

[Quoted text hidden]

Full_image_processing_For_5_png_&_6_png_Deionising_autoencoder_code (1).ipynb
391K

Kishor Durve <kishor.durve@outlook.com> Thu, Jun 19, 2025 at 1:42 PM
To: Hindvee Vispute <hindveevispute@gmail.com>

I have seen the processed images. I think it is good at this point to quantify 'sharpness' of an image. It will give us a measure to compare different methods / models.

There can be several approaches to do this. We use Opencv::Laplacian() to judge image sharpness. But you can try other approaches.

Kishor Durve

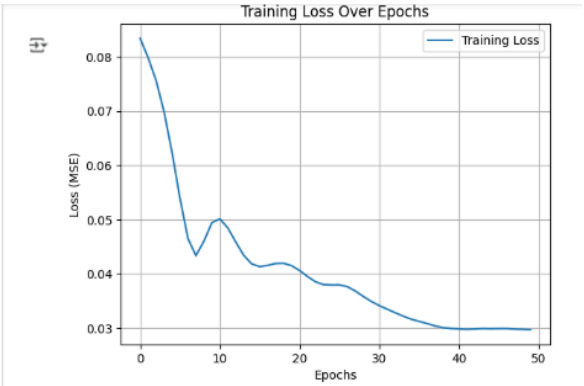
From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, June 19, 2025 2:52 PM
[Quoted text hidden]

[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Fri, Jun 20, 2025 at 12:29 PM
To: Kishor Durve <kishor.durve@outlook.com>

Hello sir!
Based on the latest code I calculated the sharpness using Laplacian variance. I obtained the following results as those attached as image.
I think that while denoising, though the noise was removed, but the model blurred or over smoothened the image. I may have lost some meaningful edge details in the process. But the sharpness value of Clean image is lower than the noisy image (because it has no random noise), but high enough to retain clear edges and structure. So I am working next on increasing the sharpness value while keeping the noise low. Please do share your views with me.

Warm regards
Hindvee




```
[19] # using laplacian variance to measure sharpness
     # high variance means sharp edges
     # low variance means blurred image

     import cv2
     import numpy as np

     def measure_sharpness(image):
         """
         image: 2D numpy array (grayscale, float32 or uint8)
         returns: scalar sharpness value
         """
         # Convert image to uint8 for compatibility with cv2.Laplacian
         # Assume image is currently float32 between 0.0 and 1.0
         image_uint8 = (image * 255).astype(np.uint8)

         laplacian = cv2.Laplacian(image_uint8, cv2.CV_64F)
         variance = laplacian.var()
         return variance
```

```
noisy = np.squeeze(x_train_noisy)
clean = np.squeeze(x_train_clean)
denoised = np.squeeze(decoded_img)

sharp_noisy = measure_sharpness(noisy)
sharp_clean = measure_sharpness(clean)
sharp_denoised = measure_sharpness(denoised)

print(f"Sharpness (Noisy): {sharp_noisy:.2f}")
print(f"Sharpness (Denoised): {sharp_denoised:.2f}")
print(f"Sharpness (Clean): {sharp_clean:.2f}")
```

```
Sharpness (Noisy): 33.72
Sharpness (Denoised): 6.92
Sharpness (Clean): 26.49
```

[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Tue, Jun 24, 2025 at 2:43 AM
To: Kishor Durve <kishor.durve@outlook.com>

Hello Durve sir,

I wanted to share an update on the denoising output. I noticed earlier that while the model was doing a good job removing noise, it was also making the image a bit too smooth — the sharpness had dropped quite a bit:

- Noisy Image: 33.72
- Denoised Image: 6.92
- Clean Image (Ground Truth): 26.49

To improve this, I added a sharpening filter after the autoencoder output and tested two variants:

1. A **strong sharpening kernel**, which raised the denoised sharpness to **112.40,** but it felt a bit overdone and introduced some artifacts.
2. A **softer kernel**, which gave a cleaner and more natural result, with denoised sharpness at **54.04**. This seems like a balanced point where edges are clearer without compromising visual quality.

If you'd like, I can also share the output images (clean vs. denoised) for a visual comparison.

Warm regards

Hindvee

```
noisy = np.squeeze(x_train_noisy)
clean = np.squeeze(x_train_clean)
denoised = np.squeeze(decoded_img)

sharp_noisy = measure_sharpness(noisy)
sharp_clean = measure_sharpness(clean)
sharp_denoised = measure_sharpness(denoised)

print(f"Sharpness (Noisy): {sharp_noisy:.2f}")
print(f"Sharpness (Denoised): {sharp_denoised:.2f}")
print(f"Sharpness (Clean): {sharp_clean:.2f}")
```

```
Sharpness (Noisy): 33.72
Sharpness (Denoised): 54.04
Sharpness (Clean): 26.49
```

[Quoted text hidden]

Kishor Durve <kishor.durve@outlook.com> Tue, Jun 24, 2025 at 5:48 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

Please share the images for visual comparison.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, June 24, 2025 5:13 AM
[Quoted text hidden]

[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Tue, Jun 24, 2025 at 11:29 AM
To: Kishor Durve <kishor.durve@outlook.com>

Hello sir,
I am attaching the .ipynb file, it has the images

warm regards
Hindvee

[Quoted text hidden]

Full_image_processing_For_5_png_&_6_png_Deionising_autoencoder_code (2).ipynb
436K

Hindvee Vispute <hindveevispute@gmail.com> Thu, Jun 26, 2025 at 2:18 AM
To: Kishor Durve <kishor.durve@outlook.com>

Hello sir, I have attached the results for png 1,2,3 and 4. Pls share ur views and insights.

Warm Regards
Hindvee
[Quoted text hidden]

---

2 attachments

Full_image_processing_For_3_png_&_4_png_Deionising_autoencoder_code.ipynb
520K

Full_image_processing_For_1_png_&_2_png_Deionising_autoencoder_code.ipynb
537K

Kishor Durve <kishor.durve@outlook.com> Thu, Jun 26, 2025 at 8:08 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

I am unable to run the files in Colab and generate images. Can you please send me the processed images on mail - .png / .jpeg files, not .ipynb files.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, June 26, 2025 4:48 AM
[Quoted text hidden]

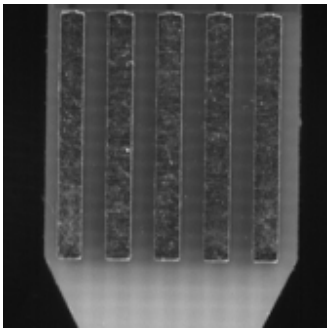
[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Thu, Jun 26, 2025 at 11:09 PM
To: Kishor Durve <kishor.durve@outlook.com>

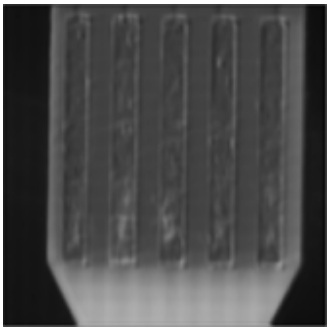
Here I am attaching the processed images.

Warm regards
Hindvee
[Quoted text hidden]

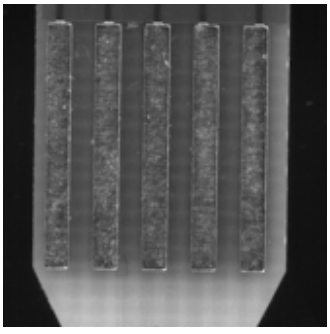
4 attachments



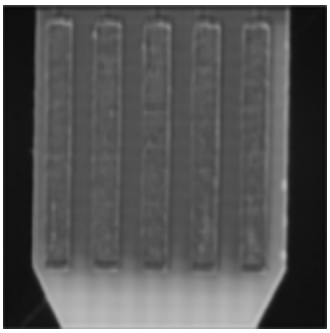
clean_image_1&2.png
125K



denoised_output_1&2.png
108K



clean_image_3&4.png
128K



denoised_output_3&4.png
103K

Hindvee Vispute <hindveevispute@gmail.com> Mon, Jun 30, 2025 at 10:33 AM
To: Kishor Durve <kishor.durve@outlook.com>

Good morning sir!

Did you get any chance to review the above images? Please share your suggestions and insights with me.

Warm regards
Hindvee
[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Mon, Jun 30, 2025 at 11:21 PM
To: Kishor Durve <kishor.durve@outlook.com>

Hello sir!

I was eager to get insights from you about the above shared images.Please do have a look as soon as possible. Also, I wanted to know whether the stipend will be credited at the end of the month or the project's end. It would be very kind of you if I could get the stipend at the end of the month.

Thanks and regards
Hindvee
[Quoted text hidden]

Kishor Durve <kishor.durve@outlook.com> Wed, Jul 2, 2025 at 3:15 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

I checked the images, but did not notice substantial difference between the before and after images. You can subtract them (after aligning) and check.
As regards payment of stipend please let me know how much we need to pay and I will transfer it on GPay.
I am travelling again and am out of office. I will be back only on 15th. My responses to your mails may be delayed.

Get Outlook for Android

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, July 1, 2025 1:51:45 AM
[Quoted text hidden]

[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Thu, Jul 3, 2025 at 12:48 AM
To: Kishor Durve <kishor.durve@outlook.com>

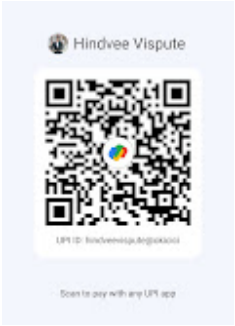
Hello sir!
I will perform the subtraction and share the results with you.
Regarding the stipend it is coming together as Rs. 9,300/- calculated from May 30th 2025 to June 30th 2025 (total - 46.5 hrs and Rs. 200/- per hour).
Also I would be happy to receive your review of my work over this month, where am I doing good and where can I perform better.

warm regards
Hindvee
[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Thu, Jul 3, 2025 at 10:53 AM
To: Kishor Durve <kishor.durve@outlook.com>

UPI ID: hindveevispute@okicici

Warm regards
Hindvee
[Quoted text hidden]



HIndvee_Gpay_QR.jpg
63K

Hindvee Vispute <hindveevispute@gmail.com> Fri, Jul 4, 2025 at 3:07 AM
To: Kishor Durve <kishor.durve@outlook.com>

Hello sir!
This is what I am been working on. I am still working on getting better edges and less whitish images.

warm regards
Hindvee
[Quoted text hidden]

Copy_of_Denoise+line_detection(Good_Bad_Images) (2).ipynb
1593K

Kishor Durve <kishor.durve@outlook.com> Fri, Jul 4, 2025 at 3:54 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

Please keep at it. Use difference image and Laplacian to gauge the effect of the algorithm. I will be able to look at your work only after 15th.
I have transferred your stipend by GPay.

Get Outlook for Android

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, July 4, 2025 5:37:35 AM
[Quoted text hidden]

[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Mon, Jul 7, 2025 at 4:39 AM
To: Kishor Durve <kishor.durve@outlook.com>

Hello sir!
I have received the stipend, thank you for that! I'd like to share a brief update on the image enhancement work, focusing on noise removal and edge preservation.
Summary of results:-
1. Previous code
---Performed basic denoising but introduced over-smoothing.
---Laplacian MAD: ~0.0175–0.0192

2. DnCNN
---Improved noise removal and structural retention.
---MAD: ~0.0186–0.0202

3. DnCNN + Sharpening (Current approach)
---Added post-processing to enhance edge clarity.
---Consistently improved MAD to ~0.0178–0.0196.
---Visual results show reduced whitening and better edge definition.

This confirms that the current setup provides better denoising while preserving fine image details. Please let me know if you'd like to proceed with further tuning using SSIM or edge-aware loss functions.

Warm regards

Hindvee

[Quoted text hidden]

DnCNN_with_Sharpening (1).ipynb
1174K

Kishor Durve <kishor.durve@outlook.com> Mon, Jul 7, 2025 at 6:23 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

I am unable to give my view of the approach as I can't view the images. I can give my opinion if you send me difference images. In any case, I think you need to have difference images as a metric in addition to the Laplacian.

Get Outlook for Android

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Monday, July 7, 2025 7:09:51 AM
[Quoted text hidden]

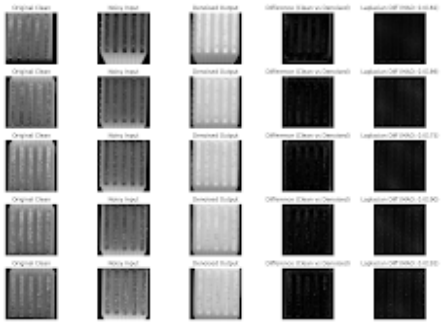
[Quoted text hidden]

Hindvee Vispute <hindveevispute@gmail.com> Mon, Jul 7, 2025 at 10:37 AM
To: Kishor Durve <kishor.durve@outlook.com>

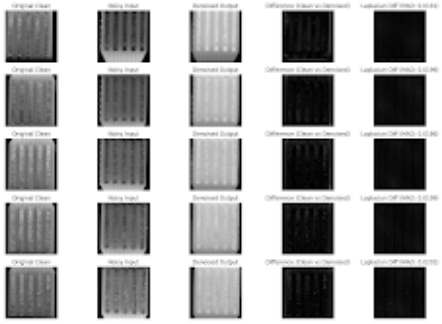
Hello sir!
I am sharing the first 5 difference images of the three codes that I mentioned previously. Please check this whenever you have time.

Warm regards
Hindvee
[Quoted text hidden]

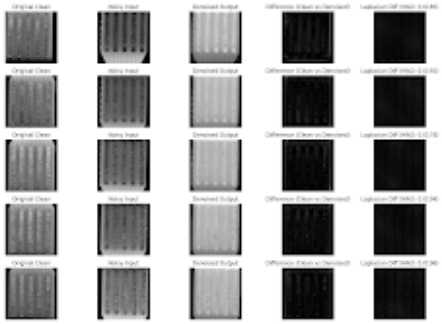
3 attachments



1)Previous code file.png
619K



2)DnCNN.png
599K



3)DnCNN+Sharpening.png
604K

Hindvee Vispute <hindveevispute@gmail.com> Wed, Jul 9, 2025 at 12:23 AM
To: Kishor Durve <kishor.durve@outlook.com>

Hello sir! I am trying the pre-trained model for detecting edges in our images; I have tried Holistically-Nested Edge Detection (HED) and DexiNed. I am attaching the output, I am going forward with HED to get better results.

Warm regards
Hindvee

On Mon, Jul 7, 2025 at 10:37 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir!
> I am sharing the first 5 difference images of the three codes that I mentioned previously. Please check this whenever you have time.
>
> Warm regards
> Hindvee
>
> On Mon, Jul 7, 2025 at 6:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > I am unable to give my view of the approach as I can't view the images. I can give my opinion if you send me difference images. In any case, I think you need to have difference images as a metric in addition to the Laplacian.
> >
> > Get Outlook for Android
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Monday, July 7, 2025 7:09:51 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Hello sir!
> > I have received the stipend, thank you for that! I'd like to share a brief update on the image enhancement work, focusing on noise removal and edge preservation.
> > Summary of results:-
> > 1. Previous code
> > ---Performed basic denoising but introduced over-smoothing.
> > ---Laplacian MAD: ~0.0175–0.0192
> >
> > 2. DnCNN
> > ---Improved noise removal and structural retention.
> > ---MAD: ~0.0186–0.0202
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> > 3. DnCNN + Sharpening (Current approach)
> > ---Added post-processing to enhance edge clarity.
> > ---Consistently improved MAD to ~0.0178–0.0196.
> > ---Visual results show reduced whitening and better edge definition.

This confirms that the current setup provides better denoising while preserving fine image details. Please let me know if you'd like to proceed with further tuning using SSIM or edge-aware loss functions.

Warm regards

Hindvee

On Fri, Jul 4, 2025 at 3:54 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Please keep at it. Use difference image and Laplacian to gauge the effect of the algorithm. I will be able to look at your work only after 15th.
> I have transferred your stipend by GPay.
>
> Get Outlook for Android
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Friday, July 4, 2025 5:37:35 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir!
> This is what I am been working on. I am still working on getting better edges and less whitish images.
>
> warm regards
> Hindvee
>
> On Thu, Jul 3, 2025 at 10:53 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > UPI ID: hindveevispute@okicici
> >
> > Warm regards
> > Hindvee
> >
> > On Wed, 2 Jul 2025, 03:16 Kishor Durve, <kishor.durve@outlook.com> wrote:
> >
> > > I checked the images, but did not notice substantial difference between the before and after images. You can subtract them (after aligning) and check.
> > > As regards payment of stipend please let me know how much we need to pay and I will transfer it on GPay.
> > > I am travelling again and am out of office. I will be back only on 15th. My responses to your mails may be delayed.
> > >
> > > Get Outlook for Android
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Tuesday, July 1, 2025 1:51:45 AM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection
> > >
> > > Hello sir!
> > >
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> > >
> > > Thanks and regards
> > > Hindvee
> > >
> > > On Mon, Jun 30, 2025 at 10:33 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > >
> > > > Good morning sir!
> > > >
> > > > Did you get any chance to review the above images? Please share your suggestions and insights with me.
> > > >
> > > > Warm regards
> > > > Hindvee
> > > >
> > > > On Thu, Jun 26, 2025 at 11:09 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Here I am attaching the processed images.

Warm regards
Hindvee

On Thu, Jun 26, 2025 at 8:08 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> I am unable to run the files in Colab and generate images. Can you please send me the processed images on mail - .png / .jpeg files, not .ipynb files.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Thursday, June 26, 2025 4:48 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir, I have attached the results for png 1,2,3 and 4. Pls share ur views and insights.
>
> Warm Regards
> Hindvee
>
> On Tue, Jun 24, 2025 at 11:29 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir,
> > I am attaching the .ipynb file, it has the images
> >
> > warm regards
> > Hindvee
> >
> > On Tue, Jun 24, 2025 at 5:48 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > Please share the images for visual comparison.
> > >
> > > Kishor Durve
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Tuesday, June 24, 2025 5:13 AM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection
> > >
> > > Hello Durve sir,
> > >
> > > I wanted to share an update on the denoising output. I noticed earlier that while the model was doing a good job removing noise, it was also making the image a bit too smooth — the sharpness had dropped quite a bit:
> > >
> > > - Noisy Image: 33.72
> > > - Denoised Image: 6.92
> > > - Clean Image (Ground Truth): 26.49
> > >
> > > To improve this, I added a sharpening filter after the autoencoder output and tested two variants:
> > >
> > > 1. A **strong sharpening kernel**, which raised the denoised sharpness to **112.40,** but it felt a bit overdone and introduced some artifacts.
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> > >
> > > If you'd like, I can also share the output images (clean vs. denoised) for a visual comparison.
> > >
> > > Warm regards
> > >
> > > Hindvee
> > >
> > > image.png

On Fri, Jun 20, 2025 at 12:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Hello sir!
Based on the latest code I calculated the sharpness using Laplacian variance. I obtained the following results as those attached as image.
I think that while denoising, though the noise was removed, but the model blurred or over smoothened the image. I may have lost some meaningful edge details in the process. But the sharpness value of Clean image is lower than the noisy image (because it has no random noise), but high enough to retain clear edges and structure. So I am working next on increasing the sharpness value while keeping the noise low. Please do share your views with me.

Warm regards
Hindvee



On Thu, 19 Jun 2025, 13:42 Kishor Durve, <kishor.durve@outlook.com> wrote:

I have seen the processed images. I think it is good at this point to quantify 'sharpness' of an image. It will give us a measure to compare different methods / models.

There can be several approaches to do this. We use Opencv::Laplacian() to judge image sharpness. But you can try other approaches.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, June 19, 2025 2:52 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir,

I've completed the full-image denoising as discussed for the images `5.png` and `6.png`. This time, the model was trained and applied directly on the complete images (cropped to a common size of **570×588**) instead of small patches, as per your suggestion. The denoising autoencoder was able to reduce background noise and enhance structural features, particularly the edges of the circles, which were initially faint or inconsistent. The model output shows a noticeable improvement in clarity while preserving the original patterns.

I've attached the processed result image along with this mail. Please let me know your thoughts on this file.

Warm regards,
Hindvee

On Wed, Jun 18, 2025 at 12:33 PM Kishor Durve <kishor.durve@outlook.com> wrote:

I have checked the file. These two images contain circles as primary features. Also, the patches in the file seem to be for a small part of the original image. If you have processed the original images (5.png and 6.png) please send me the processed images on mail. If you have processed only parts of these images, I suggest you process the full images and send me the results.

Similarly, in the previous exercise please process complete images. Most of the set are of size 500x500 pixel, so it should not consume too much time.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, June 18, 2025 1:44 PM

To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Sir, please let me know your views on the above .ipynb file.
Warm Regards
Hindvee

On Wed, Jun 18, 2025 at 1:56 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir,
> I have processed the 5.png and 6.png, I faced lower accuracy rate earlier. After fixing the earlier issues with mismatched data sizes and high Binary Crossentropy loss (~0.6098), the model is now working smoothly. It reached a much lower loss of just 0.0528 in 5 epochs, and both training and validation losses settled nicely by the 2nd or 3rd epoch without signs of overfitting. The denoised images look very clean and closely match the original ones, even when the input images had different levels of noise. I am attaching the results in the attached file.
> Warm regards
> Hindvee
>
> On Tue, Jun 17, 2025 at 7:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Sorry! Forgot the attachment.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Kishor Durve <kishor.durve@outlook.com>
> > **Sent:** Tuesday, June 17, 2025 9:46 AM
> > **To:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Good to know that you are making progress at a nice clip.
> >
> > I saw the images and the difference between before / after can be seen. Can you please process the images attached to this mail and send me the results? The images are similar to those I had sent earlier, the exception being 5.png and 6.png. In these last two images the challenge is to detect circles whose edges are not well defined and the edge quality may vary from image to image. This is to test the effectiveness of noise removal / sharpening.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Tuesday, June 17, 2025 3:06 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Hello sir,
> > I have completed the training and evaluation of the denoising autoencoder using grayscale image patches taken from the `IMG` folder. The model was trained for 20 epochs on 64×64 patches with binary cross entropy loss and a 10% validation split. During training, the loss reduced steadily and stabilized, with the final reconstruction loss reaching 0.3688. The results show that the model can effectively denoise images and recover clean patterns like faint lines. In the next step, I plan to use the output of the Super-Resolution model as a base to evaluate the accuracy and quality of the denoising process, to better understand how both methods contribute to image enhancement. I am attaching the .ipynb file for your reference and also the IMG Folder which I just mentioned.
> >
> > Warm regards
> > Hindvee
> >
> > IMG FOLDER
> >
> > Accuracy results:-

image.png

On Fri, Jun 13, 2025 at 5:33 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> The images I sent are big. OpenCV will not be able to handle them as the max size allowed is 32K x 32K. The crash you are facing could be due to this reason.
>
> You can crop out small patches from the big images and use them for trials.
>
> Kishor Durve
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 11, 2025 1:37 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Thank you for sharing more images. I have updated the work on the sheets.
>
> Warm regards
> Hindvee
>
> On Sat, Jun 7, 2025 at 2:51 PM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > I have uploaded more images. You can crop out patches from them. One of the images has many boards which will give many useful patches.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Saturday, June 7, 2025 4:36 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings sir,
> > It would be great if I get more images, that would expand my dataset. Thank you for your previous email.
> > I also wanted tk inform you that I am travelling and will be able to resume work by Tuesday. I request you to kindly allow my leave.
> >
> > Warm Regards
> > Hindvee
> >
> > On Fri, 6 Jun 2025, 11:31 Kishor Durve, <kishor.durve@outlook.com> wrote:
> >
> > > I am not going to be of much help to you as I have limited experience in this field. Also, what little I know was gained by applying available models in OpenCV's DNN module.
> > >
> > > There are two images available in the images folder. You may crop out patches from them for the Training / test sets. We use ImageJ and IrfanView for such work as MS Paint cannot handle such big images. If you need more images please let me know.
> > >
> > > Kishor Durve
> > > Duravision Systems
> > > 9823164286
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Friday, June 6, 2025 5:24 AM

To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings sir,
I hope you're doing well. I've just finished training the CNN-based line presence classifier, and I'm attaching the loss graph for your review. The model seems to have learned quite quickly, with both training and validation losses staying very close to zero throughout. There was a small spike in training loss around epoch 80, but validation stayed stable, so it might just be a minor fluctuation.

Since the dataset is quite small and the model converged very early, I'm unsure how best to move forward.According to you should I try to get more data to avoid overfitting. Please guide me towards a direction in which I can make this model more effective.

Warm regards

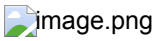
Hindvee

On Wed, Jun 4, 2025 at 6:42 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Yes, it looks ok to me. We will take a relook once you make more progress.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 4, 2025 4:26 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Additionally, Sir, I would like to confirm if the current format of the Google Sheet meets your expectations. Please let me know if you would prefer any modifications, such as a more detailed breakdown of the work completed.
>
> Warm Regards
> Hindvee
>
> On Wed, Jun 4, 2025 at 1:34 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Greetings Mr. Durve,
> >
> > Thanks for sharing the super-resolution model. I've set it up and tested it, the results inspiring. It does a good job cleaning up noise and improving contrast, especially in the faint line areas. As we discussed, I'll now add this to the bigger pipeline along with the AI modules (NB-1 to NB-5). The super-resolution output can help improve both the CNN classifier and regression models. I'll also check how it works together with NB-5, the denoising autoencoder, to see if we get even better results.
> >
> > Please let me know if you want me to run it on any specific ROI samples or start putting together comparison results.
> >
> > Warm regards
> > Hindvee
> >
> > On Tue, Jun 3, 2025 at 9:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > I have file network.py in folder 'models'. This file has the RRDBNet implementaion.
> > >
> > > Kishor Durve
> > > Duravision Systems
> > > 9823164286
> > >
> > > ---

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Monday, June 2, 2025 7:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr. Durve,
I am facing an issue again an again while running the VS 2022 code, pls help me with this, we have the module named as model_zoo not models. Pls tell me how to proceed with this.



Warm Regards
Hindvee

On Sat, May 31, 2025 at 12:46 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Thankyou sir! I will try this right away.
> Also I hope you received this google sheet, I have shared edit access to you on your gmail account
>
> Hindvee Vispute - Time data - Intern
>
> Regards
> Hindvee
>
> On Sat, May 31, 2025 at 5:43 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Sure. You need to do the following to make the project work.
> >
> > 1. Download and install VS 2022 Community edition
> > 2. Download and install Anaconda. Install an additional environment Python 3.8.16. Use this version of Python for the project.
> >
> > The project should run at this point.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, May 30, 2025 6:34 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings sir!
> > I went through the google drive folder shared by you. It would be really helpful if you could walk me through it.
> > Best regards
> > Hindvee
> >
> > On Fri, May 30, 2025 at 11:41 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > You can download the model, a working VC++ (VS 2022) project and image here.
> > >
> > > https://drive.google.com/drive/folders/1mLBDjJR3fFKs8Brr_nNj2z9YyI89nruE?usp=sharing
> > >
> > > Kishor Durve
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Friday, May 30, 2025 12:50 PM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection
> > >
> > > Greetings sir!

Pls do share the model over drive I will try it out. Also I request you to share more image database so that I can start working on it.
I will start work from today itself then.

Regards
Hindvee

On Fri, 30 May 2025, 06:11 Kishor Durve, <kishor.durve@outlook.com> wrote:

> We have found that using a super resolution Pytorch model to get a larger image results in significant noise removal. Perhaps it is a by-product.
>
> You can try it out. I have the model and can share it on Google Drive if you want to try it out. We have tried the models available in OpenCV DNN but they do not seem to give the same results.
>
> Regards,
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Thursday, May 29, 2025 5:28 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Dear Sir,
>
> Thank you for the confirmation. I will maintain the time data on a Google Sheet and ensure it is updated regularly.
>
> Looking forward to starting from Monday, 2nd June.
>
> Best regards,
> Hindvee
>
> On Thu, May 29, 2025 at 2:53 PM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Hello Hindvee,
> >
> > Welcome aboard!
> >
> > You can keep the time data on Google Sheet.
> >
> > Regards,
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Thursday, May 29, 2025 5:12 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings Mr. Durve,
> >
> > Thank you very much for giving me the opportunity to contribute to Guangdong Longtian Smart Instruments Co. Ltd. through this exciting project. I am pleased to accept the offer and look forward to working on the development of the AI-assisted line detection tool.

With your permission, I would like to begin the internship from the coming Monday, i.e., 2nd June 2025.

Additionally, I wanted to ask about the preferred method for tracking and sharing the time I spend on the project. Would you like me to maintain a shared timesheet (e.g., a Google Sheet or Excel file), or would you prefer weekly time summaries via email or any other format?

Looking forward to your guidance on this.

Thank you once again for this opportunity.

Warm regards,
Hindvee Rajesh Vispute

On Thu, May 29, 2025 at 8:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have attached the on-boarding letter.
>
> Please let me know if you have any queries.
>
> Regards,
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Saturday, May 10, 2025 3:46 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings Mr. Durve,
>
> Thank you for your detailed response and clarification. I now clearly understand that the goal is to enhance line detection using AI within known ROIs, leveraging the expected position of endpoints.
>
> To support this objective, I've explored and implemented several **AI-based approaches**, any of which can be used to complement or improve upon conventional methods:
>
> 1. **CNN-Based Line Presence Classifier (NB-1):** Identifies whether a line exists within a given patch/ROI—useful for filtering candidate regions before applying traditional detection.
> 2. **CNN Regression Model (NB-2):** Predicts line endpoints directly from the image, offering robust localization even in low-contrast or noisy scenarios.
> 3. **Denoising Autoencoder (NB-5):** Enhances faint or broken lines by cleaning up noise, thereby boosting the effectiveness of existing edge or Hough-based methods.
> 4. **Vision Transformer Model (NB-4):** Designed for fine-grained classification tasks and can be adapted for future objectives like surface defect or anomaly detection.
>
> These methods are independently deployable and can be integrated selectively depending on the specific inspection challenge or ROI structure. I'd be happy to align with your ROI-driven strategy and customize accordingly.

Regarding the stipend, I agree with the proposed Rs. 200/hour base rate and performance-based bonus, and I'm enthusiastic about the opportunity to collaborate further on defect detection and shape anomaly tasks in the future.

I have attached screenshots of the code files I executed today. I have the names of the notebooks (NB) with NB-1, NB-2,NB-4 and NB-5 respectively.

NB-1
NB-2
NB-4
NB-5

Please let me know if you'd like me to share these outputs or run a quick live demonstration.

Warm regards,
Hindvee

On Fri, May 9, 2025 at 6:03 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> I think there is a mismatch in understanding. We are already using conventional line detection techniques (various edge detection operators and Hough Transform and other methods for line fitment) .
>
> What we are looking to get from you is AI based / AI assisted line detection. Since we know the probable position of the line beforehand we can use this information to find a line. For example, say we know that the line end points are expected to be (x1, y1) and (x2, y2), we can construct a small rectangle (ROI) using these points and search for a line within this ROI.
>
> As for the stipend it depends both on the time you spend on the project and the results you produce. It will comprise of an hourly rate and a result linked amount. I suggest we have an hourly rate of Rs. 200/hour and a matching amount linked to the results at the end of the project. So, if you are able to produce satisfactory results it will come to Rs. 400/hour. Also, if the results are satisfactory, we can take this forward next year and work on other issues like surface defect detection, detection of shape anomalies etc.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Thursday, May 8, 2025 5:15 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings Mr Durve,
> I have the problem now so I will focus on faint line detection for the first step. I can proceed using Sobel, Scharr, and Canny edge detectors provided by OpenCV. First I will preprocess the images and then move to edge detection using the above 3 detectors. I will carry out line

detection using Hough Transform, and further we will filter out short or irrelevant lines based on length and their angle of orientation. Finally, we will draw the detected lines on the images using cv.line()

I can devote 14 hrs per week to this task for 2 months, we can extend the time if required. Also, I wanted to ask what stipend you can provide me. I believe I can complete the task in the above time frame. Thank you for your time.

Warm regards
Hindvee

On Thu, May 8, 2025 at 7:18 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Your analysis seems good.
>
> At this point we are not interested in detecting defects (though at a later stage we may be), and want to detect the faint lines only.
>
> Note that though I have annotated only a single line in the image I sent to you yesterday, we need to detect edge lines of all vertical stripes.
>
> I have attached 2 sets of 20 images each named 'Good Images' and 'Bad Images' - images in which it is mostly easy to detect lines by the conventional image processing method and those in which lines are faint and hard to detect. These images may help you get a better understanding of the problem.
>
> Please also check the following issues:
> 1. How much time will you be able to devote to this task?
> 2. Will you be able to complete the task within the available time?
>
> There are other issues like time taken by the AI route, integration of AI line detect code in VC++ project, further training / fine tuning of model, applicability of the method to detect faint circles etc. But we will get into them later once the basic method is proven.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, May 7, 2025 3:15 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings, Mr. Durve,
> Thank you for reviewing my proposal and for sharing the sample images. By looking at the images I came across the specific challenges faced by the inspection system, like detecting faint lines, subtle surface variations, and fine defects that can be missed by

traditional image processing. Based on these observations I suggest that we can do the following:-

1. **Hybrid approach**: Objective - Combine the speed of OpenCV line detection (example Hough Transform) with deep learning's pattern recognition capability.
Approach
- Use OpenCV to pre-locate candidate regions and feed them into a deep model for verification.
- This can speed up inference and increase resistance to false positives.

2. **Deep learning-based fine defect detection**:
Objective: Improve the detection of faint, low contrast line defects and localized surface anomalies.
Approach
- Train a model, such as YOLOv8 or U-Net, tailored to your defect patterns using the annotated images.
- To improve feature visibility, add contrast enhancement.
- Assess using detection accuracy and intersection over union (IoU) to compare to baselines based on OpenCV.

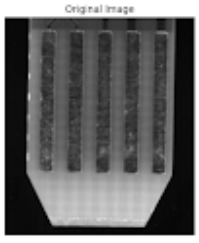
I can begin with a proof-of-concept model using the images shared by you. Thank you for sharing the opportunity. I am excited to get a chance to contribute to the advancement in inspection technology.

Warm regards
Hindvee Vispute

On Wed, 7 May 2025, 09:31 Kishor Durve, <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have gone thru the document uploaded by you.
>
> It looks ok to me.
>
> I can have attached a few images so you will get a better idea of the requirements.
>
> Kishor Durve

2 attachments





DexiNet_output.png
117K



Holistically-nested edge detection.png
82K

Hindvee Vispute <hindveevispute@gmail.com> Fri, Jul 11, 2025 at 3:29 PM
To: Kishor Durve <kishor.durve@outlook.com>

Hello sir!

Following up on my previous email regarding edge detection, I wanted to share the progress I've made since then. I have been focusing on using the Holistically-Nested Edge Detection (HED) method. I have now:

CODE FILE

1. Successfully applied the HED process to a folder containing our images to generate edge maps for the entire dataset.
2. Trained a custom autoencoder model designed to refine and correct the lines in these images.
3. Applied the trained autoencoder to the HED output images to produce corrected versions.
4. Started working on post-processing techniques.

I am currently experimenting with different parameters and methods to achieve the best results for this step. I am continuing to refine the line clarification process and will share updated results once I have finalized the post-processing approach.

warm regards

Hindvee

On Wed, Jul 9, 2025 at 12:23 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir! I am trying the pre-trained model for detecting edges in our images; I have tried Holistically-Nested Edge Detection (HED) and DexiNed. I am attaching the output, I am going forward with HED to get better results.
>
> Warm regards
> Hindvee
>
> On Mon, Jul 7, 2025 at 10:37 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir!
> > I am sharing the first 5 difference images of the three codes that I mentioned previously. Please check this whenever you have time.
> >
> > Warm regards
> > Hindvee
> >
> > On Mon, Jul 7, 2025 at 6:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > I am unable to give my view of the approach as I can't view the images. I can give my opinion if you send me difference images. In any case, I think you need to have difference images as a metric in addition to the Laplacian.
> > >
> > > Get Outlook for Android
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Monday, July 7, 2025 7:09:51 AM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection
> > >
> > > Hello sir!
> > > I have received the stipend, thank you for that! I'd like to share a brief update on the image enhancement work, focusing on noise removal and edge preservation.
> > > Summary of results:-
> > > 1. Previous code
> > > ---Performed basic denoising but introduced over-smoothing.
> > > ---Laplacian MAD: ~0.0175–0.0192
> > >
> > > 2. DnCNN
> > > ---Improved noise removal and structural retention.
> > > ---MAD: ~0.0186–0.0202
> > >
> > > 3. DnCNN + Sharpening (Current approach)
> > > ---Added post-processing to enhance edge clarity.
> > > ---Consistently improved MAD to ~0.0178–0.0196.
> > > ---Visual results show reduced whitening and better edge definition.

This confirms that the current setup provides better denoising while preserving fine image details. Please let me know if you'd like to proceed with further tuning using SSIM or edge-aware loss functions.

Warm regards

Hindvee

On Fri, Jul 4, 2025 at 3:54 AM Kishor Durve <kishor.durve@outlook.com> wrote:

Please keep at it. Use difference image and Laplacian to gauge the effect of the algorithm. I will be able to look at your work only after 15th.
I have transferred your stipend by GPay.

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From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, July 4, 2025 5:37:35 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!
This is what I am been working on. I am still working on getting better edges and less whitish images.

warm regards
Hindvee

On Thu, Jul 3, 2025 at 10:53 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

UPI ID: hindveevispute@okicici

Warm regards
Hindvee

On Wed, 2 Jul 2025, 03:16 Kishor Durve, <kishor.durve@outlook.com> wrote:

I checked the images, but did not notice substantial difference between the before and after images. You can subtract them (after aligning) and check.
As regards payment of stipend please let me know how much we need to pay and I will transfer it on GPay.
I am travelling again and am out of office. I will be back only on 15th. My responses to your mails may be delayed.

Get Outlook for Android

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, July 1, 2025 1:51:45 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!

I was eager to get insights from you about the above shared images.Please do have a look as soon as possible. Also, I wanted to know whether the stipend will be credited at the end of the month or the project's end. It would be very kind of you if I could get the stipend at the end of the month.

Thanks and regards
Hindvee

On Mon, Jun 30, 2025 at 10:33 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Good morning sir!

Did you get any chance to review the above images? Please share your suggestions and insights with me.

Warm regards
Hindvee

On Thu, Jun 26, 2025 at 11:09 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Here I am attaching the processed images.

Warm regards
Hindvee

On Thu, Jun 26, 2025 at 8:08 AM Kishor Durve <kishor.durve@outlook.com> wrote:

I am unable to run the files in Colab and generate images. Can you please send me the processed images on mail - .png / .jpeg files, not .ipynb files.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, June 26, 2025 4:48 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir, I have attached the results for png 1,2,3 and 4. Pls share ur views and insights.

Warm Regards
Hindvee

On Tue, Jun 24, 2025 at 11:29 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Hello sir,
I am attaching the .ipynb file, it has the images

warm regards
Hindvee

On Tue, Jun 24, 2025 at 5:48 AM Kishor Durve <kishor.durve@outlook.com> wrote:

Please share the images for visual comparison.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, June 24, 2025 5:13 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello Durve sir,

I wanted to share an update on the denoising output. I noticed earlier that while the model was doing a good job removing noise, it was also making the image a bit too smooth — the sharpness had dropped quite a bit:

- Noisy Image: 33.72
- Denoised Image: 6.92
- Clean Image (Ground Truth): 26.49

To improve this, I added a sharpening filter after the autoencoder output and tested two variants:

1. A **strong sharpening kernel**, which raised the denoised sharpness to **112.40,** but it felt a bit overdone and introduced some artifacts.
2. A **softer kernel**, which gave a cleaner and more natural result, with denoised sharpness at **54.04**. This seems like a balanced point where edges are clearer without compromising visual quality.

If you'd like, I can also share the output images (clean vs. denoised) for a visual comparison.

Warm regards

Hindvee

image.png

On Fri, Jun 20, 2025 at 12:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Hello sir!
Based on the latest code I calculated the sharpness using Laplacian variance. I obtained the following results as those attached as image.
I think that while denoising, though the noise was removed, but the model blurred or over smoothened the image. I may have lost some meaningful edge details in the process. But the sharpness value of Clean image is lower than the noisy image (because it has no random noise), but high enough to retain clear edges and structure. So I am working next on increasing the sharpness value while keeping the noise low. Please do share your views with me.

Warm regards
Hindvee



On Thu, 19 Jun 2025, 13:42 Kishor Durve, <kishor.durve@outlook.com> wrote:

I have seen the processed images. I think it is good at this point to quantify 'sharpness' of an image. It will give us a measure to compare different methods / models.

There can be several approaches to do this. We use Opencv::Laplacian() to judge image sharpness. But you can try other approaches.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, June 19, 2025 2:52 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir,

I've completed the full-image denoising as discussed for the images `5.png` and `6.png`. This time, the model was trained and applied directly on the complete images (cropped to a common size of **570×588**) instead of small patches, as per your suggestion. The denoising autoencoder was able to reduce background noise and enhance structural features, particularly the edges of the circles, which were initially faint or inconsistent. The model output shows a noticeable improvement in clarity while preserving the original patterns.

I've attached the processed result image along with this mail. Please let me know your thoughts on this file.

Warm regards,
Hindvee

On Wed, Jun 18, 2025 at 12:33 PM Kishor Durve <kishor.durve@outlook.com> wrote:

I have checked the file. These two images contain circles as primary features. Also, the patches in the file seem to be for a small part of the original image. If you have processed the original images (5.png and 6.png) please send me the processed images on mail. If you have processed only parts of these images, I suggest you process the full images and send me the results.

Similarly, in the previous exercise please process complete images. Most of the set are of size 500x500 pixel, so it should not consume too much time.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, June 18, 2025 1:44 PM

To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Sir, please let me know your views on the above .ipynb file.
Warm Regards
Hindvee

On Wed, Jun 18, 2025 at 1:56 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir,
> I have processed the 5.png and 6.png, I faced lower accuracy rate earlier. After fixing the earlier issues with mismatched data sizes and high Binary Crossentropy loss (~0.6098), the model is now working smoothly. It reached a much lower loss of just 0.0528 in 5 epochs, and both training and validation losses settled nicely by the 2nd or 3rd epoch without signs of overfitting. The denoised images look very clean and closely match the original ones, even when the input images had different levels of noise. I am attaching the results in the attached file.
> Warm regards
> Hindvee
>
> On Tue, Jun 17, 2025 at 7:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Sorry! Forgot the attachment.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Kishor Durve <kishor.durve@outlook.com>
> > **Sent:** Tuesday, June 17, 2025 9:46 AM
> > **To:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Good to know that you are making progress at a nice clip.
> >
> > I saw the images and the difference between before / after can be seen. Can you please process the images attached to this mail and send me the results? The images are similar to those I had sent earlier, the exception being 5.png and 6.png. In these last two images the challenge is to detect circles whose edges are not well defined and the edge quality may vary from image to image. This is to test the effectiveness of noise removal / sharpening.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Tuesday, June 17, 2025 3:06 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Hello sir,
> > I have completed the training and evaluation of the denoising autoencoder using grayscale image patches taken from the `IMG` folder. The model was trained for 20 epochs on 64×64 patches with binary cross entropy loss and a 10% validation split. During training, the loss reduced steadily and stabilized, with the final reconstruction loss reaching 0.3688. The results show that the model can effectively denoise images and recover clean patterns like faint lines. In the next step, I plan to use the output of the Super-Resolution model as a base to evaluate the accuracy and quality of the denoising process, to better understand how both methods contribute to image enhancement. I am attaching the .ipynb file for your reference and also the IMG Folder which I just mentioned.
> >
> > Warm regards
> > Hindvee
> >
> > IMG FOLDER
> >
> > Accuracy results:-



On Fri, Jun 13, 2025 at 5:33 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> The images I sent are big. OpenCV will not be able to handle them as the max size allowed is 32K x 32K. The crash you are facing could be due to this reason.
>
> You can crop out small patches from the big images and use them for trials.
>
> Kishor Durve
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 11, 2025 1:37 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Thank you for sharing more images. I have updated the work on the sheets.
>
> Warm regards
> Hindvee
>
> On Sat, Jun 7, 2025 at 2:51 PM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > I have uploaded more images. You can crop out patches from them. One of the images has many boards which will give many useful patches.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Saturday, June 7, 2025 4:36 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings sir,
> > It would be great if I get more images, that would expand my dataset. Thank you for your previous email.
> > I also wanted tk inform you that I am travelling and will be able to resume work by Tuesday. I request you to kindly allow my leave.
> >
> > Warm Regards
> > Hindvee
> >
> > On Fri, 6 Jun 2025, 11:31 Kishor Durve, <kishor.durve@outlook.com> wrote:
> >
> > > I am not going to be of much help to you as I have limited experience in this field. Also, what little I know was gained by applying available models in OpenCV's DNN module.
> > >
> > > There are two images available in the images folder. You may crop out patches from them for the Training / test sets. We use ImageJ and IrfanView for such work as MS Paint cannot handle such big images. If you need more images please let me know.
> > >
> > > Kishor Durve
> > > Duravision Systems
> > > 9823164286
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Friday, June 6, 2025 5:24 AM

To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings sir,
I hope you're doing well. I've just finished training the CNN-based line presence classifier, and I'm attaching the loss graph for your review. The model seems to have learned quite quickly, with both training and validation losses staying very close to zero throughout. There was a small spike in training loss around epoch 80, but validation stayed stable, so it might just be a minor fluctuation.

Since the dataset is quite small and the model converged very early, I'm unsure how best to move forward.According to you should I try to get more data to avoid overfitting. Please guide me towards a direction in which I can make this model more effective.

Warm regards

Hindvee

On Wed, Jun 4, 2025 at 6:42 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Yes, it looks ok to me. We will take a relook once you make more progress.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 4, 2025 4:26 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Additionally, Sir, I would like to confirm if the current format of the Google Sheet meets your expectations. Please let me know if you would prefer any modifications, such as a more detailed breakdown of the work completed.
>
> Warm Regards
> Hindvee
>
> On Wed, Jun 4, 2025 at 1:34 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Greetings Mr. Durve,
> >
> > Thanks for sharing the super-resolution model. I've set it up and tested it, the results inspiring. It does a good job cleaning up noise and improving contrast, especially in the faint line areas. As we discussed, I'll now add this to the bigger pipeline along with the AI modules (NB-1 to NB-5). The super-resolution output can help improve both the CNN classifier and regression models. I'll also check how it works together with NB-5, the denoising autoencoder, to see if we get even better results.
> >
> > Please let me know if you want me to run it on any specific ROI samples or start putting together comparison results.
> >
> > Warm regards
> > Hindvee
> >
> > On Tue, Jun 3, 2025 at 9:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > I have file network.py in folder 'models'. This file has the RRDBNet implementaion.
> > >
> > > Kishor Durve
> > > Duravision Systems
> > > 9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Monday, June 2, 2025 7:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr. Durve,
I am facing an issue again an again while running the VS 2022 code, pls help me with this, we have the module named as model_zoo not models. Pls tell me how to proceed with this.

 image.png

Warm Regards
Hindvee

On Sat, May 31, 2025 at 12:46 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Thankyou sir! I will try this right away.
> Also I hope you received this google sheet, I have shared edit access to you on your gmail account
>
> Hindvee Vispute - Time data - Intern
>
> Regards
> Hindvee
>
> On Sat, May 31, 2025 at 5:43 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Sure. You need to do the following to make the project work.
> >
> > 1. Download and install VS 2022 Community edition
> > 2. Download and install Anaconda. Install an additional environment Python 3.8.16. Use this version of Python for the project.
> >
> > The project should run at this point.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, May 30, 2025 6:34 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings sir!
> > I went through the google drive folder shared by you. It would be really helpful if you could walk me through it.
> > Best regards
> > Hindvee
> >
> > On Fri, May 30, 2025 at 11:41 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > You can download the model, a working VC++ (VS 2022) project and image here.
> > >
> > > https://drive.google.com/drive/folders/1mLBDjJR3fFKs8Brr_nNj2z9YyI89nruE?usp=sharing
> > >
> > > Kishor Durve
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Friday, May 30, 2025 12:50 PM

To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings sir!
Pls do share the model over drive I will try it out. Also I request you to share more image database so that I can start working on it.
I will start work from today itself then.

Regards
Hindvee

On Fri, 30 May 2025, 06:11 Kishor Durve, <kishor.durve@outlook.com> wrote:

> We have found that using a super resolution Pytorch model to get a larger image results in significant noise removal. Perhaps it is a by-product.
>
> You can try it out. I have the model and can share it on Google Drive if you want to try it out. We have tried the models available in OpenCV DNN but they do not seem to give the same results.
>
> Regards,
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Thursday, May 29, 2025 5:28 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Dear Sir,
>
> Thank you for the confirmation. I will maintain the time data on a Google Sheet and ensure it is updated regularly.
>
> Looking forward to starting from Monday, 2nd June.
>
> Best regards,
> Hindvee
>
> On Thu, May 29, 2025 at 2:53 PM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Hello Hindvee,
> >
> > Welcome aboard!
> >
> > You can keep the time data on Google Sheet.
> >
> > Regards,
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Thursday, May 29, 2025 5:12 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings Mr. Durve,

Thank you very much for giving me the opportunity to contribute to Guangdong Longtian Smart Instruments Co. Ltd. through this exciting project. I am pleased to accept the offer and look forward to working on the development of the AI-assisted line detection tool.

With your permission, I would like to begin the internship from the coming Monday, i.e., 2nd June 2025.

Additionally, I wanted to ask about the preferred method for tracking and sharing the time I spend on the project. Would you like me to maintain a shared timesheet (e.g., a Google Sheet or Excel file), or would you prefer weekly time summaries via email or any other format?

Looking forward to your guidance on this.

Thank you once again for this opportunity.

Warm regards,
Hindvee Rajesh Vispute

On Thu, May 29, 2025 at 8:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have attached the on-boarding letter.
>
> Please let me know if you have any queries.
>
> Regards,
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Saturday, May 10, 2025 3:46 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings Mr. Durve,
>
> Thank you for your detailed response and clarification. I now clearly understand that the goal is to enhance line detection using AI within known ROIs, leveraging the expected position of endpoints.
>
> To support this objective, I've explored and implemented several **AI-based approaches**, any of which can be used to complement or improve upon conventional methods:
>
> 1. **CNN-Based Line Presence Classifier (NB-1):** Identifies whether a line exists within a given patch/ROI—useful for filtering candidate regions before applying traditional detection.
> 2. **CNN Regression Model (NB-2):** Predicts line endpoints directly from the image, offering robust localization even in low-contrast or noisy scenarios.
> 3. **Denoising Autoencoder (NB-5):** Enhances faint or broken lines by cleaning up noise, thereby boosting the effectiveness of existing edge or Hough-based methods.
> 4. **Vision Transformer Model (NB-4):** Designed for fine-grained classification tasks and can be

adapted for future objectives like surface defect or anomaly detection.

These methods are independently deployable and can be integrated selectively depending on the specific inspection challenge or ROI structure. I'd be happy to align with your ROI-driven strategy and customize accordingly.

Regarding the stipend, I agree with the proposed Rs. 200/hour base rate and performance-based bonus, and I'm enthusiastic about the opportunity to collaborate further on defect detection and shape anomaly tasks in the future.

I have attached screenshots of the code files I executed today. I have the names of the notebooks (NB) with NB-1, NB-2,NB-4 and NB-5 respectively.

NB-1
NB-2
NB-4
NB-5

Please let me know if you'd like me to share these outputs or run a quick live demonstration.

Warm regards,
Hindvee

On Fri, May 9, 2025 at 6:03 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> I think there is a mismatch in understanding. We are already using conventional line detection techniques (various edge detection operators and Hough Transform and other methods for line fitment) .
>
> What we are looking to get from you is AI based / AI assisted line detection. Since we know the probable position of the line beforehand we can use this information to find a line. For example, say we know that the line end points are expected to be (x1, y1) and (x2, y2), we can construct a small rectangle (ROI) using these points and search for a line within this ROI.
>
> As for the stipend it depends both on the time you spend on the project and the results you produce. It will comprise of an hourly rate and a result linked amount. I suggest we have an hourly rate of Rs. 200/hour and a matching amount linked to the results at the end of the project. So, if you are able to produce satisfactory results it will come to Rs. 400/hour. Also, if the results are satisfactory, we can take this forward next year and work on other issues like surface defect detection, detection of shape anomalies etc.
>
> Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 8, 2025 5:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr Durve,
I have the problem now so I will focus on faint line detection for the first step. I can proceed using Sobel, Scharr, and Canny edge detectors provided by OpenCV. First I will preprocess the images and then move to edge detection using the above 3 detectors. I will carry out line detection using Hough Transform, and further we will filter out short or irrelevant lines based on length and their angle of orientation. Finally, we will draw the detected lines on the images using cv.line()

I can devote 14 hrs per week to this task for 2 months, we can extend the time if required. Also, I wanted to ask what stipend you can provide me. I believe I can complete the task in the above time frame. Thank you for your time.

Warm regards
Hindvee

On Thu, May 8, 2025 at 7:18 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Your analysis seems good.
>
> At this point we are not interested in detecting defects (though at a later stage we may be), and want to detect the faint lines only.
>
> Note that though I have annotated only a single line in the image I sent to you yesterday, we need to detect edge lines of all vertical stripes.
>
> I have attached 2 sets of 20 images each named 'Good Images' and 'Bad Images' - images in which it is mostly easy to detect lines by the conventional image processing method and those in which lines are faint and hard to detect. These images may help you get a better understanding of the problem.
>
> Please also check the following issues:
> 1. How much time will you be able to devote to this task?
> 2. Will you be able to complete the task within the available time?
>
> There are other issues like time taken by the AI route, integration of AI line detect code in VC++ project, further training / fine tuning of model, applicability of the method to detect faint circles etc. But we will get into them later once the basic method is proven.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, May 7, 2025 3:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings, Mr. Durve,
Thank you for reviewing my proposal and for sharing the sample images. By looking at the images I came across the specific challenges faced by the inspection system, like detecting faint lines, subtle surface variations, and fine defects that can be missed by traditional image processing. Based on these observations I suggest that we can do the following:-

1. **Hybrid approach**: Objective - Combine the speed of OpenCV line detection (example Hough Transform) with deep learning's pattern recognition capability.
Approach
- Use OpenCV to pre-locate candidate regions and feed them into a deep model for verification.
- This can speed up inference and increase resistance to false positives.

2. **Deep learning-based fine defect detection**: Objective: Improve the detection of faint, low contrast line defects and localized surface anomalies.
Approach
- Train a model, such as YOLOv8 or U-Net, tailored to your defect patterns using the annotated images.
- To improve feature visibility, add contrast enhancement.
- Assess using detection accuracy and intersection over union (IoU) to compare to baselines based on OpenCV.

I can begin with a proof-of-concept model using the images shared by you. Thank you for sharing the opportunity. I am excited to get a chance to contribute to the advancement in inspection technology.

Warm regards
Hindvee Vispute

On Wed, 7 May 2025, 09:31 Kishor Durve, <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have gone thru the document uploaded by you.
>
> It looks ok to me.
>
> I can have attached a few images so you will get a better idea of the requirements.
>
> Kishor Durve

Hindvee Vispute <hindveevispute@gmail.com> Wed, Jul 23, 2025 at 3:20 AM
To: Kishor Durve <kishor.durve@outlook.com>

Hello sir! Could you please share ur insights and views on the above mail thread, it shall be very helpful to get to know your feedback and move forward with the project.

Warm Regards
Hindvee

On Fri, Jul 11, 2025 at 3:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir!
>
> Following up on my previous email regarding edge detection, I wanted to share the progress I've made since then. I have been focusing on using the Holistically-Nested Edge Detection (HED) method. I have now:
>
> CODE FILE
>
> 1. Successfully applied the HED process to a folder containing our images to generate edge maps for the entire dataset.
>
> 2. Trained a custom autoencoder model designed to refine and correct the lines in these images.
>
> 3. Applied the trained autoencoder to the HED output images to produce corrected versions.
>
> 4. Started working on post-processing techniques.
>
> I am currently experimenting with different parameters and methods to achieve the best results for this step. I am continuing to refine the line clarification process and will share updated results once I have finalized the post-processing approach.
>
> warm regards
>
> Hindvee

On Wed, Jul 9, 2025 at 12:23 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir! I am trying the pre-trained model for detecting edges in our images; I have tried Holistically-Nested Edge Detection (HED) and DexiNed. I am attaching the output, I am going forward with HED to get better results.
>
> Warm regards
> Hindvee

On Mon, Jul 7, 2025 at 10:37 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir!
> I am sharing the first 5 difference images of the three codes that I mentioned previously. Please check this whenever you have time.
>
> Warm regards
> Hindvee

On Mon, Jul 7, 2025 at 6:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> I am unable to give my view of the approach as I can't view the images. I can give my opinion if you send me difference images. In any case, I think you need to have difference images as a metric in addition to the Laplacian.
>
> Get Outlook for Android
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Monday, July 7, 2025 7:09:51 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir!
> I have received the stipend, thank you for that! I'd like to share a brief update on the image enhancement work, focusing on noise removal and edge preservation.
> Summary of results:-
> 1. Previous code
> ---Performed basic denoising but introduced over-smoothing.
> ---Laplacian MAD: ~0.0175–0.0192
>
> 2. DnCNN
> ---Improved noise removal and structural retention.
> ---MAD: ~0.0186–0.0202

3. DnCNN + Sharpening (Current approach)
---Added post-processing to enhance edge clarity.
---Consistently improved MAD to ~0.0178–0.0196.
---Visual results show reduced whitening and better edge definition.

This confirms that the current setup provides better denoising while preserving fine image details. Please let me know if you'd like to proceed with further tuning using SSIM or edge-aware loss functions.

Warm regards

Hindvee

On Fri, Jul 4, 2025 at 3:54 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Please keep at it. Use difference image and Laplacian to gauge the effect of the algorithm. I will be able to look at your work only after 15th.
> I have transferred your stipend by GPay.
>
> Get Outlook for Android
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Friday, July 4, 2025 5:37:35 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir!
> This is what I am been working on. I am still working on getting better edges and less whitish images.
>
> warm regards
> Hindvee
>
> On Thu, Jul 3, 2025 at 10:53 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > UPI ID: hindveevispute@okicici
> >
> > Warm regards
> > Hindvee
> >
> > On Wed, 2 Jul 2025, 03:16 Kishor Durve, <kishor.durve@outlook.com> wrote:
> >
> > > I checked the images, but did not notice substantial difference between the before and after images. You can subtract them (after aligning) and check.
> > > As regards payment of stipend please let me know how much we need to pay and I will transfer it on GPay.
> > > I am travelling again and am out of office. I will be back only on 15th. My responses to your mails may be delayed.
> > >
> > > Get Outlook for Android
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Tuesday, July 1, 2025 1:51:45 AM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection
> > >
> > > Hello sir!
> > >
> > > I was eager to get insights from you about the above shared images.Please do have a look as soon as possible. Also, I wanted to know whether the stipend will be credited at the end of the month or the project's end. It would be very kind of you if I could get the stipend at the end of the month.
> > >
> > > Thanks and regards
> > > Hindvee
> > >
> > > On Mon, Jun 30, 2025 at 10:33 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > >
> > > > Good morning sir!
> > > >
> > > > Did you get any chance to review the above images? Please share your suggestions and insights with me.

Warm regards
Hindvee

On Thu, Jun 26, 2025 at 11:09 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Here I am attaching the processed images.
>
> Warm regards
> Hindvee
>
> On Thu, Jun 26, 2025 at 8:08 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > I am unable to run the files in Colab and generate images. Can you please send me the processed images on mail - .png / .jpeg files, not .ipynb files.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Thursday, June 26, 2025 4:48 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Hello sir, I have attached the results for png 1,2,3 and 4. Pls share ur views and insights.
> >
> > Warm Regards
> > Hindvee
> >
> > On Tue, Jun 24, 2025 at 11:29 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> >
> > > Hello sir,
> > > I am attaching the .ipynb file, it has the images
> > >
> > > warm regards
> > > Hindvee
> > >
> > > On Tue, Jun 24, 2025 at 5:48 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> > >
> > > > Please share the images for visual comparison.
> > > >
> > > > Kishor Durve
> > > >
> > > > ---
> > > >
> > > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > > **Sent:** Tuesday, June 24, 2025 5:13 AM
> > > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > > **Subject:** Re: AI assisted edge detection
> > > >
> > > > Hello Durve sir,
> > > >
> > > > I wanted to share an update on the denoising output. I noticed earlier that while the model was doing a good job removing noise, it was also making the image a bit too smooth — the sharpness had dropped quite a bit:
> > > >
> > > > - Noisy Image: 33.72
> > > > - Denoised Image: 6.92
> > > > - Clean Image (Ground Truth): 26.49
> > > >
> > > > To improve this, I added a sharpening filter after the autoencoder output and tested two variants:
> > > >
> > > > 1. A **strong sharpening kernel**, which raised the denoised sharpness to **112.40,** but it felt a bit overdone and introduced some artifacts.
> > > > 2. A **softer kernel**, which gave a cleaner and more natural result, with denoised sharpness at **54.04**. This seems like a balanced point where edges are clearer without compromising visual quality.
> > > >
> > > > If you'd like, I can also share the output images (clean vs. denoised) for a visual comparison.

Warm regards

Hindvee

image.png

On Fri, Jun 20, 2025 at 12:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir!
> Based on the latest code I calculated the sharpness using Laplacian variance. I obtained the following results as those attached as image.
> I think that while denoising, though the noise was removed, but the model blurred or over smoothened the image. I may have lost some meaningful edge details in the process. But the sharpness value of Clean image is lower than the noisy image (because it has no random noise), but high enough to retain clear edges and structure. So I am working next on increasing the sharpness value while keeping the noise low. Please do share your views with me.
>
> Warm regards
> Hindvee
>
> image.png
>
> On Thu, 19 Jun 2025, 13:42 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > I have seen the processed images. I think it is good at this point to quantify 'sharpness' of an image. It will give us a measure to compare different methods / models.
> >
> > There can be several approaches to do this. We use Opencv::Laplacian() to judge image sharpness. But you can try other approaches.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Thursday, June 19, 2025 2:52 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Hello sir,
> >
> > I've completed the full-image denoising as discussed for the images `5.png` and `6.png`. This time, the model was trained and applied directly on the complete images (cropped to a common size of **570×588**) instead of small patches, as per your suggestion. The denoising autoencoder was able to reduce background noise and enhance structural features, particularly the edges of the circles, which were initially faint or inconsistent. The model output shows a noticeable improvement in clarity while preserving the original patterns.
> >
> > I've attached the processed result image along with this mail. Please let me know your thoughts on this file.
> >
> > Warm regards,
> > Hindvee
> >
> > On Wed, Jun 18, 2025 at 12:33 PM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > I have checked the file. These two images contain circles as primary features. Also, the patches in the file seem to be for a small part of the original image. If you have processed the original images (5.png and 6.png) please send me the processed images on mail. If you have processed only parts of these images, I suggest you process the full images and send me the results.

Similarly, in the previous exercise please process complete images. Most of the set are of size 500x500 pixel, so it should not consume too much time.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, June 18, 2025 1:44 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Sir, please let me know your views on the above .ipynb file.
Warm Regards
Hindvee

On Wed, Jun 18, 2025 at 1:56 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir,
> I have processed the 5.png and 6.png, I faced lower accuracy rate earlier. After fixing the earlier issues with mismatched data sizes and high Binary Crossentropy loss (~0.6098), the model is now working smoothly. It reached a much lower loss of just 0.0528 in 5 epochs, and both training and validation losses settled nicely by the 2nd or 3rd epoch without signs of overfitting. The denoised images look very clean and closely match the original ones, even when the input images had different levels of noise. I am attaching the results in the attached file.
> Warm regards
> Hindvee
>
> On Tue, Jun 17, 2025 at 7:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Sorry! Forgot the attachment.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Kishor Durve <kishor.durve@outlook.com>
> > **Sent:** Tuesday, June 17, 2025 9:46 AM
> > **To:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Good to know that you are making progress at a nice clip.
> >
> > I saw the images and the difference between before / after can be seen. Can you please process the images attached to this mail and send me the results? The images are similar to those I had sent earlier, the exception being 5.png and 6.png. In these last two images the challenge is to detect circles whose edges are not well defined and the edge quality may vary from image to image. This is to test the effectiveness of noise removal / sharpening.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Tuesday, June 17, 2025 3:06 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Hello sir,
> > I have completed the training and evaluation of the denoising autoencoder using grayscale image patches taken from the `IMG` folder. The model was trained for 20 epochs on 64×64 patches with binary cross entropy loss and a 10% validation split. During training, the loss reduced steadily and stabilized, with the final reconstruction loss reaching 0.3688. The results show that the model can effectively denoise images and recover clean patterns like faint lines. In the next step, I plan to use the output of the Super-Resolution model as a base to evaluate the accuracy and quality of the denoising process, to better understand how both

methods contribute to image enhancement. I am attaching the .ipynb file for your reference and also the IMG Folder which I just mentioned.

Warm regards
Hindvee

IMG FOLDER

Accuracy results:-

image.png

On Fri, Jun 13, 2025 at 5:33 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> The images I sent are big. OpenCV will not be able to handle them as the max size allowed is 32K x 32K. The crash you are facing could be due to this reason.
>
> You can crop out small patches from the big images and use them for trials.
>
> Kishor Durve
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 11, 2025 1:37 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Thank you for sharing more images. I have updated the work on the sheets.
>
> Warm regards
> Hindvee
>
> On Sat, Jun 7, 2025 at 2:51 PM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > I have uploaded more images. You can crop out patches from them. One of the images has many boards which will give many useful patches.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Saturday, June 7, 2025 4:36 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings sir,
> > It would be great if I get more images, that would expand my dataset. Thank you for your previous email.
> > I also wanted tk inform you that I am travelling and will be able to resume work by Tuesday. I request you to kindly allow my leave.
> >
> > Warm Regards
> > Hindvee
> >
> > On Fri, 6 Jun 2025, 11:31 Kishor Durve, <kishor.durve@outlook.com> wrote:
> >
> > > I am not going to be of much help to you as I have limited experience in this field. Also, what little I know was gained by applying available models in OpenCV's DNN module.

There are two images available in the images folder. You may crop out patches from them for the Training / test sets. We use ImageJ and IrfanView for such work as MS Paint cannot handle such big images. If you need more images please let me know.

Kishor Durve
Duravision Systems
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, June 6, 2025 5:24 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings sir,
I hope you're doing well. I've just finished training the CNN-based line presence classifier, and I'm attaching the loss graph for your review. The model seems to have learned quite quickly, with both training and validation losses staying very close to zero throughout. There was a small spike in training loss around epoch 80, but validation stayed stable, so it might just be a minor fluctuation.

Since the dataset is quite small and the model converged very early, I'm unsure how best to move forward.According to you should I try to get more data to avoid overfitting. Please guide me towards a direction in which I can make this model more effective.

Warm regards

Hindvee

On Wed, Jun 4, 2025 at 6:42 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Yes, it looks ok to me. We will take a relook once you make more progress.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 4, 2025 4:26 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Additionally, Sir, I would like to confirm if the current format of the Google Sheet meets your expectations. Please let me know if you would prefer any modifications, such as a more detailed breakdown of the work completed.
>
> Warm Regards
> Hindvee
>
> On Wed, Jun 4, 2025 at 1:34 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Greetings Mr. Durve,
> >
> > Thanks for sharing the super-resolution model. I've set it up and tested it, the results inspiring. It does a good job cleaning up noise and improving contrast, especially in the faint line areas. As we discussed, I'll now add this to the bigger pipeline along with the AI modules (NB-1 to NB-5). The super-resolution output can help improve both the CNN classifier and regression models. I'll also check how it works together with NB-5, the denoising autoencoder, to see if we get even better results.

Please let me know if you want me to run it on any specific ROI samples or start putting together comparison results.

Warm regards
Hindvee

On Tue, Jun 3, 2025 at 9:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have file network.py in folder 'models'. This file has the RRDBNet implementaion.
>
> Kishor Durve
> Duravision Systems
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Monday, June 2, 2025 7:15 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings Mr. Durve,
> I am facing an issue again an again while running the VS 2022 code, pls help me with this, we have the module named as model_zoo not models. Pls tell me how to proceed with this.
>
> image.png
>
> Warm Regards
> Hindvee
>
> On Sat, May 31, 2025 at 12:46 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Thankyou sir! I will try this right away.
> > Also I hope you received this google sheet, I have shared edit access to you on your gmail account
> >
> > Hindvee Vispute - Time data - Intern
> >
> > Regards
> > Hindvee
> >
> > On Sat, May 31, 2025 at 5:43 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > Sure. You need to do the following to make the project work.
> > >
> > > 1. Download and install VS 2022 Community edition
> > > 2. Download and install Anaconda. Install an additional environment Python 3.8.16. Use this version of Python for the project.
> > >
> > > The project should run at this point.
> > >
> > > Kishor Durve
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Friday, May 30, 2025 6:34 PM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection
> > >
> > > Greetings sir!
> > > I went through the google drive folder shared by you. It would be really helpful if you could walk me through it.
> > > Best regards
> > > Hindvee

On Fri, May 30, 2025 at 11:41 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> You can download the model, a working VC++ (VS 2022) project and image here.
>
> https://drive.google.com/drive/folders/1mLBDjJR3fFKs8Brr_nNj2z9YyI89nruE?usp=sharing
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Friday, May 30, 2025 12:50 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings sir!
> Pls do share the model over drive I will try it out. Also I request you to share more image database so that I can start working on it.
> I will start work from today itself then.
>
> Regards
> Hindvee
>
> On Fri, 30 May 2025, 06:11 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > We have found that using a super resolution Pytorch model to get a larger image results in significant noise removal. Perhaps it is a by-product.
> >
> > You can try it out. I have the model and can share it on Google Drive if you want to try it out. We have tried the models available in OpenCV DNN but they do not seem to give the same results.
> >
> > Regards,
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Thursday, May 29, 2025 5:28 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Dear Sir,
> >
> > Thank you for the confirmation. I will maintain the time data on a Google Sheet and ensure it is updated regularly.
> >
> > Looking forward to starting from Monday, 2nd June.
> >
> > Best regards,
> > Hindvee
> >
> > On Thu, May 29, 2025 at 2:53 PM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > Hello Hindvee,

Welcome aboard!

You can keep the time data on Google Sheet.

Regards,

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 29, 2025 5:12 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr. Durve,

Thank you very much for giving me the opportunity to contribute to Guangdong Longtian Smart Instruments Co. Ltd. through this exciting project. I am pleased to accept the offer and look forward to working on the development of the AI-assisted line detection tool.

With your permission, I would like to begin the internship from the coming Monday, i.e., 2nd June 2025.

Additionally, I wanted to ask about the preferred method for tracking and sharing the time I spend on the project. Would you like me to maintain a shared timesheet (e.g., a Google Sheet or Excel file), or would you prefer weekly time summaries via email or any other format?

Looking forward to your guidance on this.

Thank you once again for this opportunity.

Warm regards,
Hindvee Rajesh Vispute

On Thu, May 29, 2025 at 8:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have attached the on-boarding letter.
>
> Please let me know if you have any queries.
>
> Regards,
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Saturday, May 10, 2025 3:46 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings Mr. Durve,
>
> Thank you for your detailed response and clarification. I now clearly understand that the goal is to enhance line detection using AI within known ROIs, leveraging the expected position of endpoints.
>
> To support this objective, I've explored and implemented several **AI-based approaches**, any of

which can be used to complement or improve upon conventional methods:

1. **CNN-Based Line Presence Classifier (NB-1):** Identifies whether a line exists within a given patch/ROI—useful for filtering candidate regions before applying traditional detection.
2. **CNN Regression Model (NB-2):** Predicts line endpoints directly from the image, offering robust localization even in low-contrast or noisy scenarios.
3. **Denoising Autoencoder (NB-5):** Enhances faint or broken lines by cleaning up noise, thereby boosting the effectiveness of existing edge or Hough-based methods.
4. **Vision Transformer Model (NB-4):** Designed for fine-grained classification tasks and can be adapted for future objectives like surface defect or anomaly detection.

These methods are independently deployable and can be integrated selectively depending on the specific inspection challenge or ROI structure. I'd be happy to align with your ROI-driven strategy and customize accordingly.

Regarding the stipend, I agree with the proposed Rs. 200/hour base rate and performance-based bonus, and I'm enthusiastic about the opportunity to collaborate further on defect detection and shape anomaly tasks in the future.

I have attached screenshots of the code files I executed today. I have the names of the notebooks (NB) with NB-1, NB-2,NB-4 and NB-5 respectively.

NB-1
NB-2
NB-4
NB-5

Please let me know if you'd like me to share these outputs or run a quick live demonstration.

Warm regards,
Hindvee

On Fri, May 9, 2025 at 6:03 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> I think there is a mismatch in understanding. We are already using conventional line detection techniques (various edge detection operators and Hough Transform and other methods for line fitment) .
>
> What we are looking to get from you is AI based / AI assisted line detection. Since we know the probable position of the line beforehand we can use this information to find a line. For example, say we know that the line end points are expected to be (x1, y1) and (x2, y2), we can construct a small rectangle (ROI) using these points and search for a line within this ROI.

As for the stipend it depends both on the time you spend on the project and the results you produce. It will comprise of an hourly rate and a result linked amount. I suggest we have an hourly rate of Rs. 200/hour and a matching amount linked to the results at the end of the project. So, if you are able to produce satisfactory results it will come to Rs. 400/hour. Also, if the results are satisfactory, we can take this forward next year and work on other issues like surface defect detection, detection of shape anomalies etc.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 8, 2025 5:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr Durve,
I have the problem now so I will focus on faint line detection for the first step. I can proceed using Sobel, Scharr, and Canny edge detectors provided by OpenCV. First I will preprocess the images and then move to edge detection using the above 3 detectors. I will carry out line detection using Hough Transform, and further we will filter out short or irrelevant lines based on length and their angle of orientation. Finally, we will draw the detected lines on the images using cv.line()

I can devote 14 hrs per week to this task for 2 months, we can extend the time if required. Also, I wanted to ask what stipend you can provide me. I believe I can complete the task in the above time frame. Thank you for your time.

Warm regards
Hindvee

On Thu, May 8, 2025 at 7:18 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Your analysis seems good.
>
> At this point we are not interested in detecting defects (though at a later stage we may be), and want to detect the faint lines only.
>
> Note that though I have annotated only a single line in the image I sent to you yesterday, we need to detect edge lines of all vertical stripes.
>
> I have attached 2 sets of 20 images each named 'Good Images' and 'Bad Images' - images in which it is mostly easy to detect lines by the conventional image processing method and those in which lines are faint and hard to detect. These

images may help you get a better understanding of the problem.

Please also check the following issues:

1. How much time will you be able to devote to this task?
2. Will you be able to complete the task within the available time?

There are other issues like time taken by the AI route, integration of AI line detect code in VC++ project, further training / fine tuning of model, applicability of the method to detect faint circles etc. But we will get into them later once the basic method is proven.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, May 7, 2025 3:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings, Mr. Durve,
Thank you for reviewing my proposal and for sharing the sample images. By looking at the images I came across the specific challenges faced by the inspection system, like detecting faint lines, subtle surface variations, and fine defects that can be missed by traditional image processing. Based on these observations I suggest that we can do the following:-

1. **Hybrid approach**: Objective - Combine the speed of OpenCV line detection (example Hough Transform) with deep learning's pattern recognition capability.
Approach
- Use OpenCV to pre-locate candidate regions and feed them into a deep model for verification.
- This can speed up inference and increase resistance to false positives.

2. **Deep learning-based fine defect detection**: Objective: Improve the detection of faint, low contrast line defects and localized surface anomalies.
Approach
- Train a model, such as YOLOv8 or U-Net, tailored to your defect patterns using the annotated images.
- To improve feature visibility, add contrast enhancement.
- Assess using detection accuracy and intersection over union (IoU) to compare to baselines based on OpenCV.

I can begin with a proof-of-concept model using the images shared by you. Thank you for sharing the opportunity. I am excited to get a chance to contribute to the advancement in inspection technology.

Warm regards

Hindvee Vispute

On Wed, 7 May 2025, 09:31 Kishor Durve, <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have gone thru the document uploaded by you.
>
> It looks ok to me.
>
> I can have attached a few images so you will get a better idea of the requirements.
>
> Kishor Durve

Hindvee Vispute <hindveevispute@gmail.com> Fri, Jul 25, 2025 at 3:16 AM
To: Kishor Durve <kishor.durve@outlook.com>

Hello sir ! This is an output image obtained from the HED model and similar image I have obtained from DexiNed, could u pls tell i am moving in the right direction or should I make improvements or change in the plan of action?

warm regards
Hindvee

On Wed, Jul 23, 2025 at 3:20 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir! Could you please share ur insights and views on the above mail thread, it shall be very helpful to get to know your feedback and move forward with the project.
>
> Warm Regards
> Hindvee
>
> On Fri, Jul 11, 2025 at 3:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir!
> >
> > Following up on my previous email regarding edge detection, I wanted to share the progress I've made since then. I have been focusing on using the Holistically-Nested Edge Detection (HED) method. I have now:
> >
> > CODE FILE
> >
> > 1. Successfully applied the HED process to a folder containing our images to generate edge maps for the entire dataset.
> >
> > 2. Trained a custom autoencoder model designed to refine and correct the lines in these images.
> >
> > 3. Applied the trained autoencoder to the HED output images to produce corrected versions.
> >
> > 4. Started working on post-processing techniques.
> >
> > I am currently experimenting with different parameters and methods to achieve the best results for this step. I am continuing to refine the line clarification process and will share updated results once I have finalized the post-processing approach.
> >
> > warm regards
> >
> > Hindvee
> >
> > On Wed, Jul 9, 2025 at 12:23 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> >
> > > Hello sir! I am trying the pre-trained model for detecting edges in our images; I have tried Holistically-Nested Edge Detection (HED) and DexiNed. I am attaching the output, I am going forward with HED to get better results.
> > >
> > > Warm regards
> > > Hindvee
> > >
> > > On Mon, Jul 7, 2025 at 10:37 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > >
> > > > Hello sir!

I am sharing the first 5 difference images of the three codes that I mentioned previously. Please check this whenever you have time.

Warm regards
Hindvee

On Mon, Jul 7, 2025 at 6:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> I am unable to give my view of the approach as I can't view the images. I can give my opinion if you send me difference images. In any case, I think you need to have difference images as a metric in addition to the Laplacian.
>
> Get Outlook for Android
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Monday, July 7, 2025 7:09:51 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir!
> I have received the stipend, thank you for that! I'd like to share a brief update on the image enhancement work, focusing on noise removal and edge preservation.
> Summary of results:-
> 1. Previous code
> ---Performed basic denoising but introduced over-smoothing.
> ---Laplacian MAD: ~0.0175–0.0192
>
> 2. DnCNN
> ---Improved noise removal and structural retention.
> ---MAD: ~0.0186–0.0202
>
> 3. DnCNN + Sharpening (Current approach)
> ---Added post-processing to enhance edge clarity.
> ---Consistently improved MAD to ~0.0178–0.0196.
> ---Visual results show reduced whitening and better edge definition.
>
> This confirms that the current setup provides better denoising while preserving fine image details. Please let me know if you'd like to proceed with further tuning using SSIM or edge-aware loss functions.
>
> Warm regards
>
> Hindvee
>
> On Fri, Jul 4, 2025 at 3:54 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Please keep at it. Use difference image and Laplacian to gauge the effect of the algorithm. I will be able to look at your work only after 15th.
> > I have transferred your stipend by GPay.
> >
> > Get Outlook for Android
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, July 4, 2025 5:37:35 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Hello sir!
> > This is what I am been working on. I am still working on getting better edges and less whitish images.
> >
> > warm regards
> > Hindvee
> >
> > On Thu, Jul 3, 2025 at 10:53 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> >
> > > UPI ID: hindveevispute@okicici
> > >
> > > Warm regards
> > > Hindvee

On Wed, 2 Jul 2025, 03:16 Kishor Durve, <kishor.durve@outlook.com> wrote:

I checked the images, but did not notice substantial difference between the before and after images. You can subtract them (after aligning) and check.
As regards payment of stipend please let me know how much we need to pay and I will transfer it on GPay.
I am travelling again and am out of office. I will be back only on 15th. My responses to your mails may be delayed.

Get Outlook for Android

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, July 1, 2025 1:51:45 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!

I was eager to get insights from you about the above shared images.Please do have a look as soon as possible. Also, I wanted to know whether the stipend will be credited at the end of the month or the project's end. It would be very kind of you if I could get the stipend at the end of the month.

Thanks and regards
Hindvee

On Mon, Jun 30, 2025 at 10:33 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Good morning sir!

Did you get any chance to review the above images? Please share your suggestions and insights with me.

Warm regards
Hindvee

On Thu, Jun 26, 2025 at 11:09 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Here I am attaching the processed images.

Warm regards
Hindvee

On Thu, Jun 26, 2025 at 8:08 AM Kishor Durve <kishor.durve@outlook.com> wrote:

I am unable to run the files in Colab and generate images. Can you please send me the processed images on mail - .png / .jpeg files, not .ipynb files.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, June 26, 2025 4:48 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir, I have attached the results for png 1,2,3 and 4. Pls share ur views and insights.

Warm Regards
Hindvee

On Tue, Jun 24, 2025 at 11:29 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Hello sir,
I am attaching the .ipynb file, it has the images

warm regards
Hindvee

On Tue, Jun 24, 2025 at 5:48 AM Kishor Durve <kishor.durve@outlook.com> wrote:

Please share the images for visual comparison.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, June 24, 2025 5:13 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello Durve sir,

I wanted to share an update on the denoising output. I noticed earlier that while the model was doing a good job removing noise, it was also making the image a bit too smooth — the sharpness had dropped quite a bit:

- Noisy Image: 33.72
- Denoised Image: 6.92
- Clean Image (Ground Truth): 26.49

To improve this, I added a sharpening filter after the autoencoder output and tested two variants:

1. A **strong sharpening kernel**, which raised the denoised sharpness to **112.40,** but it felt a bit overdone and introduced some artifacts.
2. A **softer kernel**, which gave a cleaner and more natural result, with denoised sharpness at **54.04**. This seems like a balanced point where edges are clearer without compromising visual quality.

If you'd like, I can also share the output images (clean vs. denoised) for a visual comparison.

Warm regards

Hindvee

image.png

On Fri, Jun 20, 2025 at 12:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir!
> Based on the latest code I calculated the sharpness using Laplacian variance. I obtained the following results as those attached as image.
> I think that while denoising, though the noise was removed, but the model blurred or over smoothened the image. I may have lost some meaningful edge details in the process. But the sharpness value of Clean image is lower than the noisy image (because it has no random noise), but high enough to retain clear edges and structure. So I am working next on increasing the sharpness value while keeping the noise low. Please do share your views with me.
>
> Warm regards
> Hindvee
>
> image.png
>
> On Thu, 19 Jun 2025, 13:42 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > I have seen the processed images. I think it is good at this point to quantify 'sharpness' of an image. It will give us a measure to compare different methods / models.
> >
> > There can be several approaches to do this. We use Opencv::Laplacian() to judge image sharpness. But you can try other approaches.
> >
> > Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, June 19, 2025 2:52 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir,

I've completed the full-image denoising as discussed for the images `5.png` and `6.png`. This time, the model was trained and applied directly on the complete images (cropped to a common size of **570×588**) instead of small patches, as per your suggestion. The denoising autoencoder was able to reduce background noise and enhance structural features, particularly the edges of the circles, which were initially faint or inconsistent. The model output shows a noticeable improvement in clarity while preserving the original patterns.

I've attached the processed result image along with this mail. Please let me know your thoughts on this file.

Warm regards,
Hindvee

On Wed, Jun 18, 2025 at 12:33 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have checked the file. These two images contain circles as primary features. Also, the patches in the file seem to be for a small part of the original image. If you have processed the original images (5.png and 6.png) please send me the processed images on mail. If you have processed only parts of these images, I suggest you process the full images and send me the results.
>
> Similarly, in the previous exercise please process complete images. Most of the set are of size 500x500 pixel, so it should not consume too much time.
>
> Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, June 18, 2025 1:44 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Sir, please let me know your views on the above .ipynb file.
Warm Regards
Hindvee

On Wed, Jun 18, 2025 at 1:56 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir,
> I have processed the 5.png and 6.png, I faced lower accuracy rate earlier. After fixing the earlier issues with mismatched data sizes and high Binary Crossentropy loss (~0.6098), the model is now working smoothly. It reached a much lower loss of just 0.0528 in 5 epochs, and both training and validation losses settled nicely by the 2nd or 3rd epoch without signs of overfitting. The denoised images look very clean and closely match the original ones, even when the input images had different levels of noise. I am attaching the results in the attached file.
> Warm regards
> Hindvee
>
> On Tue, Jun 17, 2025 at 7:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Sorry! Forgot the attachment.
> >
> > Kishor Durve

From: Kishor Durve <kishor.durve@outlook.com>
Sent: Tuesday, June 17, 2025 9:46 AM
To: Hindvee Vispute <hindveevispute@gmail.com>
Subject: Re: AI assisted edge detection

Good to know that you are making progress at a nice clip.

I saw the images and the difference between before / after can be seen. Can you please process the images attached to this mail and send me the results? The images are similar to those I had sent earlier, the exception being 5.png and 6.png. In these last two images the challenge is to detect circles whose edges are not well defined and the edge quality may vary from image to image. This is to test the effectiveness of noise removal / sharpening.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, June 17, 2025 3:06 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir,
I have completed the training and evaluation of the denoising autoencoder using grayscale image patches taken from the `IMG` folder. The model was trained for 20 epochs on 64×64 patches with binary cross entropy loss and a 10% validation split. During training, the loss reduced steadily and stabilized, with the final reconstruction loss reaching 0.3688. The results show that the model can effectively denoise images and recover clean patterns like faint lines. In the next step, I plan to use the output of the Super-Resolution model as a base to evaluate the accuracy and quality of the denoising process, to better understand how both methods contribute to image enhancement. I am attaching the .ipynb file for your reference and also the IMG Folder which I just mentioned.

Warm regards
Hindvee

IMG FOLDER

Accuracy results:-

image.png

On Fri, Jun 13, 2025 at 5:33 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> The images I sent are big. OpenCV will not be able to handle them as the max size allowed is 32K x 32K. The crash you are facing could be due to this reason.
>
> You can crop out small patches from the big images and use them for trials.
>
> Kishor Durve
> 9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, June 11, 2025 1:37 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Thank you for sharing more images. I have updated the work on the sheets.

Warm regards
Hindvee

On Sat, Jun 7, 2025 at 2:51 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have uploaded more images. You can crop out patches from them. One of the images has many boards which will give many useful patches.
>
> Kishor Durve
> Duravision Systems
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Saturday, June 7, 2025 4:36 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings sir,
> It would be great if I get more images, that would expand my dataset. Thank you for your previous email.
> I also wanted tk inform you that I am travelling and will be able to resume work by Tuesday. I request you to kindly allow my leave.
>
> Warm Regards
> Hindvee
>
> On Fri, 6 Jun 2025, 11:31 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > I am not going to be of much help to you as I have limited experience in this field. Also, what little I know was gained by applying available models in OpenCV's DNN module.
> >
> > There are two images available in the images folder. You may crop out patches from them for the Training / test sets. We use ImageJ and IrfanView for such work as MS Paint cannot handle such big images. If you need more images please let me know.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, June 6, 2025 5:24 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings sir,
> > I hope you're doing well. I've just finished training the CNN-based line presence classifier, and I'm attaching the loss graph for your review. The model seems to have learned quite quickly, with both training and validation losses staying very close to zero throughout. There was a small spike in training loss around epoch 80, but validation stayed stable, so it might just be a minor fluctuation.
> >
> > Since the dataset is quite small and the model converged very early, I'm unsure how best to move forward.According to you should I try to get more data to avoid overfitting. Please guide me towards a direction in which I can make this model more effective.
> >
> > Warm regards
> >
> > Hindvee

On Wed, Jun 4, 2025 at 6:42 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Yes, it looks ok to me. We will take a relook once you make more progress.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 4, 2025 4:26 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Additionally, Sir, I would like to confirm if the current format of the Google Sheet meets your expectations. Please let me know if you would prefer any modifications, such as a more detailed breakdown of the work completed.
>
> Warm Regards
> Hindvee
>
> On Wed, Jun 4, 2025 at 1:34 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Greetings Mr. Durve,
> >
> > Thanks for sharing the super-resolution model. I've set it up and tested it, the results inspiring. It does a good job cleaning up noise and improving contrast, especially in the faint line areas. As we discussed, I'll now add this to the bigger pipeline along with the AI modules (NB-1 to NB-5). The super-resolution output can help improve both the CNN classifier and regression models. I'll also check how it works together with NB-5, the denoising autoencoder, to see if we get even better results.
> >
> > Please let me know if you want me to run it on any specific ROI samples or start putting together comparison results.
> >
> > Warm regards
> > Hindvee
> >
> > On Tue, Jun 3, 2025 at 9:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > I have file network.py in folder 'models'. This file has the RRDBNet implementaion.
> > >
> > > Kishor Durve
> > > Duravision Systems
> > > 9823164286
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Monday, June 2, 2025 7:15 PM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection
> > >
> > > Greetings Mr. Durve,
> > > I am facing an issue again an again while running the VS 2022 code, pls help me with this, we have the module named as model_zoo not models. Pls tell me how to proceed with this.
> > >
> > >  image.png
> > >
> > > Warm Regards
> > > Hindvee

On Sat, May 31, 2025 at 12:46 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Thankyou sir! I will try this right away.
> Also I hope you received this google sheet, I have shared edit access to you on your gmail account
>
> Hindvee Vispute - Time data - Intern
>
> Regards
> Hindvee

On Sat, May 31, 2025 at 5:43 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Sure. You need to do the following to make the project work.
>
> 1. Download and install VS 2022 Community edition
> 2. Download and install Anaconda. Install an additional environment Python 3.8.16. Use this version of Python for the project.
>
> The project should run at this point.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Friday, May 30, 2025 6:34 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings sir!
> I went through the google drive folder shared by you. It would be really helpful if you could walk me through it.
> Best regards
> Hindvee

On Fri, May 30, 2025 at 11:41 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> You can download the model, a working VC++ (VS 2022) project and image here.
>
> https://drive.google.com/drive/folders/1mLBDjJR3fFKs8Brr_nNj2z9YyI89nruE?usp=sharing
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Friday, May 30, 2025 12:50 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings sir!
> Pls do share the model over drive I will try it out. Also I request you to share more image database so that I can start working on it.
> I will start work from today itself then.
>
> Regards
> Hindvee

On Fri, 30 May 2025, 06:11 Kishor Durve, <kishor.durve@outlook.com> wrote:

> We have found that using a super resolution Pytorch model to get a larger image results in

significant noise removal. Perhaps it is a by-product.

You can try it out. I have the model and can share it on Google Drive if you want to try it out. We have tried the models available in OpenCV DNN but they do not seem to give the same results.

Regards,

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 29, 2025 5:28 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Dear Sir,

Thank you for the confirmation. I will maintain the time data on a Google Sheet and ensure it is updated regularly.

Looking forward to starting from Monday, 2nd June.

Best regards,
Hindvee

On Thu, May 29, 2025 at 2:53 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> Welcome aboard!
>
> You can keep the time data on Google Sheet.
>
> Regards,
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Thursday, May 29, 2025 5:12 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings Mr. Durve,
>
> Thank you very much for giving me the opportunity to contribute to Guangdong Longtian Smart Instruments Co. Ltd. through this exciting project. I am pleased to accept the offer and look forward to working on the development of the AI-assisted line detection tool.
>
> With your permission, I would like to begin the internship from the coming Monday, i.e., 2nd June 2025.
>
> Additionally, I wanted to ask about the preferred method for tracking and sharing the time I spend on the

project. Would you like me to maintain a shared timesheet (e.g., a Google Sheet or Excel file), or would you prefer weekly time summaries via email or any other format?

Looking forward to your guidance on this.

Thank you once again for this opportunity.

Warm regards,
Hindvee Rajesh Vispute

On Thu, May 29, 2025 at 8:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have attached the on-boarding letter.
>
> Please let me know if you have any queries.
>
> Regards,
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Saturday, May 10, 2025 3:46 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings Mr. Durve,
>
> Thank you for your detailed response and clarification. I now clearly understand that the goal is to enhance line detection using AI within known ROIs, leveraging the expected position of endpoints.
>
> To support this objective, I've explored and implemented several **AI-based approaches**, any of which can be used to complement or improve upon conventional methods:
>
> 1. **CNN-Based Line Presence Classifier (NB-1):** Identifies whether a line exists within a given patch/ROI—useful for filtering candidate regions before applying traditional detection.
> 2. **CNN Regression Model (NB-2):** Predicts line endpoints directly from the image, offering robust localization even in low-contrast or noisy scenarios.
> 3. **Denoising Autoencoder (NB-5):** Enhances faint or broken lines by cleaning up noise, thereby boosting the effectiveness of existing edge or Hough-based methods.
> 4. **Vision Transformer Model (NB-4):** Designed for fine-grained classification tasks and can be adapted for future objectives like surface defect or anomaly detection.
>
> These methods are independently deployable and can be integrated selectively depending on the specific inspection challenge or ROI structure. I'd be happy to align with your ROI-driven strategy and customize accordingly.

Regarding the stipend, I agree with the proposed Rs. 200/hour base rate and performance-based bonus, and I'm enthusiastic about the opportunity to collaborate further on defect detection and shape anomaly tasks in the future.

I have attached screenshots of the code files I executed today. I have the names of the notebooks (NB) with NB-1, NB-2,NB-4 and NB-5 respectively.

NB-1
NB-2
NB-4
NB-5

Please let me know if you'd like me to share these outputs or run a quick live demonstration.

Warm regards,
Hindvee

On Fri, May 9, 2025 at 6:03 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> I think there is a mismatch in understanding. We are already using conventional line detection techniques (various edge detection operators and Hough Transform and other methods for line fitment) .
>
> What we are looking to get from you is AI based / AI assisted line detection. Since we know the probable position of the line beforehand we can use this information to find a line. For example, say we know that the line end points are expected to be (x1, y1) and (x2, y2), we can construct a small rectangle (ROI) using these points and search for a line within this ROI.
>
> As for the stipend it depends both on the time you spend on the project and the results you produce. It will comprise of an hourly rate and a result linked amount. I suggest we have an hourly rate of Rs. 200/hour and a matching amount linked to the results at the end of the project. So, if you are able to produce satisfactory results it will come to Rs. 400/hour. Also, if the results are satisfactory, we can take this forward next year and work on other issues like surface defect detection, detection of shape anomalies etc.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Thursday, May 8, 2025 5:15 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection

Greetings Mr Durve,
I have the problem now so I will focus on faint line detection for the first step. I can proceed using Sobel, Scharr, and Canny edge detectors provided by OpenCV. First I will preprocess the images and then move to edge detection using the above 3 detectors. I will carry out line detection using Hough Transform, and further we will filter out short or irrelevant lines based on length and their angle of orientation. Finally, we will draw the detected lines on the images using cv.line()

I can devote 14 hrs per week to this task for 2 months, we can extend the time if required. Also, I wanted to ask what stipend you can provide me. I believe I can complete the task in the above time frame. Thank you for your time.

Warm regards
Hindvee

On Thu, May 8, 2025 at 7:18 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Your analysis seems good.
>
> At this point we are not interested in detecting defects (though at a later stage we may be), and want to detect the faint lines only.
>
> Note that though I have annotated only a single line in the image I sent to you yesterday, we need to detect edge lines of all vertical stripes.
>
> I have attached 2 sets of 20 images each named 'Good Images' and 'Bad Images' - images in which it is mostly easy to detect lines by the conventional image processing method and those in which lines are faint and hard to detect. These images may help you get a better understanding of the problem.
>
> Please also check the following issues:
> 1. How much time will you be able to devote to this task?
> 2. Will you be able to complete the task within the available time?
>
> There are other issues like time taken by the AI route, integration of AI line detect code in VC++ project, further training / fine tuning of model, applicability of the method to detect faint circles etc. But we will get into them later once the basic method is proven.
>
> Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, May 7, 2025 3:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings, Mr. Durve,
Thank you for reviewing my proposal and for sharing the sample images. By looking at the images I came across the specific challenges faced by the inspection system, like detecting faint lines, subtle surface variations, and fine defects that can be missed by traditional image processing. Based on these observations I suggest that we can do the following:-

1. **Hybrid approach**: Objective - Combine the speed of OpenCV line detection (example Hough Transform) with deep learning's pattern recognition capability.
Approach
- Use OpenCV to pre-locate candidate regions and feed them into a deep model for verification.
- This can speed up inference and increase resistance to false positives.

2. **Deep learning-based fine defect detection**: Objective: Improve the detection of faint, low contrast line defects and localized surface anomalies.
Approach
- Train a model, such as YOLOv8 or U-Net, tailored to your defect patterns using the annotated images.
- To improve feature visibility, add contrast enhancement.
- Assess using detection accuracy and intersection over union (IoU) to compare to baselines based on OpenCV.

I can begin with a proof-of-concept model using the images shared by you. Thank you for sharing the opportunity. I am excited to get a chance to contribute to the advancement in inspection technology.

Warm regards
Hindvee Vispute

On Wed, 7 May 2025, 09:31 Kishor Durve, <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have gone thru the document uploaded by you.
>
> It looks ok to me.
>
> I can have attached a few images so you will get a better idea of the requirements.
>
> Kishor Durve



Holistically-nested edge detection.png
82K

Kishor Durve <kishor.durve@outlook.com> Fri, Jul 25, 2025 at 6:34 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

Sorry for delay in replying. I was away for quite some time.

A couple of months back we had tried OpenCV / HED and obtained similar results.

Please go ahead and try DexiNed after which we will decide about line detection.

After that another important step would be to remove noise from image using a suitable model.

I know this is somewhat away from our goal of 'AI assisted edge detection' but it can be an important intermediate step. We are currently using a model that does this. This is the model I had sent to you initially. It is developed in a Chinese University by a professor who works with our company.

Kishor Durve
Duravision Systems
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, July 25, 2025 5:46 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir ! This is an output image obtained from the HED model and similar image I have obtained from DexiNed, could u pls tell i am moving in the right direction or should I make improvements or change in the plan of action?

warm regards
Hindvee

On Wed, Jul 23, 2025 at 3:20 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir! Could you please share ur insights and views on the above mail thread, it shall be very helpful to get to know your feedback and move forward with the project.
>
> Warm Regards
> Hindvee
>
> On Fri, Jul 11, 2025 at 3:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir!
> >
> > Following up on my previous email regarding edge detection, I wanted to share the progress I've made since then. I have been focusing on using the Holistically-Nested Edge Detection (HED) method. I have now:
> >
> > CODE FILE
> >
> > 1. Successfully applied the HED process to a folder containing our images to generate edge maps for the entire dataset.
> >
> > 2. Trained a custom autoencoder model designed to refine and correct the lines in these images.
> >
> > 3. Applied the trained autoencoder to the HED output images to produce corrected versions.
> >
> > 4. Started working on post-processing techniques.

I am currently experimenting with different parameters and methods to achieve the best results for this step. I am continuing to refine the line clarification process and will share updated results once I have finalized the post-processing approach.

warm regards

Hindvee

On Wed, Jul 9, 2025 at 12:23 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Hello sir! I am trying the pre-trained model for detecting edges in our images; I have tried Holistically-Nested Edge Detection (HED) and DexiNed. I am attaching the output, I am going forward with HED to get better results.

Warm regards
Hindvee

On Mon, Jul 7, 2025 at 10:37 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Hello sir!
I am sharing the first 5 difference images of the three codes that I mentioned previously. Please check this whenever you have time.

Warm regards
Hindvee

On Mon, Jul 7, 2025 at 6:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:

I am unable to give my view of the approach as I can't view the images. I can give my opinion if you send me difference images. In any case, I think you need to have difference images as a metric in addition to the Laplacian.

Get Outlook for Android

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Monday, July 7, 2025 7:09:51 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!
I have received the stipend, thank you for that! I'd like to share a brief update on the image enhancement work, focusing on noise removal and edge preservation.
Summary of results:-
1. Previous code
---Performed basic denoising but introduced over-smoothing.
---Laplacian MAD: ~0.0175–0.0192

2. DnCNN
---Improved noise removal and structural retention.
---MAD: ~0.0186–0.0202

3. DnCNN + Sharpening (Current approach)
---Added post-processing to enhance edge clarity.
---Consistently improved MAD to ~0.0178–0.0196.
---Visual results show reduced whitening and better edge definition.

This confirms that the current setup provides better denoising while preserving fine image details. Please let me know if you'd like to proceed with further tuning using SSIM or edge-aware loss functions.

Warm regards

Hindvee

On Fri, Jul 4, 2025 at 3:54 AM Kishor Durve <kishor.durve@outlook.com> wrote:

Please keep at it. Use difference image and Laplacian to gauge the effect of the algorithm. I will be able to look at your work only after 15th.
I have transferred your stipend by GPay.

Get Outlook for Android

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, July 4, 2025 5:37:35 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!
This is what I am been working on. I am still working on getting better edges and less whitish images.

warm regards
Hindvee

On Thu, Jul 3, 2025 at 10:53 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> UPI ID: hindveevispute@okicici
>
> Warm regards
> Hindvee
>
> On Wed, 2 Jul 2025, 03:16 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > I checked the images, but did not notice substantial difference between the before and after images. You can subtract them (after aligning) and check.
> > As regards payment of stipend please let me know how much we need to pay and I will transfer it on GPay.
> > I am travelling again and am out of office. I will be back only on 15th. My responses to your mails may be delayed.
> >
> > Get Outlook for Android
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Tuesday, July 1, 2025 1:51:45 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Hello sir!
> >
> > I was eager to get insights from you about the above shared images.Please do have a look as soon as possible. Also, I wanted to know whether the stipend will be credited at the end of the month or the project's end. It would be very kind of you if I could get the stipend at the end of the month.
> >
> > Thanks and regards
> > Hindvee
> >
> > On Mon, Jun 30, 2025 at 10:33 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> >
> > > Good morning sir!
> > >
> > > Did you get any chance to review the above images? Please share your suggestions and insights with me.
> > >
> > > Warm regards
> > > Hindvee
> > >
> > > On Thu, Jun 26, 2025 at 11:09 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > >
> > > > Here I am attaching the processed images.
> > > >
> > > > Warm regards
> > > > Hindvee
> > > >
> > > > On Thu, Jun 26, 2025 at 8:08 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> > > >
> > > > > I am unable to run the files in Colab and generate images. Can you please send me the processed images on mail - .png / .jpeg files, not .ipynb files.
> > > > >
> > > > > Kishor Durve
> > > > >
> > > > > ---
> > > > >
> > > > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > > > **Sent:** Thursday, June 26, 2025 4:48 AM
> > > > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > > > **Subject:** Re: AI assisted edge detection

Hello sir, I have attached the results for png 1,2,3 and 4. Pls share ur views and insights.

Warm Regards
Hindvee

On Tue, Jun 24, 2025 at 11:29 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

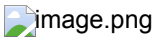
> Hello sir,
> I am attaching the .ipynb file, it has the images
>
> warm regards
> Hindvee

On Tue, Jun 24, 2025 at 5:48 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Please share the images for visual comparison.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Tuesday, June 24, 2025 5:13 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello Durve sir,
>
> I wanted to share an update on the denoising output. I noticed earlier that while the model was doing a good job removing noise, it was also making the image a bit too smooth — the sharpness had dropped quite a bit:
>
> - Noisy Image: 33.72
> - Denoised Image: 6.92
> - Clean Image (Ground Truth): 26.49
>
> To improve this, I added a sharpening filter after the autoencoder output and tested two variants:
>
> 1. A **strong sharpening kernel**, which raised the denoised sharpness to **112.40,** but it felt a bit overdone and introduced some artifacts.
> 2. A **softer kernel**, which gave a cleaner and more natural result, with denoised sharpness at **54.04**. This seems like a balanced point where edges are clearer without compromising visual quality.
>
> If you'd like, I can also share the output images (clean vs. denoised) for a visual comparison.
>
> Warm regards
>
> Hindvee
>
> image.png

On Fri, Jun 20, 2025 at 12:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir!
> Based on the latest code I calculated the sharpness using Laplacian variance. I obtained the following results as those attached as image.
> I think that while denoising, though the noise was removed, but the model blurred or over smoothened the image. I may have lost some meaningful edge details in the process. But the sharpness value of Clean image is lower than the noisy image (because it has no random noise), but high enough to retain clear edges and structure. So I am working next on increasing the sharpness value while keeping the noise low. Please do share your views with me.
>
> Warm regards
> Hindvee



On Thu, 19 Jun 2025, 13:42 Kishor Durve, <kishor.durve@outlook.com> wrote:

> I have seen the processed images. I think it is good at this point to quantify 'sharpness' of an image. It will give us a measure to compare different methods / models.
>
> There can be several approaches to do this. We use Opencv::Laplacian() to judge image sharpness. But you can try other approaches.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Thursday, June 19, 2025 2:52 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir,
>
> I've completed the full-image denoising as discussed for the images `5.png` and `6.png`. This time, the model was trained and applied directly on the complete images (cropped to a common size of **570×588**) instead of small patches, as per your suggestion. The denoising autoencoder was able to reduce background noise and enhance structural features, particularly the edges of the circles, which were initially faint or inconsistent. The model output shows a noticeable improvement in clarity while preserving the original patterns.
>
> I've attached the processed result image along with this mail. Please let me know your thoughts on this file.
>
> Warm regards,
> Hindvee
>
> On Wed, Jun 18, 2025 at 12:33 PM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > I have checked the file. These two images contain circles as primary features. Also, the patches in the file seem to be for a small part of the original image. If you have processed the original images (5.png and 6.png) please send me the processed images on mail. If you have processed only parts of these images, I suggest you process the full images and send me the results.
> >
> > Similarly, in the previous exercise please process complete images. Most of the set are of size 500x500 pixel, so it should not consume too much time.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Wednesday, June 18, 2025 1:44 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Sir, please let me know your views on the above .ipynb file.
> > Warm Regards
> > Hindvee
> >
> > On Wed, Jun 18, 2025 at 1:56 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> >
> > > Hello sir,
> > > I have processed the 5.png and 6.png, I faced lower accuracy rate earlier. After fixing the earlier issues with mismatched data sizes and high Binary Crossentropy

loss (~0.6098), the model is now working smoothly. It reached a much lower loss of just 0.0528 in 5 epochs, and both training and validation losses settled nicely by the 2nd or 3rd epoch without signs of overfitting. The denoised images look very clean and closely match the original ones, even when the input images had different levels of noise. I am attaching the results in the attached file.
Warm regards
Hindvee

On Tue, Jun 17, 2025 at 7:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Sorry! Forgot the attachment.
>
> Kishor Durve
>
> ---
>
> **From:** Kishor Durve <kishor.durve@outlook.com>
> **Sent:** Tuesday, June 17, 2025 9:46 AM
> **To:** Hindvee Vispute <hindveevispute@gmail.com>
> **Subject:** Re: AI assisted edge detection
>
> Good to know that you are making progress at a nice clip.
>
> I saw the images and the difference between before / after can be seen. Can you please process the images attached to this mail and send me the results? The images are similar to those I had sent earlier, the exception being 5.png and 6.png. In these last two images the challenge is to detect circles whose edges are not well defined and the edge quality may vary from image to image. This is to test the effectiveness of noise removal / sharpening.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Tuesday, June 17, 2025 3:06 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir,
> I have completed the training and evaluation of the denoising autoencoder using grayscale image patches taken from the `IMG` folder. The model was trained for 20 epochs on 64×64 patches with binary cross entropy loss and a 10% validation split. During training, the loss reduced steadily and stabilized, with the final reconstruction loss reaching 0.3688. The results show that the model can effectively denoise images and recover clean patterns like faint lines. In the next step, I plan to use the output of the Super-Resolution model as a base to evaluate the accuracy and quality of the denoising process, to better understand how both methods contribute to image enhancement. I am attaching the .ipynb file for your reference and also the IMG Folder which I just mentioned.
>
> Warm regards
> Hindvee
>
> IMG FOLDER
>
> Accuracy results:-
>
> image.png
>
> On Fri, Jun 13, 2025 at 5:33 AM Kishor Durve <kishor.durve@outlook.com> wrote:

The images I sent are big. OpenCV will not be able to handle them as the max size allowed is 32K x 32K. The crash you are facing could be due to this reason.

You can crop out small patches from the big images and use them for trials.

Kishor Durve
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, June 11, 2025 1:37 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Thank you for sharing more images. I have updated the work on the sheets.

Warm regards
Hindvee

On Sat, Jun 7, 2025 at 2:51 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have uploaded more images. You can crop out patches from them. One of the images has many boards which will give many useful patches.
>
> Kishor Durve
> Duravision Systems
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Saturday, June 7, 2025 4:36 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings sir,
> It would be great if I get more images, that would expand my dataset. Thank you for your previous email.
> I also wanted tk inform you that I am travelling and will be able to resume work by Tuesday. I request you to kindly allow my leave.
>
> Warm Regards
> Hindvee
>
> On Fri, 6 Jun 2025, 11:31 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > I am not going to be of much help to you as I have limited experience in this field. Also, what little I know was gained by applying available models in OpenCV's DNN module.
> >
> > There are two images available in the images folder. You may crop out patches from them for the Training / test sets. We use ImageJ and IrfanView for such work as MS Paint cannot handle such big images. If you need more images please let me know.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, June 6, 2025 5:24 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection

Greetings sir,
I hope you're doing well. I've just finished training the CNN-based line presence classifier, and I'm attaching the loss graph for your review. The model seems to have learned quite quickly, with both training and validation losses staying very close to zero throughout. There was a small spike in training loss around epoch 80, but validation stayed stable, so it might just be a minor fluctuation.

Since the dataset is quite small and the model converged very early, I'm unsure how best to move forward.According to you should I try to get more data to avoid overfitting. Please guide me towards a direction in which I can make this model more effective.

Warm regards

Hindvee

On Wed, Jun 4, 2025 at 6:42 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Yes, it looks ok to me. We will take a relook once you make more progress.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 4, 2025 4:26 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Additionally, Sir, I would like to confirm if the current format of the Google Sheet meets your expectations. Please let me know if you would prefer any modifications, such as a more detailed breakdown of the work completed.
>
> Warm Regards
> Hindvee
>
> On Wed, Jun 4, 2025 at 1:34 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Greetings Mr. Durve,
> >
> > Thanks for sharing the super-resolution model. I've set it up and tested it, the results inspiring. It does a good job cleaning up noise and improving contrast, especially in the faint line areas. As we discussed, I'll now add this to the bigger pipeline along with the AI modules (NB-1 to NB-5). The super-resolution output can help improve both the CNN classifier and regression models. I'll also check how it works together with NB-5, the denoising autoencoder, to see if we get even better results.
> >
> > Please let me know if you want me to run it on any specific ROI samples or start putting together comparison results.
> >
> > Warm regards
> > Hindvee
> >
> > On Tue, Jun 3, 2025 at 9:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > I have file network.py in folder 'models'. This file has the RRDBNet implementaion.
> > >
> > > Kishor Durve
> > > Duravision Systems
> > > 9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Monday, June 2, 2025 7:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr. Durve,
I am facing an issue again an again while running the VS 2022 code, pls help me with this, we have the module named as model_zoo not models. Pls tell me how to proceed with this.

 image.png

Warm Regards
Hindvee

On Sat, May 31, 2025 at 12:46 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Thankyou sir! I will try this right away.
> Also I hope you received this google sheet, I have shared edit access to you on your gmail account
>
> Hindvee Vispute - Time data - Intern
>
> Regards
> Hindvee
>
> On Sat, May 31, 2025 at 5:43 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Sure. You need to do the following to make the project work.
> >
> > 1. Download and install VS 2022 Community edition
> > 2. Download and install Anaconda. Install an additional environment Python 3.8.16. Use this version of Python for the project.
> >
> > The project should run at this point.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, May 30, 2025 6:34 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings sir!
> > I went through the google drive folder shared by you. It would be really helpful if you could walk me through it.
> > Best regards
> > Hindvee
> >
> > On Fri, May 30, 2025 at 11:41 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > You can download the model, a working VC++ (VS 2022) project and image here.
> > >
> > > https://drive.google.com/drive/folders/1mLBDjJR3fFKs8Brr_nNj2z9YyI89nruE?usp=sharing
> > >
> > > Kishor Durve
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>

Sent: Friday, May 30, 2025 12:50 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings sir!
Pls do share the model over drive I will try it out. Also I request you to share more image database so that I can start working on it.
I will start work from today itself then.

Regards
Hindvee

On Fri, 30 May 2025, 06:11 Kishor Durve, <kishor.durve@outlook.com> wrote:

> We have found that using a super resolution Pytorch model to get a larger image results in significant noise removal. Perhaps it is a by-product.
>
> You can try it out. I have the model and can share it on Google Drive if you want to try it out. We have tried the models available in OpenCV DNN but they do not seem to give the same results.
>
> Regards,
>
> Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 29, 2025 5:28 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Dear Sir,

Thank you for the confirmation. I will maintain the time data on a Google Sheet and ensure it is updated regularly.

Looking forward to starting from Monday, 2nd June.

Best regards,
Hindvee

On Thu, May 29, 2025 at 2:53 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> Welcome aboard!
>
> You can keep the time data on Google Sheet.
>
> Regards,
>
> Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 29, 2025 5:12 PM

To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr. Durve,

Thank you very much for giving me the opportunity to contribute to Guangdong Longtian Smart Instruments Co. Ltd. through this exciting project. I am pleased to accept the offer and look forward to working on the development of the AI-assisted line detection tool.

With your permission, I would like to begin the internship from the coming Monday, i.e., 2nd June 2025.

Additionally, I wanted to ask about the preferred method for tracking and sharing the time I spend on the project. Would you like me to maintain a shared timesheet (e.g., a Google Sheet or Excel file), or would you prefer weekly time summaries via email or any other format?

Looking forward to your guidance on this.

Thank you once again for this opportunity.

Warm regards,
Hindvee Rajesh Vispute

On Thu, May 29, 2025 at 8:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have attached the on-boarding letter.
>
> Please let me know if you have any queries.
>
> Regards,
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Saturday, May 10, 2025 3:46 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings Mr. Durve,
>
> Thank you for your detailed response and clarification. I now clearly understand that the goal is to enhance line detection using AI within known ROIs, leveraging the expected position of endpoints.
>
> To support this objective, I've explored and implemented several **AI-based approaches**, any of which can be used to complement or improve upon conventional methods:
>
> 1. **CNN-Based Line Presence Classifier (NB-1):** Identifies whether a line exists within a given patch/ROI—useful for filtering candidate regions before applying traditional detection.
> 2. **CNN Regression Model (NB-2):** Predicts line endpoints directly from the image, offering

robust localization even in low-contrast or noisy scenarios.

3. **Denoising Autoencoder (NB-5):** Enhances faint or broken lines by cleaning up noise, thereby boosting the effectiveness of existing edge or Hough-based methods.

4. **Vision Transformer Model (NB-4):** Designed for fine-grained classification tasks and can be adapted for future objectives like surface defect or anomaly detection.

These methods are independently deployable and can be integrated selectively depending on the specific inspection challenge or ROI structure. I'd be happy to align with your ROI-driven strategy and customize accordingly.

Regarding the stipend, I agree with the proposed Rs. 200/hour base rate and performance-based bonus, and I'm enthusiastic about the opportunity to collaborate further on defect detection and shape anomaly tasks in the future.

I have attached screenshots of the code files I executed today. I have the names of the notebooks (NB) with NB-1, NB-2,NB-4 and NB-5 respectively.

NB-1
NB-2
NB-4
NB-5

Please let me know if you'd like me to share these outputs or run a quick live demonstration.

Warm regards,
Hindvee

On Fri, May 9, 2025 at 6:03 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> I think there is a mismatch in understanding. We are already using conventional line detection techniques (various edge detection operators and Hough Transform and other methods for line fitment) .
>
> What we are looking to get from you is AI based / AI assisted line detection. Since we know the probable position of the line beforehand we can use this information to find a line. For example, say we know that the line end points are expected to be (x1, y1) and (x2, y2), we can construct a small rectangle (ROI) using these points and search for a line within this ROI.
>
> As for the stipend it depends both on the time you spend on the project and the results you produce. It will comprise of an hourly rate and a result linked amount. I suggest we have an hourly rate of Rs. 200/hour and a matching amount linked to the results at the end of the project.

So, if you are able to produce satisfactory results it will come to Rs. 400/hour. Also, if the results are satisfactory, we can take this forward next year and work on other issues like surface defect detection, detection of shape anomalies etc.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 8, 2025 5:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr Durve,
I have the problem now so I will focus on faint line detection for the first step. I can proceed using Sobel, Scharr, and Canny edge detectors provided by OpenCV. First I will preprocess the images and then move to edge detection using the above 3 detectors. I will carry out line detection using Hough Transform, and further we will filter out short or irrelevant lines based on length and their angle of orientation. Finally, we will draw the detected lines on the images using cv.line()

I can devote 14 hrs per week to this task for 2 months, we can extend the time if required. Also, I wanted to ask what stipend you can provide me. I believe I can complete the task in the above time frame. Thank you for your time.

Warm regards
Hindvee

On Thu, May 8, 2025 at 7:18 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Your analysis seems good.
>
> At this point we are not interested in detecting defects (though at a later stage we may be), and want to detect the faint lines only.
>
> Note that though I have annotated only a single line in the image I sent to you yesterday, we need to detect edge lines of all vertical stripes.
>
> I have attached 2 sets of 20 images each named 'Good Images' and 'Bad Images' - images in which it is mostly easy to detect lines by the conventional image processing method and those in which lines are faint and hard to detect. These images may help you get a better understanding of the problem.
>
> Please also check the following issues:

1. How much time will you be able to devote to this task?
2. Will you be able to complete the task within the available time?

There are other issues like time taken by the AI route, integration of AI line detect code in VC++ project, further training / fine tuning of model, applicability of the method to detect faint circles etc. But we will get into them later once the basic method is proven.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, May 7, 2025 3:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings, Mr. Durve,
Thank you for reviewing my proposal and for sharing the sample images. By looking at the images I came across the specific challenges faced by the inspection system, like detecting faint lines, subtle surface variations, and fine defects that can be missed by traditional image processing. Based on these observations I suggest that we can do the following:-

1. **Hybrid approach**: Objective - Combine the speed of OpenCV line detection (example Hough Transform) with deep learning's pattern recognition capability.
Approach
- Use OpenCV to pre-locate candidate regions and feed them into a deep model for verification.
- This can speed up inference and increase resistance to false positives.

2. **Deep learning-based fine defect detection**: Objective: Improve the detection of faint, low contrast line defects and localized surface anomalies.
Approach
- Train a model, such as YOLOv8 or U-Net, tailored to your defect patterns using the annotated images.
- To improve feature visibility, add contrast enhancement.
- Assess using detection accuracy and intersection over union (IoU) to compare to baselines based on OpenCV.

I can begin with a proof-of-concept model using the images shared by you. Thank you for sharing the opportunity. I am excited to get a chance to contribute to the advancement in inspection technology.

Warm regards
Hindvee Vispute

On Wed, 7 May 2025, 09:31 Kishor Durve, <kishor.durve@outlook.com> wrote:

Hello Hindvee,

I have gone thru the document uploaded by you.

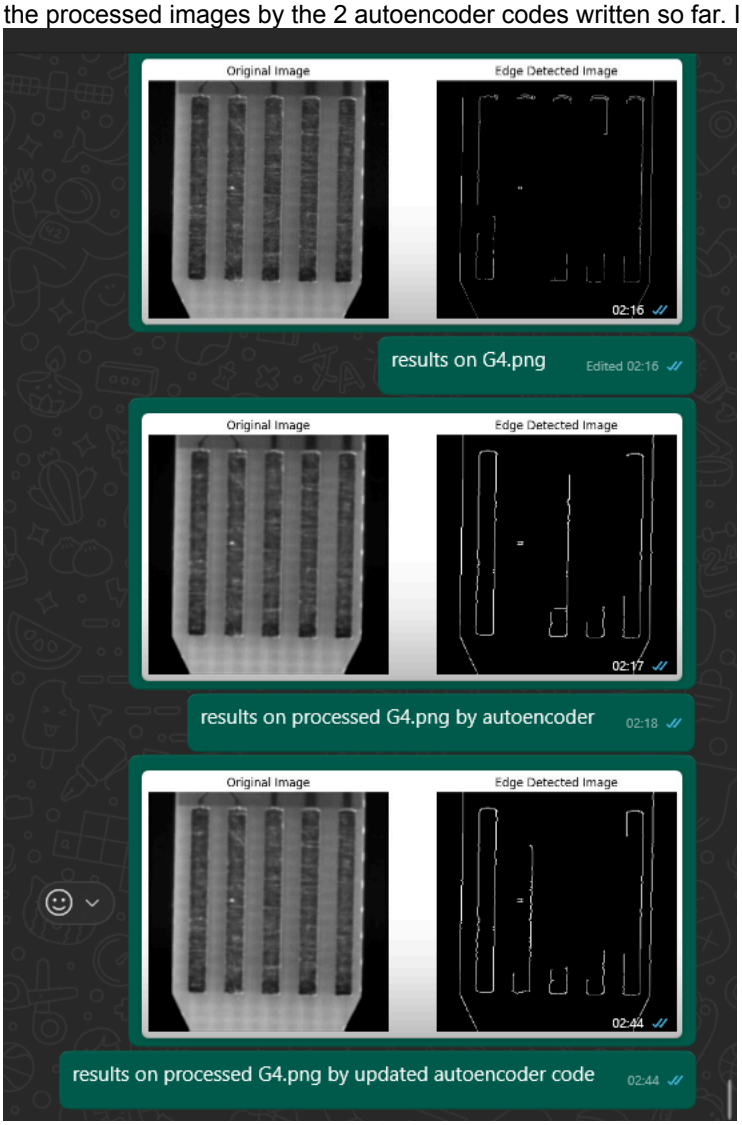
It looks ok to me.

I can have attached a few images so you will get a better idea of the requirements.

Kishor Durve

Hindvee Vispute <hindveevispute@gmail.com> Mon, Jul 28, 2025 at 2:48 AM
To: Kishor Durve <kishor.durve@outlook.com>

Noted sir. I wrote code for autoencoder to get sharper images, and I applied DexiNed on the original image and also on the processed images by the 2 autoencoder codes written so far. I am sharing the results here.



warm regards
Hindvee

On Fri, Jul 25, 2025 at 6:34 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Sorry for delay in replying. I was away for quite some time.
>
> A couple of months back we had tried OpenCV / HED and obtained similar results.
>
> Please go ahead and try DexiNed after which we will decide about line detection.

After that another important step would be to remove noise from image using a suitable model.

I know this is somewhat away from our goal of 'AI assisted edge detection' but it can be an important intermediate step. We are currently using a model that does this. This is the model I had sent to you initially. It is developed in a Chinese University by a professor who works with our company.

Kishor Durve
Duravision Systems
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, July 25, 2025 5:46 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir ! This is an output image obtained from the HED model and similar image I have obtained from DexiNed, could u pls tell i am moving in the right direction or should I make improvements or change in the plan of action?

warm regards
Hindvee

On Wed, Jul 23, 2025 at 3:20 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir! Could you please share ur insights and views on the above mail thread, it shall be very helpful to get to know your feedback and move forward with the project.
>
> Warm Regards
> Hindvee
>
> On Fri, Jul 11, 2025 at 3:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir!
> >
> > Following up on my previous email regarding edge detection, I wanted to share the progress I've made since then. I have been focusing on using the Holistically-Nested Edge Detection (HED) method. I have now:
> >
> > CODE FILE
> >
> > 1. Successfully applied the HED process to a folder containing our images to generate edge maps for the entire dataset.
> >
> > 2. Trained a custom autoencoder model designed to refine and correct the lines in these images.
> >
> > 3. Applied the trained autoencoder to the HED output images to produce corrected versions.
> >
> > 4. Started working on post-processing techniques.
> >
> > I am currently experimenting with different parameters and methods to achieve the best results for this step. I am continuing to refine the line clarification process and will share updated results once I have finalized the post-processing approach.
> >
> > warm regards
> >
> > Hindvee
> >
> > On Wed, Jul 9, 2025 at 12:23 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> >
> > > Hello sir! I am trying the pre-trained model for detecting edges in our images; I have tried Holistically-Nested Edge Detection (HED) and DexiNed. I am attaching the output, I am going forward with HED to get better results.
> > >
> > > Warm regards
> > > Hindvee
> > >
> > > On Mon, Jul 7, 2025 at 10:37 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > >
> > > > Hello sir!
> > > > I am sharing the first 5 difference images of the three codes that I mentioned previously. Please check this whenever you have time.
> > > >
> > > > Warm regards
> > > > Hindvee

On Mon, Jul 7, 2025 at 6:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:

I am unable to give my view of the approach as I can't view the images. I can give my opinion if you send me difference images. In any case, I think you need to have difference images as a metric in addition to the Laplacian.

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From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Monday, July 7, 2025 7:09:51 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!
I have received the stipend, thank you for that! I'd like to share a brief update on the image enhancement work, focusing on noise removal and edge preservation.
Summary of results:-
1. Previous code
---Performed basic denoising but introduced over-smoothing.
---Laplacian MAD: ~0.0175–0.0192

2. DnCNN
---Improved noise removal and structural retention.
---MAD: ~0.0186–0.0202

3. DnCNN + Sharpening (Current approach)
---Added post-processing to enhance edge clarity.
---Consistently improved MAD to ~0.0178–0.0196.
---Visual results show reduced whitening and better edge definition.

This confirms that the current setup provides better denoising while preserving fine image details. Please let me know if you'd like to proceed with further tuning using SSIM or edge-aware loss functions.

Warm regards

Hindvee

On Fri, Jul 4, 2025 at 3:54 AM Kishor Durve <kishor.durve@outlook.com> wrote:

Please keep at it. Use difference image and Laplacian to gauge the effect of the algorithm. I will be able to look at your work only after 15th.
I have transferred your stipend by GPay.

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From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, July 4, 2025 5:37:35 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!
This is what I am been working on. I am still working on getting better edges and less whitish images.

warm regards
Hindvee

On Thu, Jul 3, 2025 at 10:53 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

UPI ID: hindveevispute@okicici

Warm regards
Hindvee

On Wed, 2 Jul 2025, 03:16 Kishor Durve, <kishor.durve@outlook.com> wrote:

I checked the images, but did not notice substantial difference between the before and after images. You can subtract them (after aligning) and check.
As regards payment of stipend please let me know how much we need to pay and I will transfer it on GPay.

I am travelling again and am out of office. I will be back only on 15th. My responses to your mails may be delayed.

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From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, July 1, 2025 1:51:45 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!

I was eager to get insights from you about the above shared images.Please do have a look as soon as possible. Also, I wanted to know whether the stipend will be credited at the end of the month or the project's end. It would be very kind of you if I could get the stipend at the end of the month.

Thanks and regards
Hindvee

On Mon, Jun 30, 2025 at 10:33 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Good morning sir!
>
> Did you get any chance to review the above images? Please share your suggestions and insights with me.
>
> Warm regards
> Hindvee
>
> On Thu, Jun 26, 2025 at 11:09 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Here I am attaching the processed images.
> >
> > Warm regards
> > Hindvee
> >
> > On Thu, Jun 26, 2025 at 8:08 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > I am unable to run the files in Colab and generate images. Can you please send me the processed images on mail - .png / .jpeg files, not .ipynb files.
> > >
> > > Kishor Durve
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Thursday, June 26, 2025 4:48 AM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection
> > >
> > > Hello sir, I have attached the results for png 1,2,3 and 4. Pls share ur views and insights.
> > >
> > > Warm Regards
> > > Hindvee
> > >
> > > On Tue, Jun 24, 2025 at 11:29 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > >
> > > > Hello sir,
> > > > I am attaching the .ipynb file, it has the images
> > > >
> > > > warm regards
> > > > Hindvee
> > > >
> > > > On Tue, Jun 24, 2025 at 5:48 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> > > >
> > > > > Please share the images for visual comparison.
> > > > >
> > > > > Kishor Durve
> > > > >
> > > > > ---
> > > > >
> > > > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > > > **Sent:** Tuesday, June 24, 2025 5:13 AM
> > > > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > > > **Subject:** Re: AI assisted edge detection

Hello Durve sir,

I wanted to share an update on the denoising output. I noticed earlier that while the model was doing a good job removing noise, it was also making the image a bit too smooth — the sharpness had dropped quite a bit:

- Noisy Image: 33.72
- Denoised Image: 6.92
- Clean Image (Ground Truth): 26.49

To improve this, I added a sharpening filter after the autoencoder output and tested two variants:

1. A **strong sharpening kernel**, which raised the denoised sharpness to **112.40,** but it felt a bit overdone and introduced some artifacts.
2. A **softer kernel**, which gave a cleaner and more natural result, with denoised sharpness at **54.04**. This seems like a balanced point where edges are clearer without compromising visual quality.

If you'd like, I can also share the output images (clean vs. denoised) for a visual comparison.

Warm regards

Hindvee

 image.png

On Fri, Jun 20, 2025 at 12:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir!
> Based on the latest code I calculated the sharpness using Laplacian variance. I obtained the following results as those attached as image.
> I think that while denoising, though the noise was removed, but the model blurred or over smoothened the image. I may have lost some meaningful edge details in the process. But the sharpness value of Clean image is lower than the noisy image (because it has no random noise), but high enough to retain clear edges and structure. So I am working next on increasing the sharpness value while keeping the noise low. Please do share your views with me.
>
> Warm regards
> Hindvee
>
>  image.png
>
> On Thu, 19 Jun 2025, 13:42 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > I have seen the processed images. I think it is good at this point to quantify 'sharpness' of an image. It will give us a measure to compare different methods / models.
> >
> > There can be several approaches to do this. We use Opencv::Laplacian() to judge image sharpness. But you can try other approaches.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Thursday, June 19, 2025 2:52 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection

Hello sir,

I've completed the full-image denoising as discussed for the images `5.png` and `6.png`. This time, the model was trained and applied directly on the complete images (cropped to a common size of **570×588**) instead of small patches, as per your suggestion. The denoising autoencoder was able to reduce background noise and enhance structural features, particularly the edges of the circles, which were initially faint or inconsistent. The model output shows a noticeable improvement in clarity while preserving the original patterns.

I've attached the processed result image along with this mail. Please let me know your thoughts on this file.

Warm regards,
Hindvee

On Wed, Jun 18, 2025 at 12:33 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have checked the file. These two images contain circles as primary features. Also, the patches in the file seem to be for a small part of the original image. If you have processed the original images (5.png and 6.png) please send me the processed images on mail. If you have processed only parts of these images, I suggest you process the full images and send me the results.
>
> Similarly, in the previous exercise please process complete images. Most of the set are of size 500x500 pixel, so it should not consume too much time.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 18, 2025 1:44 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Sir, please let me know your views on the above .ipynb file.
> Warm Regards
> Hindvee
>
> On Wed, Jun 18, 2025 at 1:56 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir,
> > I have processed the 5.png and 6.png, I faced lower accuracy rate earlier. After fixing the earlier issues with mismatched data sizes and high Binary Crossentropy loss (~0.6098), the model is now working smoothly. It reached a much lower loss of just 0.0528 in 5 epochs, and both training and validation losses settled nicely by the 2nd or 3rd epoch without signs of overfitting. The denoised images look very clean and closely match the original ones, even when the input images had different levels of noise. I am attaching the results in the attached file.
> > Warm regards
> > Hindvee
> >
> > On Tue, Jun 17, 2025 at 7:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > Sorry! Forgot the attachment.
> > >
> > > Kishor Durve
> > >
> > > ---
> > >
> > > **From:** Kishor Durve <kishor.durve@outlook.com>
> > > **Sent:** Tuesday, June 17, 2025 9:46 AM
> > > **To:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Subject:** Re: AI assisted edge detection

Good to know that you are making progress at a nice clip.

I saw the images and the difference between before / after can be seen. Can you please process the images attached to this mail and send me the results? The images are similar to those I had sent earlier, the exception being 5.png and 6.png. In these last two images the challenge is to detect circles whose edges are not well defined and the edge quality may vary from image to image. This is to test the effectiveness of noise removal / sharpening.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, June 17, 2025 3:06 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir,
I have completed the training and evaluation of the denoising autoencoder using grayscale image patches taken from the `IMG` folder. The model was trained for 20 epochs on 64×64 patches with binary cross entropy loss and a 10% validation split. During training, the loss reduced steadily and stabilized, with the final reconstruction loss reaching 0.3688. The results show that the model can effectively denoise images and recover clean patterns like faint lines. In the next step, I plan to use the output of the Super-Resolution model as a base to evaluate the accuracy and quality of the denoising process, to better understand how both methods contribute to image enhancement. I am attaching the .ipynb file for your reference and also the IMG Folder which I just mentioned.

Warm regards
Hindvee

IMG FOLDER

Accuracy results:-

image.png

On Fri, Jun 13, 2025 at 5:33 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> The images I sent are big. OpenCV will not be able to handle them as the max size allowed is 32K x 32K. The crash you are facing could be due to this reason.
>
> You can crop out small patches from the big images and use them for trials.
>
> Kishor Durve
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 11, 2025 1:37 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Thank you for sharing more images. I have updated the work on the sheets.
>
> Warm regards
> Hindvee

On Sat, Jun 7, 2025 at 2:51 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have uploaded more images. You can crop out patches from them. One of the images has many boards which will give many useful patches.
>
> Kishor Durve
> Duravision Systems
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Saturday, June 7, 2025 4:36 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings sir,
> It would be great if I get more images, that would expand my dataset.
> Thank you for your previous email.
> I also wanted tk inform you that I am travelling and will be able to resume work by Tuesday. I request you to kindly allow my leave.
>
> Warm Regards
> Hindvee
>
> On Fri, 6 Jun 2025, 11:31 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > I am not going to be of much help to you as I have limited experience in this field. Also, what little I know was gained by applying available models in OpenCV's DNN module.
> >
> > There are two images available in the images folder. You may crop out patches from them for the Training / test sets. We use ImageJ and IrfanView for such work as MS Paint cannot handle such big images. If you need more images please let me know.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, June 6, 2025 5:24 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings sir,
> > I hope you're doing well. I've just finished training the CNN-based line presence classifier, and I'm attaching the loss graph for your review. The model seems to have learned quite quickly, with both training and validation losses staying very close to zero throughout. There was a small spike in training loss around epoch 80, but validation stayed stable, so it might just be a minor fluctuation.
> >
> > Since the dataset is quite small and the model converged very early, I'm unsure how best to move forward.According to you should I try to get more data to avoid overfitting. Please guide me towards a direction in which I can make this model more effective.
> >
> > Warm regards
> >
> > Hindvee
> >
> > .
> >
> > On Wed, Jun 4, 2025 at 6:42 AM Kishor Durve <kishor.durve@outlook.com> wrote:

Yes, it looks ok to me. We will take a relook once you make more progress.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, June 4, 2025 4:26 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Additionally, Sir, I would like to confirm if the current format of the Google Sheet meets your expectations. Please let me know if you would prefer any modifications, such as a more detailed breakdown of the work completed.

Warm Regards
Hindvee

On Wed, Jun 4, 2025 at 1:34 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Greetings Mr. Durve,
>
> Thanks for sharing the super-resolution model. I've set it up and tested it, the results inspiring. It does a good job cleaning up noise and improving contrast, especially in the faint line areas. As we discussed, I'll now add this to the bigger pipeline along with the AI modules (NB-1 to NB-5). The super-resolution output can help improve both the CNN classifier and regression models. I'll also check how it works together with NB-5, the denoising autoencoder, to see if we get even better results.
>
> Please let me know if you want me to run it on any specific ROI samples or start putting together comparison results.
>
> Warm regards
> Hindvee
>
> On Tue, Jun 3, 2025 at 9:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > I have file network.py in folder 'models'. This file has the RRDBNet implementaion.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Monday, June 2, 2025 7:15 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings Mr. Durve,
> > I am facing an issue again an again while running the VS 2022 code, pls help me with this, we have the module named as model_zoo not models. Pls tell me how to proceed with this.
> >
> > image.png
> >
> > Warm Regards
> > Hindvee
> >
> > On Sat, May 31, 2025 at 12:46 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> >
> > > Thankyou sir! I will try this right away.
> > > Also I hope you received this google sheet, I have shared edit

access to you on your gmail account

Hindvee Vispute - Time data - Intern

Regards
Hindvee

On Sat, May 31, 2025 at 5:43 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Sure. You need to do the following to make the project work.
>
> 1. Download and install VS 2022 Community edition
> 2. Download and install Anaconda. Install an additional environment Python 3.8.16. Use this version of Python for the project.
>
> The project should run at this point.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Friday, May 30, 2025 6:34 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings sir!
> I went through the google drive folder shared by you. It would be really helpful if you could walk me through it.
> Best regards
> Hindvee
>
> On Fri, May 30, 2025 at 11:41 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > You can download the model, a working VC++ (VS 2022) project and image here.
> >
> > https://drive.google.com/drive/folders/1mLBDjJR3fFKs8Brr_nNj2z9YyI89nruE?usp=sharing
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, May 30, 2025 12:50 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings sir!
> > Pls do share the model over drive I will try it out. Also I request you to share more image database so that I can start working on it.
> > I will start work from today itself then.
> >
> > Regards
> > Hindvee
> >
> > On Fri, 30 May 2025, 06:11 Kishor Durve, <kishor.durve@outlook.com> wrote:
> >
> > > We have found that using a super resolution Pytorch model to get a larger image results in significant noise removal. Perhaps it is a by-product.

You can try it out. I have the model and can share it on Google Drive if you want to try it out. We have tried the models available in OpenCV DNN but they do not seem to give the same results.

Regards,

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 29, 2025 5:28 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Dear Sir,

Thank you for the confirmation. I will maintain the time data on a Google Sheet and ensure it is updated regularly.

Looking forward to starting from Monday, 2nd June.

Best regards,
Hindvee

On Thu, May 29, 2025 at 2:53 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> Welcome aboard!
>
> You can keep the time data on Google Sheet.
>
> Regards,
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Thursday, May 29, 2025 5:12 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings Mr. Durve,
>
> Thank you very much for giving me the opportunity to contribute to Guangdong Longtian Smart Instruments Co. Ltd. through this exciting project. I am pleased to accept the offer and look forward to working on the development of the AI-assisted line detection tool.
>
> With your permission, I would like to begin the internship from the coming Monday, i.e., 2nd June 2025.
>
> Additionally, I wanted to ask about the preferred method for tracking and sharing the time I spend on the project. Would you like me to maintain a shared timesheet (e.g., a Google Sheet or Excel file), or would you prefer weekly time summaries via email or any other format?

Looking forward to your guidance on this.

Thank you once again for this opportunity.

Warm regards,
Hindvee Rajesh Vispute

On Thu, May 29, 2025 at 8:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have attached the on-boarding letter.
>
> Please let me know if you have any queries.
>
> Regards,
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Saturday, May 10, 2025 3:46 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings Mr. Durve,
>
> Thank you for your detailed response and clarification. I now clearly understand that the goal is to enhance line detection using AI within known ROIs, leveraging the expected position of endpoints.
>
> To support this objective, I've explored and implemented several **AI-based approaches**, any of which can be used to complement or improve upon conventional methods:
>
> 1. **CNN-Based Line Presence Classifier (NB-1):** Identifies whether a line exists within a given patch/ROI—useful for filtering candidate regions before applying traditional detection.
> 2. **CNN Regression Model (NB-2):** Predicts line endpoints directly from the image, offering robust localization even in low-contrast or noisy scenarios.
> 3. **Denoising Autoencoder (NB-5):** Enhances faint or broken lines by cleaning up noise, thereby boosting the effectiveness of existing edge or Hough-based methods.
> 4. **Vision Transformer Model (NB-4):** Designed for fine-grained classification tasks and can be adapted for future objectives like surface defect or anomaly detection.
>
> These methods are independently deployable and can be integrated selectively depending on the specific inspection challenge or ROI structure. I'd be happy to align with your ROI-driven strategy and customize accordingly.

Regarding the stipend, I agree with the proposed Rs. 200/hour base rate and performance-based bonus, and I'm enthusiastic about the opportunity to collaborate further on defect detection and shape anomaly tasks in the future.

I have attached screenshots of the code files I executed today. I have the names of the notebooks (NB) with NB-1, NB-2,NB-4 and NB-5 respectively.

NB-1
NB-2
NB-4
NB-5

Please let me know if you'd like me to share these outputs or run a quick live demonstration.

Warm regards,
Hindvee

On Fri, May 9, 2025 at 6:03 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> I think there is a mismatch in understanding. We are already using conventional line detection techniques (various edge detection operators and Hough Transform and other methods for line fitment) .
>
> What we are looking to get from you is AI based / AI assisted line detection. Since we know the probable position of the line beforehand we can use this information to find a line. For example, say we know that the line end points are expected to be (x1, y1) and (x2, y2), we can construct a small rectangle (ROI) using these points and search for a line within this ROI.
>
> As for the stipend it depends both on the time you spend on the project and the results you produce. It will comprise of an hourly rate and a result linked amount. I suggest we have an hourly rate of Rs. 200/hour and a matching amount linked to the results at the end of the project. So, if you are able to produce satisfactory results it will come to Rs. 400/hour. Also, if the results are satisfactory, we can take this forward next year and work on other issues like surface defect detection, detection of shape anomalies etc.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Thursday, May 8, 2025 5:15 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection

Greetings Mr Durve,
I have the problem now so I will focus on faint line detection for the first step. I can proceed using Sobel, Scharr, and Canny edge detectors provided by OpenCV. First I will preprocess the images and then move to edge detection using the above 3 detectors. I will carry out line detection using Hough Transform, and further we will filter out short or irrelevant lines based on length and their angle of orientation. Finally, we will draw the detected lines on the images using cv.line()

I can devote 14 hrs per week to this task for 2 months, we can extend the time if required. Also, I wanted to ask what stipend you can provide me. I believe I can complete the task in the above time frame. Thank you for your time.

Warm regards
Hindvee

On Thu, May 8, 2025 at 7:18 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Your analysis seems good.
>
> At this point we are not interested in detecting defects (though at a later stage we may be), and want to detect the faint lines only.
>
> Note that though I have annotated only a single line in the image I sent to you yesterday, we need to detect edge lines of all vertical stripes.
>
> I have attached 2 sets of 20 images each named 'Good Images' and 'Bad Images' - images in which it is mostly easy to detect lines by the conventional image processing method and those in which lines are faint and hard to detect. These images may help you get a better understanding of the problem.
>
> Please also check the following issues:
>
> 1. How much time will you be able to devote to this task?
> 2. Will you be able to complete the task within the available time?
>
> There are other issues like time taken by the AI route, integration of AI line detect code in VC++ project, further training / fine tuning of model, applicability of the method to detect faint circles etc. But we will get into them later once the basic method is proven.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, May 7, 2025 3:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings, Mr. Durve,
Thank you for reviewing my proposal and for sharing the sample images. By looking at the images I came across the specific challenges faced by the inspection system, like detecting faint lines, subtle surface variations, and fine defects that can be missed by traditional image processing. Based on these observations I suggest that we can do the following:-

1. **Hybrid approach**: Objective - Combine the speed of OpenCV line detection (example Hough Transform) with deep learning's pattern recognition capability.
Approach
- Use OpenCV to pre-locate candidate regions and feed them into a deep model for verification.
- This can speed up inference and increase resistance to false positives.

2. **Deep learning-based fine defect detection**: Objective: Improve the detection of faint, low contrast line defects and localized surface anomalies.
Approach
- Train a model, such as YOLOv8 or U-Net, tailored to your defect patterns using the annotated images.
- To improve feature visibility, add contrast enhancement.
- Assess using detection accuracy and intersection over union (IoU) to compare to baselines based on OpenCV.

I can begin with a proof-of-concept model using the images shared by you. Thank you for sharing the opportunity. I am excited to get a chance to contribute to the advancement in inspection technology.

Warm regards
Hindvee Vispute

On Wed, 7 May 2025, 09:31 Kishor Durve, <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have gone thru the document uploaded by you.
>
> It looks ok to me.
>
> I can have attached a few images so you will get a better idea of the requirements.
>
> Kishor Durve

Hindvee Vispute <hindveevispute@gmail.com> Tue, Jul 29, 2025 at 1:07 AM
To: Kishor Durve <kishor.durve@outlook.com>

Hello sir!
current update:-

- Using an autoencoder trained on clean (good) images to repair bad images
- Applying a **hybrid loss** of MSE and SSIM to preserve structure and perceptual quality.
- Saving and visualizing both original and reconstructed images which helps in debugging

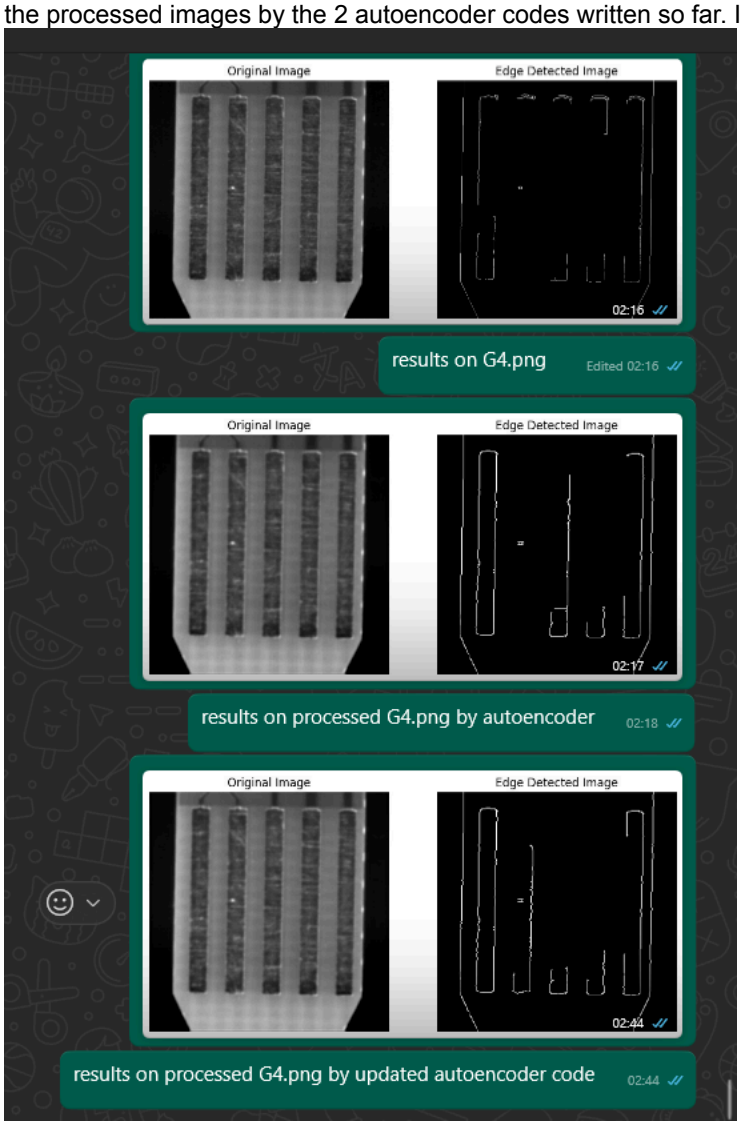
Despite this denoising strategy, DexiNED is still failing for bad images, the possible reasons are as follows

1. Contrast remains too low after reconstruction.
2. Edges are too smooth or faint due to over-smoothing by the autoencoder.
3. DexiNED is not tuned for synthetic/industrial patterns (e.g., PCB, x-ray like structures).

I will now work on these factors to improve the quality of processed images. I am attaching the processed Good and Bad Images folders.
BAD IMAGES _ PROCESSED
GOOD IMAGES_ PROCESSED

warm regards
Hindvee

On Mon, Jul 28, 2025 at 2:48 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Noted sir. I wrote code for autoencoder to get sharper images, and I applied DexiNed on the original image and also on the processed images by the 2 autoencoder codes written so far. I am sharing the results here.
>
> 
>
>
> warm regards
> Hindvee

On Fri, Jul 25, 2025 at 6:34 AM Kishor Durve <kishor.durve@outlook.com> wrote:

Sorry for delay in replying. I was away for quite some time.

A couple of months back we had tried OpenCV / HED and obtained similar results.

Please go ahead and try DexiNed after which we will decide about line detection.

After that another important step would be to remove noise from image using a suitable model.

I know this is somewhat away from our goal of 'AI assisted edge detection' but it can be an important intermediate step. We are currently using a model that does this. This is the model I had sent to you initially. It is developed in a Chinese University by a professor who works with our company.

Kishor Durve
Duravision Systems
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, July 25, 2025 5:46 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir ! This is an output image obtained from the HED model and similar image I have obtained from DexiNed, could u pls tell i am moving in the right direction or should I make improvements or change in the plan of action?

warm regards
Hindvee

On Wed, Jul 23, 2025 at 3:20 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Hello sir! Could you please share ur insights and views on the above mail thread, it shall be very helpful to get to know your feedback and move forward with the project.

Warm Regards
Hindvee

On Fri, Jul 11, 2025 at 3:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Hello sir!

Following up on my previous email regarding edge detection, I wanted to share the progress I've made since then. I have been focusing on using the Holistically-Nested Edge Detection (HED) method. I have now:

CODE FILE

1. Successfully applied the HED process to a folder containing our images to generate edge maps for the entire dataset.
2. Trained a custom autoencoder model designed to refine and correct the lines in these images.
3. Applied the trained autoencoder to the HED output images to produce corrected versions.
4. Started working on post-processing techniques.

I am currently experimenting with different parameters and methods to achieve the best results for this step. I am continuing to refine the line clarification process and will share updated results once I have finalized the post-processing approach.

warm regards

Hindvee

On Wed, Jul 9, 2025 at 12:23 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Hello sir! I am trying the pre-trained model for detecting edges in our images; I have tried Holistically-Nested Edge Detection (HED) and DexiNed. I am attaching the output, I am going forward with HED to get better results.

Warm regards

Hindvee

On Mon, Jul 7, 2025 at 10:37 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Hello sir!
I am sharing the first 5 difference images of the three codes that I mentioned previously. Please check this whenever you have time.

Warm regards
Hindvee

On Mon, Jul 7, 2025 at 6:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:

I am unable to give my view of the approach as I can't view the images. I can give my opinion if you send me difference images. In any case, I think you need to have difference images as a metric in addition to the Laplacian.

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From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Monday, July 7, 2025 7:09:51 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!
I have received the stipend, thank you for that! I'd like to share a brief update on the image enhancement work, focusing on noise removal and edge preservation.
Summary of results:-
1. Previous code
---Performed basic denoising but introduced over-smoothing.
---Laplacian MAD: ~0.0175–0.0192

2. DnCNN
---Improved noise removal and structural retention.
---MAD: ~0.0186–0.0202

3. DnCNN + Sharpening (Current approach)
---Added post-processing to enhance edge clarity.
---Consistently improved MAD to ~0.0178–0.0196.
---Visual results show reduced whitening and better edge definition.

This confirms that the current setup provides better denoising while preserving fine image details. Please let me know if you'd like to proceed with further tuning using SSIM or edge-aware loss functions.

Warm regards

Hindvee

On Fri, Jul 4, 2025 at 3:54 AM Kishor Durve <kishor.durve@outlook.com> wrote:

Please keep at it. Use difference image and Laplacian to gauge the effect of the algorithm. I will be able to look at your work only after 15th.
I have transferred your stipend by GPay.

Get Outlook for Android

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, July 4, 2025 5:37:35 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!
This is what I am been working on. I am still working on getting better edges and less whitish images.

warm regards
Hindvee

On Thu, Jul 3, 2025 at 10:53 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

UPI ID: hindveevispute@okicici

Warm regards
Hindvee

On Wed, 2 Jul 2025, 03:16 Kishor Durve, <kishor.durve@outlook.com> wrote:

I checked the images, but did not notice substantial difference between the before and after images. You can subtract them (after aligning) and check.
As regards payment of stipend please let me know how much we need to pay and I will transfer it on GPay.
I am travelling again and am out of office. I will be back only on 15th. My responses to your mails may be delayed.

Get Outlook for Android

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, July 1, 2025 1:51:45 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!

I was eager to get insights from you about the above shared images.Please do have a look as soon as possible. Also, I wanted to know whether the stipend will be credited at the end of the month or the project's end. It would be very kind of you if I could get the stipend at the end of the month.

Thanks and regards
Hindvee

On Mon, Jun 30, 2025 at 10:33 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Good morning sir!

Did you get any chance to review the above images? Please share your suggestions and insights with me.

Warm regards
Hindvee

On Thu, Jun 26, 2025 at 11:09 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Here I am attaching the processed images.

Warm regards
Hindvee

On Thu, Jun 26, 2025 at 8:08 AM Kishor Durve <kishor.durve@outlook.com> wrote:

I am unable to run the files in Colab and generate images. Can you please send me the processed images on mail - .png / .jpeg files, not .ipynb files.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, June 26, 2025 4:48 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir, I have attached the results for png 1,2,3 and 4. Pls share ur views and insights.

Warm Regards
Hindvee

On Tue, Jun 24, 2025 at 11:29 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Hello sir,
I am attaching the .ipynb file, it has the images

warm regards
Hindvee

On Tue, Jun 24, 2025 at 5:48 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Please share the images for visual comparison.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Tuesday, June 24, 2025 5:13 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello Durve sir,
>
> I wanted to share an update on the denoising output. I noticed earlier that while the model was doing a good job removing noise, it was also making the image a bit too smooth — the sharpness had dropped quite a bit:
>
> - Noisy Image: 33.72
> - Denoised Image: 6.92
> - Clean Image (Ground Truth): 26.49
>
> To improve this, I added a sharpening filter after the autoencoder output and tested two variants:
>
> 1. A **strong sharpening kernel**, which raised the denoised sharpness to **112.40,** but it felt a bit overdone and introduced some artifacts.
> 2. A **softer kernel**, which gave a cleaner and more natural result, with denoised sharpness at **54.04**. This seems like a balanced point where edges are clearer without compromising visual quality.
>
> If you'd like, I can also share the output images (clean vs. denoised) for a visual comparison.
>
> Warm regards
>
> Hindvee
>
> image.png

On Fri, Jun 20, 2025 at 12:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir!
> Based on the latest code I calculated the sharpness using Laplacian variance. I obtained the following results as those attached as image.
> I think that while denoising, though the noise was removed, but the model blurred or over smoothened the image. I may have lost some meaningful edge details in the process. But the sharpness value of Clean image is lower than the noisy image (because it has no random noise), but high enough to retain clear edges and structure. So I am working next on increasing the sharpness value while keeping the noise low. Please do share your views with me.
>
> Warm regards
> Hindvee
>
> image.png

On Thu, 19 Jun 2025, 13:42 Kishor Durve, <kishor.durve@outlook.com> wrote:

> I have seen the processed images. I think it is good at this point to quantify 'sharpness' of an image. It will give us a measure to compare different methods / models.

There can be several approaches to do this. We use Opencv::Laplacian() to judge image sharpness. But you can try other approaches.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, June 19, 2025 2:52 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir,

I've completed the full-image denoising as discussed for the images `5.png` and `6.png`. This time, the model was trained and applied directly on the complete images (cropped to a common size of **570×588**) instead of small patches, as per your suggestion. The denoising autoencoder was able to reduce background noise and enhance structural features, particularly the edges of the circles, which were initially faint or inconsistent. The model output shows a noticeable improvement in clarity while preserving the original patterns.

I've attached the processed result image along with this mail. Please let me know your thoughts on this file.

Warm regards,
Hindvee

On Wed, Jun 18, 2025 at 12:33 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have checked the file. These two images contain circles as primary features. Also, the patches in the file seem to be for a small part of the original image. If you have processed the original images (5.png and 6.png) please send me the processed images on mail. If you have processed only parts of these images, I suggest you process the full images and send me the results.
>
> Similarly, in the previous exercise please process complete images. Most of the set are of size 500x500 pixel, so it should not consume too much time.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 18, 2025 1:44 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Sir, please let me know your views on the above .ipynb file.
> Warm Regards
> Hindvee
>
> On Wed, Jun 18, 2025 at 1:56 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir,
> > I have processed the 5.png and 6.png, I faced lower accuracy rate earlier. After fixing the earlier issues with mismatched data sizes and high Binary Crossentropy loss (~0.6098), the model is now working smoothly. It reached a much lower loss of just 0.0528 in 5 epochs, and both training and validation losses settled nicely by the 2nd or 3rd epoch without signs of overfitting. The denoised images look very clean and closely match the original ones, even when the input images had different levels of noise. I am attaching the results in the attached file.
> > Warm regards
> > Hindvee

On Tue, Jun 17, 2025 at 7:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Sorry! Forgot the attachment.
>
> Kishor Durve
>
> ---
>
> **From:** Kishor Durve <kishor.durve@outlook.com>
> **Sent:** Tuesday, June 17, 2025 9:46 AM
> **To:** Hindvee Vispute <hindveevispute@gmail.com>
> **Subject:** Re: AI assisted edge detection
>
> Good to know that you are making progress at a nice clip.
>
> I saw the images and the difference between before / after can be seen. Can you please process the images attached to this mail and send me the results? The images are similar to those I had sent earlier, the exception being 5.png and 6.png. In these last two images the challenge is to detect circles whose edges are not well defined and the edge quality may vary from image to image. This is to test the effectiveness of noise removal / sharpening.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Tuesday, June 17, 2025 3:06 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir,
> I have completed the training and evaluation of the denoising autoencoder using grayscale image patches taken from the `IMG` folder. The model was trained for 20 epochs on 64×64 patches with binary cross entropy loss and a 10% validation split. During training, the loss reduced steadily and stabilized, with the final reconstruction loss reaching 0.3688. The results show that the model can effectively denoise images and recover clean patterns like faint lines. In the next step, I plan to use the output of the Super-Resolution model as a base to evaluate the accuracy and quality of the denoising process, to better understand how both methods contribute to image enhancement. I am attaching the .ipynb file for your reference and also the IMG Folder which I just mentioned.
>
> Warm regards
> Hindvee
>
> IMG FOLDER
>
> Accuracy results:-
>
> image.png

On Fri, Jun 13, 2025 at 5:33 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> The images I sent are big. OpenCV will not be able to handle them as the max size allowed is 32K x 32K. The crash you are facing could be due to this reason.
>
> You can crop out small patches from the big images and use them for trials.

Kishor Durve
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, June 11, 2025 1:37 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Thank you for sharing more images. I have updated the work on the sheets.

Warm regards
Hindvee

On Sat, Jun 7, 2025 at 2:51 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have uploaded more images. You can crop out patches from them. One of the images has many boards which will give many useful patches.
>
> Kishor Durve
> Duravision Systems
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Saturday, June 7, 2025 4:36 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings sir,
> It would be great if I get more images, that would expand my dataset.
> Thank you for your previous email.
> I also wanted tk inform you that I am travelling and will be able to resume work by Tuesday. I request you to kindly allow my leave.
>
> Warm Regards
> Hindvee
>
> On Fri, 6 Jun 2025, 11:31 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > I am not going to be of much help to you as I have limited experience in this field. Also, what little I know was gained by applying available models in OpenCV's DNN module.
> >
> > There are two images available in the images folder. You may crop out patches from them for the Training / test sets. We use ImageJ and IrfanView for such work as MS Paint cannot handle such big images. If you need more images please let me know.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, June 6, 2025 5:24 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings sir,
> > I hope you're doing well. I've just finished training the CNN-based line presence classifier, and I'm attaching the loss graph for your

review. The model seems to have learned quite quickly, with both training and validation losses staying very close to zero throughout. There was a small spike in training loss around epoch 80, but validation stayed stable, so it might just be a minor fluctuation.

Since the dataset is quite small and the model converged very early, I'm unsure how best to move forward.According to you should I try to get more data to avoid overfitting. Please guide me towards a direction in which I can make this model more effective.

Warm regards

Hindvee

On Wed, Jun 4, 2025 at 6:42 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Yes, it looks ok to me. We will take a relook once you make more progress.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 4, 2025 4:26 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Additionally, Sir, I would like to confirm if the current format of the Google Sheet meets your expectations. Please let me know if you would prefer any modifications, such as a more detailed breakdown of the work completed.
>
> Warm Regards
> Hindvee
>
> On Wed, Jun 4, 2025 at 1:34 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Greetings Mr. Durve,
> >
> > Thanks for sharing the super-resolution model. I've set it up and tested it, the results inspiring. It does a good job cleaning up noise and improving contrast, especially in the faint line areas. As we discussed, I'll now add this to the bigger pipeline along with the AI modules (NB-1 to NB-5). The super-resolution output can help improve both the CNN classifier and regression models. I'll also check how it works together with NB-5, the denoising autoencoder, to see if we get even better results.
> >
> > Please let me know if you want me to run it on any specific ROI samples or start putting together comparison results.
> >
> > Warm regards
> > Hindvee
> >
> > On Tue, Jun 3, 2025 at 9:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > I have file network.py in folder 'models'. This file has the RRDBNet implementaion.
> > >
> > > Kishor Durve
> > > Duravision Systems
> > > 9823164286
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Monday, June 2, 2025 7:15 PM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection

Greetings Mr. Durve,
I am facing an issue again an again while running the VS 2022 code, pls help me with this, we have the module named as model_zoo not models. Pls tell me how to proceed with this.

 image.png

Warm Regards
Hindvee

On Sat, May 31, 2025 at 12:46 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Thankyou sir! I will try this right away.
> Also I hope you received this google sheet, I have shared edit access to you on your gmail account
>
> Hindvee Vispute - Time data - Intern
>
> Regards
> Hindvee
>
> On Sat, May 31, 2025 at 5:43 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Sure. You need to do the following to make the project work.
> >
> > 1. Download and install VS 2022 Community edition
> > 2. Download and install Anaconda. Install an additional environment Python 3.8.16. Use this version of Python for the project.
> >
> > The project should run at this point.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, May 30, 2025 6:34 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings sir!
> > I went through the google drive folder shared by you. It would be really helpful if you could walk me through it.
> > Best regards
> > Hindvee
> >
> > On Fri, May 30, 2025 at 11:41 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > You can download the model, a working VC++ (VS 2022) project and image here.
> > >
> > > https://drive.google.com/drive/folders/1mLBDjJR3fFKs8Brr_nNj2z9YyI89nruE?usp=sharing
> > >
> > > Kishor Durve
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Friday, May 30, 2025 12:50 PM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection

Greetings sir!
Pls do share the model over drive I will try it out. Also I request you to share more image database so that I can start working on it.
I will start work from today itself then.

Regards
Hindvee

On Fri, 30 May 2025, 06:11 Kishor Durve, <kishor.durve@outlook.com> wrote:

> We have found that using a super resolution Pytorch model to get a larger image results in significant noise removal. Perhaps it is a by-product.
>
> You can try it out. I have the model and can share it on Google Drive if you want to try it out. We have tried the models available in OpenCV DNN but they do not seem to give the same results.
>
> Regards,
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Thursday, May 29, 2025 5:28 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Dear Sir,
>
> Thank you for the confirmation. I will maintain the time data on a Google Sheet and ensure it is updated regularly.
>
> Looking forward to starting from Monday, 2nd June.
>
> Best regards,
> Hindvee
>
> On Thu, May 29, 2025 at 2:53 PM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Hello Hindvee,
> >
> > Welcome aboard!
> >
> > You can keep the time data on Google Sheet.
> >
> > Regards,
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Thursday, May 29, 2025 5:12 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection

Greetings Mr. Durve,

Thank you very much for giving me the opportunity to contribute to Guangdong Longtian Smart Instruments Co. Ltd. through this exciting project. I am pleased to accept the offer and look forward to working on the development of the AI-assisted line detection tool.

With your permission, I would like to begin the internship from the coming Monday, i.e., 2nd June 2025.

Additionally, I wanted to ask about the preferred method for tracking and sharing the time I spend on the project. Would you like me to maintain a shared timesheet (e.g., a Google Sheet or Excel file), or would you prefer weekly time summaries via email or any other format?

Looking forward to your guidance on this.

Thank you once again for this opportunity.

Warm regards,
Hindvee Rajesh Vispute

On Thu, May 29, 2025 at 8:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have attached the on-boarding letter.
>
> Please let me know if you have any queries.
>
> Regards,
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Saturday, May 10, 2025 3:46 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings Mr. Durve,
>
> Thank you for your detailed response and clarification. I now clearly understand that the goal is to enhance line detection using AI within known ROIs, leveraging the expected position of endpoints.
>
> To support this objective, I've explored and implemented several **AI-based approaches**, any of which can be used to complement or improve upon conventional methods:
>
> 1. **CNN-Based Line Presence Classifier (NB-1):** Identifies whether a line exists within a given patch/ROI—useful for filtering candidate regions before applying traditional detection.

2. **CNN Regression Model (NB-2):** Predicts line endpoints directly from the image, offering robust localization even in low-contrast or noisy scenarios.

3. **Denoising Autoencoder (NB-5):** Enhances faint or broken lines by cleaning up noise, thereby boosting the effectiveness of existing edge or Hough-based methods.

4. **Vision Transformer Model (NB-4):** Designed for fine-grained classification tasks and can be adapted for future objectives like surface defect or anomaly detection.

These methods are independently deployable and can be integrated selectively depending on the specific inspection challenge or ROI structure. I'd be happy to align with your ROI-driven strategy and customize accordingly.

Regarding the stipend, I agree with the proposed Rs. 200/hour base rate and performance-based bonus, and I'm enthusiastic about the opportunity to collaborate further on defect detection and shape anomaly tasks in the future.

I have attached screenshots of the code files I executed today. I have the names of the notebooks (NB) with NB-1, NB-2,NB-4 and NB-5 respectively.

NB-1
NB-2
NB-4
NB-5

Please let me know if you'd like me to share these outputs or run a quick live demonstration.

Warm regards,
Hindvee

On Fri, May 9, 2025 at 6:03 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> I think there is a mismatch in understanding. We are already using conventional line detection techniques (various edge detection operators and Hough Transform and other methods for line fitment) .
>
> What we are looking to get from you is AI based / AI assisted line detection. Since we know the probable position of the line beforehand we can use this information to find a line. For example, say we know that the line end points are expected to be (x1, y1) and (x2, y2), we can construct a small rectangle (ROI) using these points and search for a line within this ROI.

As for the stipend it depends both on the time you spend on the project and the results you produce. It will comprise of an hourly rate and a result linked amount. I suggest we have an hourly rate of Rs. 200/hour and a matching amount linked to the results at the end of the project. So, if you are able to produce satisfactory results it will come to Rs. 400/hour. Also, if the results are satisfactory, we can take this forward next year and work on other issues like surface defect detection, detection of shape anomalies etc.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 8, 2025 5:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr Durve,
I have the problem now so I will focus on faint line detection for the first step. I can proceed using Sobel, Scharr, and Canny edge detectors provided by OpenCV. First I will preprocess the images and then move to edge detection using the above 3 detectors. I will carry out line detection using Hough Transform, and further we will filter out short or irrelevant lines based on length and their angle of orientation. Finally, we will draw the detected lines on the images using cv.line()

I can devote 14 hrs per week to this task for 2 months, we can extend the time if required. Also, I wanted to ask what stipend you can provide me. I believe I can complete the task in the above time frame. Thank you for your time.

Warm regards
Hindvee

On Thu, May 8, 2025 at 7:18 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Your analysis seems good.
>
> At this point we are not interested in detecting defects (though at a later stage we may be), and want to detect the faint lines only.
>
> Note that though I have annotated only a single line in the image I sent to you yesterday, we need to detect edge lines of all vertical stripes.
>
> I have attached 2 sets of 20 images each named 'Good Images' and

'Bad Images' - images in which it is mostly easy to detect lines by the conventional image processing method and those in which lines are faint and hard to detect. These images may help you get a better understanding of the problem.

Please also check the following issues:

1. How much time will you be able to devote to this task?
2. Will you be able to complete the task within the available time?

There are other issues like time taken by the AI route, integration of AI line detect code in VC++ project, further training / fine tuning of model, applicability of the method to detect faint circles etc. But we will get into them later once the basic method is proven.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, May 7, 2025 3:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings, Mr. Durve,
Thank you for reviewing my proposal and for sharing the sample images. By looking at the images I came across the specific challenges faced by the inspection system, like detecting faint lines, subtle surface variations, and fine defects that can be missed by traditional image processing. Based on these observations I suggest that we can do the following:-

1. **Hybrid approach**: Objective - Combine the speed of OpenCV line detection (example Hough Transform) with deep learning's pattern recognition capability.
Approach
- Use OpenCV to pre-locate candidate regions and feed them into a deep model for verification.
- This can speed up inference and increase resistance to false positives.

2. **Deep learning-based fine defect detection**: Objective: Improve the detection of faint, low contrast line defects and localized surface anomalies.
Approach
- Train a model, such as YOLOv8 or U-Net, tailored to your defect patterns using the annotated images.

- To improve feature visibility, add contrast enhancement.
- Assess using detection accuracy and intersection over union (IoU) to compare to baselines based on OpenCV.

I can begin with a proof-of-concept model using the images shared by you. Thank you for sharing the opportunity. I am excited to get a chance to contribute to the advancement in inspection technology.

Warm regards
Hindvee Vispute

On Wed, 7 May 2025, 09:31 Kishor Durve, <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have gone thru the document uploaded by you.
>
> It looks ok to me.
>
> I can have attached a few images so you will get a better idea of the requirements.
>
> Kishor Durve

Kishor Durve <kishor.durve@outlook.com> Wed, Jul 30, 2025 at 11:02 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

I have checked the images. Your conclusions are correct.

Please carry on with your efforts.

Kishor Durve
Duravision Systems
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, July 29, 2025 3:37 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!
current update:-

- Using an autoencoder trained on clean (good) images to repair bad images
- Applying a **hybrid loss** of MSE and SSIM to preserve structure and perceptual quality.
- Saving and visualizing both original and reconstructed images which helps in debugging

Despite this denoising strategy, DexiNED is still failing for bad images, the possible reasons are as follows

1. Contrast remains too low after reconstruction.
2. Edges are too smooth or faint due to over-smoothing by the autoencoder.
3. DexiNED is not tuned for synthetic/industrial patterns (e.g., PCB, x-ray like structures).

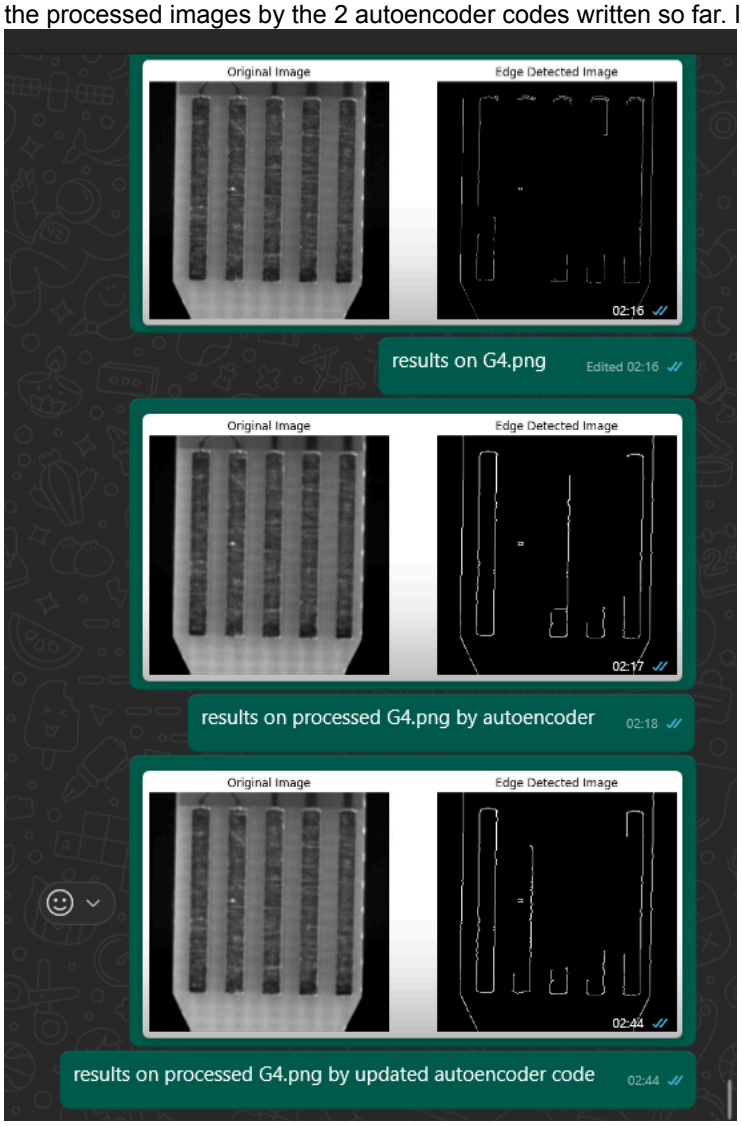
I will now work on these factors to improve the quality of processed images. I am attaching the processed Good and Bad Images folders.
BAD IMAGES _ PROCESSED

GOOD IMAGES_ PROCESSED

warm regards
Hindvee

On Mon, Jul 28, 2025 at 2:48 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Noted sir. I wrote code for autoencoder to get sharper images, and I applied DexiNed on the original image and also on the processed images by the 2 autoencoder codes written so far. I am sharing the results here.



warm regards
Hindvee

On Fri, Jul 25, 2025 at 6:34 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Sorry for delay in replying. I was away for quite some time.
>
> A couple of months back we had tried OpenCV / HED and obtained similar results.
>
> Please go ahead and try DexiNed after which we will decide about line detection.
>
> After that another important step would be to remove noise from image using a suitable model.
>
> I know this is somewhat away from our goal of 'AI assisted edge detection' but it can be an important intermediate step. We are currently using a model that does this. This is the model I had sent to you initially. It is developed in a Chinese University by a professor who works with our company.
>
> Kishor Durve
> Duravision Systems
> 9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, July 25, 2025 5:46 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir ! This is an output image obtained from the HED model and similar image I have obtained from DexiNed, could u pls tell i am moving in the right direction or should I make improvements or change in the plan of action?

warm regards
Hindvee

On Wed, Jul 23, 2025 at 3:20 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> Hello sir! Could you please share ur insights and views on the above mail thread, it shall be very helpful to get to know your feedback and move forward with the project.
>
> Warm Regards
> Hindvee
>
> On Fri, Jul 11, 2025 at 3:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > Hello sir!
> >
> > Following up on my previous email regarding edge detection, I wanted to share the progress I've made since then. I have been focusing on using the Holistically-Nested Edge Detection (HED) method. I have now:
> >
> > CODE FILE
> >
> > 1. Successfully applied the HED process to a folder containing our images to generate edge maps for the entire dataset.
> >
> > 2. Trained a custom autoencoder model designed to refine and correct the lines in these images.
> >
> > 3. Applied the trained autoencoder to the HED output images to produce corrected versions.
> >
> > 4. Started working on post-processing techniques.
> >
> > I am currently experimenting with different parameters and methods to achieve the best results for this step. I am continuing to refine the line clarification process and will share updated results once I have finalized the post-processing approach.
> >
> > warm regards
> >
> > Hindvee
> >
> > On Wed, Jul 9, 2025 at 12:23 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > > Hello sir! I am trying the pre-trained model for detecting edges in our images; I have tried Holistically-Nested Edge Detection (HED) and DexiNed. I am attaching the output, I am going forward with HED to get better results.
> > >
> > > Warm regards
> > > Hindvee
> > >
> > > On Mon, Jul 7, 2025 at 10:37 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > > > Hello sir!
> > > > I am sharing the first 5 difference images of the three codes that I mentioned previously. Please check this whenever you have time.
> > > >
> > > > Warm regards
> > > > Hindvee
> > > >
> > > > On Mon, Jul 7, 2025 at 6:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> > > > > I am unable to give my view of the approach as I can't view the images. I can give my opinion if you send me difference images. In any case, I think you need to have difference images as a metric in addition to the Laplacian.
> > > > >
> > > > > Get Outlook for Android
> > > > >
> > > > > ---
> > > > >
> > > > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > > > **Sent:** Monday, July 7, 2025 7:09:51 AM
> > > > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > > > **Subject:** Re: AI assisted edge detection

Hello sir!
I have received the stipend, thank you for that! I'd like to share a brief update on the image enhancement work, focusing on noise removal and edge preservation.
Summary of results:-
1. Previous code
---Performed basic denoising but introduced over-smoothing.
---Laplacian MAD: ~0.0175–0.0192

2. DnCNN
---Improved noise removal and structural retention.
---MAD: ~0.0186–0.0202

3. DnCNN + Sharpening (Current approach)
---Added post-processing to enhance edge clarity.
---Consistently improved MAD to ~0.0178–0.0196.
---Visual results show reduced whitening and better edge definition.

This confirms that the current setup provides better denoising while preserving fine image details. Please let me know if you'd like to proceed with further tuning using SSIM or edge-aware loss functions.

Warm regards

Hindvee

On Fri, Jul 4, 2025 at 3:54 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Please keep at it. Use difference image and Laplacian to gauge the effect of the algorithm. I will be able to look at your work only after 15th.
> I have transferred your stipend by GPay.
>
> Get Outlook for Android
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Friday, July 4, 2025 5:37:35 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir!
> This is what I am been working on. I am still working on getting better edges and less whitish images.
>
> warm regards
> Hindvee
>
> On Thu, Jul 3, 2025 at 10:53 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > UPI ID: hindveevispute@okicici
> >
> > Warm regards
> > Hindvee
> >
> > On Wed, 2 Jul 2025, 03:16 Kishor Durve, <kishor.durve@outlook.com> wrote:
> >
> > > I checked the images, but did not notice substantial difference between the before and after images. You can subtract them (after aligning) and check.
> > > As regards payment of stipend please let me know how much we need to pay and I will transfer it on GPay.
> > > I am travelling again and am out of office. I will be back only on 15th. My responses to your mails may be delayed.
> > >
> > > Get Outlook for Android
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Tuesday, July 1, 2025 1:51:45 AM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection
> > >
> > > Hello sir!

I was eager to get insights from you about the above shared images.Please do have a look as soon as possible. Also, I wanted to know whether the stipend will be credited at the end of the month or the project's end. It would be very kind of you if I could get the stipend at the end of the month.

Thanks and regards
Hindvee

On Mon, Jun 30, 2025 at 10:33 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Good morning sir!
>
> Did you get any chance to review the above images? Please share your suggestions and insights with me.
>
> Warm regards
> Hindvee
>
> On Thu, Jun 26, 2025 at 11:09 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Here I am attaching the processed images.
> >
> > Warm regards
> > Hindvee
> >
> > On Thu, Jun 26, 2025 at 8:08 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > I am unable to run the files in Colab and generate images. Can you please send me the processed images on mail - .png / .jpeg files, not .ipynb files.
> > >
> > > Kishor Durve
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Thursday, June 26, 2025 4:48 AM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection
> > >
> > > Hello sir, I have attached the results for png 1,2,3 and 4. Pls share ur views and insights.
> > >
> > > Warm Regards
> > > Hindvee
> > >
> > > On Tue, Jun 24, 2025 at 11:29 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > >
> > > > Hello sir,
> > > > I am attaching the .ipynb file, it has the images
> > > >
> > > > warm regards
> > > > Hindvee
> > > >
> > > > On Tue, Jun 24, 2025 at 5:48 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> > > >
> > > > > Please share the images for visual comparison.
> > > > >
> > > > > Kishor Durve
> > > > >
> > > > > ---
> > > > >
> > > > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > > > **Sent:** Tuesday, June 24, 2025 5:13 AM
> > > > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > > > **Subject:** Re: AI assisted edge detection
> > > > >
> > > > > Hello Durve sir,
> > > > >
> > > > > I wanted to share an update on the denoising output. I noticed earlier that while the model was doing a good job removing noise, it was also making the image a bit too smooth — the sharpness had dropped quite a bit:
> > > > >
> > > > > - Noisy Image: 33.72
> > > > > - Denoised Image: 6.92

- Clean Image (Ground Truth): 26.49

To improve this, I added a sharpening filter after the autoencoder output and tested two variants:

1. A **strong sharpening kernel**, which raised the denoised sharpness to **112.40,** but it felt a bit overdone and introduced some artifacts.
2. A **softer kernel**, which gave a cleaner and more natural result, with denoised sharpness at **54.04**. This seems like a balanced point where edges are clearer without compromising visual quality.

If you'd like, I can also share the output images (clean vs. denoised) for a visual comparison.

Warm regards

Hindvee

image.png

On Fri, Jun 20, 2025 at 12:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir!
> Based on the latest code I calculated the sharpness using Laplacian variance. I obtained the following results as those attached as image.
> I think that while denoising, though the noise was removed, but the model blurred or over smoothened the image. I may have lost some meaningful edge details in the process. But the sharpness value of Clean image is lower than the noisy image (because it has no random noise), but high enough to retain clear edges and structure. So I am working next on increasing the sharpness value while keeping the noise low. Please do share your views with me.
>
> Warm regards
> Hindvee
>
> image.png
>
> On Thu, 19 Jun 2025, 13:42 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > I have seen the processed images. I think it is good at this point to quantify 'sharpness' of an image. It will give us a measure to compare different methods / models.
> >
> > There can be several approaches to do this. We use Opencv::Laplacian() to judge image sharpness. But you can try other approaches.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Thursday, June 19, 2025 2:52 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Hello sir,
> >
> > I've completed the full-image denoising as discussed for the images `5.png` and `6.png`. This time, the model was trained and applied directly on the complete images (cropped to a common size of **570×588**) instead of small patches, as per your suggestion. The denoising autoencoder was able to reduce background noise and enhance structural features, particularly the edges of the circles, which were initially faint or inconsistent. The model output shows a noticeable improvement in clarity while preserving the original patterns.

I've attached the processed result image along with this mail. Please let me know your thoughts on this file.

Warm regards,
Hindvee

On Wed, Jun 18, 2025 at 12:33 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have checked the file. These two images contain circles as primary features. Also, the patches in the file seem to be for a small part of the original image. If you have processed the original images (5.png and 6.png) please send me the processed images on mail. If you have processed only parts of these images, I suggest you process the full images and send me the results.
>
> Similarly, in the previous exercise please process complete images. Most of the set are of size 500x500 pixel, so it should not consume too much time.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 18, 2025 1:44 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Sir, please let me know your views on the above .ipynb file.
> Warm Regards
> Hindvee
>
> On Wed, Jun 18, 2025 at 1:56 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir,
> > I have processed the 5.png and 6.png, I faced lower accuracy rate earlier. After fixing the earlier issues with mismatched data sizes and high Binary Crossentropy loss (~0.6098), the model is now working smoothly. It reached a much lower loss of just 0.0528 in 5 epochs, and both training and validation losses settled nicely by the 2nd or 3rd epoch without signs of overfitting. The denoised images look very clean and closely match the original ones, even when the input images had different levels of noise. I am attaching the results in the attached file.
> > Warm regards
> > Hindvee
> >
> > On Tue, Jun 17, 2025 at 7:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > Sorry! Forgot the attachment.
> > >
> > > Kishor Durve
> > >
> > > ---
> > >
> > > **From:** Kishor Durve <kishor.durve@outlook.com>
> > > **Sent:** Tuesday, June 17, 2025 9:46 AM
> > > **To:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Subject:** Re: AI assisted edge detection
> > >
> > > Good to know that you are making progress at a nice clip.
> > >
> > > I saw the images and the difference between before / after can be seen. Can you please process the images attached to this mail and send me the results? The images are similar to those I had sent earlier, the exception being 5.png and 6.png. In these last two images the challenge is to detect circles

whose edges are not well defined and the edge quality may vary from image to image. This is to test the effectiveness of noise removal / sharpening.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, June 17, 2025 3:06 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir,
I have completed the training and evaluation of the denoising autoencoder using grayscale image patches taken from the `IMG` folder. The model was trained for 20 epochs on 64×64 patches with binary cross entropy loss and a 10% validation split. During training, the loss reduced steadily and stabilized, with the final reconstruction loss reaching 0.3688. The results show that the model can effectively denoise images and recover clean patterns like faint lines. In the next step, I plan to use the output of the Super-Resolution model as a base to evaluate the accuracy and quality of the denoising process, to better understand how both methods contribute to image enhancement. I am attaching the .ipynb file for your reference and also the IMG Folder which I just mentioned.

Warm regards
Hindvee

IMG FOLDER

Accuracy results:-

 image.png

On Fri, Jun 13, 2025 at 5:33 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> The images I sent are big. OpenCV will not be able to handle them as the max size allowed is 32K x 32K. The crash you are facing could be due to this reason.
>
> You can crop out small patches from the big images and use them for trials.
>
> Kishor Durve
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 11, 2025 1:37 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Thank you for sharing more images. I have updated the work on the sheets.
>
> Warm regards
> Hindvee
>
> On Sat, Jun 7, 2025 at 2:51 PM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > I have uploaded more images. You can crop out patches from them. One of the images has many boards which will give many useful patches.
> >
> > Kishor Durve

Duravision Systems
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Saturday, June 7, 2025 4:36 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings sir,
It would be great if I get more images, that would expand my dataset.
Thank you for your previous email.
I also wanted tk inform you that I am travelling and will be able to resume work by Tuesday. I request you to kindly allow my leave.

Warm Regards
Hindvee

On Fri, 6 Jun 2025, 11:31 Kishor Durve, <kishor.durve@outlook.com> wrote:

> I am not going to be of much help to you as I have limited experience in this field. Also, what little I know was gained by applying available models in OpenCV's DNN module.
>
> There are two images available in the images folder. You may crop out patches from them for the Training / test sets. We use ImageJ and IrfanView for such work as MS Paint cannot handle such big images. If you need more images please let me know.
>
> Kishor Durve
> Duravision Systems
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Friday, June 6, 2025 5:24 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings sir,
> I hope you're doing well. I've just finished training the CNN-based line presence classifier, and I'm attaching the loss graph for your review. The model seems to have learned quite quickly, with both training and validation losses staying very close to zero throughout. There was a small spike in training loss around epoch 80, but validation stayed stable, so it might just be a minor fluctuation.
>
> Since the dataset is quite small and the model converged very early, I'm unsure how best to move forward.According to you should I try to get more data to avoid overfitting. Please guide me towards a direction in which I can make this model more effective.
>
> Warm regards
>
> Hindvee
>
> On Wed, Jun 4, 2025 at 6:42AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Yes, it looks ok to me. We will take a relook once you make more progress.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Wednesday, June 4, 2025 4:26 AM

To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Additionally, Sir, I would like to confirm if the current format of the Google Sheet meets your expectations. Please let me know if you would prefer any modifications, such as a more detailed breakdown of the work completed.

Warm Regards
Hindvee

On Wed, Jun 4, 2025 at 1:34 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Greetings Mr. Durve,
>
> Thanks for sharing the super-resolution model. I've set it up and tested it, the results inspiring. It does a good job cleaning up noise and improving contrast, especially in the faint line areas. As we discussed, I'll now add this to the bigger pipeline along with the AI modules (NB-1 to NB-5). The super-resolution output can help improve both the CNN classifier and regression models. I'll also check how it works together with NB-5, the denoising autoencoder, to see if we get even better results.
>
> Please let me know if you want me to run it on any specific ROI samples or start putting together comparison results.
>
> Warm regards
> Hindvee
>
> On Tue, Jun 3, 2025 at 9:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > I have file network.py in folder 'models'. This file has the RRDBNet implementaion.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Monday, June 2, 2025 7:15 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings Mr. Durve,
> > I am facing an issue again an again while running the VS 2022 code, pls help me with this, we have the module named as model_zoo not models. Pls tell me how to proceed with this.
> >
> > image.png
> >
> > Warm Regards
> > Hindvee
> >
> > On Sat, May 31, 2025 at 12:46 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> >
> > > Thankyou sir! I will try this right away.
> > > Also I hope you received this google sheet, I have shared edit access to you on your gmail account
> > >
> > > Hindvee Vispute - Time data - Intern
> > >
> > > Regards
> > > Hindvee

On Sat, May 31, 2025 at 5:43 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Sure. You need to do the following to make the project work.
>
> 1. Download and install VS 2022 Community edition
> 2. Download and install Anaconda. Install an additional environment Python 3.8.16. Use this version of Python for the project.
>
> The project should run at this point.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Friday, May 30, 2025 6:34 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings sir!
> I went through the google drive folder shared by you. It would be really helpful if you could walk me through it.
> Best regards
> Hindvee
>
> On Fri, May 30, 2025 at 11:41 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > You can download the model, a working VC++ (VS 2022) project and image here.
> >
> > https://drive.google.com/drive/folders/1mLBDjJR3fFKs8Brr_nNj2z9YyI89nruE?usp=sharing
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, May 30, 2025 12:50 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings sir!
> > Pls do share the model over drive I will try it out. Also I request you to share more image database so that I can start working on it.
> > I will start work from today itself then.
> >
> > Regards
> > Hindvee
> >
> > On Fri, 30 May 2025, 06:11 Kishor Durve, <kishor.durve@outlook.com> wrote:
> >
> > > We have found that using a super resolution Pytorch model to get a larger image results in significant noise removal. Perhaps it is a by-product.
> > >
> > > You can try it out. I have the model and can share it on Google Drive if you want to try it out. We have tried the models available in OpenCV DNN but they do not seem to give the same results.

Regards,

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 29, 2025 5:28 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Dear Sir,

Thank you for the confirmation. I will maintain the time data on a Google Sheet and ensure it is updated regularly.

Looking forward to starting from Monday, 2nd June.

Best regards,
Hindvee

On Thu, May 29, 2025 at 2:53 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> Welcome aboard!
>
> You can keep the time data on Google Sheet.
>
> Regards,
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Thursday, May 29, 2025 5:12 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings Mr. Durve,
>
> Thank you very much for giving me the opportunity to contribute to Guangdong Longtian Smart Instruments Co. Ltd. through this exciting project. I am pleased to accept the offer and look forward to working on the development of the AI-assisted line detection tool.
>
> With your permission, I would like to begin the internship from the coming Monday, i.e., 2nd June 2025.
>
> Additionally, I wanted to ask about the preferred method for tracking and sharing the time I spend on the project. Would you like me to maintain a shared timesheet (e.g., a Google Sheet or Excel file), or would you prefer weekly time summaries via email or any other format?
>
> Looking forward to your guidance on this.
>
> Thank you once again for this opportunity.

Warm regards,
Hindvee Rajesh Vispute

On Thu, May 29, 2025 at 8:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have attached the on-boarding letter.
>
> Please let me know if you have any queries.
>
> Regards,
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Saturday, May 10, 2025 3:46 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings Mr. Durve,
>
> Thank you for your detailed response and clarification. I now clearly understand that the goal is to enhance line detection using AI within known ROIs, leveraging the expected position of endpoints.
>
> To support this objective, I've explored and implemented several **AI-based approaches**, any of which can be used to complement or improve upon conventional methods:
>
> 1. **CNN-Based Line Presence Classifier (NB-1):** Identifies whether a line exists within a given patch/ROI—useful for filtering candidate regions before applying traditional detection.
> 2. **CNN Regression Model (NB-2):** Predicts line endpoints directly from the image, offering robust localization even in low-contrast or noisy scenarios.
> 3. **Denoising Autoencoder (NB-5):** Enhances faint or broken lines by cleaning up noise, thereby boosting the effectiveness of existing edge or Hough-based methods.
> 4. **Vision Transformer Model (NB-4):** Designed for fine-grained classification tasks and can be adapted for future objectives like surface defect or anomaly detection.
>
> These methods are independently deployable and can be integrated selectively depending on the specific inspection challenge or ROI structure. I'd be happy to align with your ROI-driven strategy and customize accordingly.
>
> Regarding the stipend, I agree with the proposed Rs. 200/hour base rate and performance-based bonus, and I'm enthusiastic

about the opportunity to collaborate further on defect detection and shape anomaly tasks in the future.

I have attached screenshots of the code files I executed today. I have the names of the notebooks (NB) with NB-1, NB-2,NB-4 and NB-5 respectively.

NB-1
NB-2
NB-4
NB-5

Please let me know if you'd like me to share these outputs or run a quick live demonstration.

Warm regards,
Hindvee

On Fri, May 9, 2025 at 6:03 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> I think there is a mismatch in understanding. We are already using conventional line detection techniques (various edge detection operators and Hough Transform and other methods for line fitment) .
>
> What we are looking to get from you is AI based / AI assisted line detection. Since we know the probable position of the line beforehand we can use this information to find a line. For example, say we know that the line end points are expected to be (x1, y1) and (x2, y2), we can construct a small rectangle (ROI) using these points and search for a line within this ROI.
>
> As for the stipend it depends both on the time you spend on the project and the results you produce. It will comprise of an hourly rate and a result linked amount. I suggest we have an hourly rate of Rs. 200/hour and a matching amount linked to the results at the end of the project. So, if you are able to produce satisfactory results it will come to Rs. 400/hour. Also, if the results are satisfactory, we can take this forward next year and work on other issues like surface defect detection, detection of shape anomalies etc.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Thursday, May 8, 2025 5:15 PM

To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr Durve,
I have the problem now so I will focus on faint line detection for the first step. I can proceed using Sobel, Scharr, and Canny edge detectors provided by OpenCV. First I will preprocess the images and then move to edge detection using the above 3 detectors. I will carry out line detection using Hough Transform, and further we will filter out short or irrelevant lines based on length and their angle of orientation. Finally, we will draw the detected lines on the images using cv.line()

I can devote 14 hrs per week to this task for 2 months, we can extend the time if required. Also, I wanted to ask what stipend you can provide me. I believe I can complete the task in the above time frame. Thank you for your time.

Warm regards
Hindvee

On Thu, May 8, 2025 at 7:18 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Your analysis seems good.
>
> At this point we are not interested in detecting defects (though at a later stage we may be), and want to detect the faint lines only.
>
> Note that though I have annotated only a single line in the image I sent to you yesterday, we need to detect edge lines of all vertical stripes.
>
> I have attached 2 sets of 20 images each named 'Good Images' and 'Bad Images' - images in which it is mostly easy to detect lines by the conventional image processing method and those in which lines are faint and hard to detect. These images may help you get a better understanding of the problem.
>
> Please also check the following issues:
>
> 1. How much time will you be able to devote to this task?
> 2. Will you be able to complete the task within the available time?
>
> There are other issues like time taken by the AI route, integration of AI line detect code in VC++ project, further training / fine tuning of

model, applicability of the method to detect faint circles etc. But we will get into them later once the basic method is proven.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, May 7, 2025 3:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings, Mr. Durve,
Thank you for reviewing my proposal and for sharing the sample images. By looking at the images I came across the specific challenges faced by the inspection system, like detecting faint lines, subtle surface variations, and fine defects that can be missed by traditional image processing. Based on these observations I suggest that we can do the following:-

1. **Hybrid approach**: Objective - Combine the speed of OpenCV line detection (example Hough Transform) with deep learning's pattern recognition capability.
Approach
- Use OpenCV to pre-locate candidate regions and feed them into a deep model for verification.
- This can speed up inference and increase resistance to false positives.

2. **Deep learning-based fine defect detection**: Objective: Improve the detection of faint, low contrast line defects and localized surface anomalies.
Approach
- Train a model, such as YOLOv8 or U-Net, tailored to your defect patterns using the annotated images.
- To improve feature visibility, add contrast enhancement.
- Assess using detection accuracy and intersection over union (IoU) to compare to baselines based on OpenCV.

I can begin with a proof-of-concept model using the images shared by you. Thank you for sharing the opportunity. I am excited to get a chance to contribute to the advancement in inspection technology.

Warm regards
Hindvee Vispute

On Wed, 7 May 2025, 09:31 Kishor Durve, <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have gone thru the document uploaded by you.

It looks ok to me.

I can have attached a few images so you will get a better idea of the requirements.

Kishor Durve

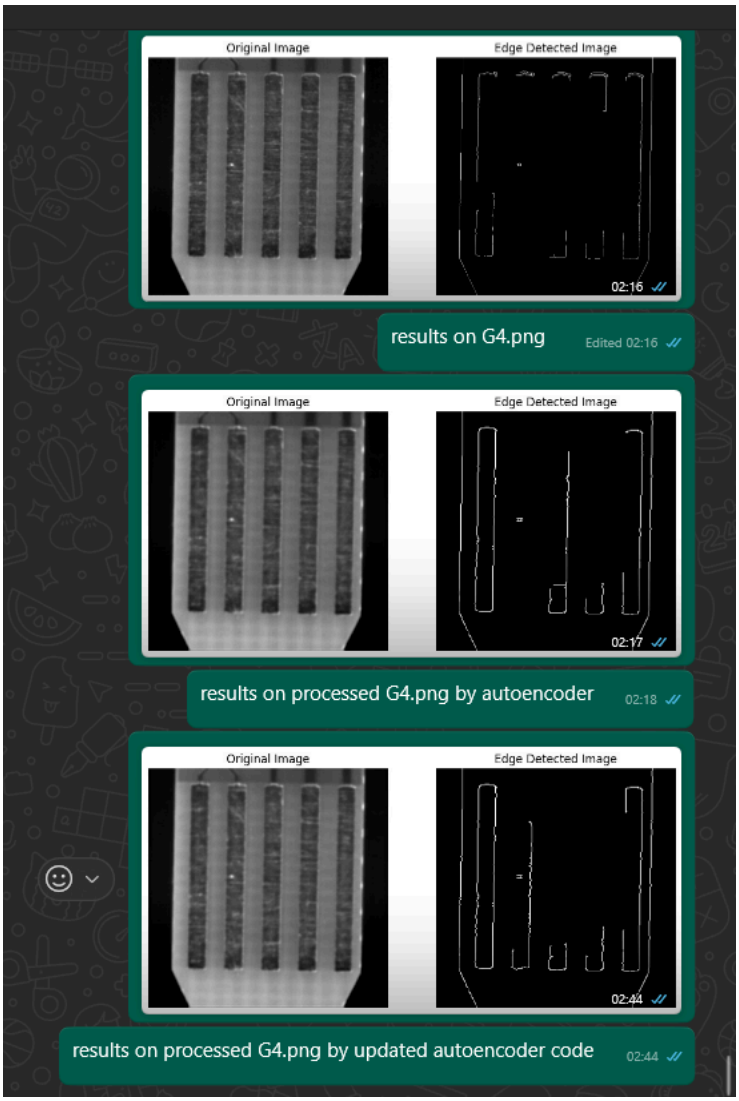
Hindvee Vispute <hindveevispute@gmail.com> Fri, Aug 1, 2025 at 2:14 AM
To: Kishor Durve <kishor.durve@outlook.com>

Thank you sir! I have worked out and increased the contrast in the processed images. I will share the file with you.
Also sir I shall request you to please process the stipend for this month. It has been 58 hrs for the month of July. If you like my work please consider increasing the per hour rate. It is just a small suggestion. The final decision depends on you and I happily accept your decision. My primary goal is to enhance my skills, and I am learning a lot from this project.

Warm regards
Hindvee

On Wed, Jul 30, 2025 at 11:02 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have checked the images. Your conclusions are correct.
>
> Please carry on with your efforts.
>
> Kishor Durve
> Duravision Systems
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Tuesday, July 29, 2025 3:37 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir!
> current update:-
>
> - Using an autoencoder trained on clean (good) images to repair bad images
> - Applying a **hybrid loss** of MSE and SSIM to preserve structure and perceptual quality.
> - Saving and visualizing both original and reconstructed images which helps in debugging
>
> Despite this denoising strategy, DexiNED is still failing for bad images, the possible reasons are as follows
>
> 1. Contrast remains too low after reconstruction.
> 2. Edges are too smooth or faint due to over-smoothing by the autoencoder.
> 3. DexiNED is not tuned for synthetic/industrial patterns (e.g., PCB, x-ray like structures).
>
> I will now work on these factors to improve the quality of processed images. I am attaching the processed Good and Bad Images folders.
> BAD IMAGES _ PROCESSED
> GOOD IMAGES_ PROCESSED
>
> warm regards
> Hindvee
>
> On Mon, Jul 28, 2025 at 2:48 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Noted sir. I wrote code for autoencoder to get sharper images, and I applied DexiNed on the original image and also on the processed images by the 2 autoencoder codes written so far. I am sharing the results here.



warm regards
Hindvee

On Fri, Jul 25, 2025 at 6:34 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Sorry for delay in replying. I was away for quite some time.
>
> A couple of months back we had tried OpenCV / HED and obtained similar results.
>
> Please go ahead and try DexiNed after which we will decide about line detection.
>
> After that another important step would be to remove noise from image using
> a suitable model.
>
> I know this is somewhat away from our goal of 'AI assisted edge detection' but it can be an important intermediate step. We are currently using a model that does this. This is the model I had sent to you initially. It is developed in a Chinese University by a professor who works with our company.
>
> Kishor Durve
> Duravision Systems
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Friday, July 25, 2025 5:46 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir ! This is an output image obtained from the HED model and similar image I have obtained from DexiNed, could u pls tell i am moving in the right direction or should I make improvements or change in the plan of action?

warm regards
Hindvee

On Wed, Jul 23, 2025 at 3:20 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> Hello sir! Could you please share ur insights and views on the above mail thread, it shall be very helpful to get to know your feedback and move forward with the project.
>
> Warm Regards
> Hindvee
>
> On Fri, Jul 11, 2025 at 3:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > Hello sir!
> >
> > Following up on my previous email regarding edge detection, I wanted to share the progress I've made since then. I have been focusing on using the Holistically-Nested Edge Detection (HED) method. I have now:
> >
> > CODE FILE
> >
> > 1. Successfully applied the HED process to a folder containing our images to generate edge maps for the entire dataset.
> >
> > 2. Trained a custom autoencoder model designed to refine and correct the lines in these images.
> >
> > 3. Applied the trained autoencoder to the HED output images to produce corrected versions.
> >
> > 4. Started working on post-processing techniques.
> >
> > I am currently experimenting with different parameters and methods to achieve the best results for this step. I am continuing to refine the line clarification process and will share updated results once I have finalized the post-processing approach.
> >
> > warm regards
> >
> > Hindvee
> >
> > On Wed, Jul 9, 2025 at 12:23 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > > Hello sir! I am trying the pre-trained model for detecting edges in our images; I have tried Holistically-Nested Edge Detection (HED) and DexiNed. I am attaching the output, I am going forward with HED to get better results.
> > >
> > > Warm regards
> > > Hindvee
> > >
> > > On Mon, Jul 7, 2025 at 10:37 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > > > Hello sir!
> > > > I am sharing the first 5 difference images of the three codes that I mentioned previously. Please check this whenever you have time.
> > > >
> > > > Warm regards
> > > > Hindvee
> > > >
> > > > On Mon, Jul 7, 2025 at 6:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> > > > > I am unable to give my view of the approach as I can't view the images. I can give my opinion if you send me difference images. In any case, I think you need to have difference images as a metric in addition to the Laplacian.
> > > > >
> > > > > Get Outlook for Android
> > > > >
> > > > > ---
> > > > >
> > > > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > > > **Sent:** Monday, July 7, 2025 7:09:51 AM
> > > > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > > > **Subject:** Re: AI assisted edge detection
> > > > >
> > > > > Hello sir!
> > > > > I have received the stipend, thank you for that! I'd like to share a brief update on the image enhancement work, focusing on noise removal and edge preservation.
> > > > > Summary of results:-
> > > > > 1. Previous code
> > > > > ---Performed basic denoising but introduced over-smoothing.

---Laplacian MAD: ~0.0175–0.0192

2. DnCNN
---Improved noise removal and structural retention.
---MAD: ~0.0186–0.0202

3. DnCNN + Sharpening (Current approach)
---Added post-processing to enhance edge clarity.
---Consistently improved MAD to ~0.0178–0.0196.
---Visual results show reduced whitening and better edge definition.

This confirms that the current setup provides better denoising while preserving fine image details.
Please let me know if you'd like to proceed with further tuning using SSIM or edge-aware loss functions.

Warm regards

Hindvee

On Fri, Jul 4, 2025 at 3:54 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Please keep at it. Use difference image and Laplacian to gauge the effect of the algorithm. I will be able to look at your work only after 15th.
> I have transferred your stipend by GPay.
>
> Get Outlook for Android
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Friday, July 4, 2025 5:37:35 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir!
> This is what I am been working on. I am still working on getting better edges and less whitish images.
>
> warm regards
> Hindvee
>
> On Thu, Jul 3, 2025 at 10:53 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > UPI ID: hindveevispute@okicici
> >
> > Warm regards
> > Hindvee
> >
> > On Wed, 2 Jul 2025, 03:16 Kishor Durve, <kishor.durve@outlook.com> wrote:
> >
> > > I checked the images, but did not notice substantial difference between the before and after images. You can subtract them (after aligning) and check.
> > > As regards payment of stipend please let me know how much we need to pay and I will transfer it on GPay.
> > > I am travelling again and am out of office. I will be back only on 15th. My responses to your mails may be delayed.
> > >
> > > Get Outlook for Android
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Tuesday, July 1, 2025 1:51:45 AM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection
> > >
> > > Hello sir!
> > >
> > > I was eager to get insights from you about the above shared images.Please do have a look as soon as possible. Also, I wanted to know whether the stipend will be credited at the end of the month or the project's end. It would be very kind of you if I could get the stipend at the end of the month.
> > >
> > > Thanks and regards
> > > Hindvee

On Mon, Jun 30, 2025 at 10:33 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Good morning sir!

Did you get any chance to review the above images? Please share your suggestions and insights with me.

Warm regards
Hindvee

On Thu, Jun 26, 2025 at 11:09 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Here I am attaching the processed images.

Warm regards
Hindvee

On Thu, Jun 26, 2025 at 8:08 AM Kishor Durve <kishor.durve@outlook.com> wrote:

I am unable to run the files in Colab and generate images. Can you please send me the processed images on mail - .png / .jpeg files, not .ipynb files.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, June 26, 2025 4:48 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir, I have attached the results for png 1,2,3 and 4. Pls share ur views and insights.

Warm Regards
Hindvee

On Tue, Jun 24, 2025 at 11:29 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Hello sir,
I am attaching the .ipynb file, it has the images

warm regards
Hindvee

On Tue, Jun 24, 2025 at 5:48 AM Kishor Durve <kishor.durve@outlook.com> wrote:

Please share the images for visual comparison.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, June 24, 2025 5:13 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello Durve sir,

I wanted to share an update on the denoising output. I noticed earlier that while the model was doing a good job removing noise, it was also making the image a bit too smooth — the sharpness had dropped quite a bit:

- Noisy Image: 33.72
- Denoised Image: 6.92
- Clean Image (Ground Truth): 26.49

To improve this, I added a sharpening filter after the autoencoder output and tested two variants:

1. A **strong sharpening kernel**, which raised the denoised sharpness to **112.40,** but it felt a bit overdone and introduced some artifacts.

2. A **softer kernel**, which gave a cleaner and more natural result, with denoised sharpness at **54.04**. This seems like a balanced point where edges are clearer without compromising visual quality.

If you'd like, I can also share the output images (clean vs. denoised) for a visual comparison.

Warm regards

Hindvee



On Fri, Jun 20, 2025 at 12:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir!
> Based on the latest code I calculated the sharpness using Laplacian variance. I obtained the following results as those attached as image.
> I think that while denoising, though the noise was removed, but the model blurred or over smoothened the image. I may have lost some meaningful edge details in the process. But the sharpness value of Clean image is lower than the noisy image (because it has no random noise), but high enough to retain clear edges and structure. So I am working next on increasing the sharpness value while keeping the noise low. Please do share your views with me.
>
> Warm regards
> Hindvee
>
> 
>
> On Thu, 19 Jun 2025, 13:42 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > I have seen the processed images. I think it is good at this point to quantify 'sharpness' of an image. It will give us a measure to compare different methods / models.
> >
> > There can be several approaches to do this. We use Opencv::Laplacian() to judge image sharpness. But you can try other approaches.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Thursday, June 19, 2025 2:52 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Hello sir,
> >
> > I've completed the full-image denoising as discussed for the images `5.png` and `6.png`. This time, the model was trained and applied directly on the complete images (cropped to a common size of **570×588**) instead of small patches, as per your suggestion. The denoising autoencoder was able to reduce background noise and enhance structural features, particularly the edges of the circles, which were initially faint or inconsistent. The model output shows a noticeable improvement in clarity while preserving the original patterns.
> >
> > I've attached the processed result image along with this mail. Please let me know your thoughts on this file.
> >
> > Warm regards,
> > Hindvee

On Wed, Jun 18, 2025 at 12:33 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have checked the file. These two images contain circles as primary features. Also, the patches in the file seem to be for a small part of the original image. If you have processed the original images (5.png and 6.png) please send me the processed images on mail. If you have processed only parts of these images, I suggest you process the full images and send me the results.
>
> Similarly, in the previous exercise please process complete images. Most of the set are of size 500x500 pixel, so it should not consume too much time.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 18, 2025 1:44 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Sir, please let me know your views on the above .ipynb file.
> Warm Regards
> Hindvee
>
> On Wed, Jun 18, 2025 at 1:56 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir,
> > I have processed the 5.png and 6.png, I faced lower accuracy rate earlier. After fixing the earlier issues with mismatched data sizes and high Binary Crossentropy loss (~0.6098), the model is now working smoothly. It reached a much lower loss of just 0.0528 in 5 epochs, and both training and validation losses settled nicely by the 2nd or 3rd epoch without signs of overfitting. The denoised images look very clean and closely match the original ones, even when the input images had different levels of noise. I am attaching the results in the attached file.
> > Warm regards
> > Hindvee
> >
> > On Tue, Jun 17, 2025 at 7:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > Sorry! Forgot the attachment.
> > >
> > > Kishor Durve
> > >
> > > ---
> > >
> > > **From:** Kishor Durve <kishor.durve@outlook.com>
> > > **Sent:** Tuesday, June 17, 2025 9:46 AM
> > > **To:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Subject:** Re: AI assisted edge detection
> > >
> > > Good to know that you are making progress at a nice clip.
> > >
> > > I saw the images and the difference between before / after can be seen. Can you please process the images attached to this mail and send me the results? The images are similar to those I had sent earlier, the exception being 5.png and 6.png. In these last two images the challenge is to detect circles whose edges are not well defined and the edge quality may vary from image to image. This is to test the effectiveness of noise removal / sharpening.
> > >
> > > Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, June 17, 2025 3:06 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir,
I have completed the training and evaluation of the denoising autoencoder using grayscale image patches taken from the `IMG` folder. The model was trained for 20 epochs on 64×64 patches with binary cross entropy loss and a 10% validation split. During training, the loss reduced steadily and stabilized, with the final reconstruction loss reaching 0.3688. The results show that the model can effectively denoise images and recover clean patterns like faint lines. In the next step, I plan to use the output of the Super-Resolution model as a base to evaluate the accuracy and quality of the denoising process, to better understand how both methods contribute to image enhancement. I am attaching the .ipynb file for your reference and also the IMG Folder which I just mentioned.

Warm regards
Hindvee

IMG FOLDER

Accuracy results:-

image.png

On Fri, Jun 13, 2025 at 5:33 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> The images I sent are big. OpenCV will not be able to handle them as the max size allowed is 32K x 32K. The crash you are facing could be due to this reason.
>
> You can crop out small patches from the big images and use them for trials.
>
> Kishor Durve
> 9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, June 11, 2025 1:37 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Thank you for sharing more images. I have updated the work on the sheets.

Warm regards
Hindvee

On Sat, Jun 7, 2025 at 2:51 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have uploaded more images. You can crop out patches from them. One of the images has many boards which will give many useful patches.
>
> Kishor Durve
> Duravision Systems
> 9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Saturday, June 7, 2025 4:36 AM

To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings sir,
It would be great if I get more images, that would expand my dataset.
Thank you for your previous email.
I also wanted tk inform you that I am travelling and will be able to resume work by Tuesday. I request you to kindly allow my leave.

Warm Regards
Hindvee

On Fri, 6 Jun 2025, 11:31 Kishor Durve, <kishor.durve@outlook.com> wrote:

> I am not going to be of much help to you as I have limited experience in this field. Also, what little I know was gained by applying available models in OpenCV's DNN module.
>
> There are two images available in the images folder. You may crop out patches from them for the Training / test sets. We use ImageJ and IrfanView for such work as MS Paint cannot handle such big images. If you need more images please let me know.
>
> Kishor Durve
> Duravision Systems
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Friday, June 6, 2025 5:24 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings sir,
> I hope you're doing well. I've just finished training the CNN-based line presence classifier, and I'm attaching the loss graph for your review. The model seems to have learned quite quickly, with both training and validation losses staying very close to zero throughout. There was a small spike in training loss around epoch 80, but validation stayed stable, so it might just be a minor fluctuation.
>
> Since the dataset is quite small and the model converged very early, I'm unsure how best to move forward.According to you should I try to get more data to avoid overfitting. Please guide me towards a direction in which I can make this model more effective.
>
> Warm regards
>
> Hindvee
>
> On Wed, Jun 4, 2025 at 6:42AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Yes, it looks ok to me. We will take a relook once you make more progress.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Wednesday, June 4, 2025 4:26 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection

Additionally, Sir, I would like to confirm if the current format of the Google Sheet meets your expectations. Please let me know if you would prefer any modifications, such as a more detailed breakdown of the work completed.

Warm Regards
Hindvee

On Wed, Jun 4, 2025 at 1:34 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Greetings Mr. Durve,
>
> Thanks for sharing the super-resolution model. I've set it up and tested it, the results inspiring. It does a good job cleaning up noise and improving contrast, especially in the faint line areas. As we discussed, I'll now add this to the bigger pipeline along with the AI modules (NB-1 to NB-5). The super-resolution output can help improve both the CNN classifier and regression models. I'll also check how it works together with NB-5, the denoising autoencoder, to see if we get even better results.
>
> Please let me know if you want me to run it on any specific ROI samples or start putting together comparison results.
>
> Warm regards
> Hindvee
>
> On Tue, Jun 3, 2025 at 9:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > I have file network.py in folder 'models'. This file has the RRDBNet implementaion.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Monday, June 2, 2025 7:15 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings Mr. Durve,
> > I am facing an issue again an again while running the VS 2022 code, pls help me with this, we have the module named as model_zoo not models. Pls tell me how to proceed with this.
> >
> > image.png
> >
> > Warm Regards
> > Hindvee
> >
> > On Sat, May 31, 2025 at 12:46 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> >
> > > Thankyou sir! I will try this right away.
> > > Also I hope you received this google sheet, I have shared edit access to you on your gmail account
> > >
> > > Hindvee Vispute - Time data - Intern
> > >
> > > Regards
> > > Hindvee
> > >
> > > On Sat, May 31, 2025 at 5:43 AM Kishor Durve <kishor.durve@outlook.com> wrote:

Sure. You need to do the following to make the project work.

1. Download and install VS 2022 Community edition
2. Download and install Anaconda. Install an additional environment Python 3.8.16. Use this version of Python for the project.

The project should run at this point.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, May 30, 2025 6:34 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings sir!
I went through the google drive folder shared by you. It would be really helpful if you could walk me through it.
Best regards
Hindvee

On Fri, May 30, 2025 at 11:41AM Kishor Durve <kishor.durve@outlook.com> wrote:

> You can download the model, a working VC++ (VS 2022) project and image here.
>
> https://drive.google.com/drive/folders/1mLBDjJR3fFKs8Brr_nNj2z9YyI89nruE?usp=sharing
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Friday, May 30, 2025 12:50 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings sir!
> Pls do share the model over drive I will try it out. Also I request you to share more image database so that I can start working on it.
> I will start work from today itself then.
>
> Regards
> Hindvee
>
> On Fri, 30 May 2025, 06:11 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > We have found that using a super resolution Pytorch model to get a larger image results in significant noise removal. Perhaps it is a by-product.
> >
> > You can try it out. I have the model and can share it on Google Drive if you want to try it out. We have tried the models available in OpenCV DNN but they do not seem to give the same results.

Regards,

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 29, 2025 5:28 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Dear Sir,

Thank you for the confirmation. I will maintain the time data on a Google Sheet and ensure it is updated regularly.

Looking forward to starting from Monday, 2nd June.

Best regards,
Hindvee

On Thu, May 29, 2025 at 2:53 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> Welcome aboard!
>
> You can keep the time data on Google Sheet.
>
> Regards,
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Thursday, May 29, 2025 5:12 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings Mr. Durve,
>
> Thank you very much for giving me the opportunity to contribute to Guangdong Longtian Smart Instruments Co. Ltd. through this exciting project. I am pleased to accept the offer and look forward to working on the development of the AI-assisted line detection tool.
>
> With your permission, I would like to begin the internship from the coming Monday, i.e., 2nd June 2025.
>
> Additionally, I wanted to ask about the preferred method for tracking and sharing the time I spend on the project. Would you like me to maintain a shared timesheet (e.g., a Google Sheet or Excel file), or would you prefer weekly time summaries via email or any other format?
>
> Looking forward to your guidance on this.

Thank you once again for this opportunity.

Warm regards,
Hindvee Rajesh Vispute

On Thu, May 29, 2025 at 8:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have attached the on-boarding letter.
>
> Please let me know if you have any queries.
>
> Regards,
>
> Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Saturday, May 10, 2025 3:46 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr. Durve,

Thank you for your detailed response and clarification. I now clearly understand that the goal is to enhance line detection using AI within known ROIs, leveraging the expected position of endpoints.

To support this objective, I've explored and implemented several **AI-based approaches**, any of which can be used to complement or improve upon conventional methods:

1. **CNN-Based Line Presence Classifier (NB-1):** Identifies whether a line exists within a given patch/ROI—useful for filtering candidate regions before applying traditional detection.
2. **CNN Regression Model (NB-2):** Predicts line endpoints directly from the image, offering robust localization even in low-contrast or noisy scenarios.
3. **Denoising Autoencoder (NB-5):** Enhances faint or broken lines by cleaning up noise, thereby boosting the effectiveness of existing edge or Hough-based methods.
4. **Vision Transformer Model (NB-4):** Designed for fine-grained classification tasks and can be adapted for future objectives like surface defect or anomaly detection.

These methods are independently deployable and can be integrated selectively depending on the specific inspection challenge or ROI structure. I'd be happy to align with your ROI-driven strategy and customize accordingly.

Regarding the stipend, I agree with the proposed Rs. 200/hour base rate and performance-based bonus, and I'm enthusiastic about the opportunity to collaborate further on defect detection and shape anomaly tasks in the future.

I have attached screenshots of the code files I executed today. I have the names of the notebooks (NB) with NB-1, NB-2,NB-4 and NB-5 respectively.

NB-1
NB-2
NB-4
NB-5

Please let me know if you'd like me to share these outputs or run a quick live demonstration.

Warm regards,
Hindvee

On Fri, May 9, 2025 at 6:03 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> I think there is a mismatch in understanding. We are already using conventional line detection techniques (various edge detection operators and Hough Transform and other methods for line fitment) .
>
> What we are looking to get from you is AI based / AI assisted line detection. Since we know the probable position of the line beforehand we can use this information to find a line. For example, say we know that the line end points are expected to be (x1, y1) and (x2, y2), we can construct a small rectangle (ROI) using these points and search for a line within this ROI.
>
> As for the stipend it depends both on the time you spend on the project and the results you produce. It will comprise of an hourly rate and a result linked amount. I suggest we have an hourly rate of Rs. 200/hour and a matching amount linked to the results at the end of the project. So, if you are able to produce satisfactory results it will come to Rs. 400/hour. Also, if the results are satisfactory, we can take this forward next year and work on other issues like surface defect detection, detection of shape anomalies etc.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 8, 2025 5:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr Durve,
I have the problem now so I will focus on faint line detection for the first step. I can proceed using Sobel, Scharr, and Canny edge detectors provided by OpenCV. First I will preprocess the images and then move to edge detection using the above 3 detectors. I will carry out line detection using Hough Transform, and further we will filter out short or irrelevant lines based on length and their angle of orientation. Finally, we will draw the detected lines on the images using cv.line()

I can devote 14 hrs per week to this task for 2 months, we can extend the time if required. Also, I wanted to ask what stipend you can provide me. I believe I can complete the task in the above time frame. Thank you for your time.

Warm regards
Hindvee

On Thu, May 8, 2025 at 7:18 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Your analysis seems good.
>
> At this point we are not interested in detecting defects (though at a later stage we may be), and want to detect the faint lines only.
>
> Note that though I have annotated only a single line in the image I sent to you yesterday, we need to detect edge lines of all vertical stripes.
>
> I have attached 2 sets of 20 images each named 'Good Images' and 'Bad Images' - images in which it is mostly easy to detect lines by the conventional image processing method and those in which lines are faint and hard to detect. These images may help you get a better understanding of the problem.
>
> Please also check the following issues:
>
> 1. How much time will you be able to devote to this task?

2. Will you be able to complete the task within the available time?

There are other issues like time taken by the AI route, integration of AI line detect code in VC++ project, further training / fine tuning of model, applicability of the method to detect faint circles etc. But we will get into them later once the basic method is proven.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, May 7, 2025 3:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings, Mr. Durve,
Thank you for reviewing my proposal and for sharing the sample images. By looking at the images I came across the specific challenges faced by the inspection system, like detecting faint lines, subtle surface variations, and fine defects that can be missed by traditional image processing. Based on these observations I suggest that we can do the following:-

1. **Hybrid approach**: Objective - Combine the speed of OpenCV line detection (example Hough Transform) with deep learning's pattern recognition capability.
Approach
- Use OpenCV to pre-locate candidate regions and feed them into a deep model for verification.
- This can speed up inference and increase resistance to false positives.

2. **Deep learning-based fine defect detection**: Objective: Improve the detection of faint, low contrast line defects and localized surface anomalies.
Approach
- Train a model, such as YOLOv8 or U-Net, tailored to your defect patterns using the annotated images.
- To improve feature visibility, add contrast enhancement.
- Assess using detection accuracy and intersection over union (IoU) to compare to baselines based on OpenCV.

I can begin with a proof-of-concept model using the images shared by you. Thank you for sharing the opportunity. I am excited to get a chance to contribute to the advancement in inspection technology.

Warm regards
Hindvee Vispute

On Wed, 7 May 2025, 09:31 Kishor Durve, <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have gone thru the document uploaded by you.
>
> It looks ok to me.
>
> I can have attached a few images so you will get a better idea of the requirements.
>
> Kishor Durve

Hindvee Vispute <hindveevispute@gmail.com> Tue, Aug 5, 2025 at 6:11 PM
To: Kishor Durve <kishor.durve@outlook.com>

Hello sir
Just following up on my previous email regarding the processed images and the stipend for July (58 hours). I have shared a small suggestion regarding the per-hour rate.

Please let me know if you need any further details or modifications from my side.
UPDATED DEXINED OUTPUT
Looking forward to your feedback.

Warm regards,
Hindvee

On Fri, Aug 1, 2025 at 2:14 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Thank you sir! I have worked out and increased the contrast in the processed images. I will share the file with you. Also sir I shall request you to please process the stipend for this month. It has been 58 hrs for the month of July. If you like my work please consider increasing the per hour rate. It is just a small suggestion. The final decision depends on you and I happily accept your decision. My primary goal is to enhance my skills, and I am learning a lot from this project.
>
> Warm regards
> Hindvee
>
> On Wed, Jul 30, 2025 at 11:02 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > I have checked the images. Your conclusions are correct.
> >
> > Please carry on with your efforts.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Tuesday, July 29, 2025 3:37 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Hello sir!
> > current update:-
> >
> > - Using an autoencoder trained on clean (good) images to repair bad images
> > - Applying a **hybrid loss** of MSE and SSIM to preserve structure and perceptual quality.
> > - Saving and visualizing both original and reconstructed images which helps in debugging

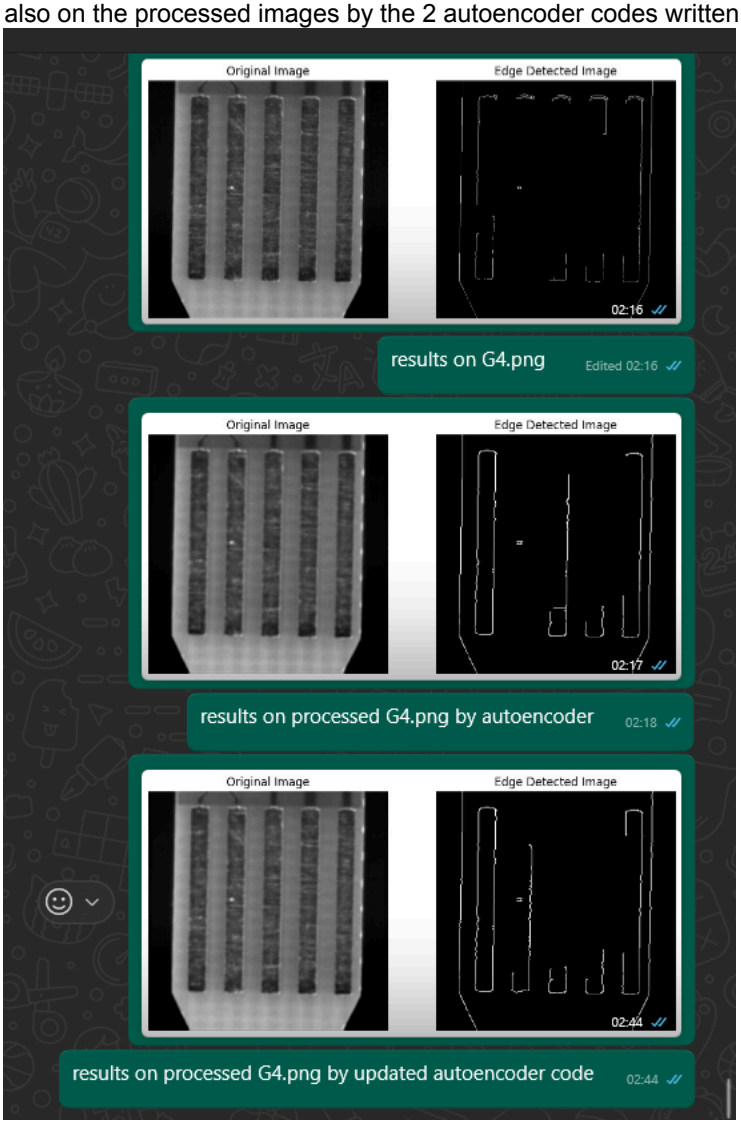
Despite this denoising strategy, DexiNED is still failing for bad images, the possible reasons are as follows

1. Contrast remains too low after reconstruction.
2. Edges are too smooth or faint due to over-smoothing by the autoencoder.
3. DexiNED is not tuned for synthetic/industrial patterns (e.g., PCB, x-ray like structures).

I will now work on these factors to improve the quality of processed images. I am attaching the processed Good and Bad Images folders.
BAD IMAGES _ PROCESSED
GOOD IMAGES_ PROCESSED

warm regards
Hindvee

On Mon, Jul 28, 2025 at 2:48 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Noted sir. I wrote code for autoencoder to get sharper images, and I applied DexiNed on the original image and also on the processed images by the 2 autoencoder codes written so far. I am sharing the results here.
>
> 
>
>
> warm regards
> Hindvee
>
> On Fri, Jul 25, 2025 at 6:34 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Sorry for delay in replying. I was away for quite some time.
> >
> > A couple of months back we had tried OpenCV / HED and obtained similar results.
> >
> > Please go ahead and try DexiNed after which we will decide about line detection.
> >
> > After that another important step would be to remove noise from image using a suitable model.

I know this is somewhat away from our goal of 'AI assisted edge detection' but it can be an important intermediate step. We are currently using a model that does this. This is the model I had sent to you initially. It is developed in a Chinese University by a professor who works with our company.

Kishor Durve
Duravision Systems
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, July 25, 2025 5:46 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir ! This is an output image obtained from the HED model and similar image I have obtained from DexiNed, could u pls tell i am moving in the right direction or should I make improvements or change in the plan of action?

warm regards
Hindvee

On Wed, Jul 23, 2025 at 3:20 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir! Could you please share ur insights and views on the above mail thread, it shall be very helpful to get to know your feedback and move forward with the project.
>
> Warm Regards
> Hindvee
>
> On Fri, Jul 11, 2025 at 3:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir!
> >
> > Following up on my previous email regarding edge detection, I wanted to share the progress I've made since then. I have been focusing on using the Holistically-Nested Edge Detection (HED) method. I have now:
> >
> > CODE FILE
> >
> > 1. Successfully applied the HED process to a folder containing our images to generate edge maps for the entire dataset.
> >
> > 2. Trained a custom autoencoder model designed to refine and correct the lines in these images.
> >
> > 3. Applied the trained autoencoder to the HED output images to produce corrected versions.
> >
> > 4. Started working on post-processing techniques.
> >
> > I am currently experimenting with different parameters and methods to achieve the best results for this step. I am continuing to refine the line clarification process and will share updated results once I have finalized the post-processing approach.
> >
> > warm regards
> >
> > Hindvee
> >
> > On Wed, Jul 9, 2025 at 12:23 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> >
> > > Hello sir! I am trying the pre-trained model for detecting edges in our images; I have tried Holistically-Nested Edge Detection (HED) and DexiNed. I am attaching the output, I am going forward with HED to get better results.
> > >
> > > Warm regards
> > > Hindvee
> > >
> > > On Mon, Jul 7, 2025 at 10:37 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > >
> > > > Hello sir!
> > > > I am sharing the first 5 difference images of the three codes that I mentioned previously. Please check this whenever you have time.
> > > >
> > > > Warm regards

Hindvee

On Mon, Jul 7, 2025 at 6:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:

I am unable to give my view of the approach as I can't view the images. I can give my opinion if you send me difference images. In any case, I think you need to have difference images as a metric in addition to the Laplacian.

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From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Monday, July 7, 2025 7:09:51 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!
I have received the stipend, thank you for that! I'd like to share a brief update on the image enhancement work, focusing on noise removal and edge preservation.
Summary of results:-
1. Previous code
---Performed basic denoising but introduced over-smoothing.
---Laplacian MAD: ~0.0175–0.0192

2. DnCNN
---Improved noise removal and structural retention.
---MAD: ~0.0186–0.0202

3. DnCNN + Sharpening (Current approach)
---Added post-processing to enhance edge clarity.
---Consistently improved MAD to ~0.0178–0.0196.
---Visual results show reduced whitening and better edge definition.

This confirms that the current setup provides better denoising while preserving fine image details. Please let me know if you'd like to proceed with further tuning using SSIM or edge-aware loss functions.

Warm regards

Hindvee

On Fri, Jul 4, 2025 at 3:54 AM Kishor Durve <kishor.durve@outlook.com> wrote:

Please keep at it. Use difference image and Laplacian to gauge the effect of the algorithm. I will be able to look at your work only after 15th.
I have transferred your stipend by GPay.

Get Outlook for Android

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, July 4, 2025 5:37:35 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!
This is what I am been working on. I am still working on getting better edges and less whitish images.

warm regards
Hindvee

On Thu, Jul 3, 2025 at 10:53 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

UPI ID: hindveevispute@okicici

Warm regards
Hindvee

On Wed, 2 Jul 2025, 03:16 Kishor Durve, <kishor.durve@outlook.com> wrote:

I checked the images, but did not notice substantial difference between the before and after images. You can subtract them (after aligning) and check.
As regards payment of stipend please let me know how much we need to pay and I will transfer it on GPay.
I am travelling again and am out of office. I will be back only on 15th. My responses to your mails may be delayed.

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From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, July 1, 2025 1:51:45 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!

I was eager to get insights from you about the above shared images.Please do have a look as soon as possible. Also, I wanted to know whether the stipend will be credited at the end of the month or the project's end. It would be very kind of you if I could get the stipend at the end of the month.

Thanks and regards
Hindvee

On Mon, Jun 30, 2025 at 10:33 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Good morning sir!
>
> Did you get any chance to review the above images? Please share your suggestions and insights with me.
>
> Warm regards
> Hindvee
>
> On Thu, Jun 26, 2025 at 11:09 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Here I am attaching the processed images.
> >
> > Warm regards
> > Hindvee
> >
> > On Thu, Jun 26, 2025 at 8:08 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > I am unable to run the files in Colab and generate images. Can you please send me the processed images on mail - .png / .jpeg files, not .ipynb files.
> > >
> > > Kishor Durve
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Thursday, June 26, 2025 4:48 AM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection
> > >
> > > Hello sir, I have attached the results for png 1,2,3 and 4. Pls share ur views and insights.
> > >
> > > Warm Regards
> > > Hindvee
> > >
> > > On Tue, Jun 24, 2025 at 11:29 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > >
> > > > Hello sir,
> > > > I am attaching the .ipynb file, it has the images
> > > >
> > > > warm regards
> > > > Hindvee
> > > >
> > > > On Tue, Jun 24, 2025 at 5:48 AM Kishor Durve <kishor.durve@outlook.com> wrote:

Please share the images for visual comparison.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, June 24, 2025 5:13 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello Durve sir,

I wanted to share an update on the denoising output. I noticed earlier that while the model was doing a good job removing noise, it was also making the image a bit too smooth — the sharpness had dropped quite a bit:

- Noisy Image: 33.72
- Denoised Image: 6.92
- Clean Image (Ground Truth): 26.49

To improve this, I added a sharpening filter after the autoencoder output and tested two variants:

1. A **strong sharpening kernel**, which raised the denoised sharpness to **112.40,** but it felt a bit overdone and introduced some artifacts.
2. A **softer kernel**, which gave a cleaner and more natural result, with denoised sharpness at **54.04**. This seems like a balanced point where edges are clearer without compromising visual quality.

If you'd like, I can also share the output images (clean vs. denoised) for a visual comparison.

Warm regards

Hindvee

image.png

On Fri, Jun 20, 2025 at 12:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir!
> Based on the latest code I calculated the sharpness using Laplacian variance. I obtained the following results as those attached as image.
> I think that while denoising, though the noise was removed, but the model blurred or over smoothened the image. I may have lost some meaningful edge details in the process. But the sharpness value of Clean image is lower than the noisy image (because it has no random noise), but high enough to retain clear edges and structure. So I am working next on increasing the sharpness value while keeping the noise low. Please do share your views with me.
>
> Warm regards
> Hindvee
>
> image.png
>
> On Thu, 19 Jun 2025, 13:42 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > I have seen the processed images. I think it is good at this point to quantify 'sharpness' of an image. It will give us a measure to compare different methods / models.
> >
> > There can be several approaches to do this. We use Opencv::Laplacian() to judge image sharpness. But you can try other approaches.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, June 19, 2025 2:52 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir,

I've completed the full-image denoising as discussed for the images `5.png` and `6.png`. This time, the model was trained and applied directly on the complete images (cropped to a common size of **570×588**) instead of small patches, as per your suggestion. The denoising autoencoder was able to reduce background noise and enhance structural features, particularly the edges of the circles, which were initially faint or inconsistent. The model output shows a noticeable improvement in clarity while preserving the original patterns.

I've attached the processed result image along with this mail. Please let me know your thoughts on this file.

Warm regards,
Hindvee

On Wed, Jun 18, 2025 at 12:33 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have checked the file. These two images contain circles as primary features. Also, the patches in the file seem to be for a small part of the original image. If you have processed the original images (5.png and 6.png) please send me the processed images on mail. If you have processed only parts of these images, I suggest you process the full images and send me the results.
>
> Similarly, in the previous exercise please process complete images. Most of the set are of size 500x500 pixel, so it should not consume too much time.
>
> Kishor Durve
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 18, 2025 1:44 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Sir, please let me know your views on the above .ipynb file.
> Warm Regards
> Hindvee
>
> On Wed, Jun 18, 2025 at 1:56 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir,
> > I have processed the 5.png and 6.png, I faced lower accuracy rate earlier. After fixing the earlier issues with mismatched data sizes and high Binary Crossentropy loss (~0.6098), the model is now working smoothly. It reached a much lower loss of just 0.0528 in 5 epochs, and both training and validation losses settled nicely by the 2nd or 3rd epoch without signs of overfitting. The denoised images look very clean and closely match the original ones, even when the input images had different levels of noise. I am attaching the results in the attached file.
> > Warm regards
> > Hindvee

On Tue, Jun 17, 2025 at 7:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Sorry! Forgot the attachment.
>
> Kishor Durve
>
> ---
>
> **From:** Kishor Durve <kishor.durve@outlook.com>
> **Sent:** Tuesday, June 17, 2025 9:46 AM
> **To:** Hindvee Vispute <hindveevispute@gmail.com>
> **Subject:** Re: AI assisted edge detection
>
> Good to know that you are making progress at a nice clip.
>
> I saw the images and the difference between before / after can be seen. Can you please process the images attached to this mail and send me the results? The images are similar to those I had sent earlier, the exception being 5.png and 6.png. In these last two images the challenge is to detect circles whose edges are not well defined and the edge quality may vary from image to image. This is to test the effectiveness of noise removal / sharpening.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Tuesday, June 17, 2025 3:06 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir,
> I have completed the training and evaluation of the denoising autoencoder using grayscale image patches taken from the `IMG` folder. The model was trained for 20 epochs on 64×64 patches with binary cross entropy loss and a 10% validation split. During training, the loss reduced steadily and stabilized, with the final reconstruction loss reaching 0.3688. The results show that the model can effectively denoise images and recover clean patterns like faint lines. In the next step, I plan to use the output of the Super-Resolution model as a base to evaluate the accuracy and quality of the denoising process, to better understand how both methods contribute to image enhancement. I am attaching the .ipynb file for your reference and also the IMG Folder which I just mentioned.
>
> Warm regards
> Hindvee
>
> IMG FOLDER
>
> Accuracy results:-
>
> image.png
>
> On Fri, Jun 13, 2025 at 5:33 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > The images I sent are big. OpenCV will not be able to handle them as the max size allowed is 32K x 32K. The crash you are facing could be due to this reason.

You can crop out small patches from the big images and use them for trials.

Kishor Durve
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, June 11, 2025 1:37 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Thank you for sharing more images. I have updated the work on the sheets.

Warm regards
Hindvee

On Sat, Jun 7, 2025 at 2:51 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have uploaded more images. You can crop out patches from them. One of the images has many boards which will give many useful patches.
>
> Kishor Durve
> Duravision Systems
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Saturday, June 7, 2025 4:36 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings sir,
> It would be great if I get more images, that would expand my dataset. Thank you for your previous email.
> I also wanted tk inform you that I am travelling and will be able to resume work by Tuesday. I request you to kindly allow my leave.
>
> Warm Regards
> Hindvee
>
> On Fri, 6 Jun 2025, 11:31 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > I am not going to be of much help to you as I have limited experience in this field. Also, what little I know was gained by applying available models in OpenCV's DNN module.
> >
> > There are two images available in the images folder. You may crop out patches from them for the Training / test sets. We use ImageJ and IrfanView for such work as MS Paint cannot handle such big images. If you need more images please let me know.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, June 6, 2025 5:24 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection

Greetings sir,
I hope you're doing well. I've just finished training the CNN-based line presence classifier, and I'm attaching the loss graph for your review. The model seems to have learned quite quickly, with both training and validation losses staying very close to zero throughout. There was a small spike in training loss around epoch 80, but validation stayed stable, so it might just be a minor fluctuation.

Since the dataset is quite small and the model converged very early, I'm unsure how best to move forward.According to you should I try to get more data to avoid overfitting. Please guide me towards a direction in which I can make this model more effective.

Warm regards

Hindvee

On Wed, Jun 4, 2025 at 6:42 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Yes, it looks ok to me. We will take a relook once you make more progress.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 4, 2025 4:26 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Additionally, Sir, I would like to confirm if the current format of the Google Sheet meets your expectations. Please let me know if you would prefer any modifications, such as a more detailed breakdown of the work completed.
>
> Warm Regards
> Hindvee
>
> On Wed, Jun 4, 2025 at 1:34 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Greetings Mr. Durve,
> >
> > Thanks for sharing the super-resolution model. I've set it up and tested it, the results inspiring. It does a good job cleaning up noise and improving contrast, especially in the faint line areas. As we discussed, I'll now add this to the bigger pipeline along with the AI modules (NB-1 to NB-5). The super-resolution output can help improve both the CNN classifier and regression models. I'll also check how it works together with NB-5, the denoising autoencoder, to see if we get even better results.
> >
> > Please let me know if you want me to run it on any specific ROI samples or start putting together comparison results.
> >
> > Warm regards
> > Hindvee
> >
> > On Tue, Jun 3, 2025 at 9:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > I have file network.py in folder 'models'. This file has the RRDBNet implementaion.
> > >
> > > Kishor Durve
> > > Duravision Systems

9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Monday, June 2, 2025 7:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr. Durve,
I am facing an issue again an again while running the VS 2022 code, pls help me with this, we have the module named as model_zoo not models. Pls tell me how to proceed with this.

 image.png

Warm Regards
Hindvee

On Sat, May 31, 2025 at 12:46 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Thankyou sir! I will try this right away.
> Also I hope you received this google sheet, I have shared edit access to you on your gmail account
>
> Hindvee Vispute - Time data - Intern
>
> Regards
> Hindvee
>
> On Sat, May 31, 2025 at 5:43 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Sure. You need to do the following to make the project work.
> >
> > 1. Download and install VS 2022 Community edition
> > 2. Download and install Anaconda. Install an additional environment Python 3.8.16. Use this version of Python for the project.
> >
> > The project should run at this point.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, May 30, 2025 6:34 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings sir!
> > I went through the google drive folder shared by you. It would be really helpful if you could walk me through it.
> > Best regards
> > Hindvee
> >
> > On Fri, May 30, 2025 at 11:41 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > You can download the model, a working VC++ (VS 2022) project and image here.
> > >
> > > https://drive.google.com/drive/folders/1mLBDjJR3fFKs8Brr_nNj2z9YyI89nruE?

usp=sharing

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, May 30, 2025 12:50 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings sir!
Pls do share the model over drive I will try it out.
Also I request you to share more image database so that I can start working on it.
I will start work from today itself then.

Regards
Hindvee

On Fri, 30 May 2025, 06:11 Kishor Durve, <kishor.durve@outlook.com> wrote:

> We have found that using a super resolution Pytorch model to get a larger image results in significant noise removal. Perhaps it is a by-product.
>
> You can try it out. I have the model and can share it on Google Drive if you want to try it out. We have tried the models available in OpenCV DNN but they do not seem to give the same results.
>
> Regards,
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Thursday, May 29, 2025 5:28 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Dear Sir,
>
> Thank you for the confirmation. I will maintain the time data on a Google Sheet and ensure it is updated regularly.
>
> Looking forward to starting from Monday, 2nd June.
>
> Best regards,
> Hindvee
>
> On Thu, May 29, 2025 at 2:53 PM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Hello Hindvee,
> >
> > Welcome aboard!

You can keep the time data on Google Sheet.

Regards,

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 29, 2025 5:12 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr. Durve,

Thank you very much for giving me the opportunity to contribute to Guangdong Longtian Smart Instruments Co. Ltd. through this exciting project. I am pleased to accept the offer and look forward to working on the development of the AI-assisted line detection tool.

With your permission, I would like to begin the internship from the coming Monday, i.e., 2nd June 2025.

Additionally, I wanted to ask about the preferred method for tracking and sharing the time I spend on the project. Would you like me to maintain a shared timesheet (e.g., a Google Sheet or Excel file), or would you prefer weekly time summaries via email or any other format?

Looking forward to your guidance on this.

Thank you once again for this opportunity.

Warm regards,
Hindvee Rajesh Vispute

On Thu, May 29, 2025 at 8:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have attached the on-boarding letter.
>
> Please let me know if you have any queries.
>
> Regards,
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Saturday, May 10, 2025 3:46 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection

Greetings Mr. Durve,

Thank you for your detailed response and clarification. I now clearly understand that the goal is to enhance line detection using AI within known ROIs, leveraging the expected position of endpoints.

To support this objective, I've explored and implemented several **AI-based approaches**, any of which can be used to complement or improve upon conventional methods:

1. **CNN-Based Line Presence Classifier (NB-1):** Identifies whether a line exists within a given patch/ROI—useful for filtering candidate regions before applying traditional detection.
2. **CNN Regression Model (NB-2):** Predicts line endpoints directly from the image, offering robust localization even in low-contrast or noisy scenarios.
3. **Denoising Autoencoder (NB-5):** Enhances faint or broken lines by cleaning up noise, thereby boosting the effectiveness of existing edge or Hough-based methods.
4. **Vision Transformer Model (NB-4):** Designed for fine-grained classification tasks and can be adapted for future objectives like surface defect or anomaly detection.

These methods are independently deployable and can be integrated selectively depending on the specific inspection challenge or ROI structure. I'd be happy to align with your ROI-driven strategy and customize accordingly.

Regarding the stipend, I agree with the proposed Rs. 200/hour base rate and performance-based bonus, and I'm enthusiastic about the opportunity to collaborate further on defect detection and shape anomaly tasks in the future.

I have attached screenshots of the code files I executed today. I have the names of the notebooks (NB) with NB-1, NB-2,NB-4 and NB-5 respectively.

NB-1
NB-2
NB-4
NB-5

Please let me know if you'd like me to share these outputs or run a quick live demonstration.

Warm regards,
Hindvee

On Fri, May 9, 2025 at 6:03 AM Kishor Durve <kishor.durve@outlook.com> wrote:

I think there is a mismatch in understanding. We are already using conventional line detection techniques (various edge detection operators and Hough Transform and other methods for line fitment) .

What we are looking to get from you is AI based / AI assisted line detection. Since we know the probable position of the line beforehand we can use this information to find a line. For example, say we know that the line end points are expected to be (x1, y1) and (x2, y2), we can construct a small rectangle (ROI) using these points and search for a line within this ROI.

As for the stipend it depends both on the time you spend on the project and the results you produce. It will comprise of an hourly rate and a result linked amount. I suggest we have an hourly rate of Rs. 200/hour and a matching amount linked to the results at the end of the project. So, if you are able to produce satisfactory results it will come to Rs. 400/hour. Also, if the results are satisfactory, we can take this forward next year and work on other issues like surface defect detection, detection of shape anomalies etc.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 8, 2025 5:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr Durve,
I have the problem now so I will focus on faint line detection for the first step. I can proceed using Sobel, Scharr, and Canny edge detectors provided by OpenCV. First I will preprocess the images and then move to edge detection using the above 3 detectors. I will carry out line detection using Hough Transform, and further we will filter out short or irrelevant lines based on length and their angle of orientation. Finally, we will draw the detected lines on the images using

cv.line()

I can devote 14 hrs per week to this task for 2 months, we can extend the time if required. Also, I wanted to ask what stipend you can provide me. I believe I can complete the task in the above time frame. Thank you for your time.

Warm regards
Hindvee

On Thu, May 8, 2025 at 7:18 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Your analysis seems good.
>
> At this point we are not interested in detecting defects (though at a later stage we may be), and want to detect the faint lines only.
>
> Note that though I have annotated only a single line in the image I sent to you yesterday, we need to detect edge lines of all vertical stripes.
>
> I have attached 2 sets of 20 images each named 'Good Images' and 'Bad Images' - images in which it is mostly easy to detect lines by the conventional image processing method and those in which lines are faint and hard to detect. These images may help you get a better understanding of the problem.
>
> Please also check the following issues:
>
> 1. How much time will you be able to devote to this task?
> 2. Will you be able to complete the task within the available time?
>
> There are other issues like time taken by the AI route, integration of AI line detect code in VC++ project, further training / fine tuning of model, applicability of the method to detect faint circles etc. But we will get into them later once the basic method is proven.
>
> Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, May 7, 2025 3:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings, Mr. Durve,
Thank you for reviewing my proposal and for sharing the sample images. By looking at the images I came across the specific challenges faced by the inspection system, like detecting faint lines, subtle surface variations, and fine defects that can be missed by traditional image processing. Based on these observations I suggest that we can do the following:-

1. **Hybrid approach**: Objective - Combine the speed of OpenCV line detection (example Hough Transform) with deep learning's pattern recognition capability.
Approach
- Use OpenCV to pre-locate candidate regions and feed them into a deep model for verification.
- This can speed up inference and increase resistance to false positives.

2. **Deep learning-based fine defect detection**: Objective: Improve the detection of faint, low contrast line defects and localized surface anomalies.
Approach
- Train a model, such as YOLOv8 or U-Net, tailored to your defect patterns using the annotated images.
- To improve feature visibility, add contrast enhancement.
- Assess using detection accuracy and intersection over union (IoU) to compare to baselines based on OpenCV.

I can begin with a proof-of-concept model using the images shared by you. Thank you for sharing the opportunity. I am excited to get a chance to contribute to the advancement in inspection technology.

Warm regards
Hindvee Vispute

On Wed, 7 May 2025, 09:31 Kishor Durve, <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have gone thru the document uploaded by you.
>
> It looks ok to me.

I can have attached a few images so you will get a better idea of the requirements.

Kishor Durve

Kishor Durve <kishor.durve@outlook.com> Wed, Aug 6, 2025 at 8:10 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

I have transferred Rs. 11,600 (200 x 58) to you. Regarding the suggested increase in per-hour rate: You have certainly made efforts to solve the problem. However, as of now the result is not yet usable to us. We will consider an increase when it is usable in our software.

Kishor Durve
Duravision Systems
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, August 5, 2025 8:41 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir
Just following up on my previous email regarding the processed images and the stipend for July (58 hours). I have shared a small suggestion regarding the per-hour rate.

Please let me know if you need any further details or modifications from my side.
UPDATED DEXINED OUTPUT
Looking forward to your feedback.

Warm regards,
Hindvee

On Fri, Aug 1, 2025 at 2:14 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Thank you sir! I have worked out and increased the contrast in the processed images. I will share the file with you. Also sir I shall request you to please process the stipend for this month. It has been 58 hrs for the month of July. If you like my work please consider increasing the per hour rate. It is just a small suggestion. The final decision depends on you and I happily accept your decision. My primary goal is to enhance my skills, and I am learning a lot from this project.
>
> Warm regards
> Hindvee
>
> On Wed, Jul 30, 2025 at 11:02 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > I have checked the images. Your conclusions are correct.
> >
> > Please carry on with your efforts.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Tuesday, July 29, 2025 3:37 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Hello sir!
> > current update:-
> >
> > - Using an autoencoder trained on clean (good) images to repair bad images
> > - Applying a **hybrid loss** of MSE and SSIM to preserve structure and perceptual quality.

- Saving and visualizing both original and reconstructed images which helps in debugging

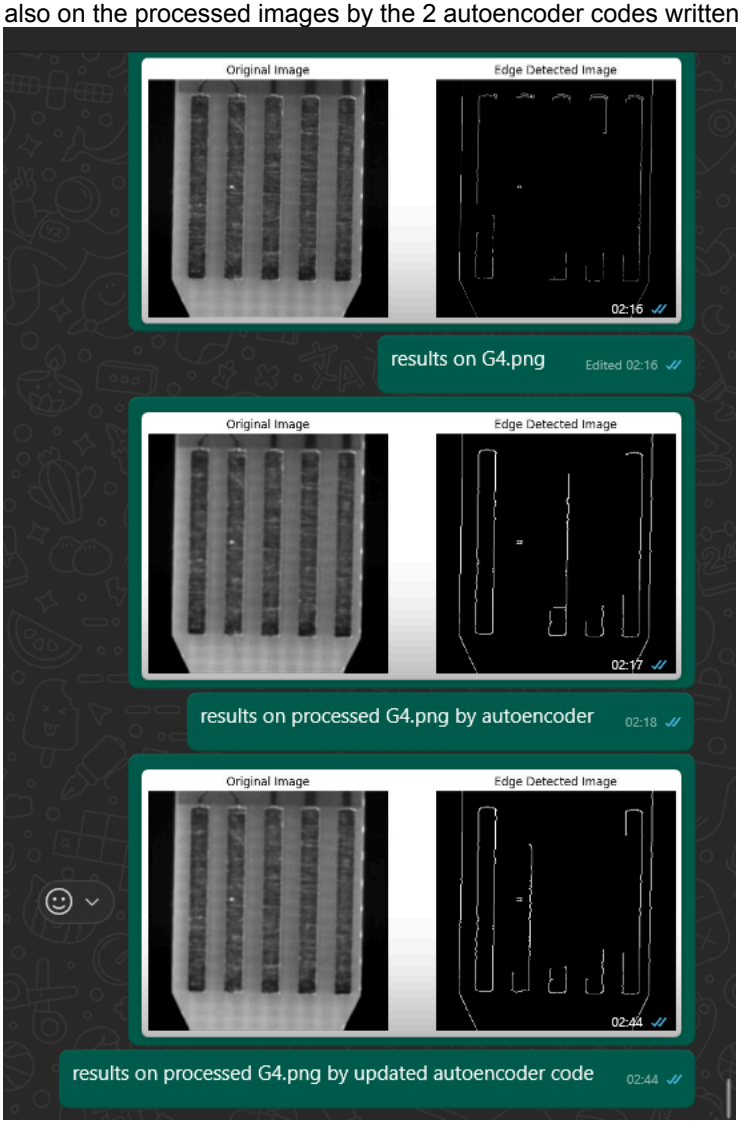
Despite this denoising strategy, DexiNED is still failing for bad images, the possible reasons are as follows

1. Contrast remains too low after reconstruction.
2. Edges are too smooth or faint due to over-smoothing by the autoencoder.
3. DexiNED is not tuned for synthetic/industrial patterns (e.g., PCB, x-ray like structures).

I will now work on these factors to improve the quality of processed images. I am attaching the processed Good and Bad Images folders.
BAD IMAGES _ PROCESSED
GOOD IMAGES_ PROCESSED

warm regards
Hindvee

On Mon, Jul 28, 2025 at 2:48 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Noted sir. I wrote code for autoencoder to get sharper images, and I applied DexiNed on the original image and also on the processed images by the 2 autoencoder codes written so far. I am sharing the results here.
>
> 
>
>
> warm regards
> Hindvee
>
> On Fri, Jul 25, 2025 at 6:34 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Sorry for delay in replying. I was away for quite some time.
> >
> > A couple of months back we had tried OpenCV / HED and obtained similar results.
> >
> > Please go ahead and try DexiNed after which we will decide about line detection.

After that another important step would be to remove noise from image using a suitable model.

I know this is somewhat away from our goal of 'AI assisted edge detection' but it can be an important intermediate step. We are currently using a model that does this. This is the model I had sent to you initially. It is developed in a Chinese University by a professor who works with our company.

Kishor Durve
Duravision Systems
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, July 25, 2025 5:46 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir ! This is an output image obtained from the HED model and similar image I have obtained from DexiNed, could u pls tell i am moving in the right direction or should I make improvements or change in the plan of action?

warm regards
Hindvee

On Wed, Jul 23, 2025 at 3:20 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir! Could you please share ur insights and views on the above mail thread, it shall be very helpful to get to know your feedback and move forward with the project.
>
> Warm Regards
> Hindvee
>
> On Fri, Jul 11, 2025 at 3:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir!
> >
> > Following up on my previous email regarding edge detection, I wanted to share the progress I've made since then. I have been focusing on using the Holistically-Nested Edge Detection (HED) method. I have now:
> >
> > CODE FILE
> >
> > 1. Successfully applied the HED process to a folder containing our images to generate edge maps for the entire dataset.
> >
> > 2. Trained a custom autoencoder model designed to refine and correct the lines in these images.
> >
> > 3. Applied the trained autoencoder to the HED output images to produce corrected versions.
> >
> > 4. Started working on post-processing techniques.
> >
> > I am currently experimenting with different parameters and methods to achieve the best results for this step. I am continuing to refine the line clarification process and will share updated results once I have finalized the post-processing approach.
> >
> > warm regards
> >
> > Hindvee
> >
> > On Wed, Jul 9, 2025 at 12:23 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> >
> > > Hello sir! I am trying the pre-trained model for detecting edges in our images; I have tried Holistically-Nested Edge Detection (HED) and DexiNed. I am attaching the output, I am going forward with HED to get better results.
> > >
> > > Warm regards
> > > Hindvee
> > >
> > > On Mon, Jul 7, 2025 at 10:37 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > >
> > > > Hello sir!
> > > > I am sharing the first 5 difference images of the three codes that I mentioned previously. Please check this whenever you have time.

Warm regards
Hindvee

On Mon, Jul 7, 2025 at 6:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> I am unable to give my view of the approach as I can't view the images. I can give my opinion if you send me difference images. In any case, I think you need to have difference images as a metric in addition to the Laplacian.
>
> Get Outlook for Android
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Monday, July 7, 2025 7:09:51 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir!
> I have received the stipend, thank you for that! I'd like to share a brief update on the image enhancement work, focusing on noise removal and edge preservation.
> Summary of results:-
> 1. Previous code
> ---Performed basic denoising but introduced over-smoothing.
> ---Laplacian MAD: ~0.0175–0.0192
>
> 2. DnCNN
> ---Improved noise removal and structural retention.
> ---MAD: ~0.0186–0.0202
>
> 3. DnCNN + Sharpening (Current approach)
> ---Added post-processing to enhance edge clarity.
> ---Consistently improved MAD to ~0.0178–0.0196.
> ---Visual results show reduced whitening and better edge definition.
>
> This confirms that the current setup provides better denoising while preserving fine image details. Please let me know if you'd like to proceed with further tuning using SSIM or edge-aware loss functions.
>
> Warm regards
>
> Hindvee
>
> On Fri, Jul 4, 2025 at 3:54 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Please keep at it. Use difference image and Laplacian to gauge the effect of the algorithm. I will be able to look at your work only after 15th.
> > I have transferred your stipend by GPay.
> >
> > Get Outlook for Android
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, July 4, 2025 5:37:35 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Hello sir!
> > This is what I am been working on. I am still working on getting better edges and less whitish images.
> >
> > warm regards
> > Hindvee
> >
> > On Thu, Jul 3, 2025 at 10:53 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> >
> > > UPI ID: hindveevispute@okicici
> > >
> > > Warm regards
> > > Hindvee

On Wed, 2 Jul 2025, 03:16 Kishor Durve, <kishor.durve@outlook.com> wrote:

I checked the images, but did not notice substantial difference between the before and after images. You can subtract them (after aligning) and check.
As regards payment of stipend please let me know how much we need to pay and I will transfer it on GPay.
I am travelling again and am out of office. I will be back only on 15th. My responses to your mails may be delayed.

Get Outlook for Android

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, July 1, 2025 1:51:45 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!

I was eager to get insights from you about the above shared images.Please do have a look as soon as possible. Also, I wanted to know whether the stipend will be credited at the end of the month or the project's end. It would be very kind of you if I could get the stipend at the end of the month.

Thanks and regards
Hindvee

On Mon, Jun 30, 2025 at 10:33 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Good morning sir!

Did you get any chance to review the above images? Please share your suggestions and insights with me.

Warm regards
Hindvee

On Thu, Jun 26, 2025 at 11:09 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Here I am attaching the processed images.

Warm regards
Hindvee

On Thu, Jun 26, 2025 at 8:08 AM Kishor Durve <kishor.durve@outlook.com> wrote:

I am unable to run the files in Colab and generate images. Can you please send me the processed images on mail - .png / .jpeg files, not .ipynb files.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, June 26, 2025 4:48 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir, I have attached the results for png 1,2,3 and 4. Pls share ur views and insights.

Warm Regards
Hindvee

On Tue, Jun 24, 2025 at 11:29 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Hello sir,
I am attaching the .ipynb file, it has the images

warm regards
Hindvee

On Tue, Jun 24, 2025 at 5:48 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Please share the images for visual comparison.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Tuesday, June 24, 2025 5:13 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello Durve sir,
>
> I wanted to share an update on the denoising output. I noticed earlier that while the model was doing a good job removing noise, it was also making the image a bit too smooth — the sharpness had dropped quite a bit:
>
> - Noisy Image: 33.72
> - Denoised Image: 6.92
> - Clean Image (Ground Truth): 26.49
>
> To improve this, I added a sharpening filter after the autoencoder output and tested two variants:
>
> 1. A **strong sharpening kernel**, which raised the denoised sharpness to **112.40,** but it felt a bit overdone and introduced some artifacts.
> 2. A **softer kernel**, which gave a cleaner and more natural result, with denoised sharpness at **54.04**. This seems like a balanced point where edges are clearer without compromising visual quality.
>
> If you'd like, I can also share the output images (clean vs. denoised) for a visual comparison.
>
> Warm regards
>
> Hindvee
>
> image.png
>
> On Fri, Jun 20, 2025 at 12:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir!
> > Based on the latest code I calculated the sharpness using Laplacian variance. I obtained the following results as those attached as image.
> > I think that while denoising, though the noise was removed, but the model blurred or over smoothened the image. I may have lost some meaningful edge details in the process. But the sharpness value of Clean image is lower than the noisy image (because it has no random noise), but high enough to retain clear edges and structure. So I am working next on increasing the sharpness value while keeping the noise low. Please do share your views with me.
> >
> > Warm regards
> > Hindvee
> >
> > image.png
> >
> > On Thu, 19 Jun 2025, 13:42 Kishor Durve, <kishor.durve@outlook.com> wrote:
> >
> > > I have seen the processed images. I think it is good at this point to quantify 'sharpness' of an image. It will give us a measure to compare different methods / models.
> > >
> > > There can be several approaches to do this. We use Opencv::Laplacian() to judge image sharpness. But you can try

other approaches.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, June 19, 2025 2:52 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir,

I've completed the full-image denoising as discussed for the images `5.png` and `6.png`. This time, the model was trained and applied directly on the complete images (cropped to a common size of **570×588**) instead of small patches, as per your suggestion. The denoising autoencoder was able to reduce background noise and enhance structural features, particularly the edges of the circles, which were initially faint or inconsistent. The model output shows a noticeable improvement in clarity while preserving the original patterns.

I've attached the processed result image along with this mail. Please let me know your thoughts on this file.

Warm regards,
Hindvee

On Wed, Jun 18, 2025 at 12:33 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have checked the file. These two images contain circles as primary features. Also, the patches in the file seem to be for a small part of the original image. If you have processed the original images (5.png and 6.png) please send me the processed images on mail. If you have processed only parts of these images, I suggest you process the full images and send me the results.
>
> Similarly, in the previous exercise please process complete images. Most of the set are of size 500x500 pixel, so it should not consume too much time.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 18, 2025 1:44 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Sir, please let me know your views on the above .ipynb file.
> Warm Regards
> Hindvee
>
> On Wed, Jun 18, 2025 at 1:56 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir,
> > I have processed the 5.png and 6.png, I faced lower accuracy rate earlier. After fixing the earlier issues with mismatched data sizes and high Binary Crossentropy loss (~0.6098), the model is now working smoothly. It reached a much lower loss of just 0.0528 in 5 epochs, and both training and validation losses settled nicely by the 2nd or 3rd epoch without signs of overfitting. The denoised images look very clean and closely match the original ones, even when the input images had different levels of noise. I am attaching the results in the attached file.
> > Warm regards
> > Hindvee

On Tue, Jun 17, 2025 at 7:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Sorry! Forgot the attachment.
>
> Kishor Durve
>
> ---
>
> **From:** Kishor Durve <kishor.durve@outlook.com>
> **Sent:** Tuesday, June 17, 2025 9:46 AM
> **To:** Hindvee Vispute <hindveevispute@gmail.com>
> **Subject:** Re: AI assisted edge detection
>
> Good to know that you are making progress at a nice clip.
>
> I saw the images and the difference between before / after can be seen. Can you please process the images attached to this mail and send me the results? The images are similar to those I had sent earlier, the exception being 5.png and 6.png. In these last two images the challenge is to detect circles whose edges are not well defined and the edge quality may vary from image to image. This is to test the effectiveness of noise removal / sharpening.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Tuesday, June 17, 2025 3:06 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir,
> I have completed the training and evaluation of the denoising autoencoder using grayscale image patches taken from the `IMG` folder. The model was trained for 20 epochs on 64×64 patches with binary cross entropy loss and a 10% validation split. During training, the loss reduced steadily and stabilized, with the final reconstruction loss reaching 0.3688. The results show that the model can effectively denoise images and recover clean patterns like faint lines. In the next step, I plan to use the output of the Super-Resolution model as a base to evaluate the accuracy and quality of the denoising process, to better understand how both methods contribute to image enhancement. I am attaching the .ipynb file for your reference and also the IMG Folder which I just mentioned.
>
> Warm regards
> Hindvee
>
> IMG FOLDER
>
> Accuracy results:-
>
> image.png
>
> On Fri, Jun 13, 2025 at 5:33 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > The images I sent are big. OpenCV will not be able to handle them as the max size allowed is 32K x 32K. The crash you are facing could be due to this reason.

You can crop out small patches from the big images and use them for trials.

Kishor Durve
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, June 11, 2025 1:37 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Thank you for sharing more images. I have updated the work on the sheets.

Warm regards
Hindvee

On Sat, Jun 7, 2025 at 2:51 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have uploaded more images. You can crop out patches from them. One of the images has many boards which will give many useful patches.
>
> Kishor Durve
> Duravision Systems
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Saturday, June 7, 2025 4:36 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings sir,
> It would be great if I get more images, that would expand my dataset. Thank you for your previous email.
> I also wanted tk inform you that I am travelling and will be able to resume work by Tuesday. I request you to kindly allow my leave.
>
> Warm Regards
> Hindvee
>
> On Fri, 6 Jun 2025, 11:31 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > I am not going to be of much help to you as I have limited experience in this field. Also, what little I know was gained by applying available models in OpenCV's DNN module.
> >
> > There are two images available in the images folder. You may crop out patches from them for the Training / test sets. We use ImageJ and IrfanView for such work as MS Paint cannot handle such big images. If you need more images please let me know.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, June 6, 2025 5:24 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection

Greetings sir,
I hope you're doing well. I've just finished training the CNN-based line presence classifier, and I'm attaching the loss graph for your review. The model seems to have learned quite quickly, with both training and validation losses staying very close to zero throughout. There was a small spike in training loss around epoch 80, but validation stayed stable, so it might just be a minor fluctuation.

Since the dataset is quite small and the model converged very early, I'm unsure how best to move forward.According to you should I try to get more data to avoid overfitting. Please guide me towards a direction in which I can make this model more effective.

Warm regards

Hindvee

On Wed, Jun 4, 2025 at 6:42 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Yes, it looks ok to me. We will take a relook once you make more progress.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 4, 2025 4:26 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Additionally, Sir, I would like to confirm if the current format of the Google Sheet meets your expectations. Please let me know if you would prefer any modifications, such as a more detailed breakdown of the work completed.
>
> Warm Regards
> Hindvee
>
> On Wed, Jun 4, 2025 at 1:34 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Greetings Mr. Durve,
> >
> > Thanks for sharing the super-resolution model. I've set it up and tested it, the results inspiring. It does a good job cleaning up noise and improving contrast, especially in the faint line areas. As we discussed, I'll now add this to the bigger pipeline along with the AI modules (NB-1 to NB-5). The super-resolution output can help improve both the CNN classifier and regression models. I'll also check how it works together with NB-5, the denoising autoencoder, to see if we get even better results.
> >
> > Please let me know if you want me to run it on any specific ROI samples or start putting together comparison results.
> >
> > Warm regards
> > Hindvee
> >
> > On Tue, Jun 3, 2025 at 9:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > I have file network.py in folder 'models'. This file has the RRDBNet implementaion.
> > >
> > > Kishor Durve
> > > Duravision Systems

9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Monday, June 2, 2025 7:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr. Durve,
I am facing an issue again an again while running the VS 2022 code, pls help me with this, we have the module named as model_zoo not models. Pls tell me how to proceed with this.

 image.png

Warm Regards
Hindvee

On Sat, May 31, 2025 at 12:46 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Thankyou sir! I will try this right away.
> Also I hope you received this google sheet, I have shared edit access to you on your gmail account
>
> Hindvee Vispute - Time data - Intern
>
> Regards
> Hindvee
>
> On Sat, May 31, 2025 at 5:43 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Sure. You need to do the following to make the project work.
> >
> > 1. Download and install VS 2022 Community edition
> > 2. Download and install Anaconda. Install an additional environment Python 3.8.16. Use this version of Python for the project.
> >
> > The project should run at this point.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, May 30, 2025 6:34 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings sir!
> > I went through the google drive folder shared by you. It would be really helpful if you could walk me through it.
> > Best regards
> > Hindvee
> >
> > On Fri, May 30, 2025 at 11:41 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > You can download the model, a working VC++ (VS 2022) project and image here.
> > >
> > > https://drive.google.com/drive/folders/1mLBDjJR3fFKs8Brr_nNj2z9YyI89nruE?

usp=sharing

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, May 30, 2025 12:50 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings sir!
Pls do share the model over drive I will try it out.
Also I request you to share more image database so that I can start working on it.
I will start work from today itself then.

Regards
Hindvee

On Fri, 30 May 2025, 06:11 Kishor Durve, <kishor.durve@outlook.com> wrote:

> We have found that using a super resolution Pytorch model to get a larger image results in significant noise removal. Perhaps it is a by-product.
>
> You can try it out. I have the model and can share it on Google Drive if you want to try it out. We have tried the models available in OpenCV DNN but they do not seem to give the same results.
>
> Regards,
>
> Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 29, 2025 5:28 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Dear Sir,

Thank you for the confirmation. I will maintain the time data on a Google Sheet and ensure it is updated regularly.

Looking forward to starting from Monday, 2nd June.

Best regards,
Hindvee

On Thu, May 29, 2025 at 2:53 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> Welcome aboard!

You can keep the time data on Google Sheet.

Regards,

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 29, 2025 5:12 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr. Durve,

Thank you very much for giving me the opportunity to contribute to Guangdong Longtian Smart Instruments Co. Ltd. through this exciting project. I am pleased to accept the offer and look forward to working on the development of the AI-assisted line detection tool.

With your permission, I would like to begin the internship from the coming Monday, i.e., 2nd June 2025.

Additionally, I wanted to ask about the preferred method for tracking and sharing the time I spend on the project. Would you like me to maintain a shared timesheet (e.g., a Google Sheet or Excel file), or would you prefer weekly time summaries via email or any other format?

Looking forward to your guidance on this.

Thank you once again for this opportunity.

Warm regards,
Hindvee Rajesh Vispute

On Thu, May 29, 2025 at 8:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have attached the on-boarding letter.
>
> Please let me know if you have any queries.
>
> Regards,
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Saturday, May 10, 2025 3:46 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection

Greetings Mr. Durve,

Thank you for your detailed response and clarification. I now clearly understand that the goal is to enhance line detection using AI within known ROIs, leveraging the expected position of endpoints.

To support this objective, I've explored and implemented several **AI-based approaches**, any of which can be used to complement or improve upon conventional methods:

1. **CNN-Based Line Presence Classifier (NB-1):** Identifies whether a line exists within a given patch/ROI—useful for filtering candidate regions before applying traditional detection.
2. **CNN Regression Model (NB-2):** Predicts line endpoints directly from the image, offering robust localization even in low-contrast or noisy scenarios.
3. **Denoising Autoencoder (NB-5):** Enhances faint or broken lines by cleaning up noise, thereby boosting the effectiveness of existing edge or Hough-based methods.
4. **Vision Transformer Model (NB-4):** Designed for fine-grained classification tasks and can be adapted for future objectives like surface defect or anomaly detection.

These methods are independently deployable and can be integrated selectively depending on the specific inspection challenge or ROI structure. I'd be happy to align with your ROI-driven strategy and customize accordingly.

Regarding the stipend, I agree with the proposed Rs. 200/hour base rate and performance-based bonus, and I'm enthusiastic about the opportunity to collaborate further on defect detection and shape anomaly tasks in the future.

I have attached screenshots of the code files I executed today. I have the names of the notebooks (NB) with NB-1, NB-2,NB-4 and NB-5 respectively.

NB-1
NB-2
NB-4
NB-5

Please let me know if you'd like me to share these outputs or run a quick live demonstration.

Warm regards,
Hindvee

On Fri, May 9, 2025 at 6:03 AM Kishor Durve <kishor.durve@outlook.com> wrote:

I think there is a mismatch in understanding. We are already using conventional line detection techniques (various edge detection operators and Hough Transform and other methods for line fitment) .

What we are looking to get from you is AI based / AI assisted line detection. Since we know the probable position of the line beforehand we can use this information to find a line. For example, say we know that the line end points are expected to be (x1, y1) and (x2, y2), we can construct a small rectangle (ROI) using these points and search for a line within this ROI.

As for the stipend it depends both on the time you spend on the project and the results you produce. It will comprise of an hourly rate and a result linked amount. I suggest we have an hourly rate of Rs. 200/hour and a matching amount linked to the results at the end of the project. So, if you are able to produce satisfactory results it will come to Rs. 400/hour. Also, if the results are satisfactory, we can take this forward next year and work on other issues like surface defect detection, detection of shape anomalies etc.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 8, 2025 5:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr Durve,
I have the problem now so I will focus on faint line detection for the first step. I can proceed using Sobel, Scharr, and Canny edge detectors provided by OpenCV. First I will preprocess the images and then move to edge detection using the above 3 detectors. I will carry out line detection using Hough Transform, and further we will filter out short or irrelevant lines based on length and their angle of orientation. Finally, we will draw the detected lines on the images using

cv.line()

I can devote 14 hrs per week to this task for 2 months, we can extend the time if required. Also, I wanted to ask what stipend you can provide me. I believe I can complete the task in the above time frame. Thank you for your time.

Warm regards
Hindvee

On Thu, May 8, 2025 at 7:18 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Your analysis seems good.
>
> At this point we are not interested in detecting defects (though at a later stage we may be), and want to detect the faint lines only.
>
> Note that though I have annotated only a single line in the image I sent to you yesterday, we need to detect edge lines of all vertical stripes.
>
> I have attached 2 sets of 20 images each named 'Good Images' and 'Bad Images' - images in which it is mostly easy to detect lines by the conventional image processing method and those in which lines are faint and hard to detect. These images may help you get a better understanding of the problem.
>
> Please also check the following issues:
>
> 1. How much time will you be able to devote to this task?
> 2. Will you be able to complete the task within the available time?
>
> There are other issues like time taken by the AI route, integration of AI line detect code in VC++ project, further training / fine tuning of model, applicability of the method to detect faint circles etc. But we will get into them later once the basic method is proven.
>
> Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, May 7, 2025 3:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings, Mr. Durve,
Thank you for reviewing my proposal and for sharing the sample images. By looking at the images I came across the specific challenges faced by the inspection system, like detecting faint lines, subtle surface variations, and fine defects that can be missed by traditional image processing. Based on these observations I suggest that we can do the following:-

1. **Hybrid approach**: Objective - Combine the speed of OpenCV line detection (example Hough Transform) with deep learning's pattern recognition capability.
Approach
- Use OpenCV to pre-locate candidate regions and feed them into a deep model for verification.
- This can speed up inference and increase resistance to false positives.

2. **Deep learning-based fine defect detection**: Objective: Improve the detection of faint, low contrast line defects and localized surface anomalies.
Approach
- Train a model, such as YOLOv8 or U-Net, tailored to your defect patterns using the annotated images.
- To improve feature visibility, add contrast enhancement.
- Assess using detection accuracy and intersection over union (IoU) to compare to baselines based on OpenCV.

I can begin with a proof-of-concept model using the images shared by you. Thank you for sharing the opportunity. I am excited to get a chance to contribute to the advancement in inspection technology.

Warm regards
Hindvee Vispute

On Wed, 7 May 2025, 09:31 Kishor Durve, <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have gone thru the document uploaded by you.
>
> It looks ok to me.

I can have attached a few images so you will get a better idea of the requirements.

Kishor Durve

Hindvee Vispute <hindveevispute@gmail.com> Fri, Aug 8, 2025 at 2:36 AM
To: Kishor Durve <kishor.durve@outlook.com>

Hello sir! I completely understand your point. I have received the stipend with thanks.
Sir, I had shared a drive link with you in the previous email. Could you pls go through those pictures and share your reviews with me.
Warm regards
Hindvee

On Wed, Aug 6, 2025 at 8:10 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have transferred Rs. 11,600 (200 x 58) to you. Regarding the suggested increase in per-hour rate: You have certainly made efforts to solve the problem. However, as of now the result is not yet usable to us. We will consider an increase when it is usable in our software.
>
> Kishor Durve
> Duravision Systems
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Tuesday, August 5, 2025 8:41 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir
> Just following up on my previous email regarding the processed images and the stipend for July (58 hours). I have shared a small suggestion regarding the per-hour rate.
>
> Please let me know if you need any further details or modifications from my side.
> UPDATED DEXINED OUTPUT
> Looking forward to your feedback.
>
> Warm regards,
> Hindvee
>
> On Fri, Aug 1, 2025 at 2:14 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Thank you sir! I have worked out and increased the contrast in the processed images. I will share the file with you. Also sir I shall request you to please process the stipend for this month. It has been 58 hrs for the month of July. If you like my work please consider increasing the per hour rate. It is just a small suggestion. The final decision depends on you and I happily accept your decision. My primary goal is to enhance my skills, and I am learning a lot from this project.
> >
> > Warm regards
> > Hindvee
> >
> > On Wed, Jul 30, 2025 at 11:02 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > I have checked the images. Your conclusions are correct.
> > >
> > > Please carry on with your efforts.
> > >
> > > Kishor Durve
> > > Duravision Systems
> > > 9823164286
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Tuesday, July 29, 2025 3:37 AM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection

Hello sir!
current update:-

- Using an autoencoder trained on clean (good) images to repair bad images
- Applying a **hybrid loss** of MSE and SSIM to preserve structure and perceptual quality.
- Saving and visualizing both original and reconstructed images which helps in debugging

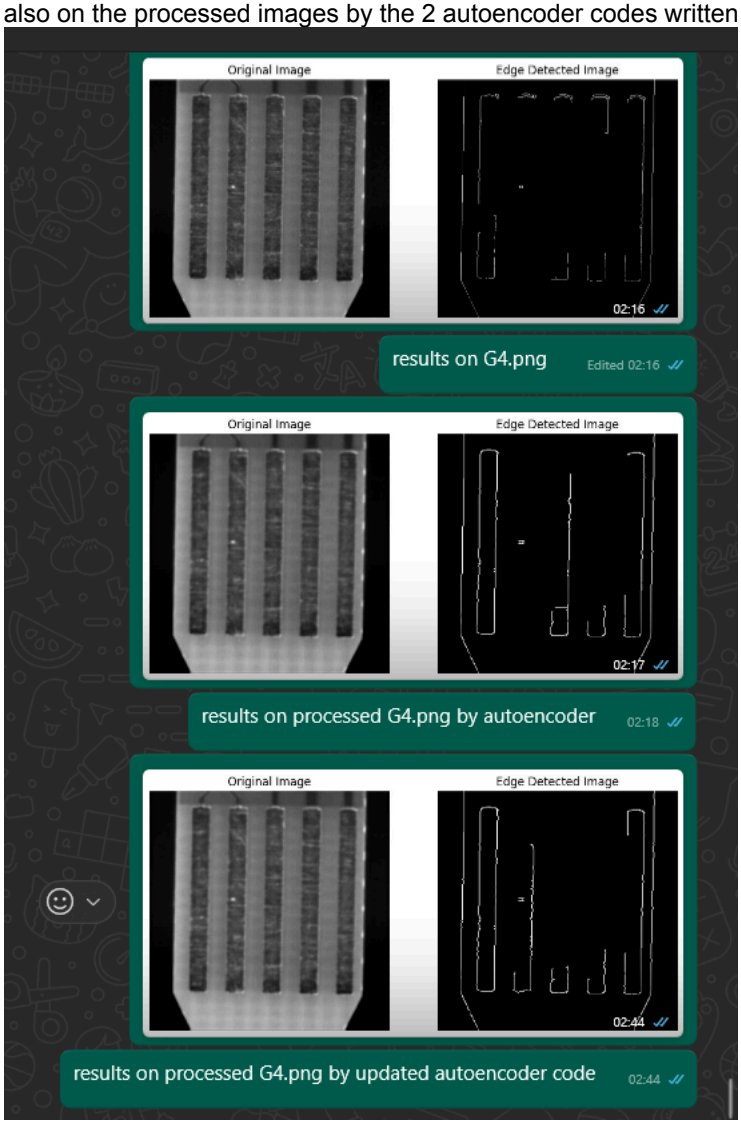
Despite this denoising strategy, DexiNED is still failing for bad images, the possible reasons are as follows

1. Contrast remains too low after reconstruction.
2. Edges are too smooth or faint due to over-smoothing by the autoencoder.
3. DexiNED is not tuned for synthetic/industrial patterns (e.g., PCB, x-ray like structures).

I will now work on these factors to improve the quality of processed images. I am attaching the processed Good and Bad Images folders.
BAD IMAGES _ PROCESSED
GOOD IMAGES_ PROCESSED

warm regards
Hindvee

On Mon, Jul 28, 2025 at 2:48 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Noted sir. I wrote code for autoencoder to get sharper images, and I applied DexiNed on the original image and also on the processed images by the 2 autoencoder codes written so far. I am sharing the results here.
>
> 
>
>
> warm regards
> Hindvee
>
> On Fri, Jul 25, 2025 at 6:34 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Sorry for delay in replying. I was away for quite some time.

A couple of months back we had tried OpenCV / HED and obtained similar results.

Please go ahead and try DexiNed after which we will decide about line detection.

After that another important step would be to remove noise from image using a suitable model.

I know this is somewhat away from our goal of 'AI assisted edge detection' but it can be an important intermediate step. We are currently using a model that does this. This is the model I had sent to you initially. It is developed in a Chinese University by a professor who works with our company.

Kishor Durve
Duravision Systems
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, July 25, 2025 5:46 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir ! This is an output image obtained from the HED model and similar image I have obtained from DexiNed, could u pls tell i am moving in the right direction or should I make improvements or change in the plan of action?

warm regards
Hindvee

On Wed, Jul 23, 2025 at 3:20 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir! Could you please share ur insights and views on the above mail thread, it shall be very helpful to get to know your feedback and move forward with the project.
>
> Warm Regards
> Hindvee
>
> On Fri, Jul 11, 2025 at 3:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir!
> >
> > Following up on my previous email regarding edge detection, I wanted to share the progress I've made since then. I have been focusing on using the Holistically-Nested Edge Detection (HED) method. I have now:
> >
> > CODE FILE
> >
> > 1. Successfully applied the HED process to a folder containing our images to generate edge maps for the entire dataset.
> >
> > 2. Trained a custom autoencoder model designed to refine and correct the lines in these images.
> >
> > 3. Applied the trained autoencoder to the HED output images to produce corrected versions.
> >
> > 4. Started working on post-processing techniques.
> >
> > I am currently experimenting with different parameters and methods to achieve the best results for this step. I am continuing to refine the line clarification process and will share updated results once I have finalized the post-processing approach.
> >
> > warm regards
> >
> > Hindvee
> >
> > On Wed, Jul 9, 2025 at 12:23 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> >
> > > Hello sir! I am trying the pre-trained model for detecting edges in our images; I have tried Holistically-Nested Edge Detection (HED) and DexiNed. I am attaching the output, I am going forward with HED to get better results.
> > >
> > > Warm regards

Hindvee

On Mon, Jul 7, 2025 at 10:37 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Hello sir!
I am sharing the first 5 difference images of the three codes that I mentioned previously. Please check this whenever you have time.

Warm regards
Hindvee

On Mon, Jul 7, 2025 at 6:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:

I am unable to give my view of the approach as I can't view the images. I can give my opinion if you send me difference images. In any case, I think you need to have difference images as a metric in addition to the Laplacian.

Get Outlook for Android

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Monday, July 7, 2025 7:09:51 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!
I have received the stipend, thank you for that! I'd like to share a brief update on the image enhancement work, focusing on noise removal and edge preservation.
Summary of results:-
1. Previous code
---Performed basic denoising but introduced over-smoothing.
---Laplacian MAD: ~0.0175–0.0192

2. DnCNN
---Improved noise removal and structural retention.
---MAD: ~0.0186–0.0202

3. DnCNN + Sharpening (Current approach)
---Added post-processing to enhance edge clarity.
---Consistently improved MAD to ~0.0178–0.0196.
---Visual results show reduced whitening and better edge definition.

This confirms that the current setup provides better denoising while preserving fine image details. Please let me know if you'd like to proceed with further tuning using SSIM or edge-aware loss functions.

Warm regards

Hindvee

On Fri, Jul 4, 2025 at 3:54 AM Kishor Durve <kishor.durve@outlook.com> wrote:

Please keep at it. Use difference image and Laplacian to gauge the effect of the algorithm. I will be able to look at your work only after 15th.
I have transferred your stipend by GPay.

Get Outlook for Android

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, July 4, 2025 5:37:35 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!
This is what I am been working on. I am still working on getting better edges and less whitish images.

warm regards
Hindvee

On Thu, Jul 3, 2025 at 10:53 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

UPI ID: hindveevispute@okicici

Warm regards
Hindvee

On Wed, 2 Jul 2025, 03:16 Kishor Durve, <kishor.durve@outlook.com> wrote:

> I checked the images, but did not notice substantial difference between the before and after images. You can subtract them (after aligning) and check.
> As regards payment of stipend please let me know how much we need to pay and I will transfer it on GPay.
> I am travelling again and am out of office. I will be back only on 15th. My responses to your mails may be delayed.
>
> Get Outlook for Android
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Tuesday, July 1, 2025 1:51:45 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir!
>
> I was eager to get insights from you about the above shared images.Please do have a look as soon as possible. Also, I wanted to know whether the stipend will be credited at the end of the month or the project's end. It would be very kind of you if I could get the stipend at the end of the month.
>
> Thanks and regards
> Hindvee
>
> On Mon, Jun 30, 2025 at 10:33 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Good morning sir!
> >
> > Did you get any chance to review the above images? Please share your suggestions and insights with me.
> >
> > Warm regards
> > Hindvee
> >
> > On Thu, Jun 26, 2025 at 11:09 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> >
> > > Here I am attaching the processed images.
> > >
> > > Warm regards
> > > Hindvee
> > >
> > > On Thu, Jun 26, 2025 at 8:08 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> > >
> > > > I am unable to run the files in Colab and generate images. Can you please send me the processed images on mail - .png / .jpeg files, not .ipynb files.
> > > >
> > > > Kishor Durve
> > > >
> > > > ---
> > > >
> > > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > > **Sent:** Thursday, June 26, 2025 4:48 AM
> > > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > > **Subject:** Re: AI assisted edge detection
> > > >
> > > > Hello sir, I have attached the results for png 1,2,3 and 4. Pls share ur views and insights.
> > > >
> > > > Warm Regards
> > > > Hindvee
> > > >
> > > > On Tue, Jun 24, 2025 at 11:29 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > > >
> > > > > Hello sir,
> > > > > I am attaching the .ipynb file, it has the images

warm regards
Hindvee

On Tue, Jun 24, 2025 at 5:48 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Please share the images for visual comparison.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Tuesday, June 24, 2025 5:13 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello Durve sir,
>
> I wanted to share an update on the denoising output. I noticed earlier that while the model was doing a good job removing noise, it was also making the image a bit too smooth — the sharpness had dropped quite a bit:
>
> - Noisy Image: 33.72
> - Denoised Image: 6.92
> - Clean Image (Ground Truth): 26.49
>
> To improve this, I added a sharpening filter after the autoencoder output and tested two variants:
>
> 1. A **strong sharpening kernel**, which raised the denoised sharpness to **112.40,** but it felt a bit overdone and introduced some artifacts.
> 2. A **softer kernel**, which gave a cleaner and more natural result, with denoised sharpness at **54.04**. This seems like a balanced point where edges are clearer without compromising visual quality.
>
> If you'd like, I can also share the output images (clean vs. denoised) for a visual comparison.
>
> Warm regards
>
> Hindvee
>
> image.png

On Fri, Jun 20, 2025 at 12:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir!
> Based on the latest code I calculated the sharpness using Laplacian variance. I obtained the following results as those attached as image.
> I think that while denoising, though the noise was removed, but the model blurred or over smoothened the image. I may have lost some meaningful edge details in the process. But the sharpness value of Clean image is lower than the noisy image (because it has no random noise), but high enough to retain clear edges and structure. So I am working next on increasing the sharpness value while keeping the noise low. Please do share your views with me.
>
> Warm regards
> Hindvee
>
> image.png

On Thu, 19 Jun 2025, 13:42 Kishor Durve, <kishor.durve@outlook.com> wrote:

> I have seen the processed images. I think it is good at this point to quantify 'sharpness' of an image. It will give us a measure to compare different methods / models.
>
> There can be several approaches to do this. We use Opencv::Laplacian() to judge image sharpness. But you can try other approaches.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Thursday, June 19, 2025 2:52 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir,
>
> I've completed the full-image denoising as discussed for the images `5.png` and `6.png`. This time, the model was trained and applied directly on the complete images (cropped to a common size of **570×588**) instead of small patches, as per your suggestion. The denoising autoencoder was able to reduce background noise and enhance structural features, particularly the edges of the circles, which were initially faint or inconsistent. The model output shows a noticeable improvement in clarity while preserving the original patterns.
>
> I've attached the processed result image along with this mail. Please let me know your thoughts on this file.
>
> Warm regards,
> Hindvee
>
> On Wed, Jun 18, 2025 at 12:33 PM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > I have checked the file. These two images contain circles as primary features. Also, the patches in the file seem to be for a small part of the original image. If you have processed the original images (5.png and 6.png) please send me the processed images on mail. If you have processed only parts of these images, I suggest you process the full images and send me the results.
> >
> > Similarly, in the previous exercise please process complete images. Most of the set are of size 500x500 pixel, so it should not consume too much time.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Wednesday, June 18, 2025 1:44 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Sir, please let me know your views on the above .ipynb file.
> > Warm Regards
> > Hindvee
> >
> > On Wed, Jun 18, 2025 at 1:56 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> >
> > > Hello sir,
> > > I have processed the 5.png and 6.png, I faced lower accuracy rate earlier. After fixing the earlier issues with mismatched data sizes and high Binary Crossentropy loss (~0.6098), the model is now working

smoothly. It reached a much lower loss of just 0.0528 in 5 epochs, and both training and validation losses settled nicely by the 2nd or 3rd epoch without signs of overfitting. The denoised images look very clean and closely match the original ones, even when the input images had different levels of noise. I am attaching the results in the attached file.
Warm regards
Hindvee

On Tue, Jun 17, 2025 at 7:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Sorry! Forgot the attachment.
>
> Kishor Durve
>
> ---
>
> **From:** Kishor Durve <kishor.durve@outlook.com>
> **Sent:** Tuesday, June 17, 2025 9:46 AM
> **To:** Hindvee Vispute <hindveevispute@gmail.com>
> **Subject:** Re: AI assisted edge detection
>
> Good to know that you are making progress at a nice clip.
>
> I saw the images and the difference between before / after can be seen. Can you please process the images attached to this mail and send me the results? The images are similar to those I had sent earlier, the exception being 5.png and 6.png. In these last two images the challenge is to detect circles whose edges are not well defined and the edge quality may vary from image to image. This is to test the effectiveness of noise removal / sharpening.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Tuesday, June 17, 2025 3:06 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir,
> I have completed the training and evaluation of the denoising autoencoder using grayscale image patches taken from the `IMG` folder. The model was trained for 20 epochs on 64×64 patches with binary cross entropy loss and a 10% validation split. During training, the loss reduced steadily and stabilized, with the final reconstruction loss reaching 0.3688. The results show that the model can effectively denoise images and recover clean patterns like faint lines. In the next step, I plan to use the output of the Super-Resolution model as a base to evaluate the accuracy and quality of the denoising process, to better understand how both methods contribute to image enhancement. I am attaching the .ipynb file for your reference and also the IMG Folder which I just mentioned.
>
> Warm regards
> Hindvee
>
> IMG FOLDER
>
> Accuracy results:-
>
> image.png

On Fri, Jun 13, 2025 at 5:33 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> The images I sent are big. OpenCV will not be able to handle them as the max size allowed is 32K x 32K. The crash you are facing could be due to this reason.
>
> You can crop out small patches from the big images and use them for trials.
>
> Kishor Durve
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 11, 2025 1:37 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Thank you for sharing more images. I have updated the work on the sheets.
>
> Warm regards
> Hindvee
>
> On Sat, Jun 7, 2025 at 2:51 PM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > I have uploaded more images. You can crop out patches from them. One of the images has many boards which will give many useful patches.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Saturday, June 7, 2025 4:36 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings sir,
> > It would be great if I get more images, that would expand my dataset. Thank you for your previous email.
> > I also wanted tk inform you that I am travelling and will be able to resume work by Tuesday. I request you to kindly allow my leave.
> >
> > Warm Regards
> > Hindvee
> >
> > On Fri, 6 Jun 2025, 11:31 Kishor Durve, <kishor.durve@outlook.com> wrote:
> >
> > > I am not going to be of much help to you as I have limited experience in this field. Also, what little I know was gained by applying available models in OpenCV's DNN module.
> > >
> > > There are two images available in the images folder. You may crop out patches from them for the Training / test sets. We use ImageJ and IrfanView for such work as MS Paint cannot handle such big images. If you need more images please let me know.
> > >
> > > Kishor Durve
> > > Duravision Systems

9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, June 6, 2025 5:24 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings sir,
I hope you're doing well. I've just finished training the CNN-based line presence classifier, and I'm attaching the loss graph for your review. The model seems to have learned quite quickly, with both training and validation losses staying very close to zero throughout. There was a small spike in training loss around epoch 80, but validation stayed stable, so it might just be a minor fluctuation.

Since the dataset is quite small and the model converged very early, I'm unsure how best to move forward.According to you should I try to get more data to avoid overfitting. Please guide me towards a direction in which I can make this model more effective.

Warm regards

Hindvee

On Wed, Jun 4, 2025 at 6:42 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Yes, it looks ok to me. We will take a relook once you make more progress.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 4, 2025 4:26 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Additionally, Sir, I would like to confirm if the current format of the Google Sheet meets your expectations. Please let me know if you would prefer any modifications, such as a more detailed breakdown of the work completed.
>
> Warm Regards
> Hindvee
>
> On Wed, Jun 4, 2025 at 1:34 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Greetings Mr. Durve,
> >
> > Thanks for sharing the super-resolution model. I've set it up and tested it, the results inspiring. It does a good job cleaning up noise and improving contrast, especially in the faint line areas. As we discussed, I'll now add this to the bigger pipeline along with the AI modules (NB-1 to NB-5). The super-resolution output can help improve both the CNN classifier and regression models. I'll also check how it works together with NB-5, the denoising autoencoder, to see if we get even better results.
> >
> > Please let me know if you want me to run it on any specific ROI samples or start putting together comparison results.
> >
> > Warm regards
> > Hindvee

On Tue, Jun 3, 2025 at 9:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have file network.py in folder 'models'. This file has the RRDBNet implementaion.
>
> Kishor Durve
> Duravision Systems
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Monday, June 2, 2025 7:15 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings Mr. Durve,
> I am facing an issue again an again while running the VS 2022 code, pls help me with this, we have the module named as model_zoo not models. Pls tell me how to proceed with this.
>
> image.png
>
> Warm Regards
> Hindvee
>
> On Sat, May 31, 2025 at 12:46 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Thankyou sir! I will try this right away.
> > Also I hope you received this google sheet, I have shared edit access to you on your gmail account
> >
> > Hindvee Vispute - Time data - Intern
> >
> > Regards
> > Hindvee
> >
> > On Sat, May 31, 2025 at 5:43 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > Sure. You need to do the following to make the project work.
> > >
> > > 1. Download and install VS 2022 Community edition
> > > 2. Download and install Anaconda. Install an additional environment Python 3.8.16. Use this version of Python for the project.
> > >
> > > The project should run at this point.
> > >
> > > Kishor Durve
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Friday, May 30, 2025 6:34 PM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection
> > >
> > > Greetings sir!
> > > I went through the google drive folder shared by you. It would be really helpful if you could walk me through it.

Best regards
Hindvee

On Fri, May 30, 2025 at 11:41 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> You can download the model, a working VC++ (VS 2022) project and image here.
>
> https://drive.google.com/drive/folders/1mLBDjJR3fFKs8Brr_nNj2z9YyI89nruE?usp=sharing
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Friday, May 30, 2025 12:50 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings sir!
> Pls do share the model over drive I will try it out.
> Also I request you to share more image database so that I can start working on it.
> I will start work from today itself then.
>
> Regards
> Hindvee
>
> On Fri, 30 May 2025, 06:11 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > We have found that using a super resolution Pytorch model to get a larger image results in significant noise removal. Perhaps it is a by-product.
> >
> > You can try it out. I have the model and can share it on Google Drive if you want to try it out. We have tried the models available in OpenCV DNN but they do not seem to give the same results.
> >
> > Regards,
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Thursday, May 29, 2025 5:28 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Dear Sir,
> >
> > Thank you for the confirmation. I will maintain the time data on a Google Sheet and ensure it is updated regularly.
> >
> > Looking forward to starting from Monday, 2nd June.

Best regards,
Hindvee

On Thu, May 29, 2025 at 2:53 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> Welcome aboard!
>
> You can keep the time data on Google Sheet.
>
> Regards,
>
> Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 29, 2025 5:12 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr. Durve,

Thank you very much for giving me the opportunity to contribute to Guangdong Longtian Smart Instruments Co. Ltd. through this exciting project. I am pleased to accept the offer and look forward to working on the development of the AI-assisted line detection tool.

With your permission, I would like to begin the internship from the coming Monday, i.e., 2nd June 2025.

Additionally, I wanted to ask about the preferred method for tracking and sharing the time I spend on the project. Would you like me to maintain a shared timesheet (e.g., a Google Sheet or Excel file), or would you prefer weekly time summaries via email or any other format?

Looking forward to your guidance on this.

Thank you once again for this opportunity.

Warm regards,
Hindvee Rajesh Vispute

On Thu, May 29, 2025 at 8:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have attached the on-boarding letter.
>
> Please let me know if you have any queries.

Regards,

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Saturday, May 10, 2025 3:46 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr. Durve,

Thank you for your detailed response and clarification. I now clearly understand that the goal is to enhance line detection using AI within known ROIs, leveraging the expected position of endpoints.

To support this objective, I've explored and implemented several **AI-based approaches**, any of which can be used to complement or improve upon conventional methods:

1. **CNN-Based Line Presence Classifier (NB-1):** Identifies whether a line exists within a given patch/ROI—useful for filtering candidate regions before applying traditional detection.
2. **CNN Regression Model (NB-2):** Predicts line endpoints directly from the image, offering robust localization even in low-contrast or noisy scenarios.
3. **Denoising Autoencoder (NB-5):** Enhances faint or broken lines by cleaning up noise, thereby boosting the effectiveness of existing edge or Hough-based methods.
4. **Vision Transformer Model (NB-4):** Designed for fine-grained classification tasks and can be adapted for future objectives like surface defect or anomaly detection.

These methods are independently deployable and can be integrated selectively depending on the specific inspection challenge or ROI structure. I'd be happy to align with your ROI-driven strategy and customize accordingly.

Regarding the stipend, I agree with the proposed Rs. 200/hour base rate and performance-based bonus, and I'm enthusiastic about the opportunity to collaborate further on defect detection and shape anomaly tasks in the future.

I have attached screenshots of the code files I executed today. I have the names

of the notebooks (NB) with NB-1, NB-2,NB-4 and NB-5 respectively.

NB-1
NB-2
NB-4
NB-5

Please let me know if you'd like me to share these outputs or run a quick live demonstration.

Warm regards,
Hindvee

On Fri, May 9, 2025 at 6:03 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> I think there is a mismatch in understanding. We are already using conventional line detection techniques (various edge detection operators and Hough Transform and other methods for line fitment) .
>
> What we are looking to get from you is AI based / AI assisted line detection. Since we know the probable position of the line beforehand we can use this information to find a line. For example, say we know that the line end points are expected to be (x1, y1) and (x2, y2), we can construct a small rectangle (ROI) using these points and search for a line within this ROI.
>
> As for the stipend it depends both on the time you spend on the project and the results you produce. It will comprise of an hourly rate and a result linked amount. I suggest we have an hourly rate of Rs. 200/hour and a matching amount linked to the results at the end of the project. So, if you are able to produce satisfactory results it will come to Rs. 400/hour. Also, if the results are satisfactory, we can take this forward next year and work on other issues like surface defect detection, detection of shape anomalies etc.
>
> Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 8, 2025 5:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr Durve,
I have the problem now so I will focus on faint line detection for the first step. I can proceed using Sobel, Scharr, and Canny edge detectors provided by OpenCV. First I will preprocess the images and then move to edge detection using the above 3 detectors. I will carry out line detection using Hough Transform, and further we will filter out short or irrelevant lines based on length and their angle of orientation. Finally, we will draw the detected lines on the images using cv.line()

I can devote 14 hrs per week to this task for 2 months, we can extend the time if required. Also, I wanted to ask what stipend you can provide me. I believe I can complete the task in the above time frame. Thank you for your time.

Warm regards
Hindvee

On Thu, May 8, 2025 at 7:18 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Your analysis seems good.
>
> At this point we are not interested in detecting defects (though at a later stage we may be), and want to detect the faint lines only.
>
> Note that though I have annotated only a single line in the image I sent to you yesterday, we need to detect edge lines of all vertical stripes.
>
> I have attached 2 sets of 20 images each named 'Good Images' and 'Bad Images' - images in which it is mostly easy to detect lines by the conventional image processing method and those in which lines are faint and hard to detect. These images may help you get a better understanding of the problem.

Please also check the following issues:

1. How much time will you be able to devote to this task?
2. Will you be able to complete the task within the available time?

There are other issues like time taken by the AI route, integration of AI line detect code in VC++ project, further training / fine tuning of model, applicability of the method to detect faint circles etc. But we will get into them later once the basic method is proven.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, May 7, 2025 3:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings, Mr. Durve,
Thank you for reviewing my proposal and for sharing the sample images. By looking at the images I came across the specific challenges faced by the inspection system, like detecting faint lines, subtle surface variations, and fine defects that can be missed by traditional image processing. Based on these observations I suggest that we can do the following:-

1. **Hybrid approach**: Objective - Combine the speed of OpenCV line detection (example Hough Transform) with deep learning's pattern recognition capability.
Approach
- Use OpenCV to pre-locate candidate regions and feed them into a deep model for verification.
- This can speed up inference and increase resistance to false positives.

2. **Deep learning-based fine defect detection**: Objective: Improve the detection of faint, low contrast line defects and localized surface anomalies.
Approach
- Train a model, such as YOLOv8 or U-Net, tailored to your defect patterns using the annotated images.
- To improve feature visibility, add contrast enhancement.

- Assess using detection accuracy and intersection over union (IoU) to compare to baselines based on OpenCV.

I can begin with a proof-of-concept model using the images shared by you. Thank you for sharing the opportunity. I am excited to get a chance to contribute to the advancement in inspection technology.

Warm regards
Hindvee Vispute

On Wed, 7 May 2025, 09:31 Kishor Durve, <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have gone thru the document uploaded by you.
>
> It looks ok to me.
>
> I can have attached a few images so you will get a better idea of the requirements.
>
> Kishor Durve

Hindvee Vispute <hindveevispute@gmail.com> Wed, Aug 20, 2025 at 10:34 PM
To: Kishor Durve <kishor.durve@outlook.com>

Hello sir!
Could you pls share your reviews on the link I shared earlier with you? Your views will determine how to proceed further.

Warm Regards
Hindvee

On Fri, Aug 8, 2025 at 2:36 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

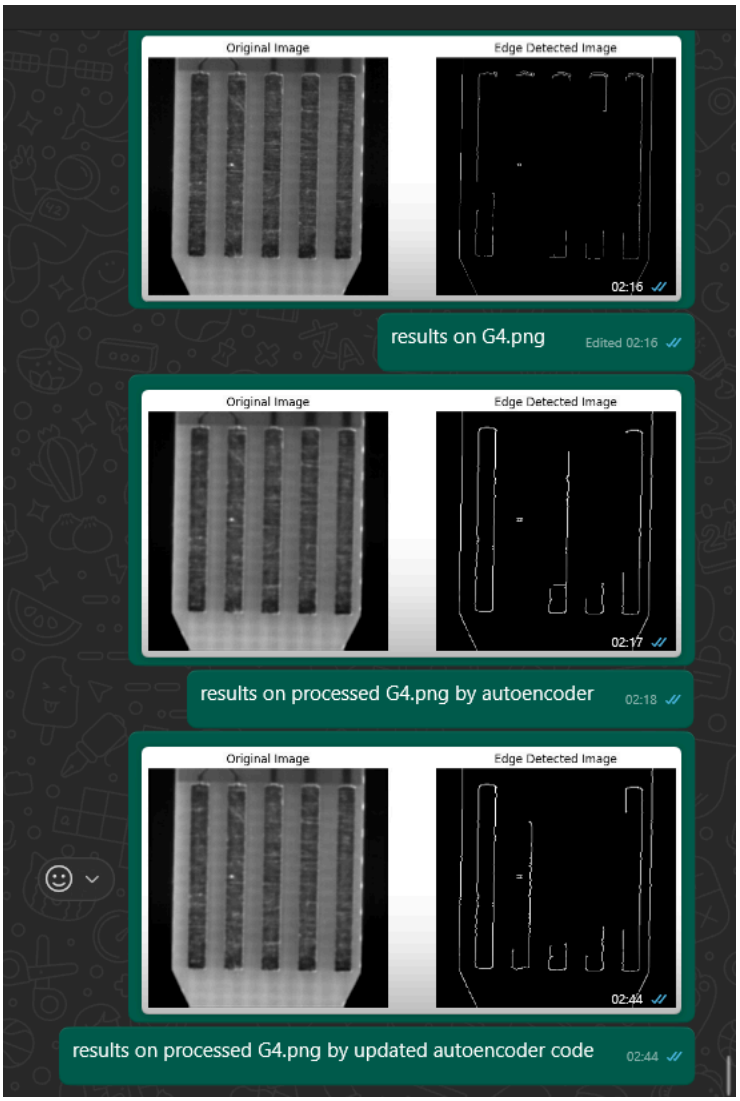
> Hello sir! I completely understand your point. I have received the stipend with thanks.
> Sir, I had shared a drive link with you in the previous email. Could you pls go through those pictures and share your reviews with me.
> Warm regards
> Hindvee
>
> On Wed, Aug 6, 2025 at 8:10 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > I have transferred Rs. 11,600 (200 x 58) to you. Regarding the suggested increase in per-hour rate: You have certainly made efforts to solve the problem. However, as of now the result is not yet usable to us. We will consider an increase when it is usable in our software.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Tuesday, August 5, 2025 8:41 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Hello sir
> > Just following up on my previous email regarding the processed images and the stipend for July (58 hours). I have shared a small suggestion regarding the per-hour rate.

Please let me know if you need any further details or modifications from my side.
UPDATED DEXINED OUTPUT
Looking forward to your feedback.

Warm regards,
Hindvee

On Fri, Aug 1, 2025 at 2:14 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Thank you sir! I have worked out and increased the contrast in the processed images. I will share the file with you. Also sir I shall request you to please process the stipend for this month. It has been 58 hrs for the month of July. If you like my work please consider increasing the per hour rate. It is just a small suggestion. The final decision depends on you and I happily accept your decision. My primary goal is to enhance my skills, and I am learning a lot from this project.
>
> Warm regards
> Hindvee
>
> On Wed, Jul 30, 2025 at 11:02 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > I have checked the images. Your conclusions are correct.
> >
> > Please carry on with your efforts.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Tuesday, July 29, 2025 3:37 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Hello sir!
> > current update:-
> >
> > - Using an autoencoder trained on clean (good) images to repair bad images
> > - Applying a **hybrid loss** of MSE and SSIM to preserve structure and perceptual quality.
> > - Saving and visualizing both original and reconstructed images which helps in debugging
> >
> > Despite this denoising strategy, DexiNED is still failing for bad images, the possible reasons are as follows
> >
> > 1. Contrast remains too low after reconstruction.
> > 2. Edges are too smooth or faint due to over-smoothing by the autoencoder.
> > 3. DexiNED is not tuned for synthetic/industrial patterns (e.g., PCB, x-ray like structures).
> >
> > I will now work on these factors to improve the quality of processed images. I am attaching the processed Good and Bad Images folders.
> > BAD IMAGES _ PROCESSED
> > GOOD IMAGES_ PROCESSED
> >
> > warm regards
> > Hindvee
> >
> > On Mon, Jul 28, 2025 at 2:48 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> >
> > > Noted sir. I wrote code for autoencoder to get sharper images, and I applied DexiNed on the original image and also on the processed images by the 2 autoencoder codes written so far. I am sharing the results here.



warm regards
Hindvee

On Fri, Jul 25, 2025 at 6:34 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Sorry for delay in replying. I was away for quite some time.
>
> A couple of months back we had tried OpenCV / HED and obtained similar results.
>
> Please go ahead and try DexiNed after which we will decide about line detection.
>
> After that another important step would be to remove noise from image using
> a suitable model.
>
> I know this is somewhat away from our goal of 'AI assisted edge detection' but it can be an important intermediate step. We are currently using a model that does this. This is the model I had sent to you initially. It is developed in a Chinese University by a professor who works with our company.
>
> Kishor Durve
> Duravision Systems
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Friday, July 25, 2025 5:46 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir ! This is an output image obtained from the HED model and similar image I have obtained from DexiNed, could u pls tell i am moving in the right direction or should I make improvements or change in the

plan of action?

warm regards
Hindvee

On Wed, Jul 23, 2025 at 3:20 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Hello sir! Could you please share ur insights and views on the above mail thread, it shall be very helpful to get to know your feedback and move forward with the project.

Warm Regards
Hindvee

On Fri, Jul 11, 2025 at 3:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Hello sir!

Following up on my previous email regarding edge detection, I wanted to share the progress I've made since then. I have been focusing on using the Holistically-Nested Edge Detection (HED) method. I have now:

CODE FILE

1. Successfully applied the HED process to a folder containing our images to generate edge maps for the entire dataset.

2. Trained a custom autoencoder model designed to refine and correct the lines in these images.

3. Applied the trained autoencoder to the HED output images to produce corrected versions.

4. Started working on post-processing techniques.

I am currently experimenting with different parameters and methods to achieve the best results for this step. I am continuing to refine the line clarification process and will share updated results once I have finalized the post-processing approach.

warm regards

Hindvee

On Wed, Jul 9, 2025 at 12:23 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Hello sir! I am trying the pre-trained model for detecting edges in our images; I have tried Holistically-Nested Edge Detection (HED) and DexiNed. I am attaching the output, I am going forward with HED to get better results.

Warm regards
Hindvee

On Mon, Jul 7, 2025 at 10:37 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Hello sir!
I am sharing the first 5 difference images of the three codes that I mentioned previously. Please check this whenever you have time.

Warm regards
Hindvee

On Mon, Jul 7, 2025 at 6:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:

I am unable to give my view of the approach as I can't view the images. I can give my opinion if you send me difference images. In any case, I think you need to have difference images as a metric in addition to the Laplacian.

Get Outlook for Android

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Monday, July 7, 2025 7:09:51 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!
I have received the stipend, thank you for that! I'd like to share a brief update on the image enhancement work, focusing on noise removal and edge preservation.
Summary of results:-

1. Previous code
---Performed basic denoising but introduced over-smoothing.
---Laplacian MAD: ~0.0175–0.0192

2. DnCNN
---Improved noise removal and structural retention.
---MAD: ~0.0186–0.0202

3. DnCNN + Sharpening (Current approach)
---Added post-processing to enhance edge clarity.
---Consistently improved MAD to ~0.0178–0.0196.
---Visual results show reduced whitening and better edge definition.

This confirms that the current setup provides better denoising while preserving fine image details. Please let me know if you'd like to proceed with further tuning using SSIM or edge-aware loss functions.

Warm regards

Hindvee

On Fri, Jul 4, 2025 at 3:54 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Please keep at it. Use difference image and Laplacian to gauge the effect of the algorithm. I will be able to look at your work only after 15th.
> I have transferred your stipend by GPay.
>
> Get Outlook for Android
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Friday, July 4, 2025 5:37:35 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir!
> This is what I am been working on. I am still working on getting better edges and less whitish images.
>
> warm regards
> Hindvee
>
> On Thu, Jul 3, 2025 at 10:53 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > UPI ID: hindveevispute@okicici
> >
> > Warm regards
> > Hindvee
> >
> > On Wed, 2 Jul 2025, 03:16 Kishor Durve, <kishor.durve@outlook.com> wrote:
> >
> > > I checked the images, but did not notice substantial difference between the before and after images. You can subtract them (after aligning) and check.
> > > As regards payment of stipend please let me know how much we need to pay and I will transfer it on GPay.
> > > I am travelling again and am out of office. I will be back only on 15th. My responses to your mails may be delayed.
> > >
> > > Get Outlook for Android
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Tuesday, July 1, 2025 1:51:45 AM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection
> > >
> > > Hello sir!
> > >
> > > I was eager to get insights from you about the above shared images.Please do have a look as soon as possible. Also, I wanted to know whether the stipend will be credited at

the end of the month or the project's end. It would be very kind of you if I could get the stipend at the end of the month.

Thanks and regards
Hindvee

On Mon, Jun 30, 2025 at 10:33 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Good morning sir!
>
> Did you get any chance to review the above images? Please share your suggestions and insights with me.
>
> Warm regards
> Hindvee
>
> On Thu, Jun 26, 2025 at 11:09 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Here I am attaching the processed images.
> >
> > Warm regards
> > Hindvee
> >
> > On Thu, Jun 26, 2025 at 8:08 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > I am unable to run the files in Colab and generate images. Can you please send me the processed images on mail - .png / .jpeg files, not .ipynb files.
> > >
> > > Kishor Durve
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Thursday, June 26, 2025 4:48 AM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection
> > >
> > > Hello sir, I have attached the results for png 1,2,3 and 4. Pls share ur views and insights.
> > >
> > > Warm Regards
> > > Hindvee
> > >
> > > On Tue, Jun 24, 2025 at 11:29 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > >
> > > > Hello sir,
> > > > I am attaching the .ipynb file, it has the images
> > > >
> > > > warm regards
> > > > Hindvee
> > > >
> > > > On Tue, Jun 24, 2025 at 5:48 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> > > >
> > > > > Please share the images for visual comparison.
> > > > >
> > > > > Kishor Durve
> > > > >
> > > > > ---
> > > > >
> > > > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > > > **Sent:** Tuesday, June 24, 2025 5:13 AM
> > > > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > > > **Subject:** Re: AI assisted edge detection
> > > > >
> > > > > Hello Durve sir,
> > > > >
> > > > > I wanted to share an update on the denoising output. I noticed earlier that while the model was doing a good job removing noise, it was also making the image a bit too smooth — the sharpness had dropped quite a bit:
> > > > >
> > > > > - Noisy Image: 33.72

- Denoised Image: 6.92
- Clean Image (Ground Truth): 26.49

To improve this, I added a sharpening filter after the autoencoder output and tested two variants:

1. A **strong sharpening kernel**, which raised the denoised sharpness to **112.40,** but it felt a bit overdone and introduced some artifacts.
2. A **softer kernel**, which gave a cleaner and more natural result, with denoised sharpness at **54.04**. This seems like a balanced point where edges are clearer without compromising visual quality.

If you'd like, I can also share the output images (clean vs. denoised) for a visual comparison.

Warm regards

Hindvee

 image.png

On Fri, Jun 20, 2025 at 12:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir!
> Based on the latest code I calculated the sharpness using Laplacian variance. I obtained the following results as those attached as image.
> I think that while denoising, though the noise was removed, but the model blurred or over smoothened the image. I may have lost some meaningful edge details in the process. But the sharpness value of Clean image is lower than the noisy image (because it has no random noise), but high enough to retain clear edges and structure. So I am working next on increasing the sharpness value while keeping the noise low. Please do share your views with me.
>
> Warm regards
> Hindvee
>
>  image.png
>
> On Thu, 19 Jun 2025, 13:42 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > I have seen the processed images. I think it is good at this point to quantify 'sharpness' of an image. It will give us a measure to compare different methods / models.
> >
> > There can be several approaches to do this. We use Opencv::Laplacian() to judge image sharpness. But you can try other approaches.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Thursday, June 19, 2025 2:52 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Hello sir,
> >
> > I've completed the full-image denoising as discussed for the images `5.png` and `6.png`. This time, the model was trained and applied directly on the complete images (cropped to a common size of **570×588**) instead of small patches, as per your suggestion. The denoising autoencoder was able to reduce background noise and enhance structural features,

particularly the edges of the circles, which were initially faint or inconsistent. The model output shows a noticeable improvement in clarity while preserving the original patterns.

I've attached the processed result image along with this mail. Please let me know your thoughts on this file.

Warm regards,
Hindvee

On Wed, Jun 18, 2025 at 12:33 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have checked the file. These two images contain circles as primary features. Also, the patches in the file seem to be for a small part of the original image. If you have processed the original images (5.png and 6.png) please send me the processed images on mail. If you have processed only parts of these images, I suggest you process the full images and send me the results.
>
> Similarly, in the previous exercise please process complete images. Most of the set are of size 500x500 pixel, so it should not consume too much time.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 18, 2025 1:44 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Sir, please let me know your views on the above .ipynb file.
> Warm Regards
> Hindvee
>
> On Wed, Jun 18, 2025 at 1:56 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir,
> > I have processed the 5.png and 6.png, I faced lower accuracy rate earlier. After fixing the earlier issues with mismatched data sizes and high Binary Crossentropy loss (~0.6098), the model is now working smoothly. It reached a much lower loss of just 0.0528 in 5 epochs, and both training and validation losses settled nicely by the 2nd or 3rd epoch without signs of overfitting. The denoised images look very clean and closely match the original ones, even when the input images had different levels of noise. I am attaching the results in the attached file.
> > Warm regards
> > Hindvee
> >
> > On Tue, Jun 17, 2025 at 7:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > Sorry! Forgot the attachment.
> > >
> > > Kishor Durve
> > >
> > > ---
> > >
> > > **From:** Kishor Durve <kishor.durve@outlook.com>
> > > **Sent:** Tuesday, June 17, 2025 9:46 AM
> > > **To:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Subject:** Re: AI assisted edge detection
> > >
> > > Good to know that you are making progress at a nice clip.

I saw the images and the difference between before / after can be seen. Can you please process the images attached to this mail and send me the results? The images are similar to those I had sent earlier, the exception being 5.png and 6.png. In these last two images the challenge is to detect circles whose edges are not well defined and the edge quality may vary from image to image. This is to test the effectiveness of noise removal / sharpening.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, June 17, 2025 3:06 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir,
I have completed the training and evaluation of the denoising autoencoder using grayscale image patches taken from the `IMG` folder. The model was trained for 20 epochs on 64×64 patches with binary cross entropy loss and a 10% validation split. During training, the loss reduced steadily and stabilized, with the final reconstruction loss reaching 0.3688. The results show that the model can effectively denoise images and recover clean patterns like faint lines. In the next step, I plan to use the output of the Super-Resolution model as a base to evaluate the accuracy and quality of the denoising process, to better understand how both methods contribute to image enhancement. I am attaching the .ipynb file for your reference and also the IMG Folder which I just mentioned.

Warm regards
Hindvee

IMG FOLDER

Accuracy results:-

image.png

On Fri, Jun 13, 2025 at 5:33 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> The images I sent are big. OpenCV will not be able to handle them as the max size allowed is 32K x 32K. The crash you are facing could be due to this reason.
>
> You can crop out small patches from the big images and use them for trials.
>
> Kishor Durve
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 11, 2025 1:37 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Thank you for sharing more images. I have updated the work on the sheets.

Warm regards
Hindvee

On Sat, Jun 7, 2025 at 2:51 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have uploaded more images. You can crop out patches from them. One of the images has many boards which will give many useful patches.
>
> Kishor Durve
> Duravision Systems
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Saturday, June 7, 2025 4:36 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings sir,
> It would be great if I get more images, that would expand my dataset. Thank you for your previous email.
> I also wanted tk inform you that I am travelling and will be able to resume work by Tuesday. I request you to kindly allow my leave.
>
> Warm Regards
> Hindvee
>
> On Fri, 6 Jun 2025, 11:31 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > I am not going to be of much help to you as I have limited experience in this field. Also, what little I know was gained by applying available models in OpenCV's DNN module.
> >
> > There are two images available in the images folder. You may crop out patches from them for the Training / test sets. We use ImageJ and IrfanView for such work as MS Paint cannot handle such big images. If you need more images please let me know.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, June 6, 2025 5:24 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings sir,
> > I hope you're doing well. I've just finished training the CNN-based line presence classifier, and I'm attaching the loss graph for your review. The model seems to have learned quite quickly, with both training and validation losses staying very close to zero throughout. There was a small spike in training loss around epoch 80, but validation stayed stable, so it might just be a minor fluctuation.
> >
> > Since the dataset is quite small and the model converged very early, I'm unsure how best to move forward.According to you should I try to get more data to avoid overfitting. Please guide me towards a direction in which I can make

this model more effective.

Warm regards

Hindvee

On Wed, Jun 4, 2025 at 6:42 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Yes, it looks ok to me. We will take a relook once you make more progress.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 4, 2025 4:26 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Additionally, Sir, I would like to confirm if the current format of the Google Sheet meets your expectations. Please let me know if you would prefer any modifications, such as a more detailed breakdown of the work completed.
>
> Warm Regards
> Hindvee
>
> On Wed, Jun 4, 2025 at 1:34 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Greetings Mr. Durve,
> >
> > Thanks for sharing the super-resolution model. I've set it up and tested it, the results inspiring. It does a good job cleaning up noise and improving contrast, especially in the faint line areas. As we discussed, I'll now add this to the bigger pipeline along with the AI modules (NB-1 to NB-5). The super-resolution output can help improve both the CNN classifier and regression models. I'll also check how it works together with NB-5, the denoising autoencoder, to see if we get even better results.
> >
> > Please let me know if you want me to run it on any specific ROI samples or start putting together comparison results.
> >
> > Warm regards
> > Hindvee
> >
> > On Tue, Jun 3, 2025 at 9:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > I have file network.py in folder 'models'. This file has the RRDBNet implementaion.
> > >
> > > Kishor Durve
> > > Duravision Systems
> > > 9823164286
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Monday, June 2, 2025 7:15 PM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection

Greetings Mr. Durve,
I am facing an issue again an again while running the VS 2022 code, pls help me with this, we have the module named as model_zoo not models. Pls tell me how to proceed with this.

 image.png

Warm Regards
Hindvee

On Sat, May 31, 2025 at 12:46 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Thankyou sir! I will try this right away.
> Also I hope you received this google sheet, I have shared edit access to you on your gmail account
>
> Hindvee Vispute - Time data - Intern
>
> Regards
> Hindvee
>
> On Sat, May 31, 2025 at 5:43 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Sure. You need to do the following to make the project work.
> >
> > 1. Download and install VS 2022 Community edition
> > 2. Download and install Anaconda. Install an additional environment Python 3.8.16. Use this version of Python for the project.
> >
> > The project should run at this point.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, May 30, 2025 6:34 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings sir!
> > I went through the google drive folder shared by you. It would be really helpful if you could walk me through it.
> > Best regards
> > Hindvee
> >
> > On Fri, May 30, 2025 at 11:41 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > You can download the model, a working VC++ (VS 2022) project and image here.
> > >
> > > https://drive.google.com/drive/folders/1mLBDjJR3fFKs8Brr_nNj2z9YyI89nruE?usp=sharing
> > >
> > > Kishor Durve
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>

Sent: Friday, May 30, 2025 12:50 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings sir!
Pls do share the model over drive I will try it out. Also I request you to share more image database so that I can start working on it.
I will start work from today itself then.

Regards
Hindvee

On Fri, 30 May 2025, 06:11 Kishor Durve, <kishor.durve@outlook.com> wrote:

> We have found that using a super resolution Pytorch model to get a larger image results in significant noise removal. Perhaps it is a by-product.
>
> You can try it out. I have the model and can share it on Google Drive if you want to try it out. We have tried the models available in OpenCV DNN but they do not seem to give the same results.
>
> Regards,
>
> Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 29, 2025 5:28 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Dear Sir,

Thank you for the confirmation. I will maintain the time data on a Google Sheet and ensure it is updated regularly.

Looking forward to starting from Monday, 2nd June.

Best regards,
Hindvee

On Thu, May 29, 2025 at 2:53 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> Welcome aboard!
>
> You can keep the time data on Google Sheet.

Regards,

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 29, 2025 5:12 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr. Durve,

Thank you very much for giving me the opportunity to contribute to Guangdong Longtian Smart Instruments Co. Ltd. through this exciting project. I am pleased to accept the offer and look forward to working on the development of the AI-assisted line detection tool.

With your permission, I would like to begin the internship from the coming Monday, i.e., 2nd June 2025.

Additionally, I wanted to ask about the preferred method for tracking and sharing the time I spend on the project. Would you like me to maintain a shared timesheet (e.g., a Google Sheet or Excel file), or would you prefer weekly time summaries via email or any other format?

Looking forward to your guidance on this.

Thank you once again for this opportunity.

Warm regards,
Hindvee Rajesh Vispute

On Thu, May 29, 2025 at 8:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have attached the on-boarding letter.
>
> Please let me know if you have any queries.
>
> Regards,
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Saturday, May 10, 2025 3:46 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection

Greetings Mr. Durve,

Thank you for your detailed response and clarification. I now clearly understand that the goal is to enhance line detection using AI within known ROIs, leveraging the expected position of endpoints.

To support this objective, I've explored and implemented several **AI-based approaches**, any of which can be used to complement or improve upon conventional methods:

1. **CNN-Based Line Presence Classifier (NB-1):** Identifies whether a line exists within a given patch/ROI—useful for filtering candidate regions before applying traditional detection.
2. **CNN Regression Model (NB-2):** Predicts line endpoints directly from the image, offering robust localization even in low-contrast or noisy scenarios.
3. **Denoising Autoencoder (NB-5):** Enhances faint or broken lines by cleaning up noise, thereby boosting the effectiveness of existing edge or Hough-based methods.
4. **Vision Transformer Model (NB-4):** Designed for fine-grained classification tasks and can be adapted for future objectives like surface defect or anomaly detection.

These methods are independently deployable and can be integrated selectively depending on the specific inspection challenge or ROI structure. I'd be happy to align with your ROI-driven strategy and customize accordingly.

Regarding the stipend, I agree with the proposed Rs. 200/hour base rate and performance-based bonus, and I'm enthusiastic about the opportunity to collaborate further on defect detection and shape anomaly tasks in the future.

I have attached screenshots of the code files I executed today. I have the names of the notebooks (NB) with NB-1, NB-2,NB-4 and NB-5 respectively.

NB-1
NB-2
NB-4
NB-5

Please let me know if you'd like me to share these outputs or run a quick live demonstration.

Warm regards,
Hindvee

On Fri, May 9, 2025 at 6:03 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> I think there is a mismatch in understanding. We are already using conventional line detection techniques (various edge detection operators and Hough Transform and other methods for line fitment) .
>
> What we are looking to get from you is AI based / AI assisted line detection. Since we know the probable position of the line beforehand we can use this information to find a line. For example, say we know that the line end points are expected to be (x1, y1) and (x2, y2), we can construct a small rectangle (ROI) using these points and search for a line within this ROI.
>
> As for the stipend it depends both on the time you spend on the project and the results you produce. It will comprise of an hourly rate and a result linked amount. I suggest we have an hourly rate of Rs. 200/hour and a matching amount linked to the results at the end of the project. So, if you are able to produce satisfactory results it will come to Rs. 400/hour. Also, if the results are satisfactory, we can take this forward next year and work on other issues like surface defect detection, detection of shape anomalies etc.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Thursday, May 8, 2025 5:15 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings Mr Durve,
> I have the problem now so I will

focus on faint line detection for the first step. I can proceed using Sobel, Scharr, and Canny edge detectors provided by OpenCV. First I will preprocess the images and then move to edge detection using the above 3 detectors. I will carry out line detection using Hough Transform, and further we will filter out short or irrelevant lines based on length and their angle of orientation. Finally, we will draw the detected lines on the images using cv.line()

I can devote 14 hrs per week to this task for 2 months, we can extend the time if required. Also, I wanted to ask what stipend you can provide me. I believe I can complete the task in the above time frame. Thank you for your time.

Warm regards
Hindvee

On Thu, May 8, 2025 at 7:18 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Your analysis seems good.
>
> At this point we are not interested in detecting defects (though at a later stage we may be), and want to detect the faint lines only.
>
> Note that though I have annotated only a single line in the image I sent to you yesterday, we need to detect edge lines of all vertical stripes.
>
> I have attached 2 sets of 20 images each named 'Good Images' and 'Bad Images' - images in which it is mostly easy to detect lines by the conventional image processing method and those in which lines are faint and hard to detect. These images may help you get a better understanding of the problem.
>
> Please also check the following issues:
>
> 1. How much time will you be able to devote to this task?
> 2. Will you be able to complete the task

within the available time?

There are other issues like time taken by the AI route, integration of AI line detect code in VC++ project, further training / fine tuning of model, applicability of the method to detect faint circles etc. But we will get into them later once the basic method is proven.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, May 7, 2025 3:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings, Mr. Durve,
Thank you for reviewing my proposal and for sharing the sample images. By looking at the images I came across the specific challenges faced by the inspection system, like detecting faint lines, subtle surface variations, and fine defects that can be missed by traditional image processing. Based on these observations I suggest that we can do the following:-

1. **Hybrid approach**: Objective - Combine the speed of OpenCV line detection (example Hough Transform) with deep learning's pattern recognition capability.
Approach
- Use OpenCV to pre-locate candidate regions and feed them into a deep model for verification.
- This can speed up inference and increase resistance to false positives.

2. **Deep learning-based fine defect detection**: Objective: Improve the detection of faint, low contrast line defects and localized surface anomalies.
Approach
- Train a model, such as YOLOv8 or U-Net, tailored to your defect patterns using the annotated images.
- To improve feature visibility, add contrast enhancement.
- Assess using detection accuracy and intersection over union (IoU)

to compare to baselines based on OpenCV.

I can begin with a proof-of-concept model using the images shared by you. Thank you for sharing the opportunity. I am excited to get a chance to contribute to the advancement in inspection technology.

Warm regards
Hindvee Vispute

On Wed, 7 May 2025, 09:31 Kishor Durve, <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have gone thru the document uploaded by you.
>
> It looks ok to me.
>
> I can have attached a few images so you will get a better idea of the requirements.
>
> Kishor Durve

Kishor Durve <kishor.durve@outlook.com> Thu, Aug 21, 2025 at 5:31 AM
To: Hindvee Vispute <hindveevispute@gmail.com>

Sorry! I didn't realize you were waiting for my feedback.

I have checked the images. I have a few questions:

1. What are the 'corrected' images? How are they different from the original?
2. How long does it take (in seconds) to process an image with DexiNed? Please use a high res timer and check the time taken. I am aware that it will depend on the image size. Let us consider only the images that you are working with.
3. Is there any way to fine tune DexiNed for our images?

Kishor Durve
Duravision Systems
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, August 21, 2025 1:04 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!
Could you pls share your reviews on the link I shared earlier with you? Your views will determine how to proceed further.

Warm Regards
Hindvee

On Fri, Aug 8, 2025 at 2:36 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir! I completely understand your point. I have received the stipend with thanks.
> Sir, I had shared a drive link with you in the previous email. Could you pls go through those pictures and share your reviews with me.

Warm regards
Hindvee

On Wed, Aug 6, 2025 at 8:10 AM Kishor Durve <kishor.durve@outlook.com> wrote:

I have transferred Rs. 11,600 (200 x 58) to you. Regarding the suggested increase in per-hour rate: You have certainly made efforts to solve the problem. However, as of now the result is not yet usable to us. We will consider an increase when it is usable in our software.

Kishor Durve
Duravision Systems
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, August 5, 2025 8:41 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir
Just following up on my previous email regarding the processed images and the stipend for July (58 hours). I have shared a small suggestion regarding the per-hour rate.

Please let me know if you need any further details or modifications from my side.
UPDATED DEXINED OUTPUT
Looking forward to your feedback.

Warm regards,
Hindvee

On Fri, Aug 1, 2025 at 2:14 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

Thank you sir! I have worked out and increased the contrast in the processed images. I will share the file with you. Also sir I shall request you to please process the stipend for this month. It has been 58 hrs for the month of July. If you like my work please consider increasing the per hour rate. It is just a small suggestion. The final decision depends on you and I happily accept your decision. My primary goal is to enhance my skills, and I am learning a lot from this project.

Warm regards
Hindvee

On Wed, Jul 30, 2025 at 11:02 AM Kishor Durve <kishor.durve@outlook.com> wrote:

I have checked the images. Your conclusions are correct.

Please carry on with your efforts.

Kishor Durve
Duravision Systems
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, July 29, 2025 3:37 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!
current update:-

- Using an autoencoder trained on clean (good) images to repair bad images
- Applying a **hybrid loss** of MSE and SSIM to preserve structure and perceptual quality.
- Saving and visualizing both original and reconstructed images which helps in debugging

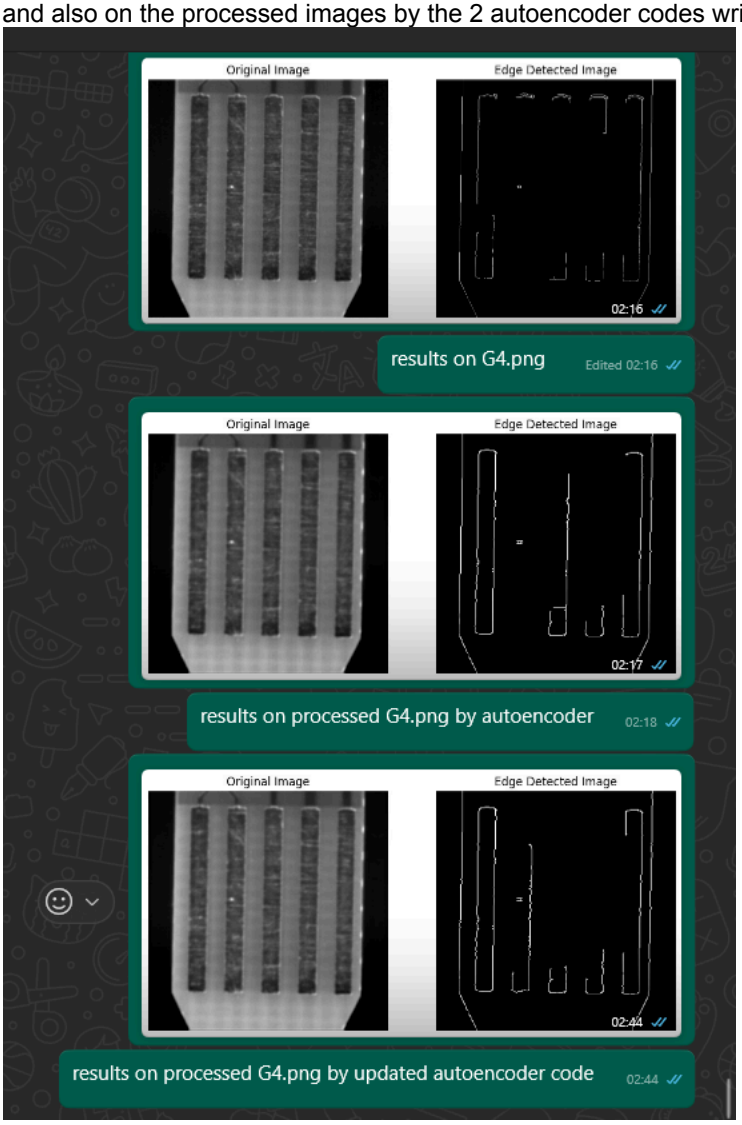
Despite this denoising strategy, DexiNED is still failing for bad images, the possible reasons are as follows

1. Contrast remains too low after reconstruction.
2. Edges are too smooth or faint due to over-smoothing by the autoencoder.
3. DexiNED is not tuned for synthetic/industrial patterns (e.g., PCB, x-ray like structures).

I will now work on these factors to improve the quality of processed images. I am attaching the processed Good and Bad Images folders.
BAD IMAGES _ PROCESSED
GOOD IMAGES_ PROCESSED

warm regards
Hindvee

On Mon, Jul 28, 2025 at 2:48 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Noted sir. I wrote code for autoencoder to get sharper images, and I applied DexiNed on the original image and also on the processed images by the 2 autoencoder codes written so far. I am sharing the results here.
>
> 
>
>
> warm regards
> Hindvee
>
> On Fri, Jul 25, 2025 at 6:34 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Sorry for delay in replying. I was away for quite some time.
> >
> > A couple of months back we had tried OpenCV / HED and obtained similar results.
> >
> > Please go ahead and try DexiNed after which we will decide about line detection.
> >
> > After that another important step would be to remove noise from image using a suitable model.
> >
> > I know this is somewhat away from our goal of 'AI assisted edge detection' but it can be an important intermediate step. We are currently using a model that does this. This is the model I had sent to you initially. It is developed in a Chinese University by a professor who works with our company.
> >
> > Kishor Durve

Duravision Systems
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, July 25, 2025 5:46 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir ! This is an output image obtained from the HED model and similar image I have obtained from DexiNed, could u pls tell i am moving in the right direction or should I make improvements or change in the plan of action?

warm regards
Hindvee

On Wed, Jul 23, 2025 at 3:20 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir! Could you please share ur insights and views on the above mail thread, it shall be very helpful to get to know your feedback and move forward with the project.
>
> Warm Regards
> Hindvee
>
> On Fri, Jul 11, 2025 at 3:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir!
> >
> > Following up on my previous email regarding edge detection, I wanted to share the progress I've made since then. I have been focusing on using the Holistically-Nested Edge Detection (HED) method. I have now:
> >
> > CODE FILE
> >
> > 1. Successfully applied the HED process to a folder containing our images to generate edge maps for the entire dataset.
> >
> > 2. Trained a custom autoencoder model designed to refine and correct the lines in these images.
> >
> > 3. Applied the trained autoencoder to the HED output images to produce corrected versions.
> >
> > 4. Started working on post-processing techniques.
> >
> > I am currently experimenting with different parameters and methods to achieve the best results for this step. I am continuing to refine the line clarification process and will share updated results once I have finalized the post-processing approach.
> >
> > warm regards
> >
> > Hindvee
> >
> > On Wed, Jul 9, 2025 at 12:23 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> >
> > > Hello sir! I am trying the pre-trained model for detecting edges in our images; I have tried Holistically-Nested Edge Detection (HED) and DexiNed. I am attaching the output, I am going forward with HED to get better results.
> > >
> > > Warm regards
> > > Hindvee
> > >
> > > On Mon, Jul 7, 2025 at 10:37 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > >
> > > > Hello sir!
> > > > I am sharing the first 5 difference images of the three codes that I mentioned previously. Please check this whenever you have time.
> > > >
> > > > Warm regards
> > > > Hindvee
> > > >
> > > > On Mon, Jul 7, 2025 at 6:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> > > >
> > > > > I am unable to give my view of the approach as I can't view the images. I can give my opinion if you send me difference images. In any case, I think you need to have difference images as a metric in addition to the Laplacian.
> > > > >
> > > > > Get Outlook for Android

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Monday, July 7, 2025 7:09:51 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!
I have received the stipend, thank you for that! I'd like to share a brief update on the image enhancement work, focusing on noise removal and edge preservation.
Summary of results:-
1. Previous code
---Performed basic denoising but introduced over-smoothing.
---Laplacian MAD: ~0.0175–0.0192

2. DnCNN
---Improved noise removal and structural retention.
---MAD: ~0.0186–0.0202

3. DnCNN + Sharpening (Current approach)
---Added post-processing to enhance edge clarity.
---Consistently improved MAD to ~0.0178–0.0196.
---Visual results show reduced whitening and better edge definition.

This confirms that the current setup provides better denoising while preserving fine image details. Please let me know if you'd like to proceed with further tuning using SSIM or edge-aware loss functions.

Warm regards

Hindvee

On Fri, Jul 4, 2025 at 3:54 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Please keep at it. Use difference image and Laplacian to gauge the effect of the algorithm. I will be able to look at your work only after 15th.
> I have transferred your stipend by GPay.
>
> Get Outlook for Android

> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Friday, July 4, 2025 5:37:35 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir!
> This is what I am been working on. I am still working on getting better edges and less whitish images.
>
> warm regards
> Hindvee
>
> On Thu, Jul 3, 2025 at 10:53 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > UPI ID: hindveevispute@okicici
> >
> > Warm regards
> > Hindvee
> >
> > On Wed, 2 Jul 2025, 03:16 Kishor Durve, <kishor.durve@outlook.com> wrote:
> >
> > > I checked the images, but did not notice substantial difference between the before and after images. You can subtract them (after aligning) and check.
> > > As regards payment of stipend please let me know how much we need to pay and I will transfer it on GPay.
> > > I am travelling again and am out of office. I will be back only on 15th. My responses to your mails may be delayed.
> > >
> > > Get Outlook for Android

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, July 1, 2025 1:51:45 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!

I was eager to get insights from you about the above shared images.Please do have a look as soon as possible. Also, I wanted to know whether the stipend will be credited at the end of the month or the project's end. It would be very kind of you if I could get the stipend at the end of the month.

Thanks and regards
Hindvee

On Mon, Jun 30, 2025 at 10:33 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Good morning sir!
>
> Did you get any chance to review the above images? Please share your suggestions and insights with me.
>
> Warm regards
> Hindvee
>
> On Thu, Jun 26, 2025 at 11:09 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Here I am attaching the processed images.
> >
> > Warm regards
> > Hindvee
> >
> > On Thu, Jun 26, 2025 at 8:08 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > I am unable to run the files in Colab and generate images. Can you please send me the processed images on mail - .png / .jpeg files, not .ipynb files.
> > >
> > > Kishor Durve
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Thursday, June 26, 2025 4:48 AM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection
> > >
> > > Hello sir, I have attached the results for png 1,2,3 and 4. Pls share ur views and insights.
> > >
> > > Warm Regards
> > > Hindvee
> > >
> > > On Tue, Jun 24, 2025 at 11:29 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > >
> > > > Hello sir,
> > > > I am attaching the .ipynb file, it has the images
> > > >
> > > > warm regards
> > > > Hindvee
> > > >
> > > > On Tue, Jun 24, 2025 at 5:48 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> > > >
> > > > > Please share the images for visual comparison.
> > > > >
> > > > > Kishor Durve
> > > > >
> > > > > ---
> > > > >
> > > > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > > > **Sent:** Tuesday, June 24, 2025 5:13 AM

To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello Durve sir,

I wanted to share an update on the denoising output. I noticed earlier that while the model was doing a good job removing noise, it was also making the image a bit too smooth — the sharpness had dropped quite a bit:

- Noisy Image: 33.72
- Denoised Image: 6.92
- Clean Image (Ground Truth): 26.49

To improve this, I added a sharpening filter after the autoencoder output and tested two variants:

1. A **strong sharpening kernel**, which raised the denoised sharpness to **112.40,** but it felt a bit overdone and introduced some artifacts.
2. A **softer kernel**, which gave a cleaner and more natural result, with denoised sharpness at **54.04**. This seems like a balanced point where edges are clearer without compromising visual quality.

If you'd like, I can also share the output images (clean vs. denoised) for a visual comparison.

Warm regards

Hindvee

 image.png

On Fri, Jun 20, 2025 at 12:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir!
> Based on the latest code I calculated the sharpness using Laplacian variance. I obtained the following results as those attached as image.
> I think that while denoising, though the noise was removed, but the model blurred or over smoothened the image. I may have lost some meaningful edge details in the process. But the sharpness value of Clean image is lower than the noisy image (because it has no random noise), but high enough to retain clear edges and structure. So I am working next on increasing the sharpness value while keeping the noise low. Please do share your views with me.
>
> Warm regards
> Hindvee
>
>  image.png
>
> On Thu, 19 Jun 2025, 13:42 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > I have seen the processed images. I think it is good at this point to quantify 'sharpness' of an image. It will give us a measure to compare different methods / models.
> >
> > There can be several approaches to do this. We use Opencv::Laplacian() to judge image sharpness. But you can try other approaches.
> >
> > Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, June 19, 2025 2:52 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir,

I've completed the full-image denoising as discussed for the images `5.png` and `6.png`. This time, the model was trained and applied directly on the complete images (cropped to a common size of **570×588**) instead of small patches, as per your suggestion. The denoising autoencoder was able to reduce background noise and enhance structural features, particularly the edges of the circles, which were initially faint or inconsistent. The model output shows a noticeable improvement in clarity while preserving the original patterns.

I've attached the processed result image along with this mail. Please let me know your thoughts on this file.

Warm regards,
Hindvee

On Wed, Jun 18, 2025 at 12:33 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have checked the file. These two images contain circles as primary features. Also, the patches in the file seem to be for a small part of the original image. If you have processed the original images (5.png and 6.png) please send me the processed images on mail. If you have processed only parts of these images, I suggest you process the full images and send me the results.
>
> Similarly, in the previous exercise please process complete images. Most of the set are of size 500x500 pixel, so it should not consume too much time.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 18, 2025 1:44 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Sir, please let me know your views on the above .ipynb file.
> Warm Regards
> Hindvee
>
> On Wed, Jun 18, 2025 at 1:56 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir,
> > I have processed the 5.png and 6.png, I faced lower accuracy rate earlier. After fixing the earlier issues with mismatched data sizes and high Binary Crossentropy loss (~0.6098), the model is now working smoothly. It reached a much lower loss of just 0.0528 in 5 epochs, and both training and validation losses settled nicely by the 2nd or 3rd epoch without signs of overfitting. The denoised images look very clean and closely match the original ones, even when the input images had different levels of noise. I am attaching the results in the attached file.
> > Warm regards
> > Hindvee
> >
> > On Tue, Jun 17, 2025 at 7:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

Sorry! Forgot the attachment.

Kishor Durve

From: Kishor Durve <kishor.durve@outlook.com>
Sent: Tuesday, June 17, 2025 9:46 AM
To: Hindvee Vispute <hindveevispute@gmail.com>
Subject: Re: AI assisted edge detection

Good to know that you are making progress at a nice clip.

I saw the images and the difference between before / after can be seen. Can you please process the images attached to this mail and send me the results? The images are similar to those I had sent earlier, the exception being 5.png and 6.png. In these last two images the challenge is to detect circles whose edges are not well defined and the edge quality may vary from image to image. This is to test the effectiveness of noise removal / sharpening.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, June 17, 2025 3:06 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir,
I have completed the training and evaluation of the denoising autoencoder using grayscale image patches taken from the `IMG` folder. The model was trained for 20 epochs on 64×64 patches with binary cross entropy loss and a 10% validation split. During training, the loss reduced steadily and stabilized, with the final reconstruction loss reaching 0.3688. The results show that the model can effectively denoise images and recover clean patterns like faint lines. In the next step, I plan to use the output of the Super-Resolution model as a base to evaluate the accuracy and quality of the denoising process, to better understand how both methods contribute to image enhancement. I am attaching the .ipynb file for your reference and also the IMG Folder which I just mentioned.

Warm regards
Hindvee

IMG FOLDER

Accuracy results:-

 image.png

On Fri, Jun 13, 2025 at 5:33 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> The images I sent are big. OpenCV will not be able to handle them as the max size allowed is 32K x 32K. The crash you are facing could be due to this reason.
>
> You can crop out small patches from the big images and use them for trials.

Kishor Durve
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, June 11, 2025 1:37 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Thank you for sharing more images. I have updated the work on the sheets.

Warm regards
Hindvee

On Sat, Jun 7, 2025 at 2:51 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have uploaded more images. You can crop out patches from them. One of the images has many boards which will give many useful patches.
>
> Kishor Durve
> Duravision Systems
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Saturday, June 7, 2025 4:36 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings sir,
> It would be great if I get more images, that would expand my dataset. Thank you for your previous email.
> I also wanted tk inform you that I am travelling and will be able to resume work by Tuesday. I request you to kindly allow my leave.
>
> Warm Regards
> Hindvee
>
> On Fri, 6 Jun 2025, 11:31 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > I am not going to be of much help to you as I have limited experience in this field. Also, what little I know was gained by applying available models in OpenCV's DNN module.
> >
> > There are two images available in the images folder. You may crop out patches from them for the Training / test sets. We use ImageJ and IrfanView for such work as MS Paint cannot handle such big images. If you need more images please let me know.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, June 6, 2025 5:24 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection

Greetings sir,
I hope you're doing well. I've just finished training the CNN-based line presence classifier, and I'm attaching the loss graph for your review. The model seems to have learned quite quickly, with both training and validation losses staying very close to zero throughout. There was a small spike in training loss around epoch 80, but validation stayed stable, so it might just be a minor fluctuation.

Since the dataset is quite small and the model converged very early, I'm unsure how best to move forward.According to you should I try to get more data to avoid overfitting. Please guide me towards a direction in which I can make this model more effective.

Warm regards

Hindvee

On Wed, Jun 4, 2025 at 6:42 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Yes, it looks ok to me. We will take a relook once you make more progress.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 4, 2025 4:26 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Additionally, Sir, I would like to confirm if the current format of the Google Sheet meets your expectations. Please let me know if you would prefer any modifications, such as a more detailed breakdown of the work completed.
>
> Warm Regards
> Hindvee
>
> On Wed, Jun 4, 2025 at 1:34 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Greetings Mr. Durve,
> >
> > Thanks for sharing the super-resolution model. I've set it up and tested it, the results inspiring. It does a good job cleaning up noise and improving contrast, especially in the faint line areas. As we discussed, I'll now add this to the bigger pipeline along with the AI modules (NB-1 to NB-5). The super-resolution output can help improve both the CNN classifier and regression models. I'll also check how it works together with NB-5, the denoising autoencoder, to see if we get even better results.
> >
> > Please let me know if you want me to run it on any specific ROI samples or start putting together comparison results.
> >
> > Warm regards
> > Hindvee
> >
> > On Tue, Jun 3, 2025 at 9:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > I have file network.py in folder 'models'. This file has the RRDBNet implementaion.

Kishor Durve
Duravision Systems
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Monday, June 2, 2025 7:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr. Durve,
I am facing an issue again an again while running the VS 2022 code, pls help me with this, we have the module named as model_zoo not models. Pls tell me how to proceed with this.



Warm Regards
Hindvee

On Sat, May 31, 2025 at 12:46 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Thankyou sir! I will try this right away.
> Also I hope you received this google sheet, I have shared edit access to you on your gmail account
>
> Hindvee Vispute - Time data - Intern
>
> Regards
> Hindvee
>
> On Sat, May 31, 2025 at 5:43 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Sure. You need to do the following to make the project work.
> >
> > 1. Download and install VS 2022 Community edition
> > 2. Download and install Anaconda. Install an additional environment Python 3.8.16. Use this version of Python for the project.
> >
> > The project should run at this point.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, May 30, 2025 6:34 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings sir!
> > I went through the google drive folder shared by you. It would be really helpful if you could walk me through it.
> > Best regards
> > Hindvee
> >
> > On Fri, May 30, 2025 at 11:41 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > You can download the model, a working VC++ (VS 2022) project and

image here.

https://drive.google.com/drive/folders/1mLBDjJR3fFKs8Brr_nNj2z9YyI89nruE?usp=sharing

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, May 30, 2025 12:50 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings sir!
Pls do share the model over drive I will try it out. Also I request you to share more image database so that I can start working on it.
I will start work from today itself then.

Regards
Hindvee

On Fri, 30 May 2025, 06:11 Kishor Durve, <kishor.durve@outlook.com> wrote:

> We have found that using a super resolution Pytorch model to get a larger image results in significant noise removal. Perhaps it is a by-product.
>
> You can try it out. I have the model and can share it on Google Drive if you want to try it out. We have tried the models available in OpenCV DNN but they do not seem to give the same results.
>
> Regards,
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Thursday, May 29, 2025 5:28 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Dear Sir,
>
> Thank you for the confirmation. I will maintain the time data on a Google Sheet and ensure it is updated regularly.
>
> Looking forward to starting from Monday, 2nd June.
>
> Best regards,
> Hindvee

On Thu, May 29, 2025 at 2:53 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> Welcome aboard!
>
> You can keep the time data on Google Sheet.
>
> Regards,
>
> Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 29, 2025 5:12 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr. Durve,

Thank you very much for giving me the opportunity to contribute to Guangdong Longtian Smart Instruments Co. Ltd. through this exciting project. I am pleased to accept the offer and look forward to working on the development of the AI-assisted line detection tool.

With your permission, I would like to begin the internship from the coming Monday, i.e., 2nd June 2025.

Additionally, I wanted to ask about the preferred method for tracking and sharing the time I spend on the project. Would you like me to maintain a shared timesheet (e.g., a Google Sheet or Excel file), or would you prefer weekly time summaries via email or any other format?

Looking forward to your guidance on this.

Thank you once again for this opportunity.

Warm regards,
Hindvee Rajesh Vispute

On Thu, May 29, 2025 at 8:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have attached the on-boarding letter.
>
> Please let me know if you have any queries.
>
> Regards,
>
> Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Saturday, May 10, 2025 3:46 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr. Durve,

Thank you for your detailed response and clarification. I now clearly understand that the goal is to enhance line detection using AI within known ROIs, leveraging the expected position of endpoints.

To support this objective, I've explored and implemented several **AI-based approaches**, any of which can be used to complement or improve upon conventional methods:

1. **CNN-Based Line Presence Classifier (NB-1):** Identifies whether a line exists within a given patch/ROI—useful for filtering candidate regions before applying traditional detection.
2. **CNN Regression Model (NB-2):** Predicts line endpoints directly from the image, offering robust localization even in low-contrast or noisy scenarios.
3. **Denoising Autoencoder (NB-5):** Enhances faint or broken lines by cleaning up noise, thereby boosting the effectiveness of existing edge or Hough-based methods.
4. **Vision Transformer Model (NB-4):** Designed for fine-grained classification tasks and can be adapted for future objectives like surface defect or anomaly detection.

These methods are independently deployable and can be integrated selectively depending on the specific inspection challenge or ROI structure. I'd be happy to align with your ROI-driven strategy and customize accordingly.

Regarding the stipend, I agree with the proposed Rs. 200/hour base rate and performance-based bonus, and I'm enthusiastic about the opportunity to collaborate further on defect detection and shape anomaly tasks in the future.

I have attached screenshots of the code files I executed today. I have the

names of the notebooks (NB) with NB-1, NB-2,NB-4 and NB-5 respectively.

NB-1
NB-2
NB-4
NB-5

Please let me know if you'd like me to share these outputs or run a quick live demonstration.

Warm regards,
Hindvee

On Fri, May 9, 2025 at 6:03 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> I think there is a mismatch in understanding. We are already using conventional line detection techniques (various edge detection operators and Hough Transform and other methods for line fitment) .
>
> What we are looking to get from you is AI based / AI assisted line detection. Since we know the probable position of the line beforehand we can use this information to find a line. For example, say we know that the line end points are expected to be (x1, y1) and (x2, y2), we can construct a small rectangle (ROI) using these points and search for a line within this ROI.
>
> As for the stipend it depends both on the time you spend on the project and the results you produce. It will comprise of an hourly rate and a result linked amount. I suggest we have an hourly rate of Rs. 200/hour and a matching amount linked to the results at the end of the project. So, if you are able to produce satisfactory results it will come to Rs. 400/hour. Also, if the results are satisfactory, we can take this forward next year and work on other issues like surface defect detection, detection of shape anomalies etc.
>
> Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 8, 2025 5:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr Durve,
I have the problem now so I will focus on faint line detection for the first step. I can proceed using Sobel, Scharr, and Canny edge detectors provided by OpenCV. First I will preprocess the images and then move to edge detection using the above 3 detectors. I will carry out line detection using Hough Transform, and further we will filter out short or irrelevant lines based on length and their angle of orientation. Finally, we will draw the detected lines on the images using cv.line()

I can devote 14 hrs per week to this task for 2 months, we can extend the time if required. Also, I wanted to ask what stipend you can provide me. I believe I can complete the task in the above time frame. Thank you for your time.

Warm regards
Hindvee

On Thu, May 8, 2025 at 7:18 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Your analysis seems good.
>
> At this point we are not interested in detecting defects (though at a later stage we may be), and want to detect the faint lines only.
>
> Note that though I have annotated only a single line in the image I sent to you yesterday, we need to detect edge lines of all vertical stripes.
>
> I have attached 2 sets of 20 images each named 'Good Images' and 'Bad Images' - images in which it is mostly easy to detect lines by the conventional image processing method and those in which lines are faint and hard to detect. These images may help you get a

better understanding of the problem.

Please also check the following issues:

1. How much time will you be able to devote to this task?
2. Will you be able to complete the task within the available time?

There are other issues like time taken by the AI route, integration of AI line detect code in VC++ project, further training / fine tuning of model, applicability of the method to detect faint circles etc. But we will get into them later once the basic method is proven.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, May 7, 2025 3:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings, Mr. Durve,
Thank you for reviewing my proposal and for sharing the sample images. By looking at the images I came across the specific challenges faced by the inspection system, like detecting faint lines, subtle surface variations, and fine defects that can be missed by traditional image processing. Based on these observations I suggest that we can do the following:-

1. **Hybrid approach**: Objective - Combine the speed of OpenCV line detection (example Hough Transform) with deep learning's pattern recognition capability.
Approach
- Use OpenCV to pre-locate candidate regions and feed them into a deep model for verification.
- This can speed up inference and increase resistance to false positives.

2. **Deep learning-based fine defect detection**: Objective: Improve the detection of faint, low

contrast line defects and localized surface anomalies.
Approach
- Train a model, such as YOLOv8 or U-Net, tailored to your defect patterns using the annotated images.
- To improve feature visibility, add contrast enhancement.
- Assess using detection accuracy and intersection over union (IoU) to compare to baselines based on OpenCV.

I can begin with a proof-of-concept model using the images shared by you. Thank you for sharing the opportunity. I am excited to get a chance to contribute to the advancement in inspection technology.

Warm regards
Hindvee Vispute

On Wed, 7 May 2025, 09:31 Kishor Durve, <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have gone thru the document uploaded by you.
>
> It looks ok to me.
>
> I can have attached a few images so you will get a better idea of the requirements.
>
> Kishor Durve

Hindvee Vispute <hindveevispute@gmail.com> Mon, Sep 15, 2025 at 11:17 AM
To: Kishor Durve <kishor.durve@outlook.com>

Hello sir, Sorry for the late reply. I was caught up in placement preparations and could not get back to you. I have worked for 31 hours last month. I will share the updates in the the next mail. I need one help from you sir. I am mentioning this internship in my placement cv, but they need an onboarding letter with you signature/company stamp. I have to show the proof in this week so pls can you help me at the earliest.

Warm regards
Hindvee

On Thu, Aug 21, 2025 at 5:31 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Sorry! I didn't realize you were waiting for my feedback.
>
> I have checked the images. I have a few questions:
>
> 1. What are the 'corrected' images? How are they different from the original?
> 2. How long does it take (in seconds) to process an image with DexiNed? Please use a high res timer and check the time taken. I am aware that it will depend on the image size. Let us consider only the images that you are working with.
> 3. Is there any way to fine tune DexiNed for our images?

Kishor Durve
Duravision Systems
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, August 21, 2025 1:04 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!
Could you pls share your reviews on the link I shared earlier with you? Your views will determine how to proceed further.

Warm Regards
Hindvee

On Fri, Aug 8, 2025 at 2:36 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir! I completely understand your point. I have received the stipend with thanks.
> Sir, I had shared a drive link with you in the previous email. Could you pls go through those pictures and share your reviews with me.
> Warm regards
> Hindvee
>
> On Wed, Aug 6, 2025 at 8:10 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > I have transferred Rs. 11,600 (200 x 58) to you. Regarding the suggested increase in per-hour rate: You have certainly made efforts to solve the problem. However, as of now the result is not yet usable to us. We will consider an increase when it is usable in our software.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Tuesday, August 5, 2025 8:41 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Hello sir
> > Just following up on my previous email regarding the processed images and the stipend for July (58 hours). I have shared a small suggestion regarding the per-hour rate.
> >
> > Please let me know if you need any further details or modifications from my side.
> > UPDATED DEXINED OUTPUT
> > Looking forward to your feedback.
> >
> > Warm regards,
> > Hindvee
> >
> > On Fri, Aug 1, 2025 at 2:14 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> >
> > > Thank you sir! I have worked out and increased the contrast in the processed images. I will share the file with you.
> > > Also sir I shall request you to please process the stipend for this month. It has been 58 hrs for the month of July. If you like my work please consider increasing the per hour rate. It is just a small suggestion. The final decision depends on you and I happily accept your decision. My primary goal is to enhance my skills, and I am learning a lot from this project.
> > >
> > > Warm regards
> > > Hindvee
> > >
> > > On Wed, Jul 30, 2025 at 11:02 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> > >
> > > > I have checked the images. Your conclusions are correct.
> > > >
> > > > Please carry on with your efforts.
> > > >
> > > > Kishor Durve
> > > > Duravision Systems
> > > > 9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, July 29, 2025 3:37 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!
current update:-

- Using an autoencoder trained on clean (good) images to repair bad images
- Applying a **hybrid loss** of MSE and SSIM to preserve structure and perceptual quality.
- Saving and visualizing both original and reconstructed images which helps in debugging

Despite this denoising strategy, DexiNED is still failing for bad images, the possible reasons are as follows

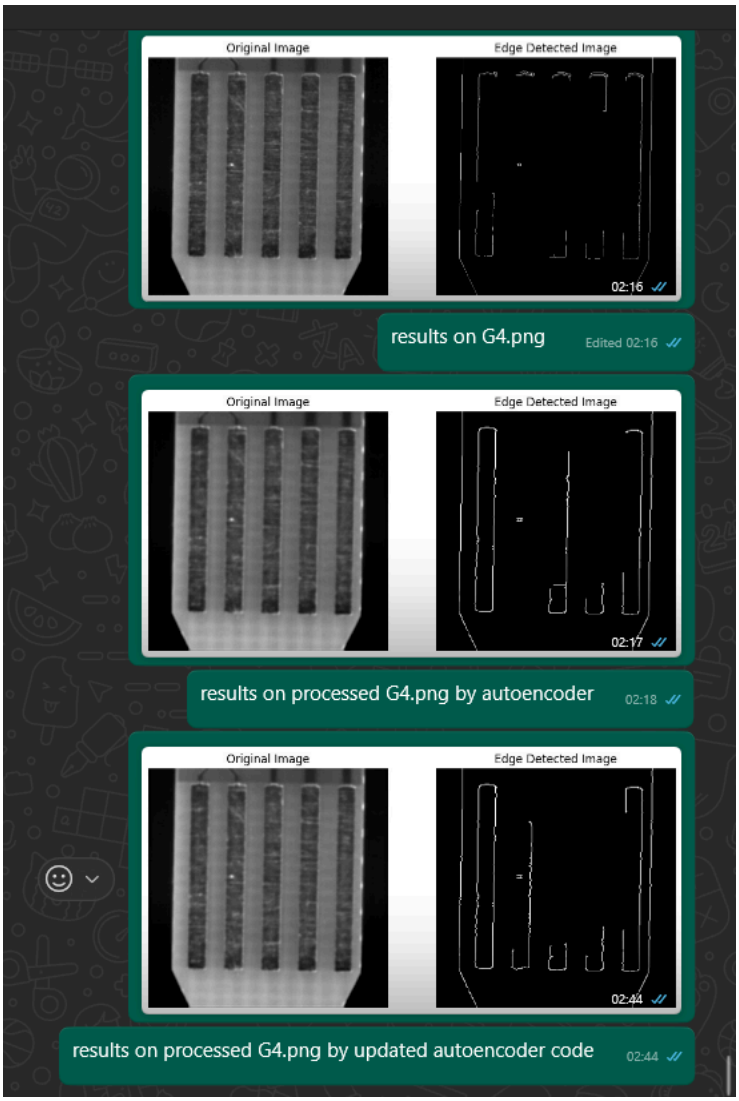
1. Contrast remains too low after reconstruction.
2. Edges are too smooth or faint due to over-smoothing by the autoencoder.
3. DexiNED is not tuned for synthetic/industrial patterns (e.g., PCB, x-ray like structures).

I will now work on these factors to improve the quality of processed images. I am attaching the processed Good and Bad Images folders.
BAD IMAGES _ PROCESSED
GOOD IMAGES_ PROCESSED

warm regards
Hindvee

On Mon, Jul 28, 2025 at 2:48 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Noted sir. I wrote code for autoencoder to get sharper images, and I applied DexiNed on the original image and also on the processed images by the 2 autoencoder codes written so far. I am sharing the results here.



warm regards
Hindvee

On Fri, Jul 25, 2025 at 6:34 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Sorry for delay in replying. I was away for quite some time.
>
> A couple of months back we had tried OpenCV / HED and obtained similar results.
>
> Please go ahead and try DexiNed after which we will decide about line detection.
>
> After that another important step would be to remove noise from image using
> a suitable model.
>
> I know this is somewhat away from our goal of 'AI assisted edge detection' but it can be an important intermediate step. We are currently using a model that does this. This is the model I had sent to you initially. It is developed in a Chinese University by a professor who works with our company.
>
> Kishor Durve
> Duravision Systems
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Friday, July 25, 2025 5:46 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir ! This is an output image obtained from the HED model and similar image I have obtained from DexiNed, could u pls tell i am moving in the right direction or should I make improvements or change in the

plan of action?

warm regards
Hindvee

On Wed, Jul 23, 2025 at 3:20 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir! Could you please share ur insights and views on the above mail thread, it shall be very helpful to get to know your feedback and move forward with the project.
>
> Warm Regards
> Hindvee
>
> On Fri, Jul 11, 2025 at 3:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir!
> >
> > Following up on my previous email regarding edge detection, I wanted to share the progress I've made since then. I have been focusing on using the Holistically-Nested Edge Detection (HED) method. I have now:
> >
> > CODE FILE
> >
> > 1. Successfully applied the HED process to a folder containing our images to generate edge maps for the entire dataset.
> >
> > 2. Trained a custom autoencoder model designed to refine and correct the lines in these images.
> >
> > 3. Applied the trained autoencoder to the HED output images to produce corrected versions.
> >
> > 4. Started working on post-processing techniques.
> >
> > I am currently experimenting with different parameters and methods to achieve the best results for this step. I am continuing to refine the line clarification process and will share updated results once I have finalized the post-processing approach.
> >
> > warm regards
> >
> > Hindvee
> >
> > On Wed, Jul 9, 2025 at 12:23 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> >
> > > Hello sir! I am trying the pre-trained model for detecting edges in our images; I have tried Holistically-Nested Edge Detection (HED) and DexiNed. I am attaching the output, I am going forward with HED to get better results.
> > >
> > > Warm regards
> > > Hindvee
> > >
> > > On Mon, Jul 7, 2025 at 10:37 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > >
> > > > Hello sir!
> > > > I am sharing the first 5 difference images of the three codes that I mentioned previously. Please check this whenever you have time.
> > > >
> > > > Warm regards
> > > > Hindvee
> > > >
> > > > On Mon, Jul 7, 2025 at 6:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> > > >
> > > > > I am unable to give my view of the approach as I can't view the images. I can give my opinion if you send me difference images. In any case, I think you need to have difference images as a metric in addition to the Laplacian.
> > > > >
> > > > > Get Outlook for Android
> > > > >
> > > > > ---
> > > > >
> > > > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > > > **Sent:** Monday, July 7, 2025 7:09:51 AM
> > > > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > > > **Subject:** Re: AI assisted edge detection
> > > > >
> > > > > Hello sir!
> > > > > I have received the stipend, thank you for that! I'd like to share a brief update on the image enhancement work, focusing on noise removal and edge preservation.
> > > > > Summary of results:-

1. Previous code
---Performed basic denoising but introduced over-smoothing.
---Laplacian MAD: ~0.0175–0.0192

2. DnCNN
---Improved noise removal and structural retention.
---MAD: ~0.0186–0.0202

3. DnCNN + Sharpening (Current approach)
---Added post-processing to enhance edge clarity.
---Consistently improved MAD to ~0.0178–0.0196.
---Visual results show reduced whitening and better edge definition.

This confirms that the current setup provides better denoising while preserving fine image details. Please let me know if you'd like to proceed with further tuning using SSIM or edge-aware loss functions.

Warm regards

Hindvee

On Fri, Jul 4, 2025 at 3:54 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Please keep at it. Use difference image and Laplacian to gauge the effect of the algorithm. I will be able to look at your work only after 15th.
> I have transferred your stipend by GPay.
>
> Get Outlook for Android
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Friday, July 4, 2025 5:37:35 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir!
> This is what I am been working on. I am still working on getting better edges and less whitish images.
>
> warm regards
> Hindvee
>
> On Thu, Jul 3, 2025 at 10:53 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > UPI ID: hindveevispute@okicici
> >
> > Warm regards
> > Hindvee
> >
> > On Wed, 2 Jul 2025, 03:16 Kishor Durve, <kishor.durve@outlook.com> wrote:
> >
> > > I checked the images, but did not notice substantial difference between the before and after images. You can subtract them (after aligning) and check.
> > > As regards payment of stipend please let me know how much we need to pay and I will transfer it on GPay.
> > > I am travelling again and am out of office. I will be back only on 15th. My responses to your mails may be delayed.
> > >
> > > Get Outlook for Android
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Tuesday, July 1, 2025 1:51:45 AM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection
> > >
> > > Hello sir!
> > >
> > > I was eager to get insights from you about the above shared images.Please do have a look as soon as possible. Also, I wanted to know whether the stipend will be credited at

the end of the month or the project's end. It would be very kind of you if I could get the stipend at the end of the month.

Thanks and regards
Hindvee

On Mon, Jun 30, 2025 at 10:33 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Good morning sir!
>
> Did you get any chance to review the above images? Please share your suggestions and insights with me.
>
> Warm regards
> Hindvee
>
> On Thu, Jun 26, 2025 at 11:09 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Here I am attaching the processed images.
> >
> > Warm regards
> > Hindvee
> >
> > On Thu, Jun 26, 2025 at 8:08 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > I am unable to run the files in Colab and generate images. Can you please send me the processed images on mail - .png / .jpeg files, not .ipynb files.
> > >
> > > Kishor Durve
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Thursday, June 26, 2025 4:48 AM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection
> > >
> > > Hello sir, I have attached the results for png 1,2,3 and 4. Pls share ur views and insights.
> > >
> > > Warm Regards
> > > Hindvee
> > >
> > > On Tue, Jun 24, 2025 at 11:29 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > >
> > > > Hello sir,
> > > > I am attaching the .ipynb file, it has the images
> > > >
> > > > warm regards
> > > > Hindvee
> > > >
> > > > On Tue, Jun 24, 2025 at 5:48 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> > > >
> > > > > Please share the images for visual comparison.
> > > > >
> > > > > Kishor Durve
> > > > >
> > > > > ---
> > > > >
> > > > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > > > **Sent:** Tuesday, June 24, 2025 5:13 AM
> > > > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > > > **Subject:** Re: AI assisted edge detection
> > > > >
> > > > > Hello Durve sir,
> > > > >
> > > > > I wanted to share an update on the denoising output. I noticed earlier that while the model was doing a good job removing noise, it was also making the image a bit too smooth — the sharpness had dropped quite a bit:

- Noisy Image: 33.72
- Denoised Image: 6.92
- Clean Image (Ground Truth): 26.49

To improve this, I added a sharpening filter after the autoencoder output and tested two variants:

1. A **strong sharpening kernel**, which raised the denoised sharpness to **112.40,** but it felt a bit overdone and introduced some artifacts.
2. A **softer kernel**, which gave a cleaner and more natural result, with denoised sharpness at **54.04**. This seems like a balanced point where edges are clearer without compromising visual quality.

If you'd like, I can also share the output images (clean vs. denoised) for a visual comparison.

Warm regards

Hindvee



On Fri, Jun 20, 2025 at 12:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir!
> Based on the latest code I calculated the sharpness using Laplacian variance. I obtained the following results as those attached as image.
> I think that while denoising, though the noise was removed, but the model blurred or over smoothened the image. I may have lost some meaningful edge details in the process. But the sharpness value of Clean image is lower than the noisy image (because it has no random noise), but high enough to retain clear edges and structure. So I am working next on increasing the sharpness value while keeping the noise low. Please do share your views with me.
>
> Warm regards
> Hindvee
>
> 
>
> On Thu, 19 Jun 2025, 13:42 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > I have seen the processed images. I think it is good at this point to quantify 'sharpness' of an image. It will give us a measure to compare different methods / models.
> >
> > There can be several approaches to do this. We use Opencv::Laplacian() to judge image sharpness. But you can try other approaches.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Thursday, June 19, 2025 2:52 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Hello sir,
> >
> > I've completed the full-image denoising as discussed for the images `5.png` and `6.png`. This time, the model was trained and applied directly on the complete images (cropped to a common size of **570×588**) instead

of small patches, as per your suggestion. The denoising autoencoder was able to reduce background noise and enhance structural features, particularly the edges of the circles, which were initially faint or inconsistent. The model output shows a noticeable improvement in clarity while preserving the original patterns.

I've attached the processed result image along with this mail. Please let me know your thoughts on this file.

Warm regards,
Hindvee

On Wed, Jun 18, 2025 at 12:33 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have checked the file. These two images contain circles as primary features. Also, the patches in the file seem to be for a small part of the original image. If you have processed the original images (5.png and 6.png) please send me the processed images on mail. If you have processed only parts of these images, I suggest you process the full images and send me the results.
>
> Similarly, in the previous exercise please process complete images. Most of the set are of size 500x500 pixel, so it should not consume too much time.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 18, 2025 1:44 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Sir, please let me know your views on the above .ipynb file.
> Warm Regards
> Hindvee
>
> On Wed, Jun 18, 2025 at 1:56 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir,
> > I have processed the 5.png and 6.png, I faced lower accuracy rate earlier. After fixing the earlier issues with mismatched data sizes and high Binary Crossentropy loss (~0.6098), the model is now working smoothly. It reached a much lower loss of just 0.0528 in 5 epochs, and both training and validation losses settled nicely by the 2nd or 3rd epoch without signs of overfitting. The denoised images look very clean and closely match the original ones, even when the input images had different levels of noise. I am attaching the results in the attached file.
> > Warm regards
> > Hindvee
> >
> > On Tue, Jun 17, 2025 at 7:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > Sorry! Forgot the attachment.
> > >
> > > Kishor Durve
> > >
> > > ---
> > >
> > > **From:** Kishor Durve <kishor.durve@outlook.com>
> > > **Sent:** Tuesday, June 17, 2025 9:46 AM
> > > **To:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Subject:** Re: AI assisted edge detection

Good to know that you are making progress at a nice clip.

I saw the images and the difference between before / after can be seen. Can you please process the images attached to this mail and send me the results? The images are similar to those I had sent earlier, the exception being 5.png and 6.png. In these last two images the challenge is to detect circles whose edges are not well defined and the edge quality may vary from image to image. This is to test the effectiveness of noise removal / sharpening.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, June 17, 2025 3:06 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir,
I have completed the training and evaluation of the denoising autoencoder using grayscale image patches taken from the `IMG` folder. The model was trained for 20 epochs on 64×64 patches with binary cross entropy loss and a 10% validation split. During training, the loss reduced steadily and stabilized, with the final reconstruction loss reaching 0.3688. The results show that the model can effectively denoise images and recover clean patterns like faint lines. In the next step, I plan to use the output of the Super-Resolution model as a base to evaluate the accuracy and quality of the denoising process, to better understand how both methods contribute to image enhancement. I am attaching the .ipynb file for your reference and also the IMG Folder which I just mentioned.

Warm regards
Hindvee

IMG FOLDER

Accuracy results:-

 image.png

On Fri, Jun 13, 2025 at 5:33 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> The images I sent are big. OpenCV will not be able to handle them as the max size allowed is 32K x 32K. The crash you are facing could be due to this reason.
>
> You can crop out small patches from the big images and use them for trials.
>
> Kishor Durve
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 11, 2025 1:37 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection

Thank you for sharing more images. I have updated the work on the sheets.

Warm regards
Hindvee

On Sat, Jun 7, 2025 at 2:51 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have uploaded more images. You can crop out patches from them. One of the images has many boards which will give many useful patches.
>
> Kishor Durve
> Duravision Systems
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Saturday, June 7, 2025 4:36 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings sir,
> It would be great if I get more images, that would expand my dataset. Thank you for your previous email.
> I also wanted tk inform you that I am travelling and will be able to resume work by Tuesday. I request you to kindly allow my leave.
>
> Warm Regards
> Hindvee
>
> On Fri, 6 Jun 2025, 11:31 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > I am not going to be of much help to you as I have limited experience in this field. Also, what little I know was gained by applying available models in OpenCV's DNN module.
> >
> > There are two images available in the images folder. You may crop out patches from them for the Training / test sets. We use ImageJ and IrfanView for such work as MS Paint cannot handle such big images. If you need more images please let me know.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, June 6, 2025 5:24 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings sir,
> > I hope you're doing well. I've just finished training the CNN-based line presence classifier, and I'm attaching the loss graph for your review. The model seems to have learned quite quickly, with both training and validation losses staying very close to zero throughout. There was a small spike in training loss around epoch 80, but validation stayed stable, so it might just be a minor fluctuation.

Since the dataset is quite small and the model converged very early, I'm unsure how best to move forward.According to you should I try to get more data to avoid overfitting. Please guide me towards a direction in which I can make this model more effective.

Warm regards

Hindvee

On Wed, Jun 4, 2025 at 6:42 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Yes, it looks ok to me. We will take a relook once you make more progress.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 4, 2025 4:26 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Additionally, Sir, I would like to confirm if the current format of the Google Sheet meets your expectations. Please let me know if you would prefer any modifications, such as a more detailed breakdown of the work completed.
>
> Warm Regards
> Hindvee
>
> On Wed, Jun 4, 2025 at 1:34 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Greetings Mr. Durve,
> >
> > Thanks for sharing the super-resolution model. I've set it up and tested it, the results inspiring. It does a good job cleaning up noise and improving contrast, especially in the faint line areas. As we discussed, I'll now add this to the bigger pipeline along with the AI modules (NB-1 to NB-5). The super-resolution output can help improve both the CNN classifier and regression models. I'll also check how it works together with NB-5, the denoising autoencoder, to see if we get even better results.
> >
> > Please let me know if you want me to run it on any specific ROI samples or start putting together comparison results.
> >
> > Warm regards
> > Hindvee
> >
> > On Tue, Jun 3, 2025 at 9:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > I have file network.py in folder 'models'. This file has the RRDBNet implementaion.
> > >
> > > Kishor Durve
> > > Duravision Systems
> > > 9823164286
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Monday, June 2, 2025 7:15 PM

To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr. Durve,
I am facing an issue again an again while running the VS 2022 code, pls help me with this, we have the module named as model_zoo not models. Pls tell me how to proceed with this.

 image.png

Warm Regards
Hindvee

On Sat, May 31, 2025 at 12:46 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Thankyou sir! I will try this right away.
> Also I hope you received this google sheet, I have shared edit access to you on your gmail account
>
> Hindvee Vispute - Time data - Intern
>
> Regards
> Hindvee
>
> On Sat, May 31, 2025 at 5:43 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Sure. You need to do the following to make the project work.
> >
> > 1. Download and install VS 2022 Community edition
> > 2. Download and install Anaconda. Install an additional environment Python 3.8.16. Use this version of Python for the project.
> >
> > The project should run at this point.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, May 30, 2025 6:34 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings sir!
> > I went through the google drive folder shared by you. It would be really helpful if you could walk me through it.
> > Best regards
> > Hindvee
> >
> > On Fri, May 30, 2025 at 11:41 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > You can download the model, a working VC++ (VS 2022) project and image here.
> > >
> > > https://drive.google.com/drive/folders/1mLBDjJR3fFKs8Brr_nNj2z9YyI89nruE?usp=sharing

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, May 30, 2025 12:50 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings sir!
Pls do share the model over drive I will try it out. Also I request you to share more image database so that I can start working on it.
I will start work from today itself then.

Regards
Hindvee

On Fri, 30 May 2025, 06:11 Kishor Durve, <kishor.durve@outlook.com> wrote:

> We have found that using a super resolution Pytorch model to get a larger image results in significant noise removal. Perhaps it is a by-product.
>
> You can try it out. I have the model and can share it on Google Drive if you want to try it out. We have tried the models available in OpenCV DNN but they do not seem to give the same results.
>
> Regards,
>
> Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 29, 2025 5:28 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Dear Sir,

Thank you for the confirmation. I will maintain the time data on a Google Sheet and ensure it is updated regularly.

Looking forward to starting from Monday, 2nd June.

Best regards,
Hindvee

On Thu, May 29, 2025 at 2:53 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> Welcome aboard!

You can keep the time data on Google Sheet.

Regards,

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 29, 2025 5:12 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr. Durve,

Thank you very much for giving me the opportunity to contribute to Guangdong Longtian Smart Instruments Co. Ltd. through this exciting project. I am pleased to accept the offer and look forward to working on the development of the AI-assisted line detection tool.

With your permission, I would like to begin the internship from the coming Monday, i.e., 2nd June 2025.

Additionally, I wanted to ask about the preferred method for tracking and sharing the time I spend on the project. Would you like me to maintain a shared timesheet (e.g., a Google Sheet or Excel file), or would you prefer weekly time summaries via email or any other format?

Looking forward to your guidance on this.

Thank you once again for this opportunity.

Warm regards,
Hindvee Rajesh Vispute

On Thu, May 29, 2025 at 8:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have attached the on-boarding letter.
>
> Please let me know if you have any queries.
>
> Regards,
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Saturday, May 10, 2025

3:46 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr. Durve,

Thank you for your detailed response and clarification. I now clearly understand that the goal is to enhance line detection using AI within known ROIs, leveraging the expected position of endpoints.

To support this objective, I've explored and implemented several **AI-based approaches**, any of which can be used to complement or improve upon conventional methods:

1. **CNN-Based Line Presence Classifier (NB-1):** Identifies whether a line exists within a given patch/ROI—useful for filtering candidate regions before applying traditional detection.
2. **CNN Regression Model (NB-2):** Predicts line endpoints directly from the image, offering robust localization even in low-contrast or noisy scenarios.
3. **Denoising Autoencoder (NB-5):** Enhances faint or broken lines by cleaning up noise, thereby boosting the effectiveness of existing edge or Hough-based methods.
4. **Vision Transformer Model (NB-4):** Designed for fine-grained classification tasks and can be adapted for future objectives like surface defect or anomaly detection.

These methods are independently deployable and can be integrated selectively depending on the specific inspection challenge or ROI structure. I'd be happy to align with your ROI-driven strategy and customize accordingly.

Regarding the stipend, I agree with the proposed Rs. 200/hour base rate and performance-based bonus, and I'm enthusiastic about the opportunity to collaborate further on defect detection and shape anomaly tasks in the future.

I have attached screenshots of the code files I executed today. I have the names of the notebooks (NB) with NB-1, NB-2,NB-4 and NB-5 respectively.

NB-1
NB-2

NB-4
NB-5

Please let me know if you'd like me to share these outputs or run a quick live demonstration.

Warm regards,
Hindvee

On Fri, May 9, 2025 at 6:03 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> I think there is a mismatch in understanding. We are already using conventional line detection techniques (various edge detection operators and Hough Transform and other methods for line fitment) .
>
> What we are looking to get from you is AI based / AI assisted line detection. Since we know the probable position of the line beforehand we can use this information to find a line. For example, say we know that the line end points are expected to be (x1, y1) and (x2, y2), we can construct a small rectangle (ROI) using these points and search for a line within this ROI.
>
> As for the stipend it depends both on the time you spend on the project and the results you produce. It will comprise of an hourly rate and a result linked amount. I suggest we have an hourly rate of Rs. 200/hour and a matching amount linked to the results at the end of the project. So, if you are able to produce satisfactory results it will come to Rs. 400/hour. Also, if the results are satisfactory, we can take this forward next year and work on other issues like surface defect detection, detection of shape anomalies etc.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>

Sent: Thursday, May 8, 2025 5:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr Durve,
I have the problem now so I will focus on faint line detection for the first step. I can proceed using Sobel, Scharr, and Canny edge detectors provided by OpenCV. First I will preprocess the images and then move to edge detection using the above 3 detectors. I will carry out line detection using Hough Transform, and further we will filter out short or irrelevant lines based on length and their angle of orientation. Finally, we will draw the detected lines on the images using cv.line()

I can devote 14 hrs per week to this task for 2 months, we can extend the time if required. Also, I wanted to ask what stipend you can provide me. I believe I can complete the task in the above time frame. Thank you for your time.

Warm regards
Hindvee

On Thu, May 8, 2025 at 7:18 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Your analysis seems good.
>
> At this point we are not interested in detecting defects (though at a later stage we may be), and want to detect the faint lines only.
>
> Note that though I have annotated only a single line in the image I sent to you yesterday, we need to detect edge lines of all vertical stripes.
>
> I have attached 2 sets of 20 images each named 'Good Images' and 'Bad Images' - images in which it is mostly easy to detect lines by the conventional image processing method and those in which lines are faint and hard to detect. These images may help you get a better

understanding of the problem.

Please also check the following issues:

1. How much time will you be able to devote to this task?
2. Will you be able to complete the task within the available time?

There are other issues like time taken by the AI route, integration of AI line detect code in VC++ project, further training / fine tuning of model, applicability of the method to detect faint circles etc. But we will get into them later once the basic method is proven.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, May 7, 2025 3:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings, Mr. Durve,
Thank you for reviewing my proposal and for sharing the sample images. By looking at the images I came across the specific challenges faced by the inspection system, like detecting faint lines, subtle surface variations, and fine defects that can be missed by traditional image processing. Based on these observations I suggest that we can do the following:-

1. **Hybrid approach**: Objective - Combine the speed of OpenCV line detection (example Hough Transform) with deep learning's pattern recognition capability.
Approach
- Use OpenCV to pre-locate candidate regions and feed them into a deep model for verification.
- This can speed up inference and increase resistance to false positives.

2. **Deep learning-based fine defect detection**: Objective: Improve the detection of faint, low

contrast line defects and localized surface anomalies.
Approach
- Train a model, such as YOLOv8 or U-Net, tailored to your defect patterns using the annotated images.
- To improve feature visibility, add contrast enhancement.
- Assess using detection accuracy and intersection over union (IoU) to compare to baselines based on OpenCV.

I can begin with a proof-of-concept model using the images shared by you. Thank you for sharing the opportunity. I am excited to get a chance to contribute to the advancement in inspection technology.

Warm regards
Hindvee Vispute

On Wed, 7 May 2025, 09:31 Kishor Durve, <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have gone thru the document uploaded by you.
>
> It looks ok to me.
>
> I can have attached a few images so you will get a better idea of the requirements.
>
> Kishor Durve

Hindvee Vispute <hindveevispute@gmail.com> Tue, Sep 16, 2025 at 5:50 AM
To: Kishor Durve <kishor.durve@outlook.com>

Hello sir,

I'd like to update you on the progress in our image correction and edge detection pipeline:

1. Corrected Image Results:

"Corrected" images refer to those processed through our denoising and enhancement pipeline (such as DnCNN or autoencoder-based methods).

Compared to the original PCBs/X-rays, these images have reduced noise and enhanced contrast, making edges and structural features more discernible. This prepares them better for accurate edge detection.

2. Quantitative Analysis – Bad Images:

Recent batch results for "bad images" yielded MSE values between 76.51 and 91.94, PSNR between 28.50 and 29.29 dB, and Laplacian Variance (sharpness) mostly in the 30-40 range.

Images with PSNR over 30 dB and higher sharpness are preferable for robust edge detection; this gives us a benchmark for further improvements.

3. DexiNed Processing Speed:

For our dataset, the total processing time was 92.24 seconds.

Average time per image: 4.61 seconds (measured using a high-precision timer).

4. Fine-tuning DexiNed:

DexiNed can be fine-tuned on our images if we have accurate edge mask labels for our PCBs/X-rays.

Fine-tuning will further adapt edge detection to our specific data, likely improving accuracy and reliability.

Please let me know if you would like additional details, I apologise for the delay sir.

Best regards,
Hindvee

On Thu, Aug 21, 2025 at 5:31 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Sorry! I didn't realize you were waiting for my feedback.
>
> I have checked the images. I have a few questions:
>
> 1. What are the 'corrected' images? How are they different from the original?
> 2. How long does it take (in seconds) to process an image with DexiNed? Please use a high res timer and check the time taken. I am aware that it will depend on the image size. Let us consider only the images that you are working with.
> 3. Is there any way to fine tune DexiNed for our images?
>
> Kishor Durve
> Duravision Systems
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Thursday, August 21, 2025 1:04 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir!
> Could you pls share your reviews on the link I shared earlier with you? Your views will determine how to proceed further.
>
> Warm Regards
> Hindvee
>
> On Fri, Aug 8, 2025 at 2:36 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir! I completely understand your point. I have received the stipend with thanks.
> > Sir, I had shared a drive link with you in the previous email. Could you pls go through those pictures and share your reviews with me.
> > Warm regards
> > Hindvee
> >
> > On Wed, Aug 6, 2025 at 8:10 AM Kishor Durve <kishor.durve@outlook.com> wrote:

I have transferred Rs. 11,600 (200 x 58) to you. Regarding the suggested increase in per-hour rate: You have certainly made efforts to solve the problem. However, as of now the result is not yet usable to us. We will consider an increase when it is usable in our software.

Kishor Durve
Duravision Systems
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, August 5, 2025 8:41 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir
Just following up on my previous email regarding the processed images and the stipend for July (58 hours). I have shared a small suggestion regarding the per-hour rate.

Please let me know if you need any further details or modifications from my side.
UPDATED DEXINED OUTPUT
Looking forward to your feedback.

Warm regards,
Hindvee

On Fri, Aug 1, 2025 at 2:14 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

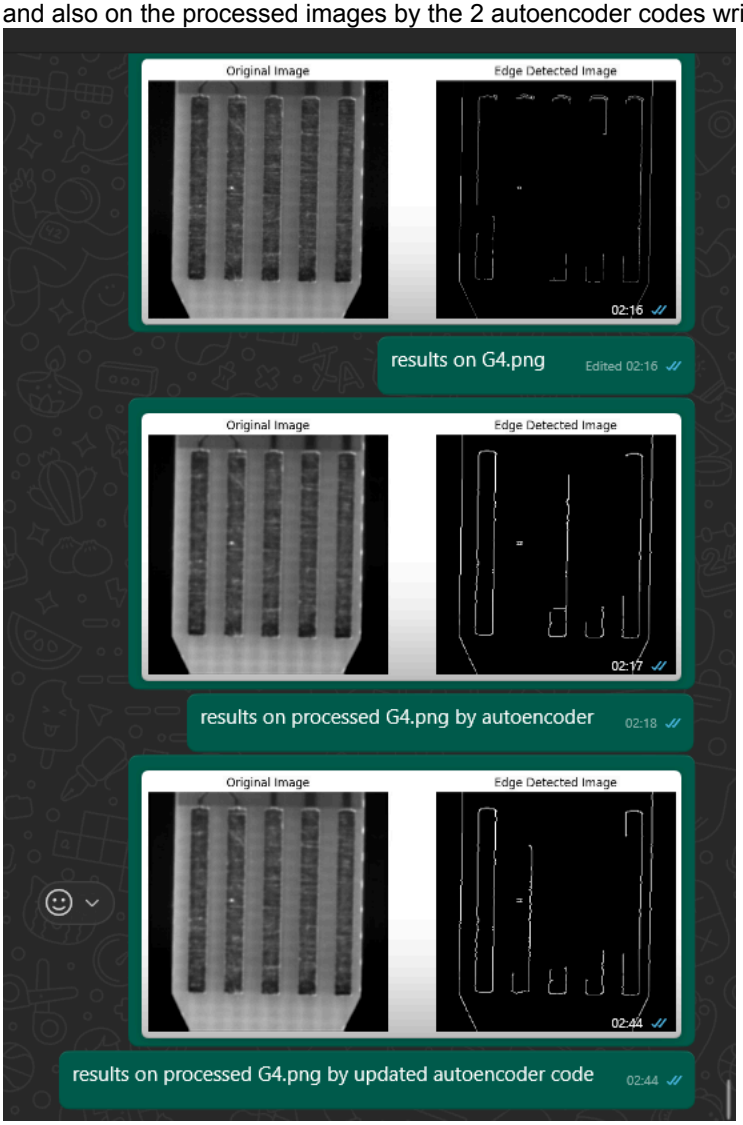
> Thank you sir! I have worked out and increased the contrast in the processed images. I will share the file with you.
> Also sir I shall request you to please process the stipend for this month. It has been 58 hrs for the month of July. If you like my work please consider increasing the per hour rate. It is just a small suggestion. The final decision depends on you and I happily accept your decision. My primary goal is to enhance my skills, and I am learning a lot from this project.
>
> Warm regards
> Hindvee
>
> On Wed, Jul 30, 2025 at 11:02 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > I have checked the images. Your conclusions are correct.
> >
> > Please carry on with your efforts.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Tuesday, July 29, 2025 3:37 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Hello sir!
> > current update:-
> >
> > - Using an autoencoder trained on clean (good) images to repair bad images
> > - Applying a **hybrid loss** of MSE and SSIM to preserve structure and perceptual quality.
> > - Saving and visualizing both original and reconstructed images which helps in debugging
> >
> > Despite this denoising strategy, DexiNED is still failing for bad images, the possible reasons are as follows
> >
> > 1. Contrast remains too low after reconstruction.
> > 2. Edges are too smooth or faint due to over-smoothing by the autoencoder.
> > 3. DexiNED is not tuned for synthetic/industrial patterns (e.g., PCB, x-ray like structures).
> >
> > I will now work on these factors to improve the quality of processed images. I am attaching the processed Good and Bad Images folders.

BAD IMAGES _ PROCESSED
GOOD IMAGES_ PROCESSED

warm regards
Hindvee

On Mon, Jul 28, 2025 at 2:48 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Noted sir. I wrote code for autoencoder to get sharper images, and I applied DexiNed on the original image and also on the processed images by the 2 autoencoder codes written so far. I am sharing the results here.



warm regards
Hindvee

On Fri, Jul 25, 2025 at 6:34 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Sorry for delay in replying. I was away for quite some time.
>
> A couple of months back we had tried OpenCV / HED and obtained similar results.
>
> Please go ahead and try DexiNed after which we will decide about line detection.
>
> After that another important step would be to remove noise from image using a suitable model.
>
> I know this is somewhat away from our goal of 'AI assisted edge detection' but it can be an important intermediate step. We are currently using a model that does this. This is the model I had sent to you initially. It is developed in a Chinese University by a professor who works with our company.
>
> Kishor Durve
> Duravision Systems

9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, July 25, 2025 5:46 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir ! This is an output image obtained from the HED model and similar image I have obtained from DexiNed, could u pls tell i am moving in the right direction or should I make improvements or change in the plan of action?

warm regards
Hindvee

On Wed, Jul 23, 2025 at 3:20 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir! Could you please share ur insights and views on the above mail thread, it shall be very helpful to get to know your feedback and move forward with the project.
>
> Warm Regards
> Hindvee
>
> On Fri, Jul 11, 2025 at 3:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir!
> >
> > Following up on my previous email regarding edge detection, I wanted to share the progress I've made since then. I have been focusing on using the Holistically-Nested Edge Detection (HED) method. I have now:
> >
> > CODE FILE
> >
> > 1. Successfully applied the HED process to a folder containing our images to generate edge maps for the entire dataset.
> >
> > 2. Trained a custom autoencoder model designed to refine and correct the lines in these images.
> >
> > 3. Applied the trained autoencoder to the HED output images to produce corrected versions.
> >
> > 4. Started working on post-processing techniques.
> >
> > I am currently experimenting with different parameters and methods to achieve the best results for this step. I am continuing to refine the line clarification process and will share updated results once I have finalized the post-processing approach.
> >
> > warm regards
> >
> > Hindvee
> >
> > On Wed, Jul 9, 2025 at 12:23 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> >
> > > Hello sir! I am trying the pre-trained model for detecting edges in our images; I have tried Holistically-Nested Edge Detection (HED) and DexiNed. I am attaching the output, I am going forward with HED to get better results.
> > >
> > > Warm regards
> > > Hindvee
> > >
> > > On Mon, Jul 7, 2025 at 10:37 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > >
> > > > Hello sir!
> > > > I am sharing the first 5 difference images of the three codes that I mentioned previously. Please check this whenever you have time.
> > > >
> > > > Warm regards
> > > > Hindvee
> > > >
> > > > On Mon, Jul 7, 2025 at 6:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> > > >
> > > > > I am unable to give my view of the approach as I can't view the images. I can give my opinion if you send me difference images. In any case, I think you need to have difference images as a metric in addition to the Laplacian.
> > > > >
> > > > > Get Outlook for Android

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Monday, July 7, 2025 7:09:51 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!
I have received the stipend, thank you for that! I'd like to share a brief update on the image enhancement work, focusing on noise removal and edge preservation.
Summary of results:-
1. Previous code
---Performed basic denoising but introduced over-smoothing.
---Laplacian MAD: ~0.0175–0.0192

2. DnCNN
---Improved noise removal and structural retention.
---MAD: ~0.0186–0.0202

3. DnCNN + Sharpening (Current approach)
---Added post-processing to enhance edge clarity.
---Consistently improved MAD to ~0.0178–0.0196.
---Visual results show reduced whitening and better edge definition.

This confirms that the current setup provides better denoising while preserving fine image details. Please let me know if you'd like to proceed with further tuning using SSIM or edge-aware loss functions.

Warm regards

Hindvee

On Fri, Jul 4, 2025 at 3:54 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Please keep at it. Use difference image and Laplacian to gauge the effect of the algorithm. I will be able to look at your work only after 15th.
> I have transferred your stipend by GPay.
>
> Get Outlook for Android
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Friday, July 4, 2025 5:37:35 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Hello sir!
> This is what I am been working on. I am still working on getting better edges and less whitish images.
>
> warm regards
> Hindvee
>
> On Thu, Jul 3, 2025 at 10:53 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > UPI ID: hindveevispute@okicici
> >
> > Warm regards
> > Hindvee
> >
> > On Wed, 2 Jul 2025, 03:16 Kishor Durve, <kishor.durve@outlook.com> wrote:
> >
> > > I checked the images, but did not notice substantial difference between the before and after images. You can subtract them (after aligning) and check.
> > > As regards payment of stipend please let me know how much we need to pay and I will transfer it on GPay.
> > > I am travelling again and am out of office. I will be back only on 15th. My responses to your mails may be delayed.
> > >
> > > Get Outlook for Android
> > >
> > > ---

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, July 1, 2025 1:51:45 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir!

I was eager to get insights from you about the above shared images.Please do have a look as soon as possible. Also, I wanted to know whether the stipend will be credited at the end of the month or the project's end. It would be very kind of you if I could get the stipend at the end of the month.

Thanks and regards
Hindvee

On Mon, Jun 30, 2025 at 10:33 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Good morning sir!
>
> Did you get any chance to review the above images? Please share your suggestions and insights with me.
>
> Warm regards
> Hindvee
>
> On Thu, Jun 26, 2025 at 11:09 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Here I am attaching the processed images.
> >
> > Warm regards
> > Hindvee
> >
> > On Thu, Jun 26, 2025 at 8:08 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > I am unable to run the files in Colab and generate images. Can you please send me the processed images on mail - .png / .jpeg files, not .ipynb files.
> > >
> > > Kishor Durve
> > >
> > > ---
> > >
> > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > **Sent:** Thursday, June 26, 2025 4:48 AM
> > > **To:** Kishor Durve <kishor.durve@outlook.com>
> > > **Subject:** Re: AI assisted edge detection
> > >
> > > Hello sir, I have attached the results for png 1,2,3 and 4. Pls share ur views and insights.
> > >
> > > Warm Regards
> > > Hindvee
> > >
> > > On Tue, Jun 24, 2025 at 11:29 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
> > >
> > > > Hello sir,
> > > > I am attaching the .ipynb file, it has the images
> > > >
> > > > warm regards
> > > > Hindvee
> > > >
> > > > On Tue, Jun 24, 2025 at 5:48 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> > > >
> > > > > Please share the images for visual comparison.
> > > > >
> > > > > Kishor Durve
> > > > >
> > > > > ---
> > > > >
> > > > > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > > > > **Sent:** Tuesday, June 24, 2025 5:13 AM

To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello Durve sir,

I wanted to share an update on the denoising output. I noticed earlier that while the model was doing a good job removing noise, it was also making the image a bit too smooth — the sharpness had dropped quite a bit:

- Noisy Image: 33.72
- Denoised Image: 6.92
- Clean Image (Ground Truth): 26.49

To improve this, I added a sharpening filter after the autoencoder output and tested two variants:

1. A **strong sharpening kernel**, which raised the denoised sharpness to **112.40,** but it felt a bit overdone and introduced some artifacts.
2. A **softer kernel**, which gave a cleaner and more natural result, with denoised sharpness at **54.04**. This seems like a balanced point where edges are clearer without compromising visual quality.

If you'd like, I can also share the output images (clean vs. denoised) for a visual comparison.

Warm regards

Hindvee

 image.png

On Fri, Jun 20, 2025 at 12:29 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Hello sir!
> Based on the latest code I calculated the sharpness using Laplacian variance. I obtained the following results as those attached as image.
> I think that while denoising, though the noise was removed, but the model blurred or over smoothened the image. I may have lost some meaningful edge details in the process. But the sharpness value of Clean image is lower than the noisy image (because it has no random noise), but high enough to retain clear edges and structure. So I am working next on increasing the sharpness value while keeping the noise low. Please do share your views with me.
>
> Warm regards
> Hindvee
>
>  image.png
>
> On Thu, 19 Jun 2025, 13:42 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > I have seen the processed images. I think it is good at this point to quantify 'sharpness' of an image. It will give us a measure to compare different methods / models.
> >
> > There can be several approaches to do this. We use Opencv::Laplacian() to judge image sharpness. But you can try other approaches.
> >
> > Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, June 19, 2025 2:52 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir,

I've completed the full-image denoising as discussed for the images `5.png` and `6.png`. This time, the model was trained and applied directly on the complete images (cropped to a common size of **570×588**) instead of small patches, as per your suggestion. The denoising autoencoder was able to reduce background noise and enhance structural features, particularly the edges of the circles, which were initially faint or inconsistent. The model output shows a noticeable improvement in clarity while preserving the original patterns.

I've attached the processed result image along with this mail. Please let me know your thoughts on this file.

Warm regards,
Hindvee

On Wed, Jun 18, 2025 at 12:33 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have checked the file. These two images contain circles as primary features. Also, the patches in the file seem to be for a small part of the original image. If you have processed the original images (5.png and 6.png) please send me the processed images on mail. If you have processed only parts of these images, I suggest you process the full images and send me the results.
>
> Similarly, in the previous exercise please process complete images. Most of the set are of size 500x500 pixel, so it should not consume too much time.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 18, 2025 1:44 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Sir, please let me know your views on the above .ipynb file.
> Warm Regards
> Hindvee
>
> On Wed, Jun 18, 2025 at 1:56 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Hello sir,
> > I have processed the 5.png and 6.png, I faced lower accuracy rate earlier. After fixing the earlier issues with mismatched data sizes and high Binary Crossentropy loss (~0.6098), the model is now working smoothly. It reached a much lower loss of just 0.0528 in 5 epochs, and both training and validation losses settled nicely by the 2nd or 3rd epoch without signs of overfitting. The denoised images look very clean and closely match the original ones, even when the input images had different levels of noise. I am attaching the results in the attached file.
> > Warm regards
> > Hindvee
> >
> > On Tue, Jun 17, 2025 at 7:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

Sorry! Forgot the attachment.

Kishor Durve

From: Kishor Durve <kishor.durve@outlook.com>
Sent: Tuesday, June 17, 2025 9:46 AM
To: Hindvee Vispute <hindveevispute@gmail.com>
Subject: Re: AI assisted edge detection

Good to know that you are making progress at a nice clip.

I saw the images and the difference between before / after can be seen. Can you please process the images attached to this mail and send me the results? The images are similar to those I had sent earlier, the exception being 5.png and 6.png. In these last two images the challenge is to detect circles whose edges are not well defined and the edge quality may vary from image to image. This is to test the effectiveness of noise removal / sharpening.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Tuesday, June 17, 2025 3:06 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Hello sir,
I have completed the training and evaluation of the denoising autoencoder using grayscale image patches taken from the `IMG` folder. The model was trained for 20 epochs on 64×64 patches with binary cross entropy loss and a 10% validation split. During training, the loss reduced steadily and stabilized, with the final reconstruction loss reaching 0.3688. The results show that the model can effectively denoise images and recover clean patterns like faint lines. In the next step, I plan to use the output of the Super-Resolution model as a base to evaluate the accuracy and quality of the denoising process, to better understand how both methods contribute to image enhancement. I am attaching the .ipynb file for your reference and also the IMG Folder which I just mentioned.

Warm regards
Hindvee

IMG FOLDER

Accuracy results:-

 image.png

On Fri, Jun 13, 2025 at 5:33 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> The images I sent are big. OpenCV will not be able to handle them as the max size allowed is 32K x 32K. The crash you are facing could be due to this reason.
>
> You can crop out small patches from the big images and use them for trials.

Kishor Durve
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, June 11, 2025 1:37 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Thank you for sharing more images. I have updated the work on the sheets.

Warm regards
Hindvee

On Sat, Jun 7, 2025 at 2:51 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> I have uploaded more images. You can crop out patches from them. One of the images has many boards which will give many useful patches.
>
> Kishor Durve
> Duravision Systems
> 9823164286
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Saturday, June 7, 2025 4:36 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Greetings sir,
> It would be great if I get more images, that would expand my dataset. Thank you for your previous email.
> I also wanted tk inform you that I am travelling and will be able to resume work by Tuesday. I request you to kindly allow my leave.
>
> Warm Regards
> Hindvee
>
> On Fri, 6 Jun 2025, 11:31 Kishor Durve, <kishor.durve@outlook.com> wrote:
>
> > I am not going to be of much help to you as I have limited experience in this field. Also, what little I know was gained by applying available models in OpenCV's DNN module.
> >
> > There are two images available in the images folder. You may crop out patches from them for the Training / test sets. We use ImageJ and IrfanView for such work as MS Paint cannot handle such big images. If you need more images please let me know.
> >
> > Kishor Durve
> > Duravision Systems
> > 9823164286
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, June 6, 2025 5:24 AM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection

Greetings sir,
I hope you're doing well. I've just finished training the CNN-based line presence classifier, and I'm attaching the loss graph for your review. The model seems to have learned quite quickly, with both training and validation losses staying very close to zero throughout. There was a small spike in training loss around epoch 80, but validation stayed stable, so it might just be a minor fluctuation.

Since the dataset is quite small and the model converged very early, I'm unsure how best to move forward.According to you should I try to get more data to avoid overfitting. Please guide me towards a direction in which I can make this model more effective.

Warm regards

Hindvee

On Wed, Jun 4, 2025 at 6:42 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Yes, it looks ok to me. We will take a relook once you make more progress.
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Wednesday, June 4, 2025 4:26 AM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Additionally, Sir, I would like to confirm if the current format of the Google Sheet meets your expectations. Please let me know if you would prefer any modifications, such as a more detailed breakdown of the work completed.
>
> Warm Regards
> Hindvee
>
> On Wed, Jun 4, 2025 at 1:34 AM Hindvee Vispute <hindveevispute@gmail.com> wrote:
>
> > Greetings Mr. Durve,
> >
> > Thanks for sharing the super-resolution model. I've set it up and tested it, the results inspiring. It does a good job cleaning up noise and improving contrast, especially in the faint line areas. As we discussed, I'll now add this to the bigger pipeline along with the AI modules (NB-1 to NB-5). The super-resolution output can help improve both the CNN classifier and regression models. I'll also check how it works together with NB-5, the denoising autoencoder, to see if we get even better results.
> >
> > Please let me know if you want me to run it on any specific ROI samples or start putting together comparison results.
> >
> > Warm regards
> > Hindvee
> >
> > On Tue, Jun 3, 2025 at 9:23 AM Kishor Durve <kishor.durve@outlook.com> wrote:
> >
> > > I have file network.py in folder 'models'. This file has the RRDBNet implementaion.

Kishor Durve
Duravision Systems
9823164286

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Monday, June 2, 2025 7:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr. Durve,
I am facing an issue again an again while running the VS 2022 code, pls help me with this, we have the module named as model_zoo not models. Pls tell me how to proceed with this.

 image.png

Warm Regards
Hindvee

On Sat, May 31, 2025 at 12:46 PM Hindvee Vispute <hindveevispute@gmail.com> wrote:

> Thankyou sir! I will try this right away.
> Also I hope you received this google sheet, I have shared edit access to you on your gmail account
>
> Hindvee Vispute - Time data - Intern
>
> Regards
> Hindvee
>
> On Sat, May 31, 2025 at 5:43 AM Kishor Durve <kishor.durve@outlook.com> wrote:
>
> > Sure. You need to do the following to make the project work.
> >
> > 1. Download and install VS 2022 Community edition
> > 2. Download and install Anaconda. Install an additional environment Python 3.8.16. Use this version of Python for the project.
> >
> > The project should run at this point.
> >
> > Kishor Durve
> >
> > ---
> >
> > **From:** Hindvee Vispute <hindveevispute@gmail.com>
> > **Sent:** Friday, May 30, 2025 6:34 PM
> > **To:** Kishor Durve <kishor.durve@outlook.com>
> > **Subject:** Re: AI assisted edge detection
> >
> > Greetings sir!
> > I went through the google drive folder shared by you. It would be really helpful if you could walk me through it.
> > Best regards
> > Hindvee
> >
> > On Fri, May 30, 2025 at 11:41 AM Kishor Durve <kishor.durve@outlook.com> wrote:

You can download the model, a working VC++ (VS 2022) project and image here.

https://drive.google.com/drive/folders/1mLBDjJR3fFKs8Brr_nNj2z9YyI89nruE?usp=sharing

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Friday, May 30, 2025 12:50 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings sir!
Pls do share the model over drive I will try it out. Also I request you to share more image database so that I can start working on it.
I will start work from today itself then.

Regards
Hindvee

On Fri, 30 May 2025, 06:11 Kishor Durve, <kishor.durve@outlook.com> wrote:

> We have found that using a super resolution Pytorch model to get a larger image results in significant noise removal. Perhaps it is a by-product.
>
> You can try it out. I have the model and can share it on Google Drive if you want to try it out. We have tried the models available in OpenCV DNN but they do not seem to give the same results.
>
> Regards,
>
> Kishor Durve
>
> ---
>
> **From:** Hindvee Vispute <hindveevispute@gmail.com>
> **Sent:** Thursday, May 29, 2025 5:28 PM
> **To:** Kishor Durve <kishor.durve@outlook.com>
> **Subject:** Re: AI assisted edge detection
>
> Dear Sir,
>
> Thank you for the confirmation. I will maintain the time data on a Google Sheet and ensure it is updated regularly.
>
> Looking forward to starting from Monday, 2nd June.
>
> Best regards,
> Hindvee

On Thu, May 29, 2025 at 2:53 PM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> Welcome aboard!
>
> You can keep the time data on Google Sheet.
>
> Regards,
>
> Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 29, 2025 5:12 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr. Durve,

Thank you very much for giving me the opportunity to contribute to Guangdong Longtian Smart Instruments Co. Ltd. through this exciting project. I am pleased to accept the offer and look forward to working on the development of the AI-assisted line detection tool.

With your permission, I would like to begin the internship from the coming Monday, i.e., 2nd June 2025.

Additionally, I wanted to ask about the preferred method for tracking and sharing the time I spend on the project. Would you like me to maintain a shared timesheet (e.g., a Google Sheet or Excel file), or would you prefer weekly time summaries via email or any other format?

Looking forward to your guidance on this.

Thank you once again for this opportunity.

Warm regards,
Hindvee Rajesh Vispute

On Thu, May 29, 2025 at 8:21 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have attached the on-boarding letter.

Please let me know if you have any queries.

Regards,

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Saturday, May 10, 2025 3:46 AM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr. Durve,

Thank you for your detailed response and clarification. I now clearly understand that the goal is to enhance line detection using AI within known ROIs, leveraging the expected position of endpoints.

To support this objective, I've explored and implemented several **AI-based approaches**, any of which can be used to complement or improve upon conventional methods:

1. **CNN-Based Line Presence Classifier (NB-1):** Identifies whether a line exists within a given patch/ROI—useful for filtering candidate regions before applying traditional detection.
2. **CNN Regression Model (NB-2):** Predicts line endpoints directly from the image, offering robust localization even in low-contrast or noisy scenarios.
3. **Denoising Autoencoder (NB-5):** Enhances faint or broken lines by cleaning up noise, thereby boosting the effectiveness of existing edge or Hough-based methods.
4. **Vision Transformer Model (NB-4):** Designed for fine-grained classification tasks and can be adapted for future objectives like surface defect or anomaly detection.

These methods are independently deployable and can be integrated selectively depending on the specific inspection challenge or ROI structure. I'd be happy to align with your ROI-driven strategy and customize accordingly.

Regarding the stipend, I agree with the proposed Rs. 200/hour base rate and performance-based bonus, and

I'm enthusiastic about the opportunity to collaborate further on defect detection and shape anomaly tasks in the future.

I have attached screenshots of the code files I executed today. I have the names of the notebooks (NB) with NB-1, NB-2,NB-4 and NB-5 respectively.

NB-1
NB-2
NB-4
NB-5

Please let me know if you'd like me to share these outputs or run a quick live demonstration.

Warm regards,
Hindvee

On Fri, May 9, 2025 at 6:03 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> I think there is a mismatch in understanding. We are already using conventional line detection techniques (various edge detection operators and Hough Transform and other methods for line fitment) .
>
> What we are looking to get from you is AI based / AI assisted line detection. Since we know the probable position of the line beforehand we can use this information to find a line. For example, say we know that the line end points are expected to be (x1, y1) and (x2, y2), we can construct a small rectangle (ROI) using these points and search for a line within this ROI.
>
> As for the stipend it depends both on the time you spend on the project and the results you produce. It will comprise of an hourly rate and a result linked amount. I suggest we have an hourly rate of Rs. 200/hour and a matching amount linked to the results at the end of the project. So, if you are able to produce satisfactory results it will come to Rs. 400/hour. Also, if the results are satisfactory,

we can take this forward next year and work on other issues like surface defect detection, detection of shape anomalies etc.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Thursday, May 8, 2025 5:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings Mr Durve,
I have the problem now so I will focus on faint line detection for the first step. I can proceed using Sobel, Scharr, and Canny edge detectors provided by OpenCV. First I will preprocess the images and then move to edge detection using the above 3 detectors. I will carry out line detection using Hough Transform, and further we will filter out short or irrelevant lines based on length and their angle of orientation. Finally, we will draw the detected lines on the images using cv.line()

I can devote 14 hrs per week to this task for 2 months, we can extend the time if required. Also, I wanted to ask what stipend you can provide me. I believe I can complete the task in the above time frame. Thank you for your time.

Warm regards
Hindvee

On Thu, May 8, 2025 at 7:18 AM Kishor Durve <kishor.durve@outlook.com> wrote:

> Your analysis seems good.
>
> At this point we are not interested in detecting defects (though at a later stage we may be), and want to detect the faint lines only.
>
> Note that though I have annotated only a single line in the image I sent to you yesterday, we need to detect edge lines of all vertical stripes.

I have attached 2 sets of 20 images each named 'Good Images' and 'Bad Images' - images in which it is mostly easy to detect lines by the conventional image processing method and those in which lines are faint and hard to detect. These images may help you get a better understanding of the problem.

Please also check the following issues:

1. How much time will you be able to devote to this task?
2. Will you be able to complete the task within the available time?

There are other issues like time taken by the AI route, integration of AI line detect code in VC++ project, further training / fine tuning of model, applicability of the method to detect faint circles etc. But we will get into them later once the basic method is proven.

Kishor Durve

From: Hindvee Vispute <hindveevispute@gmail.com>
Sent: Wednesday, May 7, 2025 3:15 PM
To: Kishor Durve <kishor.durve@outlook.com>
Subject: Re: AI assisted edge detection

Greetings, Mr. Durve,
Thank you for reviewing my proposal and for sharing the sample images. By looking at the images I came across the specific challenges faced by the inspection system, like detecting faint lines, subtle surface variations, and fine defects that can be missed by traditional image processing. Based on these observations I suggest that we can do the following:-

1. **Hybrid approach**: Objective - Combine the speed of OpenCV line detection (example Hough Transform) with deep

learning's pattern recognition capability.
Approach
- Use OpenCV to pre-locate candidate regions and feed them into a deep model for verification.
- This can speed up inference and increase resistance to false positives.

2. **Deep learning-based fine defect detection**: Objective: Improve the detection of faint, low contrast line defects and localized surface anomalies.
Approach
- Train a model, such as YOLOv8 or U-Net, tailored to your defect patterns using the annotated images.
- To improve feature visibility, add contrast enhancement.
- Assess using detection accuracy and intersection over union (IoU) to compare to baselines based on OpenCV.

I can begin with a proof-of-concept model using the images shared by you. Thank you for sharing the opportunity. I am excited to get a chance to contribute to the advancement in inspection technology.

Warm regards
Hindvee Vispute

On Wed, 7 May 2025, 09:31 Kishor Durve, <kishor.durve@outlook.com> wrote:

> Hello Hindvee,
>
> I have gone thru the document uploaded by you.
>
> It looks ok to me.
>
> I can have attached a few images so you will get a better idea of the requirements.
>
> Kishor Durve